

The Sextant Navigator's Almanac 2026

& Sight-Reduction Handbook

A self-contained nautical almanac and paper sight-reduction manual,
computed entirely from a validated, open-source celestial-navigation engine.

Greenwich Hour Angle and Declination of the Sun,
GHA of Aries, and the 58 navigational stars, for the year 2026 (UT).

Generated from SEXTANTNAVIGATION
<https://github.com/mthiel74/SextantNavigation>

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FOR EDUCATION AND PRACTICE ONLY — NOT FOR SOLE-MEANS NAVIGATION

IMPORTANT — READ BEFORE USE

This almanac and handbook are an **educational deliverable**. Every number in it was computed offline from the open-source ephemeris engine at <https://github.com/mthiel74/SextantNavigation>, not copied from an official publication.

Accuracy. The underlying engine agrees with JPL for the Sun's geographic position to about $0.14'$, reproduces Meeus' lunar worked example to $0.002'$ (geocentric), and precesses the star catalogue with IAU–1976 precession plus proper motion. Realistic end-to-end accuracy for a reduced sight is on the order of **one arcminute (about one nautical mile)**. That is good enough to *learn and practise* celestial navigation, and to check your paper arithmetic — but it is **not** certified for life-safety use.

Do not use this document as your sole means of navigation at sea. For real voyages, use an official *Nautical Almanac* (HM Nautical Almanac Office / US Naval Observatory) and cross-check every figure. The authors accept no liability for any use of this material.

Sources & method. The positions in this book are *computed with the Wolfram Language*, not reproduced from any published almanac. The Sun's geographic position is from the low-precision solar formulae of the *Astronomical Almanac*; the Moon from Meeus's abridged ELP theory (*Astronomical Algorithms*, ch. 47); Greenwich sidereal time from the IAU GMST formula; and the star places are propagated from *Hipparcos* catalogue (ESA) J2000 mean positions using IAU–1976 precession and catalogue proper motions to epoch 2026.5. Full source code and tests: <https://github.com/mthiel74/SextantNavigation>.

Acknowledgements. All computations and tables were produced with the *Wolfram Language* (Wolfram Research). This almanac gratefully builds on published astronomy: the star catalogue derives from the ESA *Hipparcos* mission; the lunar theory follows Jean Meeus, *Astronomical Algorithms*; and the solar and sidereal-time formulae follow the conventions of the IAU and the *Astronomical Almanac* (HM Nautical Almanac Office / US Naval Observatory). It is an independent, open-source companion to the article cited above, and is not affiliated with, or endorsed by, any of these organisations.

How to use this almanac

All times in the data tables are **Universal Time (UT)**, the time kept by your chronometer set to Greenwich. All angles are given in degrees and decimal minutes to $0.1'$, the same precision as a commercial nautical almanac.

To reduce a sight you need four numbers for the instant of your observation: the body's *Greenwich Hour Angle* (GHA) and *Declination* (Dec), your *assumed position*, and your *observed altitude* H_o . This book supplies the first two:

- **Sun.** Turn to Part II, find the month, then the row for your day and the whole hour of UT. Read GHA and Dec directly. For the odd minutes and seconds, add the increment from Part IV (the Sun's GHA grows a steady 15° per hour). Declination changes so slowly you may interpolate it by eye.
- **A star.** Turn to Part III. Read **GHA Aries** for your day and hour, then add the star's **SHA** (Sidereal Hour Angle) from the star table: $\text{GHA}_* = \text{GHA Aries} + \text{SHA}$. The star's declination is taken straight from the table. Add the Part IV increment for minutes and seconds (Aries runs at 15.041° per hour).

Then work the triangle. Part I explains, step by step and with fully worked 2026 examples, how to correct your sextant reading and turn these numbers into a line of position on your chart.

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Part I

Finding Yourself with a Sextant

Chapter 1

The idea: circles of equal altitude

When you measure the altitude of a celestial body — the angle between it and the horizon — you learn one thing about where you are: your distance from the body’s *geographic position* (GP), the point on the Earth’s surface directly beneath it. If the Sun’s measured altitude is H_o , then your *zenith distance* is

$$z = 90^\circ - H_o,$$

and z , expressed in arcminutes, is exactly your great-circle distance from the GP in nautical miles (one minute of arc on the Earth = one nautical mile). You must therefore lie somewhere on a circle of radius z centred on the GP: a **circle of equal altitude**.

One sight gives a circle. A *second* sight of another body (or the same body later) gives a second circle, and two circles cross at two points. A *third* sight resolves which of the two intersections is really you. In practice we never draw these enormous circles; near your estimated position each circle is indistinguishable from its tangent straight line, the **line of position** (LOP), and we plot those instead. That is the method of this chapter.

1.1 What you need

- A **sextant** to measure altitude H_s above the sea horizon.
- An accurate **clock on UT** (a chronometer). An error of 4 seconds of time moves your fix about one nautical mile in longitude.
- This **almanac**, for GHA and Dec of the body.
- A **dead-reckoning (DR) position**: your best estimate of where you are, used to pick an “assumed position” close by.
- Plotting sheet, dividers, parallel rule, and a pocket calculator with trigonometric functions (or the concise sight-reduction tables).

Chapter 2

Correcting the sight

The raw sextant altitude H_s is not yet the true altitude of the body's centre. Apply, in order:

1. **Index error** I . Your own sextant's zero error, found by sighting the horizon. Subtract if "on the arc", add if "off the arc".
2. **Dip** of the horizon. Because your eye is above the sea, the visible horizon is depressed below the true horizontal by

$$\text{Dip} = 1.76\sqrt{h} \text{ arcminutes}, \quad h = \text{height of eye in metres.}$$

Always subtract. (See the table in Part IV.)

3. **Refraction** R . The atmosphere bends the light down, raising the apparent altitude; subtract R . This almanac uses Bennett's formula, $R = \cot(H_a + 7.31/(H_a + 4.4))$ arcminutes, evaluated at the apparent altitude H_a (altitude after index error and dip). Refraction is about $1'$ at 45° but $10'$ near the horizon.
4. **Semidiameter** (Sun and Moon). You bring the *limb* (edge), not the centre, to the horizon. Add the semidiameter (SD) for a lower-limb sight, subtract it for an upper-limb sight. The Sun's SD is about $16'$; the exact monthly value is tabulated in Part IV.
5. **Parallax** (Moon, and slightly the Sun). A near body is seen against the stars from a slightly different direction than it would be from the Earth's centre. The correction, always *added*, is

$$P = \arcsin(\sin(\text{HP}) \cos H) \approx \text{HP} \cos H,$$

where HP is the horizontal parallax. For the Moon HP is about $57'$, so the parallax reaches nearly a full degree; for the Sun it is a negligible $0.1'$. Stars are effectively infinitely far away: no parallax, no semidiameter.

The result is the **observed altitude**

$$H_o = H_s - I - \text{Dip} - R (\pm \text{SD}) (+P).$$

Chapter 3

Sight reduction: the Marcq St-Hilaire intercept

We cannot solve for position directly, so we *guess* an assumed position (AP) close to our DR and ask: *if I were at the AP, what altitude and bearing would the body have?* Comparing that computed altitude H_c with the observed H_o tells us how far, and in which direction, to shift from the AP onto our true LOP.

3.1 The navigational (PZX) triangle

On the celestial sphere, three points form a spherical triangle: the elevated pole P , your zenith Z , and the body X . Its sides are the co-latitude $90^\circ - \varphi$, the co-declination $90^\circ - \delta$, and the co-altitude $90^\circ - H_c$; the angle at the pole is the **local hour angle** $LHA = GHA - (\text{west longitude})$ (equivalently $GHA + \text{east longitude}$). Applying the spherical law of cosines to the side $90^\circ - H_c$:

$$\sin H_c = \sin \varphi \sin \delta + \cos \varphi \cos \delta \cos(LHA) \quad (3.1)$$

where φ is the assumed latitude and δ the declination. The body's **azimuth** Z_n (true bearing, measured clockwise from north) comes from

$$\cos Z = \frac{\sin \delta - \sin H_c \sin \varphi}{\cos H_c \cos \varphi}, \quad Z_n = \begin{cases} Z & \text{if } \sin(LHA) < 0 \text{ (LHA } 180^\circ\text{--}360^\circ), \\ 360^\circ - Z & \text{if } \sin(LHA) > 0 \text{ (LHA } 0^\circ\text{--}180^\circ). \end{cases} \quad (3.2)$$

3.2 The intercept, and how to plot it

The heart of the method is a single subtraction. Because one minute of altitude equals one nautical mile of distance from the GP,

$$a = 60 (H_o - H_c) \text{ nautical miles} \quad (3.3)$$

(with H_o, H_c in degrees). If $H_o > H_c$ you are *closer* to the body than the AP is, so step the intercept **Toward** the body along the azimuth Z_n ; if $H_o < H_c$, step it **Away** (along $Z_n + 180^\circ$). A useful mnemonic is “*Ho Mo To*” — H_o More means Toward.

To plot: mark the AP on your sheet. Draw the azimuth line through it in the direction Z_n . Measure a nautical miles along that line, Toward or Away. Through that point draw the LOP *perpendicular* to the azimuth — that perpendicular is your line of position. Repeat for a second body; where the two LOPs cross is your fix. A third LOP turns the small “cocked hat” triangle into a confident position.

3.3 Choosing the assumed position

To keep the arithmetic clean, navigators choose the AP cleverly: take the *assumed latitude* as the whole degree of latitude nearest your DR, and the *assumed longitude* with whatever minutes make the LHA come out to a whole number of degrees. Both examples below use this trick.

Chapter 4

Worked example 1: a Sun sight

At sea near 40°N , 30°W on 2026 June 29, you take a lower-limb Sun sight. Your chronometer reads exactly **15:00:00 UT**. Height of eye is 3.0 m and your sextant's index error is $2.0'$ on the arc. Work as follows (all almanac figures are read from Parts II and IV of this book).

Sight and corrections	
Body	Sun (lower limb)
UT	2026 June 29, 15:00:00
DR position	40°N , 30°W (approx.)
Sextant altitude H_s	$69^\circ 22.1'$
Index error (on the arc)	$-2.0'$
Dip (height of eye 3.0 m)	$-3.0'$
Apparent altitude H_a	$69^\circ 17.1'$
Refraction	$-0.4'$
Semidiameter (lower limb)	$+15.7'$
Observed altitude H_o	$69^\circ 32.4'$
From the almanac (15:00 UT, 29 June)	
GHA Sun	$44^\circ 06.9'$
Declination	$\text{N}23^\circ 12.0'$
Assumed longitude (W)	$30^\circ 06.9'$
Local hour angle LHA	$14^\circ 00.0'$
Assumed latitude	$\text{N}40^\circ 00.0'$
Sight reduction	
Computed altitude H_c	$69^\circ 27.4'$
Azimuth Z_n	$219^\circ 19.2'$
Intercept $a = 60(H_o - H_c)$	5.0 nm Toward

Reading the result. The observed altitude exceeds the computed altitude by $H_o - H_c = 5.0'$, so the intercept is **5.0 nautical miles Toward** the Sun. Plot the assumed position (40°N , $30^\circ 06.9'\text{W}$), draw the azimuth line toward $Z_n = 219^\circ 19.2'$, step 5.0 nm along it Toward the Sun, and draw your line of position at right angles to that azimuth. A single Sun sight gives one LOP; a second sight a few hours later (when the Sun's azimuth has swung round) crosses it to give a running fix.

Check it yourself: $\sin H_c = \sin 40^\circ \sin(23^\circ 12.0') + \cos 40^\circ \cos(23^\circ 12.0') \cos(14^\circ)$ gives $H_c = 69^\circ 27.4'$, exactly the value in the workform. The whole-degree LHA of $14^\circ 00.0'$ is what the assumed-longitude choice was designed to produce.

Chapter 5

Worked example 2: latitude by the noon Sun

The simplest sight of all needs no azimuth and almost no arithmetic. When the Sun crosses your meridian (*local apparent noon*, LAN) it bears due north or due south and reaches its greatest altitude. At that instant your latitude follows from the meridian altitude alone.

At 30°W the Sun transits your meridian when its GHA equals 30°; from Part II that occurs on 2026 June 29 at the UT shown below. Measure the Sun's maximum altitude, correct it to H_o as before, and take the zenith distance $z = 90^\circ - H_o$. Because the Sun is south of you (its declination 23°N is less than your latitude), your latitude is

$$\varphi = \delta + z \quad (\text{body to the south}).$$

Observer (near)	40°N, 30°W
Date	2026 June 29
Local apparent noon (UT)	14:03.5
Observed mer. altitude H_o	73°06.1'
Zenith distance $z = 90^\circ - H_o$	16°53.9'
Sun's declination (from almanac)	N23°12.1'
Sun bears	South
Latitude = Dec + z	N40°06.0'

The result, $\varphi = N40^\circ06.0'$, agrees with the true position to a tenth of a minute. (Had the Sun been *north* of you, the rule would be $\varphi = \delta - z$; if you straddle the sub-solar latitude, $\varphi = z - \delta$ with the Sun to the south of an overhead-passing Sun. Always sketch the meridian to get the sign right.)

Chapter 6

Worked example 3: a star sight

Stars are reduced exactly like the Sun, but their GHA is built in two pieces. On 2026 January 15 at **22:00:00 UT**, from an assumed position 30°N, you shoot the star **Sirius**. Read GHA Aries from Part III and Sirius’s SHA and Dec from the star table, then add:

Star	Sirius
UT	2026 January 15, 22:00:00
Assumed position	30°N, 40°47.1'W
GHA Aries (Part III)	85°21.8'
SHA Sirius (star table)	258°25.3'
GHA Sirius = GHA γ +SHA	343°47.1'
Declination Sirius	S16°45.3'
LHA = GHA – Long(W)	303°00.0'
Computed altitude H_c	17°54.6'
Azimuth Z_n	122°26.3'

Because a star has no semidiameter and no parallax, its sight correction is only index error, dip, and refraction — simpler than the Sun. The computed altitude $H_c = 17^\circ54.6'$ and azimuth $Z_n = 122^\circ26.3'$ are then compared with your observed H_o exactly as in Example 1 to get the intercept.

Chapter 7

A note on longitude and time

Latitude falls out of a noon sight almost for free, but **longitude is a question of time**. The Earth turns 15° of longitude per hour, so every second of chronometer error displaces your computed longitude by a quarter of a minute of arc ($\approx 0.25' \cos \varphi$ nautical miles). This is why the marine chronometer was one of history's great navigational prizes: to find longitude you must know GMT precisely and compare it with your *local* time as read from the Sun or stars. In the intercept method above, the chronometer time enters through the GHA you look up; an error in time is indistinguishable from an error in longitude. Keep your clock honest, log its rate, and never reduce a sight without the exact UT of the observation.

Part II

The Sun, 2026

Chapter 8

Daily data: meridian passage, equation of time, semidiameter

For each day of 2026 the table below gives the **meridian passage** of the Sun at Greenwich (the UT at which the Sun's GHA is 0° , i.e. it crosses the Greenwich meridian), the **equation of time** ($\text{EoT} = 12^{\text{h}} - \text{Mer.Pass.} = \text{apparent} - \text{mean solar time}$; a positive value means the sundial is ahead of the clock), and the Sun's **semidiameter**. Note the semidiameter is largest in early January (Earth at perihelion) and smallest in early July.

Table 8.1: Sun — daily Meridian Passage at Greenwich, Equation of Time, and Semidiameter, 2026

Date	Mer.Pass (UT)	EoT (m:s)	SD (')	Date	Mer.Pass (UT)	EoT (m:s)	SD (')
January 2026							
Jan 1	12:03.6	−3:34	16.3	Jan 17	12:10.1	−10:05	16.3
Jan 2	12:04.0	−4:02	16.3	Jan 18	12:10.4	−10:24	16.3
Jan 3	12:04.5	−4:30	16.3	Jan 19	12:10.7	−10:43	16.3
Jan 4	12:05.0	−4:57	16.3	Jan 20	12:11.0	−11:01	16.3
Jan 5	12:05.4	−5:24	16.3	Jan 21	12:11.3	−11:18	16.3
Jan 6	12:05.8	−5:51	16.3	Jan 22	12:11.6	−11:34	16.3
Jan 7	12:06.3	−6:17	16.3	Jan 23	12:11.8	−11:50	16.3
Jan 8	12:06.7	−6:42	16.3	Jan 24	12:12.1	−12:04	16.3
Jan 9	12:07.1	−7:07	16.3	Jan 25	12:12.3	−12:18	16.3
Jan 10	12:07.5	−7:31	16.3	Jan 26	12:12.5	−12:31	16.3
Jan 11	12:07.9	−7:55	16.3	Jan 27	12:12.7	−12:44	16.2
Jan 12	12:08.3	−8:19	16.3	Jan 28	12:12.9	−12:55	16.2
Jan 13	12:08.7	−8:41	16.3	Jan 29	12:13.1	−13:06	16.2
Jan 14	12:09.1	−9:03	16.3	Jan 30	12:13.3	−13:16	16.2
Jan 15	12:09.4	−9:24	16.3	Jan 31	12:13.4	−13:25	16.2
Jan 16	12:09.8	−9:45	16.3				
February 2026							
Feb 1	12:13.6	−13:33	16.2	Feb 15	12:14.1	−14:05	16.2
Feb 2	12:13.7	−13:41	16.2	Feb 16	12:14.0	−14:02	16.2
Feb 3	12:13.8	−13:47	16.2	Feb 17	12:14.0	−13:58	16.2
Feb 4	12:13.9	−13:53	16.2	Feb 18	12:13.9	−13:53	16.2
Feb 5	12:14.0	−13:58	16.2	Feb 19	12:13.8	−13:48	16.2
Feb 6	12:14.0	−14:03	16.2	Feb 20	12:13.7	−13:41	16.2
Feb 7	12:14.1	−14:06	16.2	Feb 21	12:13.6	−13:35	16.2
Feb 8	12:14.1	−14:09	16.2	Feb 22	12:13.5	−13:27	16.2
Feb 9	12:14.2	−14:10	16.2	Feb 23	12:13.3	−13:19	16.2
Feb 10	12:14.2	−14:12	16.2	Feb 24	12:13.2	−13:10	16.2
Feb 11	12:14.2	−14:12	16.2	Feb 25	12:13.0	−13:01	16.2
Feb 12	12:14.2	−14:11	16.2	Feb 26	12:12.9	−12:51	16.2
Feb 13	12:14.2	−14:10	16.2	Feb 27	12:12.7	−12:41	16.2
Feb 14	12:14.1	−14:08	16.2	Feb 28	12:12.5	−12:30	16.2
March 2026							
Mar 1	12:12.3	−12:19	16.1	Mar 17	12:08.3	−8:20	16.1

continued

Daily Sun data, 2026 (continued)

Date	Mer.Pass	EoT	SD	Date	Mer.Pass	EoT	SD
Mar 2	12:12.1	−12:07	16.1	Mar 18	12:08.0	−8:03	16.1
Mar 3	12:11.9	−11:54	16.1	Mar 19	12:07.8	−7:45	16.1
Mar 4	12:11.7	−11:41	16.1	Mar 20	12:07.5	−7:28	16.1
Mar 5	12:11.5	−11:28	16.1	Mar 21	12:07.2	−7:10	16.1
Mar 6	12:11.2	−11:14	16.1	Mar 22	12:06.9	−6:52	16.1
Mar 7	12:11.0	−11:00	16.1	Mar 23	12:06.6	−6:34	16.1
Mar 8	12:10.8	−10:45	16.1	Mar 24	12:06.3	−6:16	16.0
Mar 9	12:10.5	−10:30	16.1	Mar 25	12:06.0	−5:58	16.0
Mar 10	12:10.3	−10:15	16.1	Mar 26	12:05.7	−5:40	16.0
Mar 11	12:10.0	−10:00	16.1	Mar 27	12:05.4	−5:22	16.0
Mar 12	12:09.7	−9:44	16.1	Mar 28	12:05.1	−5:04	16.0
Mar 13	12:09.5	−9:27	16.1	Mar 29	12:04.8	−4:46	16.0
Mar 14	12:09.2	−9:11	16.1	Mar 30	12:04.5	−4:28	16.0
Mar 15	12:08.9	−8:54	16.1	Mar 31	12:04.2	−4:10	16.0
Mar 16	12:08.6	−8:37	16.1				
April 2026							
Apr 1	12:03.9	−3:52	16.0	Apr 16	11:59.8	+0:12	15.9
Apr 2	12:03.6	−3:34	16.0	Apr 17	11:59.6	+0:26	15.9
Apr 3	12:03.3	−3:17	16.0	Apr 18	11:59.3	+0:39	15.9
Apr 4	12:03.0	−2:59	16.0	Apr 19	11:59.1	+0:52	15.9
Apr 5	12:02.7	−2:42	16.0	Apr 20	11:58.9	+1:05	15.9
Apr 6	12:02.4	−2:25	16.0	Apr 21	11:58.7	+1:17	15.9
Apr 7	12:02.1	−2:08	16.0	Apr 22	11:58.5	+1:29	15.9
Apr 8	12:01.9	−1:51	16.0	Apr 23	11:58.3	+1:40	15.9
Apr 9	12:01.6	−1:35	16.0	Apr 24	11:58.1	+1:51	15.9
Apr 10	12:01.3	−1:19	16.0	Apr 25	11:58.0	+2:01	15.9
Apr 11	12:01.0	−1:03	16.0	Apr 26	11:57.8	+2:11	15.9
Apr 12	12:00.8	−0:47	16.0	Apr 27	11:57.7	+2:21	15.9
Apr 13	12:00.5	−0:32	16.0	Apr 28	11:57.5	+2:30	15.9
Apr 14	12:00.3	−0:17	16.0	Apr 29	11:57.4	+2:38	15.9
Apr 15	12:00.0	−0:03	15.9	Apr 30	11:57.2	+2:46	15.9
May 2026							
May 1	11:57.1	+2:53	15.9	May 17	11:56.4	+3:35	15.8
May 2	11:57.0	+3:00	15.9	May 18	11:56.5	+3:33	15.8
May 3	11:56.9	+3:06	15.9	May 19	11:56.5	+3:30	15.8
May 4	11:56.8	+3:12	15.9	May 20	11:56.6	+3:27	15.8
May 5	11:56.7	+3:17	15.9	May 21	11:56.6	+3:23	15.8
May 6	11:56.6	+3:22	15.9	May 22	11:56.7	+3:19	15.8
May 7	11:56.6	+3:26	15.9	May 23	11:56.8	+3:14	15.8
May 8	11:56.5	+3:29	15.9	May 24	11:56.9	+3:08	15.8
May 9	11:56.5	+3:32	15.8	May 25	11:57.0	+3:02	15.8
May 10	11:56.4	+3:35	15.8	May 26	11:57.1	+2:56	15.8
May 11	11:56.4	+3:36	15.8	May 27	11:57.2	+2:49	15.8
May 12	11:56.4	+3:38	15.8	May 28	11:57.3	+2:42	15.8
May 13	11:56.4	+3:38	15.8	May 29	11:57.4	+2:34	15.8
May 14	11:56.4	+3:38	15.8	May 30	11:57.6	+2:26	15.8
May 15	11:56.4	+3:38	15.8	May 31	11:57.7	+2:17	15.8
May 16	11:56.4	+3:37	15.8				
June 2026							
Jun 1	11:57.9	+2:08	15.8	Jun 16	12:00.7	−0:44	15.8
Jun 2	11:58.0	+1:59	15.8	Jun 17	12:01.0	−0:57	15.7
Jun 3	11:58.2	+1:49	15.8	Jun 18	12:01.2	−1:10	15.7
Jun 4	11:58.4	+1:39	15.8	Jun 19	12:01.4	−1:23	15.7
Jun 5	11:58.5	+1:28	15.8	Jun 20	12:01.6	−1:37	15.7
Jun 6	11:58.7	+1:17	15.8	Jun 21	12:01.8	−1:50	15.7
Jun 7	11:58.9	+1:06	15.8	Jun 22	12:02.0	−2:03	15.7
Jun 8	11:59.1	+0:55	15.8	Jun 23	12:02.3	−2:16	15.7
Jun 9	11:59.3	+0:43	15.8	Jun 24	12:02.5	−2:28	15.7
Jun 10	11:59.5	+0:31	15.8	Jun 25	12:02.7	−2:41	15.7
Jun 11	11:59.7	+0:19	15.8	Jun 26	12:02.9	−2:54	15.7
Jun 12	11:59.9	+0:07	15.8	Jun 27	12:03.1	−3:06	15.7
Jun 13	12:00.1	−0:06	15.8	Jun 28	12:03.3	−3:19	15.7
Jun 14	12:00.3	−0:19	15.8	Jun 29	12:03.5	−3:31	15.7
Jun 15	12:00.5	−0:31	15.8	Jun 30	12:03.7	−3:43	15.7

continued

Daily Sun data, 2026 (continued)

Date	Mer.Pass	EoT	SD	Date	Mer.Pass	EoT	SD
July 2026							
Jul 1	12:03.9	−3:54	15.7	Jul 17	12:06.2	−6:12	15.7
Jul 2	12:04.1	−4:06	15.7	Jul 18	12:06.3	−6:17	15.7
Jul 3	12:04.3	−4:17	15.7	Jul 19	12:06.3	−6:21	15.7
Jul 4	12:04.5	−4:28	15.7	Jul 20	12:06.4	−6:24	15.7
Jul 5	12:04.6	−4:38	15.7	Jul 21	12:06.5	−6:27	15.7
Jul 6	12:04.8	−4:48	15.7	Jul 22	12:06.5	−6:30	15.7
Jul 7	12:05.0	−4:58	15.7	Jul 23	12:06.5	−6:32	15.7
Jul 8	12:05.1	−5:07	15.7	Jul 24	12:06.5	−6:33	15.8
Jul 9	12:05.3	−5:16	15.7	Jul 25	12:06.6	−6:34	15.8
Jul 10	12:05.4	−5:25	15.7	Jul 26	12:06.6	−6:34	15.8
Jul 11	12:05.6	−5:33	15.7	Jul 27	12:06.6	−6:33	15.8
Jul 12	12:05.7	−5:41	15.7	Jul 28	12:06.5	−6:32	15.8
Jul 13	12:05.8	−5:48	15.7	Jul 29	12:06.5	−6:30	15.8
Jul 14	12:05.9	−5:55	15.7	Jul 30	12:06.5	−6:28	15.8
Jul 15	12:06.0	−6:01	15.7	Jul 31	12:06.4	−6:25	15.8
Jul 16	12:06.1	−6:07	15.7				
August 2026							
Aug 1	12:06.4	−6:22	15.8	Aug 17	12:04.1	−4:05	15.8
Aug 2	12:06.3	−6:17	15.8	Aug 18	12:03.9	−3:51	15.8
Aug 3	12:06.2	−6:13	15.8	Aug 19	12:03.6	−3:38	15.8
Aug 4	12:06.1	−6:07	15.8	Aug 20	12:03.4	−3:24	15.8
Aug 5	12:06.0	−6:01	15.8	Aug 21	12:03.1	−3:09	15.8
Aug 6	12:05.9	−5:55	15.8	Aug 22	12:02.9	−2:54	15.8
Aug 7	12:05.8	−5:48	15.8	Aug 23	12:02.6	−2:38	15.8
Aug 8	12:05.7	−5:40	15.8	Aug 24	12:02.4	−2:22	15.8
Aug 9	12:05.5	−5:32	15.8	Aug 25	12:02.1	−2:06	15.8
Aug 10	12:05.4	−5:23	15.8	Aug 26	12:01.8	−1:49	15.8
Aug 11	12:05.2	−5:13	15.8	Aug 27	12:01.5	−1:32	15.8
Aug 12	12:05.1	−5:03	15.8	Aug 28	12:01.2	−1:14	15.8
Aug 13	12:04.9	−4:53	15.8	Aug 29	12:00.9	−0:56	15.8
Aug 14	12:04.7	−4:41	15.8	Aug 30	12:00.6	−0:38	15.8
Aug 15	12:04.5	−4:30	15.8	Aug 31	12:00.3	−0:19	15.8
Aug 16	12:04.3	−4:17	15.8				
September 2026							
Sep 1	12:00.0	−0:00	15.9	Sep 16	11:54.8	+5:09	15.9
Sep 2	11:59.7	+0:19	15.9	Sep 17	11:54.5	+5:31	15.9
Sep 3	11:59.4	+0:39	15.9	Sep 18	11:54.1	+5:52	15.9
Sep 4	11:59.0	+0:59	15.9	Sep 19	11:53.8	+6:14	15.9
Sep 5	11:58.7	+1:19	15.9	Sep 20	11:53.4	+6:35	15.9
Sep 6	11:58.4	+1:39	15.9	Sep 21	11:53.1	+6:56	15.9
Sep 7	11:58.0	+1:59	15.9	Sep 22	11:52.7	+7:18	15.9
Sep 8	11:57.7	+2:20	15.9	Sep 23	11:52.4	+7:39	15.9
Sep 9	11:57.3	+2:41	15.9	Sep 24	11:52.0	+8:00	15.9
Sep 10	11:57.0	+3:02	15.9	Sep 25	11:51.7	+8:20	16.0
Sep 11	11:56.6	+3:23	15.9	Sep 26	11:51.3	+8:41	16.0
Sep 12	11:56.3	+3:44	15.9	Sep 27	11:51.0	+9:02	16.0
Sep 13	11:55.9	+4:05	15.9	Sep 28	11:50.6	+9:22	16.0
Sep 14	11:55.6	+4:27	15.9	Sep 29	11:50.3	+9:42	16.0
Sep 15	11:55.2	+4:48	15.9	Sep 30	11:50.0	+10:02	16.0
October 2026							
Oct 1	11:49.6	+10:21	16.0	Oct 17	11:45.3	+14:39	16.1
Oct 2	11:49.3	+10:40	16.0	Oct 18	11:45.1	+14:51	16.1
Oct 3	11:49.0	+10:59	16.0	Oct 19	11:45.0	+15:03	16.1
Oct 4	11:48.7	+11:18	16.0	Oct 20	11:44.8	+15:13	16.1
Oct 5	11:48.4	+11:36	16.0	Oct 21	11:44.6	+15:23	16.1
Oct 6	11:48.1	+11:54	16.0	Oct 22	11:44.5	+15:32	16.1
Oct 7	11:47.8	+12:11	16.0	Oct 23	11:44.3	+15:41	16.1
Oct 8	11:47.5	+12:28	16.0	Oct 24	11:44.2	+15:49	16.1
Oct 9	11:47.3	+12:45	16.0	Oct 25	11:44.1	+15:56	16.1
Oct 10	11:47.0	+13:01	16.0	Oct 26	11:44.0	+16:03	16.1
Oct 11	11:46.7	+13:16	16.0	Oct 27	11:43.9	+16:08	16.1
Oct 12	11:46.5	+13:32	16.0	Oct 28	11:43.8	+16:13	16.1
Oct 13	11:46.2	+13:46	16.0	Oct 29	11:43.7	+16:17	16.1

continued

Daily Sun data, 2026 (continued)

Date	Mer.Pass	EoT	SD	Date	Mer.Pass	EoT	SD
Oct 14	11:46.0	+14:00	16.0	Oct 30	11:43.7	+16:21	16.1
Oct 15	11:45.8	+14:14	16.0	Oct 31	11:43.6	+16:24	16.1
Oct 16	11:45.5	+14:27	16.0				
November 2026							
Nov 1	11:43.6	+16:26	16.1	Nov 16	11:44.8	+15:15	16.2
Nov 2	11:43.6	+16:27	16.1	Nov 17	11:44.9	+15:03	16.2
Nov 3	11:43.6	+16:27	16.1	Nov 18	11:45.2	+14:51	16.2
Nov 4	11:43.6	+16:26	16.1	Nov 19	11:45.4	+14:38	16.2
Nov 5	11:43.6	+16:25	16.1	Nov 20	11:45.6	+14:24	16.2
Nov 6	11:43.6	+16:23	16.1	Nov 21	11:45.9	+14:09	16.2
Nov 7	11:43.7	+16:20	16.1	Nov 22	11:46.1	+13:53	16.2
Nov 8	11:43.7	+16:16	16.1	Nov 23	11:46.4	+13:37	16.2
Nov 9	11:43.8	+16:11	16.2	Nov 24	11:46.7	+13:20	16.2
Nov 10	11:43.9	+16:06	16.2	Nov 25	11:47.0	+13:02	16.2
Nov 11	11:44.0	+15:59	16.2	Nov 26	11:47.3	+12:43	16.2
Nov 12	11:44.1	+15:52	16.2	Nov 27	11:47.6	+12:24	16.2
Nov 13	11:44.3	+15:44	16.2	Nov 28	11:47.9	+12:04	16.2
Nov 14	11:44.4	+15:35	16.2	Nov 29	11:48.3	+11:43	16.2
Nov 15	11:44.6	+15:25	16.2	Nov 30	11:48.6	+11:22	16.2
December 2026							
Dec 1	11:49.0	+10:59	16.2	Dec 17	11:56.1	+3:55	16.3
Dec 2	11:49.4	+10:37	16.2	Dec 18	11:56.6	+3:25	16.3
Dec 3	11:49.8	+10:13	16.2	Dec 19	11:57.1	+2:56	16.3
Dec 4	11:50.2	+9:49	16.2	Dec 20	11:57.6	+2:26	16.3
Dec 5	11:50.6	+9:25	16.2	Dec 21	11:58.1	+1:56	16.3
Dec 6	11:51.0	+8:59	16.2	Dec 22	11:58.6	+1:26	16.3
Dec 7	11:51.4	+8:34	16.2	Dec 23	11:59.1	+0:57	16.3
Dec 8	11:51.9	+8:08	16.2	Dec 24	11:59.6	+0:27	16.3
Dec 9	11:52.3	+7:41	16.2	Dec 25	12:00.0	−0:03	16.3
Dec 10	11:52.8	+7:14	16.2	Dec 26	12:00.5	−0:33	16.3
Dec 11	11:53.2	+6:46	16.2	Dec 27	12:01.0	−1:02	16.3
Dec 12	11:53.7	+6:19	16.3	Dec 28	12:01.5	−1:32	16.3
Dec 13	11:54.2	+5:50	16.3	Dec 29	12:02.0	−2:01	16.3
Dec 14	11:54.6	+5:22	16.3	Dec 30	12:02.5	−2:30	16.3
Dec 15	11:55.1	+4:53	16.3	Dec 31	12:03.0	−2:59	16.3
Dec 16	11:55.6	+4:24	16.3				

Chapter 9

Hourly GHA and Declination of the Sun

Read the Greenwich Hour Angle and Declination of the Sun's centre for the day and whole hour of UT; interpolate the minutes and seconds with the Sun increment table in Part IV. Entries run left to right, then top to bottom, in three columns of day / hour / GHA / Dec.

Table 9.1: Sun — GHA and Declination, January 2026 (UT)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
1	0	179°10.0'	S23°00.9'	1	1	194°09.7'	S23°00.7'	1	2	209°09.4'	S23°00.5'
1	3	224°09.1'	S23°00.3'	1	4	239°08.8'	S23°00.1'	1	5	254°08.6'	S22°59.9'
1	6	269°08.3'	S22°59.7'	1	7	284°08.0'	S22°59.4'	1	8	299°07.7'	S22°59.2'
1	9	314°07.4'	S22°59.0'	1	10	329°07.1'	S22°58.8'	1	11	344°06.8'	S22°58.6'
1	12	359°06.5'	S22°58.4'	1	13	14°06.2'	S22°58.2'	1	14	29°05.9'	S22°58.0'
1	15	44°05.6'	S22°57.8'	1	16	59°05.3'	S22°57.5'	1	17	74°05.0'	S22°57.3'
1	18	89°04.7'	S22°57.1'	1	19	104°04.4'	S22°56.9'	1	20	119°04.1'	S22°56.7'
1	21	134°03.9'	S22°56.5'	1	22	149°03.6'	S22°56.2'	1	23	164°03.3'	S22°56.0'
2	0	179°03.0'	S22°55.8'	2	1	194°02.7'	S22°55.6'	2	2	209°02.4'	S22°55.4'
2	3	224°02.1'	S22°55.1'	2	4	239°01.8'	S22°54.9'	2	5	254°01.5'	S22°54.7'
2	6	269°01.2'	S22°54.5'	2	7	284°00.9'	S22°54.2'	2	8	299°00.6'	S22°54.0'
2	9	314°00.4'	S22°53.8'	2	10	329°00.1'	S22°53.6'	2	11	343°59.8'	S22°53.3'
2	12	358°59.5'	S22°53.1'	2	13	13°59.2'	S22°52.9'	2	14	28°58.9'	S22°52.6'
2	15	43°58.6'	S22°52.4'	2	16	58°58.3'	S22°52.2'	2	17	73°58.0'	S22°51.9'
2	18	88°57.7'	S22°51.7'	2	19	103°57.5'	S22°51.5'	2	20	118°57.2'	S22°51.2'
2	21	133°56.9'	S22°51.0'	2	22	148°56.6'	S22°50.7'	2	23	163°56.3'	S22°50.5'
3	0	178°56.0'	S22°50.3'	3	1	193°55.7'	S22°50.0'	3	2	208°55.4'	S22°49.8'
3	3	223°55.1'	S22°49.5'	3	4	238°54.9'	S22°49.3'	3	5	253°54.6'	S22°49.1'
3	6	268°54.3'	S22°48.8'	3	7	283°54.0'	S22°48.6'	3	8	298°53.7'	S22°48.3'
3	9	313°53.4'	S22°48.1'	3	10	328°53.1'	S22°47.8'	3	11	343°52.8'	S22°47.6'
3	12	358°52.6'	S22°47.3'	3	13	13°52.3'	S22°47.1'	3	14	28°52.0'	S22°46.8'
3	15	43°51.7'	S22°46.6'	3	16	58°51.4'	S22°46.3'	3	17	73°51.1'	S22°46.1'
3	18	88°50.8'	S22°45.8'	3	19	103°50.6'	S22°45.6'	3	20	118°50.3'	S22°45.3'
3	21	133°50.0'	S22°45.1'	3	22	148°49.7'	S22°44.8'	3	23	163°49.4'	S22°44.5'
4	0	178°49.1'	S22°44.3'	4	1	193°48.8'	S22°44.0'	4	2	208°48.6'	S22°43.8'
4	3	223°48.3'	S22°43.5'	4	4	238°48.0'	S22°43.2'	4	5	253°47.7'	S22°43.0'
4	6	268°47.4'	S22°42.7'	4	7	283°47.1'	S22°42.5'	4	8	298°46.9'	S22°42.2'
4	9	313°46.6'	S22°41.9'	4	10	328°46.3'	S22°41.7'	4	11	343°46.0'	S22°41.4'
4	12	358°45.7'	S22°41.1'	4	13	13°45.4'	S22°40.9'	4	14	28°45.2'	S22°40.6'
4	15	43°44.9'	S22°40.3'	4	16	58°44.6'	S22°40.0'	4	17	73°44.3'	S22°39.8'
4	18	88°44.0'	S22°39.5'	4	19	103°43.8'	S22°39.2'	4	20	118°43.5'	S22°38.9'
4	21	133°43.2'	S22°38.7'	4	22	148°42.9'	S22°38.4'	4	23	163°42.6'	S22°38.1'
5	0	178°42.4'	S22°37.8'	5	1	193°42.1'	S22°37.6'	5	2	208°41.8'	S22°37.3'
5	3	223°41.5'	S22°37.0'	5	4	238°41.2'	S22°36.7'	5	5	253°41.0'	S22°36.4'
5	6	268°40.7'	S22°36.2'	5	7	283°40.4'	S22°35.9'	5	8	298°40.1'	S22°35.6'
5	9	313°39.8'	S22°35.3'	5	10	328°39.6'	S22°35.0'	5	11	343°39.3'	S22°34.7'
5	12	358°39.0'	S22°34.5'	5	13	13°38.7'	S22°34.2'	5	14	28°38.4'	S22°33.9'
5	15	43°38.2'	S22°33.6'	5	16	58°37.9'	S22°33.3'	5	17	73°37.6'	S22°33.0'
5	18	88°37.3'	S22°32.7'	5	19	103°37.1'	S22°32.4'	5	20	118°36.8'	S22°32.1'
5	21	133°36.5'	S22°31.8'	5	22	148°36.2'	S22°31.5'	5	23	163°36.0'	S22°31.3'
6	0	178°35.7'	S22°31.0'	6	1	193°35.4'	S22°30.7'	6	2	208°35.1'	S22°30.4'

continued on next page

Sun — GHA and Declination, January 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
6	3	223°34.9'	S22°30.1'	6	4	238°34.6'	S22°29.8'	6	5	253°34.3'	S22°29.5'
6	6	268°34.0'	S22°29.2'	6	7	283°33.8'	S22°28.9'	6	8	298°33.5'	S22°28.6'
6	9	313°33.2'	S22°28.3'	6	10	328°32.9'	S22°28.0'	6	11	343°32.7'	S22°27.7'
6	12	358°32.4'	S22°27.3'	6	13	13°32.1'	S22°27.0'	6	14	28°31.8'	S22°26.7'
6	15	43°31.6'	S22°26.4'	6	16	58°31.3'	S22°26.1'	6	17	73°31.0'	S22°25.8'
6	18	88°30.8'	S22°25.5'	6	19	103°30.5'	S22°25.2'	6	20	118°30.2'	S22°24.9'
6	21	133°29.9'	S22°24.6'	6	22	148°29.7'	S22°24.3'	6	23	163°29.4'	S22°23.9'
7	0	178°29.1'	S22°23.6'	7	1	193°28.9'	S22°23.3'	7	2	208°28.6'	S22°23.0'
7	3	223°28.3'	S22°22.7'	7	4	238°28.0'	S22°22.4'	7	5	253°27.8'	S22°22.0'
7	6	268°27.5'	S22°21.7'	7	7	283°27.2'	S22°21.4'	7	8	298°27.0'	S22°21.1'
7	9	313°26.7'	S22°20.8'	7	10	328°26.4'	S22°20.4'	7	11	343°26.2'	S22°20.1'
7	12	358°25.9'	S22°19.8'	7	13	13°25.6'	S22°19.5'	7	14	28°25.4'	S22°19.1'
7	15	43°25.1'	S22°18.8'	7	16	58°24.8'	S22°18.5'	7	17	73°24.6'	S22°18.2'
7	18	88°24.3'	S22°17.8'	7	19	103°24.0'	S22°17.5'	7	20	118°23.8'	S22°17.2'
7	21	133°23.5'	S22°16.9'	7	22	148°23.2'	S22°16.5'	7	23	163°23.0'	S22°16.2'
8	0	178°22.7'	S22°15.9'	8	1	193°22.4'	S22°15.5'	8	2	208°22.2'	S22°15.2'
8	3	223°21.9'	S22°14.9'	8	4	238°21.6'	S22°14.5'	8	5	253°21.4'	S22°14.2'
8	6	268°21.1'	S22°13.8'	8	7	283°20.8'	S22°13.5'	8	8	298°20.6'	S22°13.2'
8	9	313°20.3'	S22°12.8'	8	10	328°20.0'	S22°12.5'	8	11	343°19.8'	S22°12.1'
8	12	358°19.5'	S22°11.8'	8	13	13°19.3'	S22°11.5'	8	14	28°19.0'	S22°11.1'
8	15	43°18.7'	S22°10.8'	8	16	58°18.5'	S22°10.4'	8	17	73°18.2'	S22°10.1'
8	18	88°18.0'	S22°09.7'	8	19	103°17.7'	S22°09.4'	8	20	118°17.4'	S22°09.0'
8	21	133°17.2'	S22°08.7'	8	22	148°16.9'	S22°08.3'	8	23	163°16.6'	S22°08.0'
9	0	178°16.4'	S22°07.6'	9	1	193°16.1'	S22°07.3'	9	2	208°15.9'	S22°06.9'
9	3	223°15.6'	S22°06.6'	9	4	238°15.3'	S22°06.2'	9	5	253°15.1'	S22°05.9'
9	6	268°14.8'	S22°05.5'	9	7	283°14.6'	S22°05.2'	9	8	298°14.3'	S22°04.8'
9	9	313°14.1'	S22°04.5'	9	10	328°13.8'	S22°04.1'	9	11	343°13.5'	S22°03.7'
9	12	358°13.3'	S22°03.4'	9	13	13°13.0'	S22°03.0'	9	14	28°12.8'	S22°02.7'
9	15	43°12.5'	S22°02.3'	9	16	58°12.3'	S22°01.9'	9	17	73°12.0'	S22°01.6'
9	18	88°11.7'	S22°01.2'	9	19	103°11.5'	S22°00.8'	9	20	118°11.2'	S22°00.5'
9	21	133°11.0'	S22°00.1'	9	22	148°10.7'	S21°59.7'	9	23	163°10.5'	S21°59.4'
10	0	178°10.2'	S21°59.0'	10	1	193°10.0'	S21°58.6'	10	2	208°09.7'	S21°58.3'
10	3	223°09.5'	S21°57.9'	10	4	238°09.2'	S21°57.5'	10	5	253°08.9'	S21°57.1'
10	6	268°08.7'	S21°56.8'	10	7	283°08.4'	S21°56.4'	10	8	298°08.2'	S21°56.0'
10	9	313°07.9'	S21°55.7'	10	10	328°07.7'	S21°55.3'	10	11	343°07.4'	S21°54.9'
10	12	358°07.2'	S21°54.5'	10	13	13°06.9'	S21°54.1'	10	14	28°06.7'	S21°53.8'
10	15	43°06.4'	S21°53.4'	10	16	58°06.2'	S21°53.0'	10	17	73°05.9'	S21°52.6'
10	18	88°05.7'	S21°52.2'	10	19	103°05.4'	S21°51.9'	10	20	118°05.2'	S21°51.5'
10	21	133°04.9'	S21°51.1'	10	22	148°04.7'	S21°50.7'	10	23	163°04.4'	S21°50.3'
11	0	178°04.2'	S21°49.9'	11	1	193°03.9'	S21°49.5'	11	2	208°03.7'	S21°49.2'
11	3	223°03.4'	S21°48.8'	11	4	238°03.2'	S21°48.4'	11	5	253°02.9'	S21°48.0'
11	6	268°02.7'	S21°47.6'	11	7	283°02.4'	S21°47.2'	11	8	298°02.2'	S21°46.8'
11	9	313°02.0'	S21°46.4'	11	10	328°01.7'	S21°46.0'	11	11	343°01.5'	S21°45.6'
11	12	358°01.2'	S21°45.2'	11	13	13°01.0'	S21°44.8'	11	14	28°00.7'	S21°44.4'
11	15	43°00.5'	S21°44.0'	11	16	58°00.2'	S21°43.6'	11	17	73°00.0'	S21°43.2'
11	18	87°59.8'	S21°42.9'	11	19	102°59.5'	S21°42.4'	11	20	117°59.3'	S21°42.0'
11	21	132°59.0'	S21°41.6'	11	22	147°58.8'	S21°41.2'	11	23	162°58.5'	S21°40.8'
12	0	177°58.3'	S21°40.4'	12	1	192°58.1'	S21°40.0'	12	2	207°57.8'	S21°39.6'
12	3	222°57.6'	S21°39.2'	12	4	237°57.3'	S21°38.8'	12	5	252°57.1'	S21°38.4'
12	6	267°56.8'	S21°38.0'	12	7	282°56.6'	S21°37.6'	12	8	297°56.4'	S21°37.2'
12	9	312°56.1'	S21°36.8'	12	10	327°55.9'	S21°36.4'	12	11	342°55.6'	S21°35.9'
12	12	357°55.4'	S21°35.5'	12	13	12°55.2'	S21°35.1'	12	14	27°54.9'	S21°34.7'
12	15	42°54.7'	S21°34.3'	12	16	57°54.5'	S21°33.9'	12	17	72°54.2'	S21°33.5'
12	18	87°54.0'	S21°33.0'	12	19	102°53.7'	S21°32.6'	12	20	117°53.5'	S21°32.2'
12	21	132°53.3'	S21°31.8'	12	22	147°53.0'	S21°31.4'	12	23	162°52.8'	S21°30.9'
13	0	177°52.6'	S21°30.5'	13	1	192°52.3'	S21°30.1'	13	2	207°52.1'	S21°29.7'
13	3	222°51.8'	S21°29.3'	13	4	237°51.6'	S21°28.8'	13	5	252°51.4'	S21°28.4'
13	6	267°51.1'	S21°28.0'	13	7	282°50.9'	S21°27.5'	13	8	297°50.7'	S21°27.1'
13	9	312°50.4'	S21°26.7'	13	10	327°50.2'	S21°26.3'	13	11	342°50.0'	S21°25.8'
13	12	357°49.7'	S21°25.4'	13	13	12°49.5'	S21°25.0'	13	14	27°49.3'	S21°24.5'
13	15	42°49.1'	S21°24.1'	13	16	57°48.8'	S21°23.7'	13	17	72°48.6'	S21°23.2'
13	18	87°48.4'	S21°22.8'	13	19	102°48.1'	S21°22.4'	13	20	117°47.9'	S21°21.9'
13	21	132°47.7'	S21°21.5'	13	22	147°47.4'	S21°21.1'	13	23	162°47.2'	S21°20.6'
14	0	177°47.0'	S21°20.2'	14	1	192°46.7'	S21°19.7'	14	2	207°46.5'	S21°19.3'

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Sun — GHA and Declination, January 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
14	3	222°46.3'	S21°18.9'	14	4	237°46.1'	S21°18.4'	14	5	252°45.8'	S21°18.0'
14	6	267°45.6'	S21°17.5'	14	7	282°45.4'	S21°17.1'	14	8	297°45.2'	S21°16.7'
14	9	312°44.9'	S21°16.2'	14	10	327°44.7'	S21°15.8'	14	11	342°44.5'	S21°15.3'
14	12	357°44.2'	S21°14.9'	14	13	12°44.0'	S21°14.4'	14	14	27°43.8'	S21°14.0'
14	15	42°43.6'	S21°13.5'	14	16	57°43.3'	S21°13.1'	14	17	72°43.1'	S21°12.6'
14	18	87°42.9'	S21°12.2'	14	19	102°42.7'	S21°11.7'	14	20	117°42.5'	S21°11.3'
14	21	132°42.2'	S21°10.8'	14	22	147°42.0'	S21°10.4'	14	23	162°41.8'	S21°09.9'
15	0	177°41.6'	S21°09.4'	15	1	192°41.3'	S21°09.0'	15	2	207°41.1'	S21°08.5'
15	3	222°40.9'	S21°08.1'	15	4	237°40.7'	S21°07.6'	15	5	252°40.5'	S21°07.2'
15	6	267°40.2'	S21°06.7'	15	7	282°40.0'	S21°06.2'	15	8	297°39.8'	S21°05.8'
15	9	312°39.6'	S21°05.3'	15	10	327°39.3'	S21°04.8'	15	11	342°39.1'	S21°04.4'
15	12	357°38.9'	S21°03.9'	15	13	12°38.7'	S21°03.5'	15	14	27°38.5'	S21°03.0'
15	15	42°38.3'	S21°02.5'	15	16	57°38.0'	S21°02.1'	15	17	72°37.8'	S21°01.6'
15	18	87°37.6'	S21°01.1'	15	19	102°37.4'	S21°00.7'	15	20	117°37.2'	S21°00.2'
15	21	132°37.0'	S20°59.7'	15	22	147°36.7'	S20°59.2'	15	23	162°36.5'	S20°58.8'
16	0	177°36.3'	S20°58.3'	16	1	192°36.1'	S20°57.8'	16	2	207°35.9'	S20°57.3'
16	3	222°35.7'	S20°56.9'	16	4	237°35.4'	S20°56.4'	16	5	252°35.2'	S20°55.9'
16	6	267°35.0'	S20°55.4'	16	7	282°34.8'	S20°55.0'	16	8	297°34.6'	S20°54.5'
16	9	312°34.4'	S20°54.0'	16	10	327°34.2'	S20°53.5'	16	11	342°34.0'	S20°53.1'
16	12	357°33.7'	S20°52.6'	16	13	12°33.5'	S20°52.1'	16	14	27°33.3'	S20°51.6'
16	15	42°33.1'	S20°51.1'	16	16	57°32.9'	S20°50.6'	16	17	72°32.7'	S20°50.2'
16	18	87°32.5'	S20°49.7'	16	19	102°32.3'	S20°49.2'	16	20	117°32.1'	S20°48.7'
16	21	132°31.9'	S20°48.2'	16	22	147°31.6'	S20°47.7'	16	23	162°31.4'	S20°47.2'
17	0	177°31.2'	S20°46.7'	17	1	192°31.0'	S20°46.3'	17	2	207°30.8'	S20°45.8'
17	3	222°30.6'	S20°45.3'	17	4	237°30.4'	S20°44.8'	17	5	252°30.2'	S20°44.3'
17	6	267°30.0'	S20°43.8'	17	7	282°29.8'	S20°43.3'	17	8	297°29.6'	S20°42.8'
17	9	312°29.4'	S20°42.3'	17	10	327°29.2'	S20°41.8'	17	11	342°29.0'	S20°41.3'
17	12	357°28.8'	S20°40.8'	17	13	12°28.5'	S20°40.3'	17	14	27°28.3'	S20°39.8'
17	15	42°28.1'	S20°39.3'	17	16	57°27.9'	S20°38.8'	17	17	72°27.7'	S20°38.3'
17	18	87°27.5'	S20°37.8'	17	19	102°27.3'	S20°37.3'	17	20	117°27.1'	S20°36.8'
17	21	132°26.9'	S20°36.3'	17	22	147°26.7'	S20°35.8'	17	23	162°26.5'	S20°35.3'
18	0	177°26.3'	S20°34.8'	18	1	192°26.1'	S20°34.3'	18	2	207°25.9'	S20°33.8'
18	3	222°25.7'	S20°33.3'	18	4	237°25.5'	S20°32.8'	18	5	252°25.3'	S20°32.3'
18	6	267°25.1'	S20°31.8'	18	7	282°24.9'	S20°31.2'	18	8	297°24.7'	S20°30.7'
18	9	312°24.5'	S20°30.2'	18	10	327°24.3'	S20°29.7'	18	11	342°24.1'	S20°29.2'
18	12	357°23.9'	S20°28.7'	18	13	12°23.7'	S20°28.2'	18	14	27°23.5'	S20°27.7'
18	15	42°23.3'	S20°27.1'	18	16	57°23.1'	S20°26.6'	18	17	72°23.0'	S20°26.1'
18	18	87°22.8'	S20°25.6'	18	19	102°22.6'	S20°25.1'	18	20	117°22.4'	S20°24.5'
18	21	132°22.2'	S20°24.0'	18	22	147°22.0'	S20°23.5'	18	23	162°21.8'	S20°23.0'
19	0	177°21.6'	S20°22.5'	19	1	192°21.4'	S20°21.9'	19	2	207°21.2'	S20°21.4'
19	3	222°21.0'	S20°20.9'	19	4	237°20.8'	S20°20.4'	19	5	252°20.6'	S20°19.8'
19	6	267°20.4'	S20°19.3'	19	7	282°20.2'	S20°18.8'	19	8	297°20.1'	S20°18.3'
19	9	312°19.9'	S20°17.7'	19	10	327°19.7'	S20°17.2'	19	11	342°19.5'	S20°16.7'
19	12	357°19.3'	S20°16.2'	19	13	12°19.1'	S20°15.6'	19	14	27°18.9'	S20°15.1'
19	15	42°18.7'	S20°14.6'	19	16	57°18.5'	S20°14.0'	19	17	72°18.4'	S20°13.5'
19	18	87°18.2'	S20°13.0'	19	19	102°18.0'	S20°12.4'	19	20	117°17.8'	S20°11.9'
19	21	132°17.6'	S20°11.4'	19	22	147°17.4'	S20°10.8'	19	23	162°17.2'	S20°10.3'
20	0	177°17.0'	S20°09.7'	20	1	192°16.9'	S20°09.2'	20	2	207°16.7'	S20°08.7'
20	3	222°16.5'	S20°08.1'	20	4	237°16.3'	S20°07.6'	20	5	252°16.1'	S20°07.0'
20	6	267°15.9'	S20°06.5'	20	7	282°15.8'	S20°06.0'	20	8	297°15.6'	S20°05.4'
20	9	312°15.4'	S20°04.9'	20	10	327°15.2'	S20°04.3'	20	11	342°15.0'	S20°03.8'
20	12	357°14.8'	S20°03.2'	20	13	12°14.7'	S20°02.7'	20	14	27°14.5'	S20°02.2'
20	15	42°14.3'	S20°01.6'	20	16	57°14.1'	S20°01.1'	20	17	72°13.9'	S20°00.5'
20	18	87°13.8'	S20°00.0'	20	19	102°13.6'	S19°59.4'	20	20	117°13.4'	S19°58.9'
20	21	132°13.2'	S19°58.3'	20	22	147°13.0'	S19°57.8'	20	23	162°12.9'	S19°57.2'
21	0	177°12.7'	S19°56.6'	21	1	192°12.5'	S19°56.1'	21	2	207°12.3'	S19°55.5'
21	3	222°12.2'	S19°55.0'	21	4	237°12.0'	S19°54.4'	21	5	252°11.8'	S19°53.9'
21	6	267°11.6'	S19°53.3'	21	7	282°11.5'	S19°52.8'	21	8	297°11.3'	S19°52.2'
21	9	312°11.1'	S19°51.6'	21	10	327°10.9'	S19°51.1'	21	11	342°10.8'	S19°50.5'
21	12	357°10.6'	S19°50.0'	21	13	12°10.4'	S19°49.4'	21	14	27°10.2'	S19°48.8'
21	15	42°10.1'	S19°48.3'	21	16	57°09.9'	S19°47.7'	21	17	72°09.7'	S19°47.1'
21	18	87°09.5'	S19°46.6'	21	19	102°09.4'	S19°46.0'	21	20	117°09.2'	S19°45.5'
21	21	132°09.0'	S19°44.9'	21	22	147°08.9'	S19°44.3'	21	23	162°08.7'	S19°43.7'
22	0	177°08.5'	S19°43.2'	22	1	192°08.4'	S19°42.6'	22	2	207°08.2'	S19°42.0'

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Sun — GHA and Declination, January 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
22	3	222°08.0'	S19°41.5'	22	4	237°07.8'	S19°40.9'	22	5	252°07.7'	S19°40.3'
22	6	267°07.5'	S19°39.8'	22	7	282°07.3'	S19°39.2'	22	8	297°07.2'	S19°38.6'
22	9	312°07.0'	S19°38.0'	22	10	327°06.8'	S19°37.5'	22	11	342°06.7'	S19°36.9'
22	12	357°06.5'	S19°36.3'	22	13	12°06.3'	S19°35.7'	22	14	27°06.2'	S19°35.2'
22	15	42°06.0'	S19°34.6'	22	16	57°05.8'	S19°34.0'	22	17	72°05.7'	S19°33.4'
22	18	87°05.5'	S19°32.8'	22	19	102°05.4'	S19°32.3'	22	20	117°05.2'	S19°31.7'
22	21	132°05.0'	S19°31.1'	22	22	147°04.9'	S19°30.5'	22	23	162°04.7'	S19°29.9'
23	0	177°04.5'	S19°29.3'	23	1	192°04.4'	S19°28.8'	23	2	207°04.2'	S19°28.2'
23	3	222°04.1'	S19°27.6'	23	4	237°03.9'	S19°27.0'	23	5	252°03.7'	S19°26.4'
23	6	267°03.6'	S19°25.8'	23	7	282°03.4'	S19°25.2'	23	8	297°03.3'	S19°24.7'
23	9	312°03.1'	S19°24.1'	23	10	327°02.9'	S19°23.5'	23	11	342°02.8'	S19°22.9'
23	12	357°02.6'	S19°22.3'	23	13	12°02.5'	S19°21.7'	23	14	27°02.3'	S19°21.1'
23	15	42°02.2'	S19°20.5'	23	16	57°02.0'	S19°19.9'	23	17	72°01.8'	S19°19.3'
23	18	87°01.7'	S19°18.7'	23	19	102°01.5'	S19°18.1'	23	20	117°01.4'	S19°17.5'
23	21	132°01.2'	S19°16.9'	23	22	147°01.1'	S19°16.3'	23	23	162°00.9'	S19°15.7'
24	0	177°00.8'	S19°15.1'	24	1	192°00.6'	S19°14.5'	24	2	207°00.5'	S19°13.9'
24	3	222°00.3'	S19°13.3'	24	4	237°00.1'	S19°12.7'	24	5	252°00.0'	S19°12.1'
24	6	266°59.8'	S19°11.5'	24	7	281°59.7'	S19°10.9'	24	8	296°59.5'	S19°10.3'
24	9	311°59.4'	S19°09.7'	24	10	326°59.2'	S19°09.1'	24	11	341°59.1'	S19°08.5'
24	12	356°58.9'	S19°07.9'	24	13	11°58.8'	S19°07.3'	24	14	26°58.6'	S19°06.7'
24	15	41°58.5'	S19°06.1'	24	16	56°58.3'	S19°05.5'	24	17	71°58.2'	S19°04.9'
24	18	86°58.1'	S19°04.3'	24	19	101°57.9'	S19°03.7'	24	20	116°57.8'	S19°03.0'
24	21	131°57.6'	S19°02.4'	24	22	146°57.5'	S19°01.8'	24	23	161°57.3'	S19°01.2'
25	0	176°57.2'	S19°00.6'	25	1	191°57.0'	S19°00.0'	25	2	206°56.9'	S18°59.4'
25	3	221°56.7'	S18°58.8'	25	4	236°56.6'	S18°58.1'	25	5	251°56.4'	S18°57.5'
25	6	266°56.3'	S18°56.9'	25	7	281°56.2'	S18°56.3'	25	8	296°56.0'	S18°55.7'
25	9	311°55.9'	S18°55.0'	25	10	326°55.7'	S18°54.4'	25	11	341°55.6'	S18°53.8'
25	12	356°55.5'	S18°53.2'	25	13	11°55.3'	S18°52.6'	25	14	26°55.2'	S18°51.9'
25	15	41°55.0'	S18°51.3'	25	16	56°54.9'	S18°50.7'	25	17	71°54.8'	S18°50.1'
25	18	86°54.6'	S18°49.5'	25	19	101°54.5'	S18°48.8'	25	20	116°54.3'	S18°48.2'
25	21	131°54.2'	S18°47.6'	25	22	146°54.1'	S18°46.9'	25	23	161°53.9'	S18°46.3'
26	0	176°53.8'	S18°45.7'	26	1	191°53.6'	S18°45.1'	26	2	206°53.5'	S18°44.4'
26	3	221°53.4'	S18°43.8'	26	4	236°53.2'	S18°43.2'	26	5	251°53.1'	S18°42.5'
26	6	266°53.0'	S18°41.9'	26	7	281°52.8'	S18°41.3'	26	8	296°52.7'	S18°40.7'
26	9	311°52.6'	S18°40.0'	26	10	326°52.4'	S18°39.4'	26	11	341°52.3'	S18°38.8'
26	12	356°52.2'	S18°38.1'	26	13	11°52.0'	S18°37.5'	26	14	26°51.9'	S18°36.8'
26	15	41°51.8'	S18°36.2'	26	16	56°51.6'	S18°35.6'	26	17	71°51.5'	S18°34.9'
26	18	86°51.4'	S18°34.3'	26	19	101°51.2'	S18°33.7'	26	20	116°51.1'	S18°33.0'
26	21	131°51.0'	S18°32.4'	26	22	146°50.9'	S18°31.7'	26	23	161°50.7'	S18°31.1'
27	0	176°50.6'	S18°30.5'	27	1	191°50.5'	S18°29.8'	27	2	206°50.3'	S18°29.2'
27	3	221°50.2'	S18°28.5'	27	4	236°50.1'	S18°27.9'	27	5	251°50.0'	S18°27.2'
27	6	266°49.8'	S18°26.6'	27	7	281°49.7'	S18°25.9'	27	8	296°49.6'	S18°25.3'
27	9	311°49.5'	S18°24.7'	27	10	326°49.3'	S18°24.0'	27	11	341°49.2'	S18°23.4'
27	12	356°49.1'	S18°22.7'	27	13	11°49.0'	S18°22.1'	27	14	26°48.8'	S18°21.4'
27	15	41°48.7'	S18°20.8'	27	16	56°48.6'	S18°20.1'	27	17	71°48.5'	S18°19.5'
27	18	86°48.3'	S18°18.8'	27	19	101°48.2'	S18°18.1'	27	20	116°48.1'	S18°17.5'
27	21	131°48.0'	S18°16.8'	27	22	146°47.8'	S18°16.2'	27	23	161°47.7'	S18°15.5'
28	0	176°47.6'	S18°14.9'	28	1	191°47.5'	S18°14.2'	28	2	206°47.4'	S18°13.6'
28	3	221°47.2'	S18°12.9'	28	4	236°47.1'	S18°12.2'	28	5	251°47.0'	S18°11.6'
28	6	266°46.9'	S18°10.9'	28	7	281°46.8'	S18°10.3'	28	8	296°46.7'	S18°09.6'
28	9	311°46.5'	S18°08.9'	28	10	326°46.4'	S18°08.3'	28	11	341°46.3'	S18°07.6'
28	12	356°46.2'	S18°07.0'	28	13	11°46.1'	S18°06.3'	28	14	26°46.0'	S18°05.6'
28	15	41°45.8'	S18°05.0'	28	16	56°45.7'	S18°04.3'	28	17	71°45.6'	S18°03.6'
28	18	86°45.5'	S18°03.0'	28	19	101°45.4'	S18°02.3'	28	20	116°45.3'	S18°01.6'
28	21	131°45.2'	S18°01.0'	28	22	146°45.0'	S18°00.3'	28	23	161°44.9'	S17°59.6'
29	0	176°44.8'	S17°59.0'	29	1	191°44.7'	S17°58.3'	29	2	206°44.6'	S17°57.6'
29	3	221°44.5'	S17°57.0'	29	4	236°44.4'	S17°56.3'	29	5	251°44.3'	S17°55.6'
29	6	266°44.2'	S17°54.9'	29	7	281°44.0'	S17°54.3'	29	8	296°43.9'	S17°53.6'
29	9	311°43.8'	S17°52.9'	29	10	326°43.7'	S17°52.2'	29	11	341°43.6'	S17°51.6'
29	12	356°43.5'	S17°50.9'	29	13	11°43.4'	S17°50.2'	29	14	26°43.3'	S17°49.5'
29	15	41°43.2'	S17°48.9'	29	16	56°43.1'	S17°48.2'	29	17	71°43.0'	S17°47.5'
29	18	86°42.9'	S17°46.8'	29	19	101°42.8'	S17°46.1'	29	20	116°42.7'	S17°45.5'
29	21	131°42.6'	S17°44.8'	29	22	146°42.4'	S17°44.1'	29	23	161°42.3'	S17°43.4'
30	0	176°42.2'	S17°42.7'	30	1	191°42.1'	S17°42.1'	30	2	206°42.0'	S17°41.4'

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Sun — GHA and Declination, January 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
30	3	221°41.9'	S17°40.7'	30	4	236°41.8'	S17°40.0'	30	5	251°41.7'	S17°39.3'
30	6	266°41.6'	S17°38.6'	30	7	281°41.5'	S17°37.9'	30	8	296°41.4'	S17°37.3'
30	9	311°41.3'	S17°36.6'	30	10	326°41.2'	S17°35.9'	30	11	341°41.1'	S17°35.2'
30	12	356°41.0'	S17°34.5'	30	13	11°40.9'	S17°33.8'	30	14	26°40.8'	S17°33.1'
30	15	41°40.7'	S17°32.4'	30	16	56°40.6'	S17°31.7'	30	17	71°40.5'	S17°31.1'
30	18	86°40.4'	S17°30.4'	30	19	101°40.3'	S17°29.7'	30	20	116°40.2'	S17°29.0'
30	21	131°40.1'	S17°28.3'	30	22	146°40.1'	S17°27.6'	30	23	161°40.0'	S17°26.9'
31	0	176°39.9'	S17°26.2'	31	1	191°39.8'	S17°25.5'	31	2	206°39.7'	S17°24.8'
31	3	221°39.6'	S17°24.1'	31	4	236°39.5'	S17°23.4'	31	5	251°39.4'	S17°22.7'
31	6	266°39.3'	S17°22.0'	31	7	281°39.2'	S17°21.3'	31	8	296°39.1'	S17°20.6'
31	9	311°39.0'	S17°19.9'	31	10	326°38.9'	S17°19.2'	31	11	341°38.8'	S17°18.5'
31	12	356°38.7'	S17°17.8'	31	13	11°38.7'	S17°17.1'	31	14	26°38.6'	S17°16.4'
31	15	41°38.5'	S17°15.7'	31	16	56°38.4'	S17°15.0'	31	17	71°38.3'	S17°14.3'
31	18	86°38.2'	S17°13.6'	31	19	101°38.1'	S17°12.9'	31	20	116°38.0'	S17°12.2'
31	21	131°37.9'	S17°11.5'	31	22	146°37.9'	S17°10.8'	31	23	161°37.8'	S17°10.1'

Table 9.2: Sun — GHA and Declination, February 2026 (UT)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
1	0	176°37.7'	S17°09.3'	1	1	191°37.6'	S17°08.6'	1	2	206°37.5'	S17°07.9'
1	3	221°37.4'	S17°07.2'	1	4	236°37.3'	S17°06.5'	1	5	251°37.3'	S17°05.8'
1	6	266°37.2'	S17°05.1'	1	7	281°37.1'	S17°04.4'	1	8	296°37.0'	S17°03.7'
1	9	311°36.9'	S17°02.9'	1	10	326°36.8'	S17°02.2'	1	11	341°36.8'	S17°01.5'
1	12	356°36.7'	S17°00.8'	1	13	11°36.6'	S17°00.1'	1	14	26°36.5'	S16°59.4'
1	15	41°36.4'	S16°58.7'	1	16	56°36.3'	S16°57.9'	1	17	71°36.3'	S16°57.2'
1	18	86°36.2'	S16°56.5'	1	19	101°36.1'	S16°55.8'	1	20	116°36.0'	S16°55.1'
1	21	131°36.0'	S16°54.3'	1	22	146°35.9'	S16°53.6'	1	23	161°35.8'	S16°52.9'
2	0	176°35.7'	S16°52.2'	2	1	191°35.6'	S16°51.5'	2	2	206°35.6'	S16°50.7'
2	3	221°35.5'	S16°50.0'	2	4	236°35.4'	S16°49.3'	2	5	251°35.3'	S16°48.6'
2	6	266°35.3'	S16°47.9'	2	7	281°35.2'	S16°47.1'	2	8	296°35.1'	S16°46.4'
2	9	311°35.0'	S16°45.7'	2	10	326°35.0'	S16°45.0'	2	11	341°34.9'	S16°44.2'
2	12	356°34.8'	S16°43.5'	2	13	11°34.7'	S16°42.8'	2	14	26°34.7'	S16°42.0'
2	15	41°34.6'	S16°41.3'	2	16	56°34.5'	S16°40.6'	2	17	71°34.4'	S16°39.9'
2	18	86°34.4'	S16°39.1'	2	19	101°34.3'	S16°38.4'	2	20	116°34.2'	S16°37.7'
2	21	131°34.2'	S16°36.9'	2	22	146°34.1'	S16°36.2'	2	23	161°34.0'	S16°35.5'
3	0	176°33.9'	S16°34.7'	3	1	191°33.9'	S16°34.0'	3	2	206°33.8'	S16°33.3'
3	3	221°33.7'	S16°32.5'	3	4	236°33.7'	S16°31.8'	3	5	251°33.6'	S16°31.1'
3	6	266°33.5'	S16°30.3'	3	7	281°33.5'	S16°29.6'	3	8	296°33.4'	S16°28.9'
3	9	311°33.3'	S16°28.1'	3	10	326°33.3'	S16°27.4'	3	11	341°33.2'	S16°26.6'
3	12	356°33.1'	S16°25.9'	3	13	11°33.1'	S16°25.2'	3	14	26°33.0'	S16°24.4'
3	15	41°32.9'	S16°23.7'	3	16	56°32.9'	S16°22.9'	3	17	71°32.8'	S16°22.2'
3	18	86°32.8'	S16°21.5'	3	19	101°32.7'	S16°20.7'	3	20	116°32.6'	S16°20.0'
3	21	131°32.6'	S16°19.2'	3	22	146°32.5'	S16°18.5'	3	23	161°32.4'	S16°17.7'
4	0	176°32.4'	S16°17.0'	4	1	191°32.3'	S16°16.3'	4	2	206°32.3'	S16°15.5'
4	3	221°32.2'	S16°14.8'	4	4	236°32.1'	S16°14.0'	4	5	251°32.1'	S16°13.3'
4	6	266°32.0'	S16°12.5'	4	7	281°32.0'	S16°11.8'	4	8	296°31.9'	S16°11.0'
4	9	311°31.9'	S16°10.3'	4	10	326°31.8'	S16°09.5'	4	11	341°31.7'	S16°08.8'
4	12	356°31.7'	S16°08.0'	4	13	11°31.6'	S16°07.3'	4	14	26°31.6'	S16°06.5'
4	15	41°31.5'	S16°05.8'	4	16	56°31.5'	S16°05.0'	4	17	71°31.4'	S16°04.3'
4	18	86°31.3'	S16°03.5'	4	19	101°31.3'	S16°02.8'	4	20	116°31.2'	S16°02.0'
4	21	131°31.2'	S16°01.2'	4	22	146°31.1'	S16°00.5'	4	23	161°31.1'	S15°59.7'
5	0	176°31.0'	S15°59.0'	5	1	191°31.0'	S15°58.2'	5	2	206°30.9'	S15°57.5'
5	3	221°30.9'	S15°56.7'	5	4	236°30.8'	S15°55.9'	5	5	251°30.8'	S15°55.2'
5	6	266°30.7'	S15°54.4'	5	7	281°30.7'	S15°53.7'	5	8	296°30.6'	S15°52.9'
5	9	311°30.6'	S15°52.1'	5	10	326°30.5'	S15°51.4'	5	11	341°30.5'	S15°50.6'
5	12	356°30.4'	S15°49.9'	5	13	11°30.4'	S15°49.1'	5	14	26°30.3'	S15°48.3'
5	15	41°30.3'	S15°47.6'	5	16	56°30.2'	S15°46.8'	5	17	71°30.2'	S15°46.0'
5	18	86°30.1'	S15°45.3'	5	19	101°30.1'	S15°44.5'	5	20	116°30.0'	S15°43.7'
5	21	131°30.0'	S15°43.0'	5	22	146°30.0'	S15°42.2'	5	23	161°29.9'	S15°41.4'
6	0	176°29.9'	S15°40.7'	6	1	191°29.8'	S15°39.9'	6	2	206°29.8'	S15°39.1'
6	3	221°29.7'	S15°38.4'	6	4	236°29.7'	S15°37.6'	6	5	251°29.6'	S15°36.8'
6	6	266°29.6'	S15°36.1'	6	7	281°29.6'	S15°35.3'	6	8	296°29.5'	S15°34.5'

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Sun — GHA and Declination, February 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
6	9	311°29.5'	S15°33.7'	6	10	326°29.4'	S15°33.0'	6	11	341°29.4'	S15°32.2'
6	12	356°29.4'	S15°31.4'	6	13	11°29.3'	S15°30.7'	6	14	26°29.3'	S15°29.9'
6	15	41°29.2'	S15°29.1'	6	16	56°29.2'	S15°28.3'	6	17	71°29.2'	S15°27.6'
6	18	86°29.1'	S15°26.8'	6	19	101°29.1'	S15°26.0'	6	20	116°29.1'	S15°25.2'
6	21	131°29.0'	S15°24.4'	6	22	146°29.0'	S15°23.7'	6	23	161°28.9'	S15°22.9'
7	0	176°28.9'	S15°22.1'	7	1	191°28.9'	S15°21.3'	7	2	206°28.8'	S15°20.6'
7	3	221°28.8'	S15°19.8'	7	4	236°28.8'	S15°19.0'	7	5	251°28.7'	S15°18.2'
7	6	266°28.7'	S15°17.4'	7	7	281°28.7'	S15°16.6'	7	8	296°28.6'	S15°15.9'
7	9	311°28.6'	S15°15.1'	7	10	326°28.6'	S15°14.3'	7	11	341°28.5'	S15°13.5'
7	12	356°28.5'	S15°12.7'	7	13	11°28.5'	S15°11.9'	7	14	26°28.4'	S15°11.2'
7	15	41°28.4'	S15°10.4'	7	16	56°28.4'	S15°09.6'	7	17	71°28.3'	S15°08.8'
7	18	86°28.3'	S15°08.0'	7	19	101°28.3'	S15°07.2'	7	20	116°28.3'	S15°06.4'
7	21	131°28.2'	S15°05.7'	7	22	146°28.2'	S15°04.9'	7	23	161°28.2'	S15°04.1'
8	0	176°28.1'	S15°03.3'	8	1	191°28.1'	S15°02.5'	8	2	206°28.1'	S15°01.7'
8	3	221°28.1'	S15°00.9'	8	4	236°28.0'	S15°00.1'	8	5	251°28.0'	S14°59.3'
8	6	266°28.0'	S14°58.5'	8	7	281°28.0'	S14°57.7'	8	8	296°27.9'	S14°57.0'
8	9	311°27.9'	S14°56.2'	8	10	326°27.9'	S14°55.4'	8	11	341°27.9'	S14°54.6'
8	12	356°27.8'	S14°53.8'	8	13	11°27.8'	S14°53.0'	8	14	26°27.8'	S14°52.2'
8	15	41°27.8'	S14°51.4'	8	16	56°27.8'	S14°50.6'	8	17	71°27.7'	S14°49.8'
8	18	86°27.7'	S14°49.0'	8	19	101°27.7'	S14°48.2'	8	20	116°27.7'	S14°47.4'
8	21	131°27.6'	S14°46.6'	8	22	146°27.6'	S14°45.8'	8	23	161°27.6'	S14°45.0'
9	0	176°27.6'	S14°44.2'	9	1	191°27.6'	S14°43.4'	9	2	206°27.6'	S14°42.6'
9	3	221°27.5'	S14°41.8'	9	4	236°27.5'	S14°41.0'	9	5	251°27.5'	S14°40.2'
9	6	266°27.5'	S14°39.4'	9	7	281°27.5'	S14°38.6'	9	8	296°27.4'	S14°37.8'
9	9	311°27.4'	S14°37.0'	9	10	326°27.4'	S14°36.2'	9	11	341°27.4'	S14°35.4'
9	12	356°27.4'	S14°34.6'	9	13	11°27.4'	S14°33.8'	9	14	26°27.4'	S14°33.0'
9	15	41°27.3'	S14°32.2'	9	16	56°27.3'	S14°31.3'	9	17	71°27.3'	S14°30.5'
9	18	86°27.3'	S14°29.7'	9	19	101°27.3'	S14°28.9'	9	20	116°27.3'	S14°28.1'
9	21	131°27.3'	S14°27.3'	9	22	146°27.2'	S14°26.5'	9	23	161°27.2'	S14°25.7'
10	0	176°27.2'	S14°24.9'	10	1	191°27.2'	S14°24.1'	10	2	206°27.2'	S14°23.3'
10	3	221°27.2'	S14°22.4'	10	4	236°27.2'	S14°21.6'	10	5	251°27.2'	S14°20.8'
10	6	266°27.2'	S14°20.0'	10	7	281°27.2'	S14°19.2'	10	8	296°27.1'	S14°18.4'
10	9	311°27.1'	S14°17.6'	10	10	326°27.1'	S14°16.7'	10	11	341°27.1'	S14°15.9'
10	12	356°27.1'	S14°15.1'	10	13	11°27.1'	S14°14.3'	10	14	26°27.1'	S14°13.5'
10	15	41°27.1'	S14°12.7'	10	16	56°27.1'	S14°11.9'	10	17	71°27.1'	S14°11.0'
10	18	86°27.1'	S14°10.2'	10	19	101°27.1'	S14°09.4'	10	20	116°27.1'	S14°08.6'
10	21	131°27.1'	S14°07.8'	10	22	146°27.1'	S14°06.9'	10	23	161°27.1'	S14°06.1'
11	0	176°27.1'	S14°05.3'	11	1	191°27.0'	S14°04.5'	11	2	206°27.0'	S14°03.7'
11	3	221°27.0'	S14°02.8'	11	4	236°27.0'	S14°02.0'	11	5	251°27.0'	S14°01.2'
11	6	266°27.0'	S14°00.4'	11	7	281°27.0'	S13°59.6'	11	8	296°27.0'	S13°58.7'
11	9	311°27.0'	S13°57.9'	11	10	326°27.0'	S13°57.1'	11	11	341°27.0'	S13°56.3'
11	12	356°27.0'	S13°55.4'	11	13	11°27.0'	S13°54.6'	11	14	26°27.0'	S13°53.8'
11	15	41°27.0'	S13°53.0'	11	16	56°27.0'	S13°52.1'	11	17	71°27.0'	S13°51.3'
11	18	86°27.0'	S13°50.5'	11	19	101°27.1'	S13°49.6'	11	20	116°27.1'	S13°48.8'
11	21	131°27.1'	S13°48.0'	11	22	146°27.1'	S13°47.2'	11	23	161°27.1'	S13°46.3'
12	0	176°27.1'	S13°45.5'	12	1	191°27.1'	S13°44.7'	12	2	206°27.1'	S13°43.8'
12	3	221°27.1'	S13°43.0'	12	4	236°27.1'	S13°42.2'	12	5	251°27.1'	S13°41.3'
12	6	266°27.1'	S13°40.5'	12	7	281°27.1'	S13°39.7'	12	8	296°27.1'	S13°38.8'
12	9	311°27.1'	S13°38.0'	12	10	326°27.1'	S13°37.2'	12	11	341°27.1'	S13°36.3'
12	12	356°27.2'	S13°35.5'	12	13	11°27.2'	S13°34.7'	12	14	26°27.2'	S13°33.8'
12	15	41°27.2'	S13°33.0'	12	16	56°27.2'	S13°32.2'	12	17	71°27.2'	S13°31.3'
12	18	86°27.2'	S13°30.5'	12	19	101°27.2'	S13°29.7'	12	20	116°27.2'	S13°28.8'
12	21	131°27.2'	S13°28.0'	12	22	146°27.3'	S13°27.1'	12	23	161°27.3'	S13°26.3'
13	0	176°27.3'	S13°25.5'	13	1	191°27.3'	S13°24.6'	13	2	206°27.3'	S13°23.8'
13	3	221°27.3'	S13°23.0'	13	4	236°27.3'	S13°22.1'	13	5	251°27.4'	S13°21.3'
13	6	266°27.4'	S13°20.4'	13	7	281°27.4'	S13°19.6'	13	8	296°27.4'	S13°18.7'
13	9	311°27.4'	S13°17.9'	13	10	326°27.4'	S13°17.1'	13	11	341°27.4'	S13°16.2'
13	12	356°27.5'	S13°15.4'	13	13	11°27.5'	S13°14.5'	13	14	26°27.5'	S13°13.7'
13	15	41°27.5'	S13°12.8'	13	16	56°27.5'	S13°12.0'	13	17	71°27.5'	S13°11.1'
13	18	86°27.6'	S13°10.3'	13	19	101°27.6'	S13°09.5'	13	20	116°27.6'	S13°08.6'
13	21	131°27.6'	S13°07.8'	13	22	146°27.6'	S13°06.9'	13	23	161°27.7'	S13°06.1'
14	0	176°27.7'	S13°05.2'	14	1	191°27.7'	S13°04.4'	14	2	206°27.7'	S13°03.5'
14	3	221°27.7'	S13°02.7'	14	4	236°27.8'	S13°01.8'	14	5	251°27.8'	S13°01.0'
14	6	266°27.8'	S13°00.1'	14	7	281°27.8'	S12°59.3'	14	8	296°27.9'	S12°58.4'

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Sun — GHA and Declination, February 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
14	9	311°27.9'	S12°57.6'	14	10	326°27.9'	S12°56.7'	14	11	341°27.9'	S12°55.9'
14	12	356°27.9'	S12°55.0'	14	13	11°28.0'	S12°54.2'	14	14	26°28.0'	S12°53.3'
14	15	41°28.0'	S12°52.5'	14	16	56°28.0'	S12°51.6'	14	17	71°28.1'	S12°50.7'
14	18	86°28.1'	S12°49.9'	14	19	101°28.1'	S12°49.0'	14	20	116°28.2'	S12°48.2'
14	21	131°28.2'	S12°47.3'	14	22	146°28.2'	S12°46.5'	14	23	161°28.2'	S12°45.6'
15	0	176°28.3'	S12°44.8'	15	1	191°28.3'	S12°43.9'	15	2	206°28.3'	S12°43.0'
15	3	221°28.3'	S12°42.2'	15	4	236°28.4'	S12°41.3'	15	5	251°28.4'	S12°40.5'
15	6	266°28.4'	S12°39.6'	15	7	281°28.5'	S12°38.8'	15	8	296°28.5'	S12°37.9'
15	9	311°28.5'	S12°37.0'	15	10	326°28.6'	S12°36.2'	15	11	341°28.6'	S12°35.3'
15	12	356°28.6'	S12°34.4'	15	13	11°28.7'	S12°33.6'	15	14	26°28.7'	S12°32.7'
15	15	41°28.7'	S12°31.9'	15	16	56°28.7'	S12°31.0'	15	17	71°28.8'	S12°30.1'
15	18	86°28.8'	S12°29.3'	15	19	101°28.9'	S12°28.4'	15	20	116°28.9'	S12°27.5'
15	21	131°28.9'	S12°26.7'	15	22	146°29.0'	S12°25.8'	15	23	161°29.0'	S12°25.0'
16	0	176°29.0'	S12°24.1'	16	1	191°29.1'	S12°23.2'	16	2	206°29.1'	S12°22.4'
16	3	221°29.1'	S12°21.5'	16	4	236°29.2'	S12°20.6'	16	5	251°29.2'	S12°19.8'
16	6	266°29.2'	S12°18.9'	16	7	281°29.3'	S12°18.0'	16	8	296°29.3'	S12°17.2'
16	9	311°29.4'	S12°16.3'	16	10	326°29.4'	S12°15.4'	16	11	341°29.4'	S12°14.5'
16	12	356°29.5'	S12°13.7'	16	13	11°29.5'	S12°12.8'	16	14	26°29.6'	S12°11.9'
16	15	41°29.6'	S12°11.1'	16	16	56°29.6'	S12°10.2'	16	17	71°29.7'	S12°09.3'
16	18	86°29.7'	S12°08.5'	16	19	101°29.8'	S12°07.6'	16	20	116°29.8'	S12°06.7'
16	21	131°29.8'	S12°05.8'	16	22	146°29.9'	S12°05.0'	16	23	161°29.9'	S12°04.1'
17	0	176°30.0'	S12°03.2'	17	1	191°30.0'	S12°02.3'	17	2	206°30.1'	S12°01.5'
17	3	221°30.1'	S12°00.6'	17	4	236°30.1'	S11°59.7'	17	5	251°30.2'	S11°58.9'
17	6	266°30.2'	S11°58.0'	17	7	281°30.3'	S11°57.1'	17	8	296°30.3'	S11°56.2'
17	9	311°30.4'	S11°55.3'	17	10	326°30.4'	S11°54.5'	17	11	341°30.5'	S11°53.6'
17	12	356°30.5'	S11°52.7'	17	13	11°30.5'	S11°51.8'	17	14	26°30.6'	S11°51.0'
17	15	41°30.6'	S11°50.1'	17	16	56°30.7'	S11°49.2'	17	17	71°30.7'	S11°48.3'
17	18	86°30.8'	S11°47.4'	17	19	101°30.8'	S11°46.6'	17	20	116°30.9'	S11°45.7'
17	21	131°30.9'	S11°44.8'	17	22	146°31.0'	S11°43.9'	17	23	161°31.0'	S11°43.0'
18	0	176°31.1'	S11°42.2'	18	1	191°31.1'	S11°41.3'	18	2	206°31.2'	S11°40.4'
18	3	221°31.2'	S11°39.5'	18	4	236°31.3'	S11°38.6'	18	5	251°31.3'	S11°37.8'
18	6	266°31.4'	S11°36.9'	18	7	281°31.4'	S11°36.0'	18	8	296°31.5'	S11°35.1'
18	9	311°31.5'	S11°34.2'	18	10	326°31.6'	S11°33.3'	18	11	341°31.7'	S11°32.4'
18	12	356°31.7'	S11°31.6'	18	13	11°31.8'	S11°30.7'	18	14	26°31.8'	S11°29.8'
18	15	41°31.9'	S11°28.9'	18	16	56°31.9'	S11°28.0'	18	17	71°32.0'	S11°27.1'
18	18	86°32.0'	S11°26.2'	18	19	101°32.1'	S11°25.4'	18	20	116°32.1'	S11°24.5'
18	21	131°32.2'	S11°23.6'	18	22	146°32.3'	S11°22.7'	18	23	161°32.3'	S11°21.8'
19	0	176°32.4'	S11°20.9'	19	1	191°32.4'	S11°20.0'	19	2	206°32.5'	S11°19.1'
19	3	221°32.5'	S11°18.3'	19	4	236°32.6'	S11°17.4'	19	5	251°32.7'	S11°16.5'
19	6	266°32.7'	S11°15.6'	19	7	281°32.8'	S11°14.7'	19	8	296°32.8'	S11°13.8'
19	9	311°32.9'	S11°12.9'	19	10	326°33.0'	S11°12.0'	19	11	341°33.0'	S11°11.1'
19	12	356°33.1'	S11°10.2'	19	13	11°33.1'	S11°09.3'	19	14	26°33.2'	S11°08.4'
19	15	41°33.3'	S11°07.6'	19	16	56°33.3'	S11°06.7'	19	17	71°33.4'	S11°05.8'
19	18	86°33.4'	S11°04.9'	19	19	101°33.5'	S11°04.0'	19	20	116°33.6'	S11°03.1'
19	21	131°33.6'	S11°02.2'	19	22	146°33.7'	S11°01.3'	19	23	161°33.8'	S11°00.4'
20	0	176°33.8'	S10°59.5'	20	1	191°33.9'	S10°58.6'	20	2	206°34.0'	S10°57.7'
20	3	221°34.0'	S10°56.8'	20	4	236°34.1'	S10°55.9'	20	5	251°34.2'	S10°55.0'
20	6	266°34.2'	S10°54.1'	20	7	281°34.3'	S10°53.2'	20	8	296°34.3'	S10°52.3'
20	9	311°34.4'	S10°51.4'	20	10	326°34.5'	S10°50.5'	20	11	341°34.5'	S10°49.6'
20	12	356°34.6'	S10°48.7'	20	13	11°34.7'	S10°47.8'	20	14	26°34.8'	S10°46.9'
20	15	41°34.8'	S10°46.0'	20	16	56°34.9'	S10°45.1'	20	17	71°35.0'	S10°44.2'
20	18	86°35.0'	S10°43.3'	20	19	101°35.1'	S10°42.4'	20	20	116°35.2'	S10°41.5'
20	21	131°35.2'	S10°40.6'	20	22	146°35.3'	S10°39.7'	20	23	161°35.4'	S10°38.8'
21	0	176°35.4'	S10°37.9'	21	1	191°35.5'	S10°37.0'	21	2	206°35.6'	S10°36.1'
21	3	221°35.7'	S10°35.2'	21	4	236°35.7'	S10°34.3'	21	5	251°35.8'	S10°33.4'
21	6	266°35.9'	S10°32.5'	21	7	281°36.0'	S10°31.6'	21	8	296°36.0'	S10°30.7'
21	9	311°36.1'	S10°29.8'	21	10	326°36.2'	S10°28.9'	21	11	341°36.2'	S10°28.0'
21	12	356°36.3'	S10°27.0'	21	13	11°36.4'	S10°26.1'	21	14	26°36.5'	S10°25.2'
21	15	41°36.5'	S10°24.3'	21	16	56°36.6'	S10°23.4'	21	17	71°36.7'	S10°22.5'
21	18	86°36.8'	S10°21.6'	21	19	101°36.8'	S10°20.7'	21	20	116°36.9'	S10°19.8'
21	21	131°37.0'	S10°18.9'	21	22	146°37.1'	S10°18.0'	21	23	161°37.2'	S10°17.1'
22	0	176°37.2'	S10°16.1'	22	1	191°37.3'	S10°15.2'	22	2	206°37.4'	S10°14.3'
22	3	221°37.5'	S10°13.4'	22	4	236°37.5'	S10°12.5'	22	5	251°37.6'	S10°11.6'
22	6	266°37.7'	S10°10.7'	22	7	281°37.8'	S10°09.8'	22	8	296°37.9'	S10°08.9'

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Sun — GHA and Declination, February 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
22	9	311°37.9'	S10°07.9'	22	10	326°38.0'	S10°07.0'	22	11	341°38.1'	S10°06.1'
22	12	356°38.2'	S10°05.2'	22	13	11°38.3'	S10°04.3'	22	14	26°38.3'	S10°03.4'
22	15	41°38.4'	S10°02.5'	22	16	56°38.5'	S10°01.6'	22	17	71°38.6'	S10°00.6'
22	18	86°38.7'	S9°59.7'	22	19	101°38.8'	S9°58.8'	22	20	116°38.8'	S9°57.9'
22	21	131°38.9'	S9°57.0'	22	22	146°39.0'	S9°56.1'	22	23	161°39.1'	S9°55.1'
23	0	176°39.2'	S9°54.2'	23	1	191°39.3'	S9°53.3'	23	2	206°39.3'	S9°52.4'
23	3	221°39.4'	S9°51.5'	23	4	236°39.5'	S9°50.6'	23	5	251°39.6'	S9°49.6'
23	6	266°39.7'	S9°48.7'	23	7	281°39.8'	S9°47.8'	23	8	296°39.9'	S9°46.9'
23	9	311°39.9'	S9°46.0'	23	10	326°40.0'	S9°45.1'	23	11	341°40.1'	S9°44.1'
23	12	356°40.2'	S9°43.2'	23	13	11°40.3'	S9°42.3'	23	14	26°40.4'	S9°41.4'
23	15	41°40.5'	S9°40.5'	23	16	56°40.5'	S9°39.5'	23	17	71°40.6'	S9°38.6'
23	18	86°40.7'	S9°37.7'	23	19	101°40.8'	S9°36.8'	23	20	116°40.9'	S9°35.8'
23	21	131°41.0'	S9°34.9'	23	22	146°41.1'	S9°34.0'	23	23	161°41.2'	S9°33.1'
24	0	176°41.3'	S9°32.2'	24	1	191°41.4'	S9°31.2'	24	2	206°41.4'	S9°30.3'
24	3	221°41.5'	S9°29.4'	24	4	236°41.6'	S9°28.5'	24	5	251°41.7'	S9°27.5'
24	6	266°41.8'	S9°26.6'	24	7	281°41.9'	S9°25.7'	24	8	296°42.0'	S9°24.8'
24	9	311°42.1'	S9°23.8'	24	10	326°42.2'	S9°22.9'	24	11	341°42.3'	S9°22.0'
24	12	356°42.4'	S9°21.1'	24	13	11°42.5'	S9°20.1'	24	14	26°42.6'	S9°19.2'
24	15	41°42.6'	S9°18.3'	24	16	56°42.7'	S9°17.4'	24	17	71°42.8'	S9°16.4'
24	18	86°42.9'	S9°15.5'	24	19	101°43.0'	S9°14.6'	24	20	116°43.1'	S9°13.7'
24	21	131°43.2'	S9°12.7'	24	22	146°43.3'	S9°11.8'	24	23	161°43.4'	S9°10.9'
25	0	176°43.5'	S9°10.0'	25	1	191°43.6'	S9°09.0'	25	2	206°43.7'	S9°08.1'
25	3	221°43.8'	S9°07.2'	25	4	236°43.9'	S9°06.2'	25	5	251°44.0'	S9°05.3'
25	6	266°44.1'	S9°04.4'	25	7	281°44.2'	S9°03.4'	25	8	296°44.3'	S9°02.5'
25	9	311°44.4'	S9°01.6'	25	10	326°44.5'	S9°00.7'	25	11	341°44.6'	S8°59.7'
25	12	356°44.7'	S8°58.8'	25	13	11°44.8'	S8°57.9'	25	14	26°44.9'	S8°56.9'
25	15	41°45.0'	S8°56.0'	25	16	56°45.1'	S8°55.1'	25	17	71°45.2'	S8°54.1'
25	18	86°45.3'	S8°53.2'	25	19	101°45.4'	S8°52.3'	25	20	116°45.5'	S8°51.3'
25	21	131°45.6'	S8°50.4'	25	22	146°45.7'	S8°49.5'	25	23	161°45.8'	S8°48.5'
26	0	176°45.9'	S8°47.6'	26	1	191°46.0'	S8°46.7'	26	2	206°46.1'	S8°45.7'
26	3	221°46.2'	S8°44.8'	26	4	236°46.3'	S8°43.9'	26	5	251°46.4'	S8°42.9'
26	6	266°46.5'	S8°42.0'	26	7	281°46.6'	S8°41.1'	26	8	296°46.7'	S8°40.1'
26	9	311°46.8'	S8°39.2'	26	10	326°46.9'	S8°38.3'	26	11	341°47.0'	S8°37.3'
26	12	356°47.1'	S8°36.4'	26	13	11°47.2'	S8°35.4'	26	14	26°47.4'	S8°34.5'
26	15	41°47.5'	S8°33.6'	26	16	56°47.6'	S8°32.6'	26	17	71°47.7'	S8°31.7'
26	18	86°47.8'	S8°30.8'	26	19	101°47.9'	S8°29.8'	26	20	116°48.0'	S8°28.9'
26	21	131°48.1'	S8°27.9'	26	22	146°48.2'	S8°27.0'	26	23	161°48.3'	S8°26.1'
27	0	176°48.4'	S8°25.1'	27	1	191°48.5'	S8°24.2'	27	2	206°48.6'	S8°23.2'
27	3	221°48.7'	S8°22.3'	27	4	236°48.9'	S8°21.4'	27	5	251°49.0'	S8°20.4'
27	6	266°49.1'	S8°19.5'	27	7	281°49.2'	S8°18.5'	27	8	296°49.3'	S8°17.6'
27	9	311°49.4'	S8°16.7'	27	10	326°49.5'	S8°15.7'	27	11	341°49.6'	S8°14.8'
27	12	356°49.7'	S8°13.8'	27	13	11°49.8'	S8°12.9'	27	14	26°50.0'	S8°12.0'
27	15	41°50.1'	S8°11.0'	27	16	56°50.2'	S8°10.1'	27	17	71°50.3'	S8°09.1'
27	18	86°50.4'	S8°08.2'	27	19	101°50.5'	S8°07.2'	27	20	116°50.6'	S8°06.3'
27	21	131°50.7'	S8°05.4'	27	22	146°50.9'	S8°04.4'	27	23	161°51.0'	S8°03.5'
28	0	176°51.1'	S8°02.5'	28	1	191°51.2'	S8°01.6'	28	2	206°51.3'	S8°00.6'
28	3	221°51.4'	S7°59.7'	28	4	236°51.5'	S7°58.7'	28	5	251°51.7'	S7°57.8'
28	6	266°51.8'	S7°56.9'	28	7	281°51.9'	S7°55.9'	28	8	296°52.0'	S7°55.0'
28	9	311°52.1'	S7°54.0'	28	10	326°52.2'	S7°53.1'	28	11	341°52.4'	S7°52.1'
28	12	356°52.5'	S7°51.2'	28	13	11°52.6'	S7°50.2'	28	14	26°52.7'	S7°49.3'
28	15	41°52.8'	S7°48.3'	28	16	56°52.9'	S7°47.4'	28	17	71°53.1'	S7°46.4'
28	18	86°53.2'	S7°45.5'	28	19	101°53.3'	S7°44.5'	28	20	116°53.4'	S7°43.6'
28	21	131°53.5'	S7°42.6'	28	22	146°53.6'	S7°41.7'	28	23	161°53.8'	S7°40.7'

Table 9.3: Sun — GHA and Declination, March 2026 (UT)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
1	0	176°53.9'	S7°39.8'	1	1	191°54.0'	S7°38.9'	1	2	206°54.1'	S7°37.9'
1	3	221°54.2'	S7°37.0'	1	4	236°54.4'	S7°36.0'	1	5	251°54.5'	S7°35.1'
1	6	266°54.6'	S7°34.1'	1	7	281°54.7'	S7°33.2'	1	8	296°54.8'	S7°32.2'
1	9	311°55.0'	S7°31.3'	1	10	326°55.1'	S7°30.3'	1	11	341°55.2'	S7°29.3'
1	12	356°55.3'	S7°28.4'	1	13	11°55.5'	S7°27.4'	1	14	26°55.6'	S7°26.5'

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Sun — GHA and Declination, March 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
1	15	41°55.7'	S7°25.5'	1	16	56°55.8'	S7°24.6'	1	17	71°55.9'	S7°23.6'
1	18	86°56.1'	S7°22.7'	1	19	101°56.2'	S7°21.7'	1	20	116°56.3'	S7°20.8'
1	21	131°56.4'	S7°19.8'	1	22	146°56.6'	S7°18.9'	1	23	161°56.7'	S7°17.9'
2	0	176°56.8'	S7°17.0'	2	1	191°56.9'	S7°16.0'	2	2	206°57.1'	S7°15.1'
2	3	221°57.2'	S7°14.1'	2	4	236°57.3'	S7°13.2'	2	5	251°57.4'	S7°12.2'
2	6	266°57.6'	S7°11.2'	2	7	281°57.7'	S7°10.3'	2	8	296°57.8'	S7°09.3'
2	9	311°57.9'	S7°08.4'	2	10	326°58.1'	S7°07.4'	2	11	341°58.2'	S7°06.5'
2	12	356°58.3'	S7°05.5'	2	13	11°58.4'	S7°04.6'	2	14	26°58.6'	S7°03.6'
2	15	41°58.7'	S7°02.6'	2	16	56°58.8'	S7°01.7'	2	17	71°59.0'	S7°00.7'
2	18	86°59.1'	S6°59.8'	2	19	101°59.2'	S6°58.8'	2	20	116°59.3'	S6°57.9'
2	21	131°59.5'	S6°56.9'	2	22	146°59.6'	S6°55.9'	2	23	161°59.7'	S6°55.0'
3	0	176°59.9'	S6°54.0'	3	1	192°00.0'	S6°53.1'	3	2	207°00.1'	S6°52.1'
3	3	222°00.2'	S6°51.2'	3	4	237°00.4'	S6°50.2'	3	5	252°00.5'	S6°49.2'
3	6	267°00.6'	S6°48.3'	3	7	282°00.8'	S6°47.3'	3	8	297°00.9'	S6°46.4'
3	9	312°01.0'	S6°45.4'	3	10	327°01.2'	S6°44.4'	3	11	342°01.3'	S6°43.5'
3	12	357°01.4'	S6°42.5'	3	13	12°01.6'	S6°41.6'	3	14	27°01.7'	S6°40.6'
3	15	42°01.8'	S6°39.6'	3	16	57°02.0'	S6°38.7'	3	17	72°02.1'	S6°37.7'
3	18	87°02.2'	S6°36.8'	3	19	102°02.4'	S6°35.8'	3	20	117°02.5'	S6°34.8'
3	21	132°02.6'	S6°33.9'	3	22	147°02.8'	S6°32.9'	3	23	162°02.9'	S6°32.0'
4	0	177°03.0'	S6°31.0'	4	1	192°03.2'	S6°30.0'	4	2	207°03.3'	S6°29.1'
4	3	222°03.4'	S6°28.1'	4	4	237°03.6'	S6°27.2'	4	5	252°03.7'	S6°26.2'
4	6	267°03.8'	S6°25.2'	4	7	282°04.0'	S6°24.3'	4	8	297°04.1'	S6°23.3'
4	9	312°04.2'	S6°22.3'	4	10	327°04.4'	S6°21.4'	4	11	342°04.5'	S6°20.4'
4	12	357°04.7'	S6°19.5'	4	13	12°04.8'	S6°18.5'	4	14	27°04.9'	S6°17.5'
4	15	42°05.1'	S6°16.6'	4	16	57°05.2'	S6°15.6'	4	17	72°05.3'	S6°14.6'
4	18	87°05.5'	S6°13.7'	4	19	102°05.6'	S6°12.7'	4	20	117°05.8'	S6°11.7'
4	21	132°05.9'	S6°10.8'	4	22	147°06.0'	S6°09.8'	4	23	162°06.2'	S6°08.8'
5	0	177°06.3'	S6°07.9'	5	1	192°06.4'	S6°06.9'	5	2	207°06.6'	S6°05.9'
5	3	222°06.7'	S6°05.0'	5	4	237°06.9'	S6°04.0'	5	5	252°07.0'	S6°03.0'
5	6	267°07.1'	S6°02.1'	5	7	282°07.3'	S6°01.1'	5	8	297°07.4'	S6°00.2'
5	9	312°07.6'	S5°59.2'	5	10	327°07.7'	S5°58.2'	5	11	342°07.8'	S5°57.3'
5	12	357°08.0'	S5°56.3'	5	13	12°08.1'	S5°55.3'	5	14	27°08.3'	S5°54.3'
5	15	42°08.4'	S5°53.4'	5	16	57°08.6'	S5°52.4'	5	17	72°08.7'	S5°51.4'
5	18	87°08.8'	S5°50.5'	5	19	102°09.0'	S5°49.5'	5	20	117°09.1'	S5°48.5'
5	21	132°09.3'	S5°47.6'	5	22	147°09.4'	S5°46.6'	5	23	162°09.6'	S5°45.6'
6	0	177°09.7'	S5°44.7'	6	1	192°09.8'	S5°43.7'	6	2	207°10.0'	S5°42.7'
6	3	222°10.1'	S5°41.8'	6	4	237°10.3'	S5°40.8'	6	5	252°10.4'	S5°39.8'
6	6	267°10.6'	S5°38.9'	6	7	282°10.7'	S5°37.9'	6	8	297°10.8'	S5°36.9'
6	9	312°11.0'	S5°35.9'	6	10	327°11.1'	S5°35.0'	6	11	342°11.3'	S5°34.0'
6	12	357°11.4'	S5°33.0'	6	13	12°11.6'	S5°32.1'	6	14	27°11.7'	S5°31.1'
6	15	42°11.9'	S5°30.1'	6	16	57°12.0'	S5°29.2'	6	17	72°12.2'	S5°28.2'
6	18	87°12.3'	S5°27.2'	6	19	102°12.5'	S5°26.2'	6	20	117°12.6'	S5°25.3'
6	21	132°12.7'	S5°24.3'	6	22	147°12.9'	S5°23.3'	6	23	162°13.0'	S5°22.4'
7	0	177°13.2'	S5°21.4'	7	1	192°13.3'	S5°20.4'	7	2	207°13.5'	S5°19.4'
7	3	222°13.6'	S5°18.5'	7	4	237°13.8'	S5°17.5'	7	5	252°13.9'	S5°16.5'
7	6	267°14.1'	S5°15.6'	7	7	282°14.2'	S5°14.6'	7	8	297°14.4'	S5°13.6'
7	9	312°14.5'	S5°12.6'	7	10	327°14.7'	S5°11.7'	7	11	342°14.8'	S5°10.7'
7	12	357°15.0'	S5°09.7'	7	13	12°15.1'	S5°08.7'	7	14	27°15.3'	S5°07.8'
7	15	42°15.4'	S5°06.8'	7	16	57°15.6'	S5°05.8'	7	17	72°15.7'	S5°04.8'
7	18	87°15.9'	S5°03.9'	7	19	102°16.0'	S5°02.9'	7	20	117°16.2'	S5°01.9'
7	21	132°16.3'	S5°00.9'	7	22	147°16.5'	S5°00.0'	7	23	162°16.6'	S4°59.0'
8	0	177°16.8'	S4°58.0'	8	1	192°16.9'	S4°57.1'	8	2	207°17.1'	S4°56.1'
8	3	222°17.2'	S4°55.1'	8	4	237°17.4'	S4°54.1'	8	5	252°17.5'	S4°53.2'
8	6	267°17.7'	S4°52.2'	8	7	282°17.8'	S4°51.2'	8	8	297°18.0'	S4°50.2'
8	9	312°18.2'	S4°49.3'	8	10	327°18.3'	S4°48.3'	8	11	342°18.5'	S4°47.3'
8	12	357°18.6'	S4°46.3'	8	13	12°18.8'	S4°45.3'	8	14	27°18.9'	S4°44.4'
8	15	42°19.1'	S4°43.4'	8	16	57°19.2'	S4°42.4'	8	17	72°19.4'	S4°41.4'
8	18	87°19.5'	S4°40.5'	8	19	102°19.7'	S4°39.5'	8	20	117°19.9'	S4°38.5'
8	21	132°20.0'	S4°37.5'	8	22	147°20.2'	S4°36.6'	8	23	162°20.3'	S4°35.6'
9	0	177°20.5'	S4°34.6'	9	1	192°20.6'	S4°33.6'	9	2	207°20.8'	S4°32.6'
9	3	222°20.9'	S4°31.7'	9	4	237°21.1'	S4°30.7'	9	5	252°21.3'	S4°29.7'
9	6	267°21.4'	S4°28.7'	9	7	282°21.6'	S4°27.8'	9	8	297°21.7'	S4°26.8'
9	9	312°21.9'	S4°25.8'	9	10	327°22.0'	S4°24.8'	9	11	342°22.2'	S4°23.8'
9	12	357°22.3'	S4°22.9'	9	13	12°22.5'	S4°21.9'	9	14	27°22.7'	S4°20.9'

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Sun — GHA and Declination, March 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
9	15	42°22.8'	S4°19.9'	9	16	57°23.0'	S4°19.0'	9	17	72°23.1'	S4°18.0'
9	18	87°23.3'	S4°17.0'	9	19	102°23.5'	S4°16.0'	9	20	117°23.6'	S4°15.0'
9	21	132°23.8'	S4°14.1'	9	22	147°23.9'	S4°13.1'	9	23	162°24.1'	S4°12.1'
10	0	177°24.2'	S4°11.1'	10	1	192°24.4'	S4°10.1'	10	2	207°24.6'	S4°09.2'
10	3	222°24.7'	S4°08.2'	10	4	237°24.9'	S4°07.2'	10	5	252°25.0'	S4°06.2'
10	6	267°25.2'	S4°05.2'	10	7	282°25.4'	S4°04.3'	10	8	297°25.5'	S4°03.3'
10	9	312°25.7'	S4°02.3'	10	10	327°25.8'	S4°01.3'	10	11	342°26.0'	S4°00.3'
10	12	357°26.2'	S3°59.4'	10	13	12°26.3'	S3°58.4'	10	14	27°26.5'	S3°57.4'
10	15	42°26.7'	S3°56.4'	10	16	57°26.8'	S3°55.4'	10	17	72°27.0'	S3°54.5'
10	18	87°27.1'	S3°53.5'	10	19	102°27.3'	S3°52.5'	10	20	117°27.5'	S3°51.5'
10	21	132°27.6'	S3°50.5'	10	22	147°27.8'	S3°49.6'	10	23	162°27.9'	S3°48.6'
11	0	177°28.1'	S3°47.6'	11	1	192°28.3'	S3°46.6'	11	2	207°28.4'	S3°45.6'
11	3	222°28.6'	S3°44.6'	11	4	237°28.8'	S3°43.7'	11	5	252°28.9'	S3°42.7'
11	6	267°29.1'	S3°41.7'	11	7	282°29.3'	S3°40.7'	11	8	297°29.4'	S3°39.7'
11	9	312°29.6'	S3°38.8'	11	10	327°29.7'	S3°37.8'	11	11	342°29.9'	S3°36.8'
11	12	357°30.1'	S3°35.8'	11	13	12°30.2'	S3°34.8'	11	14	27°30.4'	S3°33.8'
11	15	42°30.6'	S3°32.9'	11	16	57°30.7'	S3°31.9'	11	17	72°30.9'	S3°30.9'
11	18	87°31.1'	S3°29.9'	11	19	102°31.2'	S3°28.9'	11	20	117°31.4'	S3°27.9'
11	21	132°31.6'	S3°27.0'	11	22	147°31.7'	S3°26.0'	11	23	162°31.9'	S3°25.0'
12	0	177°32.1'	S3°24.0'	12	1	192°32.2'	S3°23.0'	12	2	207°32.4'	S3°22.0'
12	3	222°32.6'	S3°21.1'	12	4	237°32.7'	S3°20.1'	12	5	252°32.9'	S3°19.1'
12	6	267°33.1'	S3°18.1'	12	7	282°33.2'	S3°17.1'	12	8	297°33.4'	S3°16.1'
12	9	312°33.6'	S3°15.2'	12	10	327°33.7'	S3°14.2'	12	11	342°33.9'	S3°13.2'
12	12	357°34.1'	S3°12.2'	12	13	12°34.2'	S3°11.2'	12	14	27°34.4'	S3°10.2'
12	15	42°34.6'	S3°09.3'	12	16	57°34.7'	S3°08.3'	12	17	72°34.9'	S3°07.3'
12	18	87°35.1'	S3°06.3'	12	19	102°35.2'	S3°05.3'	12	20	117°35.4'	S3°04.3'
12	21	132°35.6'	S3°03.3'	12	22	147°35.7'	S3°02.4'	12	23	162°35.9'	S3°01.4'
13	0	177°36.1'	S3°00.4'	13	1	192°36.2'	S2°59.4'	13	2	207°36.4'	S2°58.4'
13	3	222°36.6'	S2°57.4'	13	4	237°36.7'	S2°56.4'	13	5	252°36.9'	S2°55.5'
13	6	267°37.1'	S2°54.5'	13	7	282°37.3'	S2°53.5'	13	8	297°37.4'	S2°52.5'
13	9	312°37.6'	S2°51.5'	13	10	327°37.8'	S2°50.5'	13	11	342°37.9'	S2°49.6'
13	12	357°38.1'	S2°48.6'	13	13	12°38.3'	S2°47.6'	13	14	27°38.4'	S2°46.6'
13	15	42°38.6'	S2°45.6'	13	16	57°38.8'	S2°44.6'	13	17	72°39.0'	S2°43.6'
13	18	87°39.1'	S2°42.7'	13	19	102°39.3'	S2°41.7'	13	20	117°39.5'	S2°40.7'
13	21	132°39.6'	S2°39.7'	13	22	147°39.8'	S2°38.7'	13	23	162°40.0'	S2°37.7'
14	0	177°40.2'	S2°36.7'	14	1	192°40.3'	S2°35.8'	14	2	207°40.5'	S2°34.8'
14	3	222°40.7'	S2°33.8'	14	4	237°40.8'	S2°32.8'	14	5	252°41.0'	S2°31.8'
14	6	267°41.2'	S2°30.8'	14	7	282°41.4'	S2°29.8'	14	8	297°41.5'	S2°28.8'
14	9	312°41.7'	S2°27.9'	14	10	327°41.9'	S2°26.9'	14	11	342°42.1'	S2°25.9'
14	12	357°42.2'	S2°24.9'	14	13	12°42.4'	S2°23.9'	14	14	27°42.6'	S2°22.9'
14	15	42°42.7'	S2°21.9'	14	16	57°42.9'	S2°21.0'	14	17	72°43.1'	S2°20.0'
14	18	87°43.3'	S2°19.0'	14	19	102°43.4'	S2°18.0'	14	20	117°43.6'	S2°17.0'
14	21	132°43.8'	S2°16.0'	14	22	147°44.0'	S2°15.0'	14	23	162°44.1'	S2°14.0'
15	0	177°44.3'	S2°13.1'	15	1	192°44.5'	S2°12.1'	15	2	207°44.7'	S2°11.1'
15	3	222°44.8'	S2°10.1'	15	4	237°45.0'	S2°09.1'	15	5	252°45.2'	S2°08.1'
15	6	267°45.4'	S2°07.1'	15	7	282°45.5'	S2°06.1'	15	8	297°45.7'	S2°05.2'
15	9	312°45.9'	S2°04.2'	15	10	327°46.1'	S2°03.2'	15	11	342°46.2'	S2°02.2'
15	12	357°46.4'	S2°01.2'	15	13	12°46.6'	S2°00.2'	15	14	27°46.8'	S1°59.2'
15	15	42°46.9'	S1°58.2'	15	16	57°47.1'	S1°57.3'	15	17	72°47.3'	S1°56.3'
15	18	87°47.5'	S1°55.3'	15	19	102°47.6'	S1°54.3'	15	20	117°47.8'	S1°53.3'
15	21	132°48.0'	S1°52.3'	15	22	147°48.2'	S1°51.3'	15	23	162°48.3'	S1°50.3'
16	0	177°48.5'	S1°49.4'	16	1	192°48.7'	S1°48.4'	16	2	207°48.9'	S1°47.4'
16	3	222°49.0'	S1°46.4'	16	4	237°49.2'	S1°45.4'	16	5	252°49.4'	S1°44.4'
16	6	267°49.6'	S1°43.4'	16	7	282°49.8'	S1°42.4'	16	8	297°49.9'	S1°41.5'
16	9	312°50.1'	S1°40.5'	16	10	327°50.3'	S1°39.5'	16	11	342°50.5'	S1°38.5'
16	12	357°50.6'	S1°37.5'	16	13	12°50.8'	S1°36.5'	16	14	27°51.0'	S1°35.5'
16	15	42°51.2'	S1°34.5'	16	16	57°51.4'	S1°33.5'	16	17	72°51.5'	S1°32.6'
16	18	87°51.7'	S1°31.6'	16	19	102°51.9'	S1°30.6'	16	20	117°52.1'	S1°29.6'
16	21	132°52.2'	S1°28.6'	16	22	147°52.4'	S1°27.6'	16	23	162°52.6'	S1°26.6'
17	0	177°52.8'	S1°25.6'	17	1	192°53.0'	S1°24.7'	17	2	207°53.1'	S1°23.7'
17	3	222°53.3'	S1°22.7'	17	4	237°53.5'	S1°21.7'	17	5	252°53.7'	S1°20.7'
17	6	267°53.9'	S1°19.7'	17	7	282°54.0'	S1°18.7'	17	8	297°54.2'	S1°17.7'
17	9	312°54.4'	S1°16.7'	17	10	327°54.6'	S1°15.8'	17	11	342°54.8'	S1°14.8'
17	12	357°54.9'	S1°13.8'	17	13	12°55.1'	S1°12.8'	17	14	27°55.3'	S1°11.8'

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Sun — GHA and Declination, March 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
17	15	42°55.5'	S1°10.8'	17	16	57°55.7'	S1°09.8'	17	17	72°55.8'	S1°08.8'
17	18	87°56.0'	S1°07.8'	17	19	102°56.2'	S1°06.9'	17	20	117°56.4'	S1°05.9'
17	21	132°56.6'	S1°04.9'	17	22	147°56.7'	S1°03.9'	17	23	162°56.9'	S1°02.9'
18	0	177°57.1'	S1°01.9'	18	1	192°57.3'	S1°00.9'	18	2	207°57.5'	S0°59.9'
18	3	222°57.6'	S0°58.9'	18	4	237°57.8'	S0°58.0'	18	5	252°58.0'	S0°57.0'
18	6	267°58.2'	S0°56.0'	18	7	282°58.4'	S0°55.0'	18	8	297°58.5'	S0°54.0'
18	9	312°58.7'	S0°53.0'	18	10	327°58.9'	S0°52.0'	18	11	342°59.1'	S0°51.0'
18	12	357°59.3'	S0°50.1'	18	13	12°59.5'	S0°49.1'	18	14	27°59.6'	S0°48.1'
18	15	42°59.8'	S0°47.1'	18	16	58°00.0'	S0°46.1'	18	17	73°00.2'	S0°45.1'
18	18	88°00.4'	S0°44.1'	18	19	103°00.5'	S0°43.1'	18	20	118°00.7'	S0°42.1'
18	21	133°00.9'	S0°41.2'	18	22	148°01.1'	S0°40.2'	18	23	163°01.3'	S0°39.2'
19	0	178°01.5'	S0°38.2'	19	1	193°01.6'	S0°37.2'	19	2	208°01.8'	S0°36.2'
19	3	223°02.0'	S0°35.2'	19	4	238°02.2'	S0°34.2'	19	5	253°02.4'	S0°33.2'
19	6	268°02.5'	S0°32.3'	19	7	283°02.7'	S0°31.3'	19	8	298°02.9'	S0°30.3'
19	9	313°03.1'	S0°29.3'	19	10	328°03.3'	S0°28.3'	19	11	343°03.5'	S0°27.3'
19	12	358°03.6'	S0°26.3'	19	13	13°03.8'	S0°25.3'	19	14	28°04.0'	S0°24.3'
19	15	43°04.2'	S0°23.4'	19	16	58°04.4'	S0°22.4'	19	17	73°04.6'	S0°21.4'
19	18	88°04.7'	S0°20.4'	19	19	103°04.9'	S0°19.4'	19	20	118°05.1'	S0°18.4'
19	21	133°05.3'	S0°17.4'	19	22	148°05.5'	S0°16.4'	19	23	163°05.7'	S0°15.5'
20	0	178°05.9'	S0°14.5'	20	1	193°06.0'	S0°13.5'	20	2	208°06.2'	S0°12.5'
20	3	223°06.4'	S0°11.5'	20	4	238°06.6'	S0°10.5'	20	5	253°06.8'	S0°09.5'
20	6	268°07.0'	S0°08.5'	20	7	283°07.1'	S0°07.5'	20	8	298°07.3'	S0°06.6'
20	9	313°07.5'	S0°05.6'	20	10	328°07.7'	S0°04.6'	20	11	343°07.9'	S0°03.6'
20	12	358°08.1'	S0°02.6'	20	13	13°08.2'	S0°01.6'	20	14	28°08.4'	S0°00.6'
20	15	43°08.6'	N0°00.4'	20	16	58°08.8'	N0°01.3'	20	17	73°09.0'	N0°02.3'
20	18	88°09.2'	N0°03.3'	20	19	103°09.4'	N0°04.3'	20	20	118°09.5'	N0°05.3'
20	21	133°09.7'	N0°06.3'	20	22	148°09.9'	N0°07.3'	20	23	163°10.1'	N0°08.3'
21	0	178°10.3'	N0°09.2'	21	1	193°10.5'	N0°10.2'	21	2	208°10.7'	N0°11.2'
21	3	223°10.8'	N0°12.2'	21	4	238°11.0'	N0°13.2'	21	5	253°11.2'	N0°14.2'
21	6	268°11.4'	N0°15.2'	21	7	283°11.6'	N0°16.2'	21	8	298°11.8'	N0°17.1'
21	9	313°12.0'	N0°18.1'	21	10	328°12.1'	N0°19.1'	21	11	343°12.3'	N0°20.1'
21	12	358°12.5'	N0°21.1'	21	13	13°12.7'	N0°22.1'	21	14	28°12.9'	N0°23.1'
21	15	43°13.1'	N0°24.1'	21	16	58°13.3'	N0°25.0'	21	17	73°13.4'	N0°26.0'
21	18	88°13.6'	N0°27.0'	21	19	103°13.8'	N0°28.0'	21	20	118°14.0'	N0°29.0'
21	21	133°14.2'	N0°30.0'	21	22	148°14.4'	N0°31.0'	21	23	163°14.6'	N0°32.0'
22	0	178°14.7'	N0°32.9'	22	1	193°14.9'	N0°33.9'	22	2	208°15.1'	N0°34.9'
22	3	223°15.3'	N0°35.9'	22	4	238°15.5'	N0°36.9'	22	5	253°15.7'	N0°37.9'
22	6	268°15.9'	N0°38.9'	22	7	283°16.0'	N0°39.9'	22	8	298°16.2'	N0°40.8'
22	9	313°16.4'	N0°41.8'	22	10	328°16.6'	N0°42.8'	22	11	343°16.8'	N0°43.8'
22	12	358°17.0'	N0°44.8'	22	13	13°17.2'	N0°45.8'	22	14	28°17.4'	N0°46.8'
22	15	43°17.5'	N0°47.7'	22	16	58°17.7'	N0°48.7'	22	17	73°17.9'	N0°49.7'
22	18	88°18.1'	N0°50.7'	22	19	103°18.3'	N0°51.7'	22	20	118°18.5'	N0°52.7'
22	21	133°18.7'	N0°53.7'	22	22	148°18.8'	N0°54.7'	22	23	163°19.0'	N0°55.6'
23	0	178°19.2'	N0°56.6'	23	1	193°19.4'	N0°57.6'	23	2	208°19.6'	N0°58.6'
23	3	223°19.8'	N0°59.6'	23	4	238°20.0'	N1°00.6'	23	5	253°20.2'	N1°01.6'
23	6	268°20.3'	N1°02.5'	23	7	283°20.5'	N1°03.5'	23	8	298°20.7'	N1°04.5'
23	9	313°20.9'	N1°05.5'	23	10	328°21.1'	N1°06.5'	23	11	343°21.3'	N1°07.5'
23	12	358°21.5'	N1°08.5'	23	13	13°21.7'	N1°09.4'	23	14	28°21.8'	N1°10.4'
23	15	43°22.0'	N1°11.4'	23	16	58°22.2'	N1°12.4'	23	17	73°22.4'	N1°13.4'
23	18	88°22.6'	N1°14.4'	23	19	103°22.8'	N1°15.3'	23	20	118°23.0'	N1°16.3'
23	21	133°23.2'	N1°17.3'	23	22	148°23.3'	N1°18.3'	23	23	163°23.5'	N1°19.3'
24	0	178°23.7'	N1°20.3'	24	1	193°23.9'	N1°21.3'	24	2	208°24.1'	N1°22.2'
24	3	223°24.3'	N1°23.2'	24	4	238°24.5'	N1°24.2'	24	5	253°24.7'	N1°25.2'
24	6	268°24.9'	N1°26.2'	24	7	283°25.0'	N1°27.2'	24	8	298°25.2'	N1°28.1'
24	9	313°25.4'	N1°29.1'	24	10	328°25.6'	N1°30.1'	24	11	343°25.8'	N1°31.1'
24	12	358°26.0'	N1°32.1'	24	13	13°26.2'	N1°33.1'	24	14	28°26.4'	N1°34.1'
24	15	43°26.5'	N1°35.0'	24	16	58°26.7'	N1°36.0'	24	17	73°26.9'	N1°37.0'
24	18	88°27.1'	N1°38.0'	24	19	103°27.3'	N1°39.0'	24	20	118°27.5'	N1°40.0'
24	21	133°27.7'	N1°40.9'	24	22	148°27.9'	N1°41.9'	24	23	163°28.1'	N1°42.9'
25	0	178°28.2'	N1°43.9'	25	1	193°28.4'	N1°44.9'	25	2	208°28.6'	N1°45.9'
25	3	223°28.8'	N1°46.8'	25	4	238°29.0'	N1°47.8'	25	5	253°29.2'	N1°48.8'
25	6	268°29.4'	N1°49.8'	25	7	283°29.6'	N1°50.8'	25	8	298°29.7'	N1°51.8'
25	9	313°29.9'	N1°52.7'	25	10	328°30.1'	N1°53.7'	25	11	343°30.3'	N1°54.7'
25	12	358°30.5'	N1°55.7'	25	13	13°30.7'	N1°56.7'	25	14	28°30.9'	N1°57.6'

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Sun — GHA and Declination, March 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
25	15	43°31.1'	N1°58.6'	25	16	58°31.3'	N1°59.6'	25	17	73°31.4'	N2°00.6'
25	18	88°31.6'	N2°01.6'	25	19	103°31.8'	N2°02.6'	25	20	118°32.0'	N2°03.5'
25	21	133°32.2'	N2°04.5'	25	22	148°32.4'	N2°05.5'	25	23	163°32.6'	N2°06.5'
26	0	178°32.8'	N2°07.5'	26	1	193°33.0'	N2°08.4'	26	2	208°33.1'	N2°09.4'
26	3	223°33.3'	N2°10.4'	26	4	238°33.5'	N2°11.4'	26	5	253°33.7'	N2°12.4'
26	6	268°33.9'	N2°13.3'	26	7	283°34.1'	N2°14.3'	26	8	298°34.3'	N2°15.3'
26	9	313°34.5'	N2°16.3'	26	10	328°34.7'	N2°17.3'	26	11	343°34.8'	N2°18.3'
26	12	358°35.0'	N2°19.2'	26	13	13°35.2'	N2°20.2'	26	14	28°35.4'	N2°21.2'
26	15	43°35.6'	N2°22.2'	26	16	58°35.8'	N2°23.2'	26	17	73°36.0'	N2°24.1'
26	18	88°36.2'	N2°25.1'	26	19	103°36.3'	N2°26.1'	26	20	118°36.5'	N2°27.1'
26	21	133°36.7'	N2°28.1'	26	22	148°36.9'	N2°29.0'	26	23	163°37.1'	N2°30.0'
27	0	178°37.3'	N2°31.0'	27	1	193°37.5'	N2°32.0'	27	2	208°37.7'	N2°33.0'
27	3	223°37.9'	N2°33.9'	27	4	238°38.0'	N2°34.9'	27	5	253°38.2'	N2°35.9'
27	6	268°38.4'	N2°36.9'	27	7	283°38.6'	N2°37.8'	27	8	298°38.8'	N2°38.8'
27	9	313°39.0'	N2°39.8'	27	10	328°39.2'	N2°40.8'	27	11	343°39.4'	N2°41.8'
27	12	358°39.6'	N2°42.7'	27	13	13°39.7'	N2°43.7'	27	14	28°39.9'	N2°44.7'
27	15	43°40.1'	N2°45.7'	27	16	58°40.3'	N2°46.7'	27	17	73°40.5'	N2°47.6'
27	18	88°40.7'	N2°48.6'	27	19	103°40.9'	N2°49.6'	27	20	118°41.1'	N2°50.6'
27	21	133°41.3'	N2°51.5'	27	22	148°41.4'	N2°52.5'	27	23	163°41.6'	N2°53.5'
28	0	178°41.8'	N2°54.5'	28	1	193°42.0'	N2°55.4'	28	2	208°42.2'	N2°56.4'
28	3	223°42.4'	N2°57.4'	28	4	238°42.6'	N2°58.4'	28	5	253°42.8'	N2°59.4'
28	6	268°42.9'	N3°00.3'	28	7	283°43.1'	N3°01.3'	28	8	298°43.3'	N3°02.3'
28	9	313°43.5'	N3°03.3'	28	10	328°43.7'	N3°04.2'	28	11	343°43.9'	N3°05.2'
28	12	358°44.1'	N3°06.2'	28	13	13°44.3'	N3°07.2'	28	14	28°44.5'	N3°08.1'
28	15	43°44.6'	N3°09.1'	28	16	58°44.8'	N3°10.1'	28	17	73°45.0'	N3°11.1'
28	18	88°45.2'	N3°12.0'	28	19	103°45.4'	N3°13.0'	28	20	118°45.6'	N3°14.0'
28	21	133°45.8'	N3°15.0'	28	22	148°46.0'	N3°15.9'	28	23	163°46.1'	N3°16.9'
29	0	178°46.3'	N3°17.9'	29	1	193°46.5'	N3°18.9'	29	2	208°46.7'	N3°19.8'
29	3	223°46.9'	N3°20.8'	29	4	238°47.1'	N3°21.8'	29	5	253°47.3'	N3°22.8'
29	6	268°47.5'	N3°23.7'	29	7	283°47.7'	N3°24.7'	29	8	298°47.8'	N3°25.7'
29	9	313°48.0'	N3°26.7'	29	10	328°48.2'	N3°27.6'	29	11	343°48.4'	N3°28.6'
29	12	358°48.6'	N3°29.6'	29	13	13°48.8'	N3°30.6'	29	14	28°49.0'	N3°31.5'
29	15	43°49.2'	N3°32.5'	29	16	58°49.3'	N3°33.5'	29	17	73°49.5'	N3°34.4'
29	18	88°49.7'	N3°35.4'	29	19	103°49.9'	N3°36.4'	29	20	118°50.1'	N3°37.4'
29	21	133°50.3'	N3°38.3'	29	22	148°50.5'	N3°39.3'	29	23	163°50.7'	N3°40.3'
30	0	178°50.8'	N3°41.2'	30	1	193°51.0'	N3°42.2'	30	2	208°51.2'	N3°43.2'
30	3	223°51.4'	N3°44.2'	30	4	238°51.6'	N3°45.1'	30	5	253°51.8'	N3°46.1'
30	6	268°52.0'	N3°47.1'	30	7	283°52.2'	N3°48.0'	30	8	298°52.3'	N3°49.0'
30	9	313°52.5'	N3°50.0'	30	10	328°52.7'	N3°51.0'	30	11	343°52.9'	N3°51.9'
30	12	358°53.1'	N3°52.9'	30	13	13°53.3'	N3°53.9'	30	14	28°53.5'	N3°54.8'
30	15	43°53.7'	N3°55.8'	30	16	58°53.8'	N3°56.8'	30	17	73°54.0'	N3°57.8'
30	18	88°54.2'	N3°58.7'	30	19	103°54.4'	N3°59.7'	30	20	118°54.6'	N4°00.7'
30	21	133°54.8'	N4°01.6'	30	22	148°55.0'	N4°02.6'	30	23	163°55.1'	N4°03.6'
31	0	178°55.3'	N4°04.5'	31	1	193°55.5'	N4°05.5'	31	2	208°55.7'	N4°06.5'
31	3	223°55.9'	N4°07.4'	31	4	238°56.1'	N4°08.4'	31	5	253°56.3'	N4°09.4'
31	6	268°56.5'	N4°10.3'	31	7	283°56.6'	N4°11.3'	31	8	298°56.8'	N4°12.3'
31	9	313°57.0'	N4°13.3'	31	10	328°57.2'	N4°14.2'	31	11	343°57.4'	N4°15.2'
31	12	358°57.6'	N4°16.2'	31	13	13°57.8'	N4°17.1'	31	14	28°57.9'	N4°18.1'
31	15	43°58.1'	N4°19.1'	31	16	58°58.3'	N4°20.0'	31	17	73°58.5'	N4°21.0'
31	18	88°58.7'	N4°22.0'	31	19	103°58.9'	N4°22.9'	31	20	118°59.1'	N4°23.9'
31	21	133°59.2'	N4°24.9'	31	22	148°59.4'	N4°25.8'	31	23	163°59.6'	N4°26.8'

Table 9.4: Sun — GHA and Declination, April 2026 (UT)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
1	0	178°59.8'	N4°27.8'	1	1	194°00.0'	N4°28.7'	1	2	209°00.2'	N4°29.7'
1	3	224°00.4'	N4°30.6'	1	4	239°00.5'	N4°31.6'	1	5	254°00.7'	N4°32.6'
1	6	269°00.9'	N4°33.5'	1	7	284°01.1'	N4°34.5'	1	8	299°01.3'	N4°35.5'
1	9	314°01.5'	N4°36.4'	1	10	329°01.7'	N4°37.4'	1	11	344°01.8'	N4°38.4'
1	12	359°02.0'	N4°39.3'	1	13	14°02.2'	N4°40.3'	1	14	29°02.4'	N4°41.3'
1	15	44°02.6'	N4°42.2'	1	16	59°02.8'	N4°43.2'	1	17	74°03.0'	N4°44.1'
1	18	89°03.1'	N4°45.1'	1	19	104°03.3'	N4°46.1'	1	20	119°03.5'	N4°47.0'

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Sun — GHA and Declination, April 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
1	21	134°03.7'	N4°48.0'	1	22	149°03.9'	N4°49.0'	1	23	164°04.1'	N4°49.9'
2	0	179°04.2'	N4°50.9'	2	1	194°04.4'	N4°51.8'	2	2	209°04.6'	N4°52.8'
2	3	224°04.8'	N4°53.8'	2	4	239°05.0'	N4°54.7'	2	5	254°05.2'	N4°55.7'
2	6	269°05.4'	N4°56.7'	2	7	284°05.5'	N4°57.6'	2	8	299°05.7'	N4°58.6'
2	9	314°05.9'	N4°59.5'	2	10	329°06.1'	N5°00.5'	2	11	344°06.3'	N5°01.5'
2	12	359°06.5'	N5°02.4'	2	13	14°06.6'	N5°03.4'	2	14	29°06.8'	N5°04.3'
2	15	44°07.0'	N5°05.3'	2	16	59°07.2'	N5°06.3'	2	17	74°07.4'	N5°07.2'
2	18	89°07.6'	N5°08.2'	2	19	104°07.7'	N5°09.1'	2	20	119°07.9'	N5°10.1'
2	21	134°08.1'	N5°11.1'	2	22	149°08.3'	N5°12.0'	2	23	164°08.5'	N5°13.0'
3	0	179°08.7'	N5°13.9'	3	1	194°08.8'	N5°14.9'	3	2	209°09.0'	N5°15.8'
3	3	224°09.2'	N5°16.8'	3	4	239°09.4'	N5°17.8'	3	5	254°09.6'	N5°18.7'
3	6	269°09.8'	N5°19.7'	3	7	284°09.9'	N5°20.6'	3	8	299°10.1'	N5°21.6'
3	9	314°10.3'	N5°22.5'	3	10	329°10.5'	N5°23.5'	3	11	344°10.7'	N5°24.5'
3	12	359°10.9'	N5°25.4'	3	13	14°11.0'	N5°26.4'	3	14	29°11.2'	N5°27.3'
3	15	44°11.4'	N5°28.3'	3	16	59°11.6'	N5°29.2'	3	17	74°11.8'	N5°30.2'
3	18	89°11.9'	N5°31.2'	3	19	104°12.1'	N5°32.1'	3	20	119°12.3'	N5°33.1'
3	21	134°12.5'	N5°34.0'	3	22	149°12.7'	N5°35.0'	3	23	164°12.9'	N5°35.9'
4	0	179°13.0'	N5°36.9'	4	1	194°13.2'	N5°37.8'	4	2	209°13.4'	N5°38.8'
4	3	224°13.6'	N5°39.7'	4	4	239°13.8'	N5°40.7'	4	5	254°13.9'	N5°41.7'
4	6	269°14.1'	N5°42.6'	4	7	284°14.3'	N5°43.6'	4	8	299°14.5'	N5°44.5'
4	9	314°14.7'	N5°45.5'	4	10	329°14.9'	N5°46.4'	4	11	344°15.0'	N5°47.4'
4	12	359°15.2'	N5°48.3'	4	13	14°15.4'	N5°49.3'	4	14	29°15.6'	N5°50.2'
4	15	44°15.8'	N5°51.2'	4	16	59°15.9'	N5°52.1'	4	17	74°16.1'	N5°53.1'
4	18	89°16.3'	N5°54.0'	4	19	104°16.5'	N5°55.0'	4	20	119°16.7'	N5°55.9'
4	21	134°16.8'	N5°56.9'	4	22	149°17.0'	N5°57.8'	4	23	164°17.2'	N5°58.8'
5	0	179°17.4'	N5°59.7'	5	1	194°17.6'	N6°00.7'	5	2	209°17.7'	N6°01.6'
5	3	224°17.9'	N6°02.6'	5	4	239°18.1'	N6°03.5'	5	5	254°18.3'	N6°04.5'
5	6	269°18.5'	N6°05.4'	5	7	284°18.6'	N6°06.4'	5	8	299°18.8'	N6°07.3'
5	9	314°19.0'	N6°08.3'	5	10	329°19.2'	N6°09.2'	5	11	344°19.4'	N6°10.2'
5	12	359°19.5'	N6°11.1'	5	13	14°19.7'	N6°12.1'	5	14	29°19.9'	N6°13.0'
5	15	44°20.1'	N6°14.0'	5	16	59°20.2'	N6°14.9'	5	17	74°20.4'	N6°15.9'
5	18	89°20.6'	N6°16.8'	5	19	104°20.8'	N6°17.8'	5	20	119°21.0'	N6°18.7'
5	21	134°21.1'	N6°19.7'	5	22	149°21.3'	N6°20.6'	5	23	164°21.5'	N6°21.5'
6	0	179°21.7'	N6°22.5'	6	1	194°21.8'	N6°23.4'	6	2	209°22.0'	N6°24.4'
6	3	224°22.2'	N6°25.3'	6	4	239°22.4'	N6°26.3'	6	5	254°22.6'	N6°27.2'
6	6	269°22.7'	N6°28.2'	6	7	284°22.9'	N6°29.1'	6	8	299°23.1'	N6°30.0'
6	9	314°23.3'	N6°31.0'	6	10	329°23.4'	N6°31.9'	6	11	344°23.6'	N6°32.9'
6	12	359°23.8'	N6°33.8'	6	13	14°24.0'	N6°34.8'	6	14	29°24.2'	N6°35.7'
6	15	44°24.3'	N6°36.7'	6	16	59°24.5'	N6°37.6'	6	17	74°24.7'	N6°38.5'
6	18	89°24.9'	N6°39.5'	6	19	104°25.0'	N6°40.4'	6	20	119°25.2'	N6°41.4'
6	21	134°25.4'	N6°42.3'	6	22	149°25.6'	N6°43.2'	6	23	164°25.7'	N6°44.2'
7	0	179°25.9'	N6°45.1'	7	1	194°26.1'	N6°46.1'	7	2	209°26.3'	N6°47.0'
7	3	224°26.4'	N6°48.0'	7	4	239°26.6'	N6°48.9'	7	5	254°26.8'	N6°49.8'
7	6	269°27.0'	N6°50.8'	7	7	284°27.1'	N6°51.7'	7	8	299°27.3'	N6°52.7'
7	9	314°27.5'	N6°53.6'	7	10	329°27.7'	N6°54.5'	7	11	344°27.8'	N6°55.5'
7	12	359°28.0'	N6°56.4'	7	13	14°28.2'	N6°57.3'	7	14	29°28.4'	N6°58.3'
7	15	44°28.5'	N6°59.2'	7	16	59°28.7'	N7°00.2'	7	17	74°28.9'	N7°01.1'
7	18	89°29.1'	N7°02.0'	7	19	104°29.2'	N7°03.0'	7	20	119°29.4'	N7°03.9'
7	21	134°29.6'	N7°04.8'	7	22	149°29.8'	N7°05.8'	7	23	164°29.9'	N7°06.7'
8	0	179°30.1'	N7°07.7'	8	1	194°30.3'	N7°08.6'	8	2	209°30.4'	N7°09.5'
8	3	224°30.6'	N7°10.5'	8	4	239°30.8'	N7°11.4'	8	5	254°31.0'	N7°12.3'
8	6	269°31.1'	N7°13.3'	8	7	284°31.3'	N7°14.2'	8	8	299°31.5'	N7°15.1'
8	9	314°31.7'	N7°16.1'	8	10	329°31.8'	N7°17.0'	8	11	344°32.0'	N7°17.9'
8	12	359°32.2'	N7°18.9'	8	13	14°32.3'	N7°19.8'	8	14	29°32.5'	N7°20.7'
8	15	44°32.7'	N7°21.7'	8	16	59°32.9'	N7°22.6'	8	17	74°33.0'	N7°23.5'
8	18	89°33.2'	N7°24.5'	8	19	104°33.4'	N7°25.4'	8	20	119°33.5'	N7°26.3'
8	21	134°33.7'	N7°27.3'	8	22	149°33.9'	N7°28.2'	8	23	164°34.1'	N7°29.1'
9	0	179°34.2'	N7°30.1'	9	1	194°34.4'	N7°31.0'	9	2	209°34.6'	N7°31.9'
9	3	224°34.7'	N7°32.9'	9	4	239°34.9'	N7°33.8'	9	5	254°35.1'	N7°34.7'
9	6	269°35.3'	N7°35.6'	9	7	284°35.4'	N7°36.6'	9	8	299°35.6'	N7°37.5'
9	9	314°35.8'	N7°38.4'	9	10	329°35.9'	N7°39.4'	9	11	344°36.1'	N7°40.3'
9	12	359°36.3'	N7°41.2'	9	13	14°36.4'	N7°42.1'	9	14	29°36.6'	N7°43.1'
9	15	44°36.8'	N7°44.0'	9	16	59°37.0'	N7°44.9'	9	17	74°37.1'	N7°45.9'
9	18	89°37.3'	N7°46.8'	9	19	104°37.5'	N7°47.7'	9	20	119°37.6'	N7°48.6'

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Sun — GHA and Declination, April 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
9	21	134°37.8'	N7°49.6'	9	22	149°38.0'	N7°50.5'	9	23	164°38.1'	N7°51.4'
10	0	179°38.3'	N7°52.3'	10	1	194°38.5'	N7°53.3'	10	2	209°38.6'	N7°54.2'
10	3	224°38.8'	N7°55.1'	10	4	239°39.0'	N7°56.0'	10	5	254°39.1'	N7°57.0'
10	6	269°39.3'	N7°57.9'	10	7	284°39.5'	N7°58.8'	10	8	299°39.6'	N7°59.7'
10	9	314°39.8'	N8°00.7'	10	10	329°40.0'	N8°01.6'	10	11	344°40.1'	N8°02.5'
10	12	359°40.3'	N8°03.4'	10	13	14°40.5'	N8°04.4'	10	14	29°40.6'	N8°05.3'
10	15	44°40.8'	N8°06.2'	10	16	59°41.0'	N8°07.1'	10	17	74°41.1'	N8°08.0'
10	18	89°41.3'	N8°09.0'	10	19	104°41.5'	N8°09.9'	10	20	119°41.6'	N8°10.8'
10	21	134°41.8'	N8°11.7'	10	22	149°42.0'	N8°12.6'	10	23	164°42.1'	N8°13.6'
11	0	179°42.3'	N8°14.5'	11	1	194°42.5'	N8°15.4'	11	2	209°42.6'	N8°16.3'
11	3	224°42.8'	N8°17.2'	11	4	239°43.0'	N8°18.2'	11	5	254°43.1'	N8°19.1'
11	6	269°43.3'	N8°20.0'	11	7	284°43.5'	N8°20.9'	11	8	299°43.6'	N8°21.8'
11	9	314°43.8'	N8°22.8'	11	10	329°43.9'	N8°23.7'	11	11	344°44.1'	N8°24.6'
11	12	359°44.3'	N8°25.5'	11	13	14°44.4'	N8°26.4'	11	14	29°44.6'	N8°27.3'
11	15	44°44.8'	N8°28.3'	11	16	59°44.9'	N8°29.2'	11	17	74°45.1'	N8°30.1'
11	18	89°45.3'	N8°31.0'	11	19	104°45.4'	N8°31.9'	11	20	119°45.6'	N8°32.8'
11	21	134°45.7'	N8°33.8'	11	22	149°45.9'	N8°34.7'	11	23	164°46.1'	N8°35.6'
12	0	179°46.2'	N8°36.5'	12	1	194°46.4'	N8°37.4'	12	2	209°46.6'	N8°38.3'
12	3	224°46.7'	N8°39.2'	12	4	239°46.9'	N8°40.1'	12	5	254°47.0'	N8°41.1'
12	6	269°47.2'	N8°42.0'	12	7	284°47.4'	N8°42.9'	12	8	299°47.5'	N8°43.8'
12	9	314°47.7'	N8°44.7'	12	10	329°47.8'	N8°45.6'	12	11	344°48.0'	N8°46.5'
12	12	359°48.2'	N8°47.4'	12	13	14°48.3'	N8°48.4'	12	14	29°48.5'	N8°49.3'
12	15	44°48.7'	N8°50.2'	12	16	59°48.8'	N8°51.1'	12	17	74°49.0'	N8°52.0'
12	18	89°49.1'	N8°52.9'	12	19	104°49.3'	N8°53.8'	12	20	119°49.5'	N8°54.7'
12	21	134°49.6'	N8°55.6'	12	22	149°49.8'	N8°56.5'	12	23	164°49.9'	N8°57.5'
13	0	179°50.1'	N8°58.4'	13	1	194°50.2'	N8°59.3'	13	2	209°50.4'	N9°00.2'
13	3	224°50.6'	N9°01.1'	13	4	239°50.7'	N9°02.0'	13	5	254°50.9'	N9°02.9'
13	6	269°51.0'	N9°03.8'	13	7	284°51.2'	N9°04.7'	13	8	299°51.4'	N9°05.6'
13	9	314°51.5'	N9°06.5'	13	10	329°51.7'	N9°07.4'	13	11	344°51.8'	N9°08.3'
13	12	359°52.0'	N9°09.2'	13	13	14°52.1'	N9°10.1'	13	14	29°52.3'	N9°11.0'
13	15	44°52.5'	N9°11.9'	13	16	59°52.6'	N9°12.9'	13	17	74°52.8'	N9°13.8'
13	18	89°52.9'	N9°14.7'	13	19	104°53.1'	N9°15.6'	13	20	119°53.2'	N9°16.5'
13	21	134°53.4'	N9°17.4'	13	22	149°53.6'	N9°18.3'	13	23	164°53.7'	N9°19.2'
14	0	179°53.9'	N9°20.1'	14	1	194°54.0'	N9°21.0'	14	2	209°54.2'	N9°21.9'
14	3	224°54.3'	N9°22.8'	14	4	239°54.5'	N9°23.7'	14	5	254°54.6'	N9°24.6'
14	6	269°54.8'	N9°25.5'	14	7	284°54.9'	N9°26.4'	14	8	299°55.1'	N9°27.3'
14	9	314°55.3'	N9°28.2'	14	10	329°55.4'	N9°29.1'	14	11	344°55.6'	N9°30.0'
14	12	359°55.7'	N9°30.9'	14	13	14°55.9'	N9°31.8'	14	14	29°56.0'	N9°32.7'
14	15	44°56.2'	N9°33.6'	14	16	59°56.3'	N9°34.5'	14	17	74°56.5'	N9°35.4'
14	18	89°56.6'	N9°36.3'	14	19	104°56.8'	N9°37.2'	14	20	119°56.9'	N9°38.0'
14	21	134°57.1'	N9°38.9'	14	22	149°57.2'	N9°39.8'	14	23	164°57.4'	N9°40.7'
15	0	179°57.6'	N9°41.6'	15	1	194°57.7'	N9°42.5'	15	2	209°57.9'	N9°43.4'
15	3	224°58.0'	N9°44.3'	15	4	239°58.2'	N9°45.2'	15	5	254°58.3'	N9°46.1'
15	6	269°58.5'	N9°47.0'	15	7	284°58.6'	N9°47.9'	15	8	299°58.8'	N9°48.8'
15	9	314°58.9'	N9°49.7'	15	10	329°59.1'	N9°50.6'	15	11	344°59.2'	N9°51.5'
15	12	359°59.4'	N9°52.4'	15	13	14°59.5'	N9°53.2'	15	14	29°59.7'	N9°54.1'
15	15	44°59.8'	N9°55.0'	15	16	60°00.0'	N9°55.9'	15	17	75°00.1'	N9°56.8'
15	18	90°00.3'	N9°57.7'	15	19	105°00.4'	N9°58.6'	15	20	120°00.6'	N9°59.5'
15	21	135°00.7'	N10°00.4'	15	22	150°00.9'	N10°01.3'	15	23	165°01.0'	N10°02.1'
16	0	180°01.2'	N10°03.0'	16	1	195°01.3'	N10°03.9'	16	2	210°01.5'	N10°04.8'
16	3	225°01.6'	N10°05.7'	16	4	240°01.8'	N10°06.6'	16	5	255°01.9'	N10°07.5'
16	6	270°02.0'	N10°08.4'	16	7	285°02.2'	N10°09.2'	16	8	300°02.3'	N10°10.1'
16	9	315°02.5'	N10°11.0'	16	10	330°02.6'	N10°11.9'	16	11	345°02.8'	N10°12.8'
16	12	0°02.9'	N10°13.7'	16	13	15°03.1'	N10°14.6'	16	14	30°03.2'	N10°15.4'
16	15	45°03.4'	N10°16.3'	16	16	60°03.5'	N10°17.2'	16	17	75°03.7'	N10°18.1'
16	18	90°03.8'	N10°19.0'	16	19	105°03.9'	N10°19.9'	16	20	120°04.1'	N10°20.7'
16	21	135°04.2'	N10°21.6'	16	22	150°04.4'	N10°22.5'	16	23	165°04.5'	N10°23.4'
17	0	180°04.7'	N10°24.3'	17	1	195°04.8'	N10°25.1'	17	2	210°05.0'	N10°26.0'
17	3	225°05.1'	N10°26.9'	17	4	240°05.2'	N10°27.8'	17	5	255°05.4'	N10°28.7'
17	6	270°05.5'	N10°29.5'	17	7	285°05.7'	N10°30.4'	17	8	300°05.8'	N10°31.3'
17	9	315°06.0'	N10°32.2'	17	10	330°06.1'	N10°33.1'	17	11	345°06.3'	N10°33.9'
17	12	0°06.4'	N10°34.8'	17	13	15°06.5'	N10°35.7'	17	14	30°06.7'	N10°36.6'
17	15	45°06.8'	N10°37.4'	17	16	60°07.0'	N10°38.3'	17	17	75°07.1'	N10°39.2'
17	18	90°07.2'	N10°40.1'	17	19	105°07.4'	N10°40.9'	17	20	120°07.5'	N10°41.8'

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Sun — GHA and Declination, April 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
17	21	135°07.7'	N10°42.7'	17	22	150°07.8'	N10°43.6'	17	23	165°08.0'	N10°44.4'
18	0	180°08.1'	N10°45.3'	18	1	195°08.2'	N10°46.2'	18	2	210°08.4'	N10°47.1'
18	3	225°08.5'	N10°47.9'	18	4	240°08.7'	N10°48.8'	18	5	255°08.8'	N10°49.7'
18	6	270°08.9'	N10°50.6'	18	7	285°09.1'	N10°51.4'	18	8	300°09.2'	N10°52.3'
18	9	315°09.3'	N10°53.2'	18	10	330°09.5'	N10°54.0'	18	11	345°09.6'	N10°54.9'
18	12	0°09.8'	N10°55.8'	18	13	15°09.9'	N10°56.7'	18	14	30°10.0'	N10°57.5'
18	15	45°10.2'	N10°58.4'	18	16	60°10.3'	N10°59.3'	18	17	75°10.5'	N11°00.1'
18	18	90°10.6'	N11°01.0'	18	19	105°10.7'	N11°01.9'	18	20	120°10.9'	N11°02.7'
18	21	135°11.0'	N11°03.6'	18	22	150°11.1'	N11°04.5'	18	23	165°11.3'	N11°05.3'
19	0	180°11.4'	N11°06.2'	19	1	195°11.5'	N11°07.1'	19	2	210°11.7'	N11°07.9'
19	3	225°11.8'	N11°08.8'	19	4	240°12.0'	N11°09.7'	19	5	255°12.1'	N11°10.5'
19	6	270°12.2'	N11°11.4'	19	7	285°12.4'	N11°12.3'	19	8	300°12.5'	N11°13.1'
19	9	315°12.6'	N11°14.0'	19	10	330°12.8'	N11°14.9'	19	11	345°12.9'	N11°15.7'
19	12	0°13.0'	N11°16.6'	19	13	15°13.2'	N11°17.4'	19	14	30°13.3'	N11°18.3'
19	15	45°13.4'	N11°19.2'	19	16	60°13.6'	N11°20.0'	19	17	75°13.7'	N11°20.9'
19	18	90°13.8'	N11°21.7'	19	19	105°14.0'	N11°22.6'	19	20	120°14.1'	N11°23.5'
19	21	135°14.2'	N11°24.3'	19	22	150°14.4'	N11°25.2'	19	23	165°14.5'	N11°26.0'
20	0	180°14.6'	N11°26.9'	20	1	195°14.8'	N11°27.8'	20	2	210°14.9'	N11°28.6'
20	3	225°15.0'	N11°29.5'	20	4	240°15.2'	N11°30.3'	20	5	255°15.3'	N11°31.2'
20	6	270°15.4'	N11°32.1'	20	7	285°15.5'	N11°32.9'	20	8	300°15.7'	N11°33.8'
20	9	315°15.8'	N11°34.6'	20	10	330°15.9'	N11°35.5'	20	11	345°16.1'	N11°36.3'
20	12	0°16.2'	N11°37.2'	20	13	15°16.3'	N11°38.0'	20	14	30°16.5'	N11°38.9'
20	15	45°16.6'	N11°39.7'	20	16	60°16.7'	N11°40.6'	20	17	75°16.8'	N11°41.5'
20	18	90°17.0'	N11°42.3'	20	19	105°17.1'	N11°43.2'	20	20	120°17.2'	N11°44.0'
20	21	135°17.4'	N11°44.9'	20	22	150°17.5'	N11°45.7'	20	23	165°17.6'	N11°46.6'
21	0	180°17.7'	N11°47.4'	21	1	195°17.9'	N11°48.3'	21	2	210°18.0'	N11°49.1'
21	3	225°18.1'	N11°50.0'	21	4	240°18.3'	N11°50.8'	21	5	255°18.4'	N11°51.7'
21	6	270°18.5'	N11°52.5'	21	7	285°18.6'	N11°53.4'	21	8	300°18.8'	N11°54.2'
21	9	315°18.9'	N11°55.1'	21	10	330°19.0'	N11°55.9'	21	11	345°19.1'	N11°56.8'
21	12	0°19.3'	N11°57.6'	21	13	15°19.4'	N11°58.4'	21	14	30°19.5'	N11°59.3'
21	15	45°19.6'	N12°00.1'	21	16	60°19.8'	N12°01.0'	21	17	75°19.9'	N12°01.8'
21	18	90°20.0'	N12°02.7'	21	19	105°20.1'	N12°03.5'	21	20	120°20.3'	N12°04.4'
21	21	135°20.4'	N12°05.2'	21	22	150°20.5'	N12°06.0'	21	23	165°20.6'	N12°06.9'
22	0	180°20.7'	N12°07.7'	22	1	195°20.9'	N12°08.6'	22	2	210°21.0'	N12°09.4'
22	3	225°21.1'	N12°10.3'	22	4	240°21.2'	N12°11.1'	22	5	255°21.4'	N12°11.9'
22	6	270°21.5'	N12°12.8'	22	7	285°21.6'	N12°13.6'	22	8	300°21.7'	N12°14.5'
22	9	315°21.8'	N12°15.3'	22	10	330°22.0'	N12°16.1'	22	11	345°22.1'	N12°17.0'
22	12	0°22.2'	N12°17.8'	22	13	15°22.3'	N12°18.7'	22	14	30°22.5'	N12°19.5'
22	15	45°22.6'	N12°20.3'	22	16	60°22.7'	N12°21.2'	22	17	75°22.8'	N12°22.0'
22	18	90°22.9'	N12°22.8'	22	19	105°23.0'	N12°23.7'	22	20	120°23.2'	N12°24.5'
22	21	135°23.3'	N12°25.3'	22	22	150°23.4'	N12°26.2'	22	23	165°23.5'	N12°27.0'
23	0	180°23.6'	N12°27.9'	23	1	195°23.8'	N12°28.7'	23	2	210°23.9'	N12°29.5'
23	3	225°24.0'	N12°30.4'	23	4	240°24.1'	N12°31.2'	23	5	255°24.2'	N12°32.0'
23	6	270°24.3'	N12°32.8'	23	7	285°24.5'	N12°33.7'	23	8	300°24.6'	N12°34.5'
23	9	315°24.7'	N12°35.3'	23	10	330°24.8'	N12°36.2'	23	11	345°24.9'	N12°37.0'
23	12	0°25.0'	N12°37.8'	23	13	15°25.2'	N12°38.7'	23	14	30°25.3'	N12°39.5'
23	15	45°25.4'	N12°40.3'	23	16	60°25.5'	N12°41.1'	23	17	75°25.6'	N12°42.0'
23	18	90°25.7'	N12°42.8'	23	19	105°25.9'	N12°43.6'	23	20	120°26.0'	N12°44.5'
23	21	135°26.1'	N12°45.3'	23	22	150°26.2'	N12°46.1'	23	23	165°26.3'	N12°46.9'
24	0	180°26.4'	N12°47.8'	24	1	195°26.5'	N12°48.6'	24	2	210°26.6'	N12°49.4'
24	3	225°26.8'	N12°50.2'	24	4	240°26.9'	N12°51.1'	24	5	255°27.0'	N12°51.9'
24	6	270°27.1'	N12°52.7'	24	7	285°27.2'	N12°53.5'	24	8	300°27.3'	N12°54.4'
24	9	315°27.4'	N12°55.2'	24	10	330°27.5'	N12°56.0'	24	11	345°27.7'	N12°56.8'
24	12	0°27.8'	N12°57.6'	24	13	15°27.9'	N12°58.5'	24	14	30°28.0'	N12°59.3'
24	15	45°28.1'	N13°00.1'	24	16	60°28.2'	N13°00.9'	24	17	75°28.3'	N13°01.7'
24	18	90°28.4'	N13°02.6'	24	19	105°28.5'	N13°03.4'	24	20	120°28.6'	N13°04.2'
24	21	135°28.8'	N13°05.0'	24	22	150°28.9'	N13°05.8'	24	23	165°29.0'	N13°06.6'
25	0	180°29.1'	N13°07.5'	25	1	195°29.2'	N13°08.3'	25	2	210°29.3'	N13°09.1'
25	3	225°29.4'	N13°09.9'	25	4	240°29.5'	N13°10.7'	25	5	255°29.6'	N13°11.5'
25	6	270°29.7'	N13°12.4'	25	7	285°29.8'	N13°13.2'	25	8	300°29.9'	N13°14.0'
25	9	315°30.1'	N13°14.8'	25	10	330°30.2'	N13°15.6'	25	11	345°30.3'	N13°16.4'
25	12	0°30.4'	N13°17.2'	25	13	15°30.5'	N13°18.0'	25	14	30°30.6'	N13°18.9'
25	15	45°30.7'	N13°19.7'	25	16	60°30.8'	N13°20.5'	25	17	75°30.9'	N13°21.3'
25	18	90°31.0'	N13°22.1'	25	19	105°31.1'	N13°22.9'	25	20	120°31.2'	N13°23.7'

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Sun — GHA and Declination, April 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
25	21	135°31.3'	N13°24.5'	25	22	150°31.4'	N13°25.3'	25	23	165°31.5'	N13°26.1'
26	0	180°31.6'	N13°26.9'	26	1	195°31.7'	N13°27.8'	26	2	210°31.8'	N13°28.6'
26	3	225°31.9'	N13°29.4'	26	4	240°32.0'	N13°30.2'	26	5	255°32.1'	N13°31.0'
26	6	270°32.2'	N13°31.8'	26	7	285°32.3'	N13°32.6'	26	8	300°32.4'	N13°33.4'
26	9	315°32.5'	N13°34.2'	26	10	330°32.6'	N13°35.0'	26	11	345°32.7'	N13°35.8'
26	12	0°32.9'	N13°36.6'	26	13	15°33.0'	N13°37.4'	26	14	30°33.1'	N13°38.2'
26	15	45°33.2'	N13°39.0'	26	16	60°33.3'	N13°39.8'	26	17	75°33.4'	N13°40.6'
26	18	90°33.5'	N13°41.4'	26	19	105°33.6'	N13°42.2'	26	20	120°33.7'	N13°43.0'
26	21	135°33.7'	N13°43.8'	26	22	150°33.8'	N13°44.6'	26	23	165°33.9'	N13°45.4'
27	0	180°34.0'	N13°46.2'	27	1	195°34.1'	N13°47.0'	27	2	210°34.2'	N13°47.8'
27	3	225°34.3'	N13°48.6'	27	4	240°34.4'	N13°49.4'	27	5	255°34.5'	N13°50.2'
27	6	270°34.6'	N13°51.0'	27	7	285°34.7'	N13°51.8'	27	8	300°34.8'	N13°52.6'
27	9	315°34.9'	N13°53.4'	27	10	330°35.0'	N13°54.2'	27	11	345°35.1'	N13°55.0'
27	12	0°35.2'	N13°55.8'	27	13	15°35.3'	N13°56.6'	27	14	30°35.4'	N13°57.3'
27	15	45°35.5'	N13°58.1'	27	16	60°35.6'	N13°58.9'	27	17	75°35.7'	N13°59.7'
27	18	90°35.8'	N14°00.5'	27	19	105°35.9'	N14°01.3'	27	20	120°36.0'	N14°02.1'
27	21	135°36.1'	N14°02.9'	27	22	150°36.2'	N14°03.7'	27	23	165°36.2'	N14°04.5'
28	0	180°36.3'	N14°05.3'	28	1	195°36.4'	N14°06.0'	28	2	210°36.5'	N14°06.8'
28	3	225°36.6'	N14°07.6'	28	4	240°36.7'	N14°08.4'	28	5	255°36.8'	N14°09.2'
28	6	270°36.9'	N14°10.0'	28	7	285°37.0'	N14°10.8'	28	8	300°37.1'	N14°11.5'
28	9	315°37.2'	N14°12.3'	28	10	330°37.3'	N14°13.1'	28	11	345°37.3'	N14°13.9'
28	12	0°37.4'	N14°14.7'	28	13	15°37.5'	N14°15.5'	28	14	30°37.6'	N14°16.2'
28	15	45°37.7'	N14°17.0'	28	16	60°37.8'	N14°17.8'	28	17	75°37.9'	N14°18.6'
28	18	90°38.0'	N14°19.4'	28	19	105°38.1'	N14°20.2'	28	20	120°38.2'	N14°20.9'
28	21	135°38.2'	N14°21.7'	28	22	150°38.3'	N14°22.5'	28	23	165°38.4'	N14°23.3'
29	0	180°38.5'	N14°24.1'	29	1	195°38.6'	N14°24.8'	29	2	210°38.7'	N14°25.6'
29	3	225°38.8'	N14°26.4'	29	4	240°38.9'	N14°27.2'	29	5	255°38.9'	N14°27.9'
29	6	270°39.0'	N14°28.7'	29	7	285°39.1'	N14°29.5'	29	8	300°39.2'	N14°30.3'
29	9	315°39.3'	N14°31.0'	29	10	330°39.4'	N14°31.8'	29	11	345°39.5'	N14°32.6'
29	12	0°39.5'	N14°33.4'	29	13	15°39.6'	N14°34.1'	29	14	30°39.7'	N14°34.9'
29	15	45°39.8'	N14°35.7'	29	16	60°39.9'	N14°36.5'	29	17	75°40.0'	N14°37.2'
29	18	90°40.0'	N14°38.0'	29	19	105°40.1'	N14°38.8'	29	20	120°40.2'	N14°39.5'
29	21	135°40.3'	N14°40.3'	29	22	150°40.4'	N14°41.1'	29	23	165°40.5'	N14°41.9'
30	0	180°40.5'	N14°42.6'	30	1	195°40.6'	N14°43.4'	30	2	210°40.7'	N14°44.2'
30	3	225°40.8'	N14°44.9'	30	4	240°40.9'	N14°45.7'	30	5	255°40.9'	N14°46.5'
30	6	270°41.0'	N14°47.2'	30	7	285°41.1'	N14°48.0'	30	8	300°41.2'	N14°48.8'
30	9	315°41.3'	N14°49.5'	30	10	330°41.4'	N14°50.3'	30	11	345°41.4'	N14°51.1'
30	12	0°41.5'	N14°51.8'	30	13	15°41.6'	N14°52.6'	30	14	30°41.7'	N14°53.3'
30	15	45°41.7'	N14°54.1'	30	16	60°41.8'	N14°54.9'	30	17	75°41.9'	N14°55.6'
30	18	90°42.0'	N14°56.4'	30	19	105°42.1'	N14°57.2'	30	20	120°42.1'	N14°57.9'
30	21	135°42.2'	N14°58.7'	30	22	150°42.3'	N14°59.4'	30	23	165°42.4'	N15°00.2'

Table 9.5: Sun — GHA and Declination, May 2026 (UT)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
1	0	180°42.4'	N15°00.9'	1	1	195°42.5'	N15°01.7'	1	2	210°42.6'	N15°02.5'
1	3	225°42.7'	N15°03.2'	1	4	240°42.7'	N15°04.0'	1	5	255°42.8'	N15°04.7'
1	6	270°42.9'	N15°05.5'	1	7	285°43.0'	N15°06.2'	1	8	300°43.1'	N15°07.0'
1	9	315°43.1'	N15°07.8'	1	10	330°43.2'	N15°08.5'	1	11	345°43.3'	N15°09.3'
1	12	0°43.3'	N15°10.0'	1	13	15°43.4'	N15°10.8'	1	14	30°43.5'	N15°11.5'
1	15	45°43.6'	N15°12.3'	1	16	60°43.6'	N15°13.0'	1	17	75°43.7'	N15°13.8'
1	18	90°43.8'	N15°14.5'	1	19	105°43.9'	N15°15.3'	1	20	120°43.9'	N15°16.0'
1	21	135°44.0'	N15°16.8'	1	22	150°44.1'	N15°17.5'	1	23	165°44.1'	N15°18.3'
2	0	180°44.2'	N15°19.0'	2	1	195°44.3'	N15°19.8'	2	2	210°44.4'	N15°20.5'
2	3	225°44.4'	N15°21.3'	2	4	240°44.5'	N15°22.0'	2	5	255°44.6'	N15°22.8'
2	6	270°44.6'	N15°23.5'	2	7	285°44.7'	N15°24.2'	2	8	300°44.8'	N15°25.0'
2	9	315°44.8'	N15°25.7'	2	10	330°44.9'	N15°26.5'	2	11	345°45.0'	N15°27.2'
2	12	0°45.1'	N15°28.0'	2	13	15°45.1'	N15°28.7'	2	14	30°45.2'	N15°29.5'
2	15	45°45.3'	N15°30.2'	2	16	60°45.3'	N15°30.9'	2	17	75°45.4'	N15°31.7'
2	18	90°45.5'	N15°32.4'	2	19	105°45.5'	N15°33.2'	2	20	120°45.6'	N15°33.9'
2	21	135°45.7'	N15°34.6'	2	22	150°45.7'	N15°35.4'	2	23	165°45.8'	N15°36.1'
3	0	180°45.9'	N15°36.8'	3	1	195°45.9'	N15°37.6'	3	2	210°46.0'	N15°38.3'

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Sun — GHA and Declination, May 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
3	3	225°46.0'	N15°39.1'	3	4	240°46.1'	N15°39.8'	3	5	255°46.2'	N15°40.5'
3	6	270°46.2'	N15°41.3'	3	7	285°46.3'	N15°42.0'	3	8	300°46.4'	N15°42.7'
3	9	315°46.4'	N15°43.5'	3	10	330°46.5'	N15°44.2'	3	11	345°46.6'	N15°44.9'
3	12	0°46.6'	N15°45.7'	3	13	15°46.7'	N15°46.4'	3	14	30°46.7'	N15°47.1'
3	15	45°46.8'	N15°47.9'	3	16	60°46.9'	N15°48.6'	3	17	75°46.9'	N15°49.3'
3	18	90°47.0'	N15°50.0'	3	19	105°47.0'	N15°50.8'	3	20	120°47.1'	N15°51.5'
3	21	135°47.2'	N15°52.2'	3	22	150°47.2'	N15°53.0'	3	23	165°47.3'	N15°53.7'
4	0	180°47.4'	N15°54.4'	4	1	195°47.4'	N15°55.1'	4	2	210°47.5'	N15°55.9'
4	3	225°47.5'	N15°56.6'	4	4	240°47.6'	N15°57.3'	4	5	255°47.6'	N15°58.0'
4	6	270°47.7'	N15°58.8'	4	7	285°47.8'	N15°59.5'	4	8	300°47.8'	N16°00.2'
4	9	315°47.9'	N16°00.9'	4	10	330°47.9'	N16°01.7'	4	11	345°48.0'	N16°02.4'
4	12	0°48.0'	N16°03.1'	4	13	15°48.1'	N16°03.8'	4	14	30°48.2'	N16°04.5'
4	15	45°48.2'	N16°05.3'	4	16	60°48.3'	N16°06.0'	4	17	75°48.3'	N16°06.7'
4	18	90°48.4'	N16°07.4'	4	19	105°48.4'	N16°08.1'	4	20	120°48.5'	N16°08.8'
4	21	135°48.5'	N16°09.6'	4	22	150°48.6'	N16°10.3'	4	23	165°48.7'	N16°11.0'
5	0	180°48.7'	N16°11.7'	5	1	195°48.8'	N16°12.4'	5	2	210°48.8'	N16°13.1'
5	3	225°48.9'	N16°13.9'	5	4	240°48.9'	N16°14.6'	5	5	255°49.0'	N16°15.3'
5	6	270°49.0'	N16°16.0'	5	7	285°49.1'	N16°16.7'	5	8	300°49.1'	N16°17.4'
5	9	315°49.2'	N16°18.1'	5	10	330°49.2'	N16°18.8'	5	11	345°49.3'	N16°19.5'
5	12	0°49.3'	N16°20.3'	5	13	15°49.4'	N16°21.0'	5	14	30°49.4'	N16°21.7'
5	15	45°49.5'	N16°22.4'	5	16	60°49.5'	N16°23.1'	5	17	75°49.6'	N16°23.8'
5	18	90°49.6'	N16°24.5'	5	19	105°49.7'	N16°25.2'	5	20	120°49.7'	N16°25.9'
5	21	135°49.8'	N16°26.6'	5	22	150°49.8'	N16°27.3'	5	23	165°49.9'	N16°28.0'
6	0	180°49.9'	N16°28.7'	6	1	195°50.0'	N16°29.4'	6	2	210°50.0'	N16°30.1'
6	3	225°50.1'	N16°30.9'	6	4	240°50.1'	N16°31.6'	6	5	255°50.2'	N16°32.3'
6	6	270°50.2'	N16°33.0'	6	7	285°50.3'	N16°33.7'	6	8	300°50.3'	N16°34.4'
6	9	315°50.4'	N16°35.1'	6	10	330°50.4'	N16°35.8'	6	11	345°50.4'	N16°36.5'
6	12	0°50.5'	N16°37.2'	6	13	15°50.5'	N16°37.9'	6	14	30°50.6'	N16°38.5'
6	15	45°50.6'	N16°39.2'	6	16	60°50.7'	N16°39.9'	6	17	75°50.7'	N16°40.6'
6	18	90°50.8'	N16°41.3'	6	19	105°50.8'	N16°42.0'	6	20	120°50.8'	N16°42.7'
6	21	135°50.9'	N16°43.4'	6	22	150°50.9'	N16°44.1'	6	23	165°51.0'	N16°44.8'
7	0	180°51.0'	N16°45.5'	7	1	195°51.1'	N16°46.2'	7	2	210°51.1'	N16°46.9'
7	3	225°51.1'	N16°47.6'	7	4	240°51.2'	N16°48.3'	7	5	255°51.2'	N16°49.0'
7	6	270°51.3'	N16°49.6'	7	7	285°51.3'	N16°50.3'	7	8	300°51.3'	N16°51.0'
7	9	315°51.4'	N16°51.7'	7	10	330°51.4'	N16°52.4'	7	11	345°51.5'	N16°53.1'
7	12	0°51.5'	N16°53.8'	7	13	15°51.5'	N16°54.5'	7	14	30°51.6'	N16°55.1'
7	15	45°51.6'	N16°55.8'	7	16	60°51.7'	N16°56.5'	7	17	75°51.7'	N16°57.2'
7	18	90°51.7'	N16°57.9'	7	19	105°51.8'	N16°58.6'	7	20	120°51.8'	N16°59.2'
7	21	135°51.8'	N16°59.9'	7	22	150°51.9'	N17°00.6'	7	23	165°51.9'	N17°01.3'
8	0	180°51.9'	N17°02.0'	8	1	195°52.0'	N17°02.7'	8	2	210°52.0'	N17°03.3'
8	3	225°52.1'	N17°04.0'	8	4	240°52.1'	N17°04.7'	8	5	255°52.1'	N17°05.4'
8	6	270°52.2'	N17°06.0'	8	7	285°52.2'	N17°06.7'	8	8	300°52.2'	N17°07.4'
8	9	315°52.3'	N17°08.1'	8	10	330°52.3'	N17°08.8'	8	11	345°52.3'	N17°09.4'
8	12	0°52.4'	N17°10.1'	8	13	15°52.4'	N17°10.8'	8	14	30°52.4'	N17°11.5'
8	15	45°52.5'	N17°12.1'	8	16	60°52.5'	N17°12.8'	8	17	75°52.5'	N17°13.5'
8	18	90°52.6'	N17°14.1'	8	19	105°52.6'	N17°14.8'	8	20	120°52.6'	N17°15.5'
8	21	135°52.7'	N17°16.2'	8	22	150°52.7'	N17°16.8'	8	23	165°52.7'	N17°17.5'
9	0	180°52.7'	N17°18.2'	9	1	195°52.8'	N17°18.8'	9	2	210°52.8'	N17°19.5'
9	3	225°52.8'	N17°20.2'	9	4	240°52.9'	N17°20.8'	9	5	255°52.9'	N17°21.5'
9	6	270°52.9'	N17°22.2'	9	7	285°53.0'	N17°22.8'	9	8	300°53.0'	N17°23.5'
9	9	315°53.0'	N17°24.2'	9	10	330°53.0'	N17°24.8'	9	11	345°53.1'	N17°25.5'
9	12	0°53.1'	N17°26.2'	9	13	15°53.1'	N17°26.8'	9	14	30°53.1'	N17°27.5'
9	15	45°53.2'	N17°28.1'	9	16	60°53.2'	N17°28.8'	9	17	75°53.2'	N17°29.5'
9	18	90°53.2'	N17°30.1'	9	19	105°53.3'	N17°30.8'	9	20	120°53.3'	N17°31.4'
9	21	135°53.3'	N17°32.1'	9	22	150°53.4'	N17°32.8'	9	23	165°53.4'	N17°33.4'
10	0	180°53.4'	N17°34.1'	10	1	195°53.4'	N17°34.7'	10	2	210°53.4'	N17°35.4'
10	3	225°53.5'	N17°36.0'	10	4	240°53.5'	N17°36.7'	10	5	255°53.5'	N17°37.3'
10	6	270°53.5'	N17°38.0'	10	7	285°53.6'	N17°38.7'	10	8	300°53.6'	N17°39.3'
10	9	315°53.6'	N17°40.0'	10	10	330°53.6'	N17°40.6'	10	11	345°53.7'	N17°41.3'
10	12	0°53.7'	N17°41.9'	10	13	15°53.7'	N17°42.6'	10	14	30°53.7'	N17°43.2'
10	15	45°53.7'	N17°43.9'	10	16	60°53.8'	N17°44.5'	10	17	75°53.8'	N17°45.2'
10	18	90°53.8'	N17°45.8'	10	19	105°53.8'	N17°46.5'	10	20	120°53.8'	N17°47.1'
10	21	135°53.9'	N17°47.7'	10	22	150°53.9'	N17°48.4'	10	23	165°53.9'	N17°49.0'
11	0	180°53.9'	N17°49.7'	11	1	195°53.9'	N17°50.3'	11	2	210°53.9'	N17°51.0'

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Sun — GHA and Declination, May 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
11	3	225°54.0'	N17°51.6'	11	4	240°54.0'	N17°52.3'	11	5	255°54.0'	N17°52.9'
11	6	270°54.0'	N17°53.5'	11	7	285°54.0'	N17°54.2'	11	8	300°54.0'	N17°54.8'
11	9	315°54.1'	N17°55.5'	11	10	330°54.1'	N17°56.1'	11	11	345°54.1'	N17°56.7'
11	12	0°54.1'	N17°57.4'	11	13	15°54.1'	N17°58.0'	11	14	30°54.1'	N17°58.6'
11	15	45°54.2'	N17°59.3'	11	16	60°54.2'	N17°59.9'	11	17	75°54.2'	N18°00.6'
11	18	90°54.2'	N18°01.2'	11	19	105°54.2'	N18°01.8'	11	20	120°54.2'	N18°02.5'
11	21	135°54.2'	N18°03.1'	11	22	150°54.3'	N18°03.7'	11	23	165°54.3'	N18°04.4'
12	0	180°54.3'	N18°05.0'	12	1	195°54.3'	N18°05.6'	12	2	210°54.3'	N18°06.3'
12	3	225°54.3'	N18°06.9'	12	4	240°54.3'	N18°07.5'	12	5	255°54.3'	N18°08.1'
12	6	270°54.3'	N18°08.8'	12	7	285°54.4'	N18°09.4'	12	8	300°54.4'	N18°10.0'
12	9	315°54.4'	N18°10.7'	12	10	330°54.4'	N18°11.3'	12	11	345°54.4'	N18°11.9'
12	12	0°54.4'	N18°12.5'	12	13	15°54.4'	N18°13.2'	12	14	30°54.4'	N18°13.8'
12	15	45°54.4'	N18°14.4'	12	16	60°54.4'	N18°15.0'	12	17	75°54.5'	N18°15.7'
12	18	90°54.5'	N18°16.3'	12	19	105°54.5'	N18°16.9'	12	20	120°54.5'	N18°17.5'
12	21	135°54.5'	N18°18.1'	12	22	150°54.5'	N18°18.8'	12	23	165°54.5'	N18°19.4'
13	0	180°54.5'	N18°20.0'	13	1	195°54.5'	N18°20.6'	13	2	210°54.5'	N18°21.2'
13	3	225°54.5'	N18°21.9'	13	4	240°54.5'	N18°22.5'	13	5	255°54.5'	N18°23.1'
13	6	270°54.5'	N18°23.7'	13	7	285°54.5'	N18°24.3'	13	8	300°54.5'	N18°24.9'
13	9	315°54.6'	N18°25.5'	13	10	330°54.6'	N18°26.2'	13	11	345°54.6'	N18°26.8'
13	12	0°54.6'	N18°27.4'	13	13	15°54.6'	N18°28.0'	13	14	30°54.6'	N18°28.6'
13	15	45°54.6'	N18°29.2'	13	16	60°54.6'	N18°29.8'	13	17	75°54.6'	N18°30.4'
13	18	90°54.6'	N18°31.0'	13	19	105°54.6'	N18°31.7'	13	20	120°54.6'	N18°32.3'
13	21	135°54.6'	N18°32.9'	13	22	150°54.6'	N18°33.5'	13	23	165°54.6'	N18°34.1'
14	0	180°54.6'	N18°34.7'	14	1	195°54.6'	N18°35.3'	14	2	210°54.6'	N18°35.9'
14	3	225°54.6'	N18°36.5'	14	4	240°54.6'	N18°37.1'	14	5	255°54.6'	N18°37.7'
14	6	270°54.6'	N18°38.3'	14	7	285°54.6'	N18°38.9'	14	8	300°54.6'	N18°39.5'
14	9	315°54.6'	N18°40.1'	14	10	330°54.6'	N18°40.7'	14	11	345°54.6'	N18°41.3'
14	12	0°54.6'	N18°41.9'	14	13	15°54.6'	N18°42.5'	14	14	30°54.6'	N18°43.1'
14	15	45°54.6'	N18°43.7'	14	16	60°54.6'	N18°44.3'	14	17	75°54.6'	N18°44.9'
14	18	90°54.6'	N18°45.5'	14	19	105°54.5'	N18°46.1'	14	20	120°54.5'	N18°46.7'
14	21	135°54.5'	N18°47.3'	14	22	150°54.5'	N18°47.9'	14	23	165°54.5'	N18°48.5'
15	0	180°54.5'	N18°49.1'	15	1	195°54.5'	N18°49.7'	15	2	210°54.5'	N18°50.3'
15	3	225°54.5'	N18°50.9'	15	4	240°54.5'	N18°51.4'	15	5	255°54.5'	N18°52.0'
15	6	270°54.5'	N18°52.6'	15	7	285°54.5'	N18°53.2'	15	8	300°54.5'	N18°53.8'
15	9	315°54.5'	N18°54.4'	15	10	330°54.5'	N18°55.0'	15	11	345°54.5'	N18°55.6'
15	12	0°54.4'	N18°56.2'	15	13	15°54.4'	N18°56.7'	15	14	30°54.4'	N18°57.3'
15	15	45°54.4'	N18°57.9'	15	16	60°54.4'	N18°58.5'	15	17	75°54.4'	N18°59.1'
15	18	90°54.4'	N18°59.7'	15	19	105°54.4'	N19°00.2'	15	20	120°54.4'	N19°00.8'
15	21	135°54.4'	N19°01.4'	15	22	150°54.3'	N19°02.0'	15	23	165°54.3'	N19°02.6'
16	0	180°54.3'	N19°03.1'	16	1	195°54.3'	N19°03.7'	16	2	210°54.3'	N19°04.3'
16	3	225°54.3'	N19°04.9'	16	4	240°54.3'	N19°05.5'	16	5	255°54.3'	N19°06.0'
16	6	270°54.3'	N19°06.6'	16	7	285°54.2'	N19°07.2'	16	8	300°54.2'	N19°07.8'
16	9	315°54.2'	N19°08.3'	16	10	330°54.2'	N19°08.9'	16	11	345°54.2'	N19°09.5'
16	12	0°54.2'	N19°10.1'	16	13	15°54.2'	N19°10.6'	16	14	30°54.1'	N19°11.2'
16	15	45°54.1'	N19°11.8'	16	16	60°54.1'	N19°12.3'	16	17	75°54.1'	N19°12.9'
16	18	90°54.1'	N19°13.5'	16	19	105°54.1'	N19°14.1'	16	20	120°54.0'	N19°14.6'
16	21	135°54.0'	N19°15.2'	16	22	150°54.0'	N19°15.8'	16	23	165°54.0'	N19°16.3'
17	0	180°54.0'	N19°16.9'	17	1	195°54.0'	N19°17.5'	17	2	210°53.9'	N19°18.0'
17	3	225°53.9'	N19°18.6'	17	4	240°53.9'	N19°19.1'	17	5	255°53.9'	N19°19.7'
17	6	270°53.9'	N19°20.3'	17	7	285°53.9'	N19°20.8'	17	8	300°53.8'	N19°21.4'
17	9	315°53.8'	N19°22.0'	17	10	330°53.8'	N19°22.5'	17	11	345°53.8'	N19°23.1'
17	12	0°53.8'	N19°23.6'	17	13	15°53.7'	N19°24.2'	17	14	30°53.7'	N19°24.8'
17	15	45°53.7'	N19°25.3'	17	16	60°53.7'	N19°25.9'	17	17	75°53.7'	N19°26.4'
17	18	90°53.6'	N19°27.0'	17	19	105°53.6'	N19°27.5'	17	20	120°53.6'	N19°28.1'
17	21	135°53.6'	N19°28.6'	17	22	150°53.5'	N19°29.2'	17	23	165°53.5'	N19°29.8'
18	0	180°53.5'	N19°30.3'	18	1	195°53.5'	N19°30.9'	18	2	210°53.5'	N19°31.4'
18	3	225°53.4'	N19°32.0'	18	4	240°53.4'	N19°32.5'	18	5	255°53.4'	N19°33.1'
18	6	270°53.4'	N19°33.6'	18	7	285°53.3'	N19°34.2'	18	8	300°53.3'	N19°34.7'
18	9	315°53.3'	N19°35.2'	18	10	330°53.3'	N19°35.8'	18	11	345°53.2'	N19°36.3'
18	12	0°53.2'	N19°36.9'	18	13	15°53.2'	N19°37.4'	18	14	30°53.2'	N19°38.0'
18	15	45°53.1'	N19°38.5'	18	16	60°53.1'	N19°39.1'	18	17	75°53.1'	N19°39.6'
18	18	90°53.0'	N19°40.1'	18	19	105°53.0'	N19°40.7'	18	20	120°53.0'	N19°41.2'
18	21	135°53.0'	N19°41.8'	18	22	150°52.9'	N19°42.3'	18	23	165°52.9'	N19°42.8'
19	0	180°52.9'	N19°43.4'	19	1	195°52.8'	N19°43.9'	19	2	210°52.8'	N19°44.5'

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Sun — GHA and Declination, May 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
19	3	225°52.8'	N19°45.0'	19	4	240°52.8'	N19°45.5'	19	5	255°52.7'	N19°46.1'
19	6	270°52.7'	N19°46.6'	19	7	285°52.7'	N19°47.1'	19	8	300°52.6'	N19°47.7'
19	9	315°52.6'	N19°48.2'	19	10	330°52.6'	N19°48.7'	19	11	345°52.5'	N19°49.3'
19	12	0°52.5'	N19°49.8'	19	13	15°52.5'	N19°50.3'	19	14	30°52.5'	N19°50.9'
19	15	45°52.4'	N19°51.4'	19	16	60°52.4'	N19°51.9'	19	17	75°52.4'	N19°52.5'
19	18	90°52.3'	N19°53.0'	19	19	105°52.3'	N19°53.5'	19	20	120°52.3'	N19°54.0'
19	21	135°52.2'	N19°54.6'	19	22	150°52.2'	N19°55.1'	19	23	165°52.2'	N19°55.6'
20	0	180°52.1'	N19°56.1'	20	1	195°52.1'	N19°56.7'	20	2	210°52.0'	N19°57.2'
20	3	225°52.0'	N19°57.7'	20	4	240°52.0'	N19°58.2'	20	5	255°51.9'	N19°58.7'
20	6	270°51.9'	N19°59.3'	20	7	285°51.9'	N19°59.8'	20	8	300°51.8'	N20°00.3'
20	9	315°51.8'	N20°00.8'	20	10	330°51.8'	N20°01.3'	20	11	345°51.7'	N20°01.9'
20	12	0°51.7'	N20°02.4'	20	13	15°51.7'	N20°02.9'	20	14	30°51.6'	N20°03.4'
20	15	45°51.6'	N20°03.9'	20	16	60°51.5'	N20°04.4'	20	17	75°51.5'	N20°05.0'
20	18	90°51.5'	N20°05.5'	20	19	105°51.4'	N20°06.0'	20	20	120°51.4'	N20°06.5'
20	21	135°51.3'	N20°07.0'	20	22	150°51.3'	N20°07.5'	20	23	165°51.3'	N20°08.0'
21	0	180°51.2'	N20°08.5'	21	1	195°51.2'	N20°09.1'	21	2	210°51.1'	N20°09.6'
21	3	225°51.1'	N20°10.1'	21	4	240°51.1'	N20°10.6'	21	5	255°51.0'	N20°11.1'
21	6	270°51.0'	N20°11.6'	21	7	285°50.9'	N20°12.1'	21	8	300°50.9'	N20°12.6'
21	9	315°50.9'	N20°13.1'	21	10	330°50.8'	N20°13.6'	21	11	345°50.8'	N20°14.1'
21	12	0°50.7'	N20°14.6'	21	13	15°50.7'	N20°15.1'	21	14	30°50.6'	N20°15.6'
21	15	45°50.6'	N20°16.1'	21	16	60°50.6'	N20°16.6'	21	17	75°50.5'	N20°17.1'
21	18	90°50.5'	N20°17.6'	21	19	105°50.4'	N20°18.1'	21	20	120°50.4'	N20°18.6'
21	21	135°50.3'	N20°19.1'	21	22	150°50.3'	N20°19.6'	21	23	165°50.2'	N20°20.1'
22	0	180°50.2'	N20°20.6'	22	1	195°50.2'	N20°21.1'	22	2	210°50.1'	N20°21.6'
22	3	225°50.1'	N20°22.1'	22	4	240°50.0'	N20°22.6'	22	5	255°50.0'	N20°23.1'
22	6	270°49.9'	N20°23.6'	22	7	285°49.9'	N20°24.1'	22	8	300°49.8'	N20°24.6'
22	9	315°49.8'	N20°25.0'	22	10	330°49.7'	N20°25.5'	22	11	345°49.7'	N20°26.0'
22	12	0°49.6'	N20°26.5'	22	13	15°49.6'	N20°27.0'	22	14	30°49.5'	N20°27.5'
22	15	45°49.5'	N20°28.0'	22	16	60°49.4'	N20°28.5'	22	17	75°49.4'	N20°28.9'
22	18	90°49.3'	N20°29.4'	22	19	105°49.3'	N20°29.9'	22	20	120°49.2'	N20°30.4'
22	21	135°49.2'	N20°30.9'	22	22	150°49.1'	N20°31.4'	22	23	165°49.1'	N20°31.8'
23	0	180°49.0'	N20°32.3'	23	1	195°49.0'	N20°32.8'	23	2	210°48.9'	N20°33.3'
23	3	225°48.9'	N20°33.8'	23	4	240°48.8'	N20°34.2'	23	5	255°48.8'	N20°34.7'
23	6	270°48.7'	N20°35.2'	23	7	285°48.7'	N20°35.7'	23	8	300°48.6'	N20°36.2'
23	9	315°48.6'	N20°36.6'	23	10	330°48.5'	N20°37.1'	23	11	345°48.5'	N20°37.6'
23	12	0°48.4'	N20°38.0'	23	13	15°48.4'	N20°38.5'	23	14	30°48.3'	N20°39.0'
23	15	45°48.3'	N20°39.5'	23	16	60°48.2'	N20°39.9'	23	17	75°48.1'	N20°40.4'
23	18	90°48.1'	N20°40.9'	23	19	105°48.0'	N20°41.3'	23	20	120°48.0'	N20°41.8'
23	21	135°47.9'	N20°42.3'	23	22	150°47.9'	N20°42.8'	23	23	165°47.8'	N20°43.2'
24	0	180°47.8'	N20°43.7'	24	1	195°47.7'	N20°44.2'	24	2	210°47.6'	N20°44.6'
24	3	225°47.6'	N20°45.1'	24	4	240°47.5'	N20°45.5'	24	5	255°47.5'	N20°46.0'
24	6	270°47.4'	N20°46.5'	24	7	285°47.4'	N20°46.9'	24	8	300°47.3'	N20°47.4'
24	9	315°47.2'	N20°47.9'	24	10	330°47.2'	N20°48.3'	24	11	345°47.1'	N20°48.8'
24	12	0°47.1'	N20°49.2'	24	13	15°47.0'	N20°49.7'	24	14	30°46.9'	N20°50.1'
24	15	45°46.9'	N20°50.6'	24	16	60°46.8'	N20°51.1'	24	17	75°46.8'	N20°51.5'
24	18	90°46.7'	N20°52.0'	24	19	105°46.6'	N20°52.4'	24	20	120°46.6'	N20°52.9'
24	21	135°46.5'	N20°53.3'	24	22	150°46.5'	N20°53.8'	24	23	165°46.4'	N20°54.2'
25	0	180°46.3'	N20°54.7'	25	1	195°46.3'	N20°55.1'	25	2	210°46.2'	N20°55.6'
25	3	225°46.2'	N20°56.0'	25	4	240°46.1'	N20°56.5'	25	5	255°46.0'	N20°56.9'
25	6	270°46.0'	N20°57.4'	25	7	285°45.9'	N20°57.8'	25	8	300°45.8'	N20°58.3'
25	9	315°45.8'	N20°58.7'	25	10	330°45.7'	N20°59.2'	25	11	345°45.6'	N20°59.6'
25	12	0°45.6'	N21°00.1'	25	13	15°45.5'	N21°00.5'	25	14	30°45.5'	N21°00.9'
25	15	45°45.4'	N21°01.4'	25	16	60°45.3'	N21°01.8'	25	17	75°45.3'	N21°02.3'
25	18	90°45.2'	N21°02.7'	25	19	105°45.1'	N21°03.2'	25	20	120°45.1'	N21°03.6'
25	21	135°45.0'	N21°04.0'	25	22	150°44.9'	N21°04.5'	25	23	165°44.9'	N21°04.9'
26	0	180°44.8'	N21°05.3'	26	1	195°44.7'	N21°05.8'	26	2	210°44.7'	N21°06.2'
26	3	225°44.6'	N21°06.6'	26	4	240°44.5'	N21°07.1'	26	5	255°44.5'	N21°07.5'
26	6	270°44.4'	N21°07.9'	26	7	285°44.3'	N21°08.4'	26	8	300°44.3'	N21°08.8'
26	9	315°44.2'	N21°09.2'	26	10	330°44.1'	N21°09.7'	26	11	345°44.1'	N21°10.1'
26	12	0°44.0'	N21°10.5'	26	13	15°43.9'	N21°11.0'	26	14	30°43.8'	N21°11.4'
26	15	45°43.8'	N21°11.8'	26	16	60°43.7'	N21°12.2'	26	17	75°43.6'	N21°12.7'
26	18	90°43.6'	N21°13.1'	26	19	105°43.5'	N21°13.5'	26	20	120°43.4'	N21°13.9'
26	21	135°43.4'	N21°14.4'	26	22	150°43.3'	N21°14.8'	26	23	165°43.2'	N21°15.2'
27	0	180°43.1'	N21°15.6'	27	1	195°43.1'	N21°16.0'	27	2	210°43.0'	N21°16.5'

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Sun — GHA and Declination, May 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
27	3	225°42.9'	N21°16.9'	27	4	240°42.9'	N21°17.3'	27	5	255°42.8'	N21°17.7'
27	6	270°42.7'	N21°18.1'	27	7	285°42.6'	N21°18.6'	27	8	300°42.6'	N21°19.0'
27	9	315°42.5'	N21°19.4'	27	10	330°42.4'	N21°19.8'	27	11	345°42.3'	N21°20.2'
27	12	0°42.3'	N21°20.6'	27	13	15°42.2'	N21°21.0'	27	14	30°42.1'	N21°21.4'
27	15	45°42.0'	N21°21.9'	27	16	60°42.0'	N21°22.3'	27	17	75°41.9'	N21°22.7'
27	18	90°41.8'	N21°23.1'	27	19	105°41.7'	N21°23.5'	27	20	120°41.7'	N21°23.9'
27	21	135°41.6'	N21°24.3'	27	22	150°41.5'	N21°24.7'	27	23	165°41.4'	N21°25.1'
28	0	180°41.4'	N21°25.5'	28	1	195°41.3'	N21°25.9'	28	2	210°41.2'	N21°26.3'
28	3	225°41.1'	N21°26.7'	28	4	240°41.1'	N21°27.2'	28	5	255°41.0'	N21°27.6'
28	6	270°40.9'	N21°28.0'	28	7	285°40.8'	N21°28.4'	28	8	300°40.7'	N21°28.8'
28	9	315°40.7'	N21°29.2'	28	10	330°40.6'	N21°29.6'	28	11	345°40.5'	N21°30.0'
28	12	0°40.4'	N21°30.4'	28	13	15°40.4'	N21°30.8'	28	14	30°40.3'	N21°31.1'
28	15	45°40.2'	N21°31.5'	28	16	60°40.1'	N21°31.9'	28	17	75°40.0'	N21°32.3'
28	18	90°40.0'	N21°32.7'	28	19	105°39.9'	N21°33.1'	28	20	120°39.8'	N21°33.5'
28	21	135°39.7'	N21°33.9'	28	22	150°39.6'	N21°34.3'	28	23	165°39.6'	N21°34.7'
29	0	180°39.5'	N21°35.1'	29	1	195°39.4'	N21°35.5'	29	2	210°39.3'	N21°35.9'
29	3	225°39.2'	N21°36.2'	29	4	240°39.1'	N21°36.6'	29	5	255°39.1'	N21°37.0'
29	6	270°39.0'	N21°37.4'	29	7	285°38.9'	N21°37.8'	29	8	300°38.8'	N21°38.2'
29	9	315°38.7'	N21°38.6'	29	10	330°38.7'	N21°38.9'	29	11	345°38.6'	N21°39.3'
29	12	0°38.5'	N21°39.7'	29	13	15°38.4'	N21°40.1'	29	14	30°38.3'	N21°40.5'
29	15	45°38.2'	N21°40.9'	29	16	60°38.1'	N21°41.2'	29	17	75°38.1'	N21°41.6'
29	18	90°38.0'	N21°42.0'	29	19	105°37.9'	N21°42.4'	29	20	120°37.8'	N21°42.7'
29	21	135°37.7'	N21°43.1'	29	22	150°37.6'	N21°43.5'	29	23	165°37.6'	N21°43.9'
30	0	180°37.5'	N21°44.2'	30	1	195°37.4'	N21°44.6'	30	2	210°37.3'	N21°45.0'
30	3	225°37.2'	N21°45.4'	30	4	240°37.1'	N21°45.7'	30	5	255°37.0'	N21°46.1'
30	6	270°37.0'	N21°46.5'	30	7	285°36.9'	N21°46.9'	30	8	300°36.8'	N21°47.2'
30	9	315°36.7'	N21°47.6'	30	10	330°36.6'	N21°48.0'	30	11	345°36.5'	N21°48.3'
30	12	0°36.4'	N21°48.7'	30	13	15°36.3'	N21°49.1'	30	14	30°36.3'	N21°49.4'
30	15	45°36.2'	N21°49.8'	30	16	60°36.1'	N21°50.2'	30	17	75°36.0'	N21°50.5'
30	18	90°35.9'	N21°50.9'	30	19	105°35.8'	N21°51.2'	30	20	120°35.7'	N21°51.6'
30	21	135°35.6'	N21°52.0'	30	22	150°35.5'	N21°52.3'	30	23	165°35.4'	N21°52.7'
31	0	180°35.4'	N21°53.0'	31	1	195°35.3'	N21°53.4'	31	2	210°35.2'	N21°53.8'
31	3	225°35.1'	N21°54.1'	31	4	240°35.0'	N21°54.5'	31	5	255°34.9'	N21°54.8'
31	6	270°34.8'	N21°55.2'	31	7	285°34.7'	N21°55.5'	31	8	300°34.6'	N21°55.9'
31	9	315°34.5'	N21°56.2'	31	10	330°34.4'	N21°56.6'	31	11	345°34.4'	N21°56.9'
31	12	0°34.3'	N21°57.3'	31	13	15°34.2'	N21°57.6'	31	14	30°34.1'	N21°58.0'
31	15	45°34.0'	N21°58.3'	31	16	60°33.9'	N21°58.7'	31	17	75°33.8'	N21°59.0'
31	18	90°33.7'	N21°59.4'	31	19	105°33.6'	N21°59.7'	31	20	120°33.5'	N22°00.1'
31	21	135°33.4'	N22°00.4'	31	22	150°33.3'	N22°00.8'	31	23	165°33.2'	N22°01.1'

Table 9.6: Sun — GHA and Declination, June 2026 (UT)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
1	0	180°33.1'	N22°01.4'	1	1	195°33.0'	N22°01.8'	1	2	210°33.0'	N22°02.1'
1	3	225°32.9'	N22°02.5'	1	4	240°32.8'	N22°02.8'	1	5	255°32.7'	N22°03.2'
1	6	270°32.6'	N22°03.5'	1	7	285°32.5'	N22°03.8'	1	8	300°32.4'	N22°04.2'
1	9	315°32.3'	N22°04.5'	1	10	330°32.2'	N22°04.8'	1	11	345°32.1'	N22°05.2'
1	12	0°32.0'	N22°05.5'	1	13	15°31.9'	N22°05.8'	1	14	30°31.8'	N22°06.2'
1	15	45°31.7'	N22°06.5'	1	16	60°31.6'	N22°06.8'	1	17	75°31.5'	N22°07.2'
1	18	90°31.4'	N22°07.5'	1	19	105°31.3'	N22°07.8'	1	20	120°31.2'	N22°08.2'
1	21	135°31.1'	N22°08.5'	1	22	150°31.0'	N22°08.8'	1	23	165°30.9'	N22°09.1'
2	0	180°30.8'	N22°09.5'	2	1	195°30.7'	N22°09.8'	2	2	210°30.6'	N22°10.1'
2	3	225°30.5'	N22°10.5'	2	4	240°30.4'	N22°10.8'	2	5	255°30.3'	N22°11.1'
2	6	270°30.2'	N22°11.4'	2	7	285°30.1'	N22°11.7'	2	8	300°30.0'	N22°12.1'
2	9	315°29.9'	N22°12.4'	2	10	330°29.8'	N22°12.7'	2	11	345°29.7'	N22°13.0'
2	12	0°29.6'	N22°13.3'	2	13	15°29.5'	N22°13.7'	2	14	30°29.4'	N22°14.0'
2	15	45°29.3'	N22°14.3'	2	16	60°29.2'	N22°14.6'	2	17	75°29.1'	N22°14.9'
2	18	90°29.0'	N22°15.2'	2	19	105°28.9'	N22°15.6'	2	20	120°28.8'	N22°15.9'
2	21	135°28.7'	N22°16.2'	2	22	150°28.6'	N22°16.5'	2	23	165°28.5'	N22°16.8'
3	0	180°28.4'	N22°17.1'	3	1	195°28.3'	N22°17.4'	3	2	210°28.2'	N22°17.7'
3	3	225°28.1'	N22°18.0'	3	4	240°28.0'	N22°18.4'	3	5	255°27.9'	N22°18.7'
3	6	270°27.8'	N22°19.0'	3	7	285°27.7'	N22°19.3'	3	8	300°27.6'	N22°19.6'

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Sun — GHA and Declination, June 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
3	9	315°27.5'	N22°19.9'	3	10	330°27.4'	N22°20.2'	3	11	345°27.3'	N22°20.5'
3	12	0°27.2'	N22°20.8'	3	13	15°27.1'	N22°21.1'	3	14	30°27.0'	N22°21.4'
3	15	45°26.9'	N22°21.7'	3	16	60°26.8'	N22°22.0'	3	17	75°26.7'	N22°22.3'
3	18	90°26.6'	N22°22.6'	3	19	105°26.4'	N22°22.9'	3	20	120°26.3'	N22°23.2'
3	21	135°26.2'	N22°23.5'	3	22	150°26.1'	N22°23.8'	3	23	165°26.0'	N22°24.1'
4	0	180°25.9'	N22°24.4'	4	1	195°25.8'	N22°24.7'	4	2	210°25.7'	N22°25.0'
4	3	225°25.6'	N22°25.2'	4	4	240°25.5'	N22°25.5'	4	5	255°25.4'	N22°25.8'
4	6	270°25.3'	N22°26.1'	4	7	285°25.2'	N22°26.4'	4	8	300°25.1'	N22°26.7'
4	9	315°25.0'	N22°27.0'	4	10	330°24.8'	N22°27.3'	4	11	345°24.7'	N22°27.6'
4	12	0°24.6'	N22°27.8'	4	13	15°24.5'	N22°28.1'	4	14	30°24.4'	N22°28.4'
4	15	45°24.3'	N22°28.7'	4	16	60°24.2'	N22°29.0'	4	17	75°24.1'	N22°29.3'
4	18	90°24.0'	N22°29.5'	4	19	105°23.9'	N22°29.8'	4	20	120°23.8'	N22°30.1'
4	21	135°23.7'	N22°30.4'	4	22	150°23.5'	N22°30.7'	4	23	165°23.4'	N22°30.9'
5	0	180°23.3'	N22°31.2'	5	1	195°23.2'	N22°31.5'	5	2	210°23.1'	N22°31.8'
5	3	225°23.0'	N22°32.1'	5	4	240°22.9'	N22°32.3'	5	5	255°22.8'	N22°32.6'
5	6	270°22.7'	N22°32.9'	5	7	285°22.6'	N22°33.2'	5	8	300°22.4'	N22°33.4'
5	9	315°22.3'	N22°33.7'	5	10	330°22.2'	N22°34.0'	5	11	345°22.1'	N22°34.2'
5	12	0°22.0'	N22°34.5'	5	13	15°21.9'	N22°34.8'	5	14	30°21.8'	N22°35.0'
5	15	45°21.7'	N22°35.3'	5	16	60°21.6'	N22°35.6'	5	17	75°21.4'	N22°35.8'
5	18	90°21.3'	N22°36.1'	5	19	105°21.2'	N22°36.4'	5	20	120°21.1'	N22°36.6'
5	21	135°21.0'	N22°36.9'	5	22	150°20.9'	N22°37.2'	5	23	165°20.8'	N22°37.4'
6	0	180°20.7'	N22°37.7'	6	1	195°20.5'	N22°38.0'	6	2	210°20.4'	N22°38.2'
6	3	225°20.3'	N22°38.5'	6	4	240°20.2'	N22°38.7'	6	5	255°20.1'	N22°39.0'
6	6	270°20.0'	N22°39.2'	6	7	285°19.9'	N22°39.5'	6	8	300°19.7'	N22°39.8'
6	9	315°19.6'	N22°40.0'	6	10	330°19.5'	N22°40.3'	6	11	345°19.4'	N22°40.5'
6	12	0°19.3'	N22°40.8'	6	13	15°19.2'	N22°41.0'	6	14	30°19.1'	N22°41.3'
6	15	45°18.9'	N22°41.5'	6	16	60°18.8'	N22°41.8'	6	17	75°18.7'	N22°42.0'
6	18	90°18.6'	N22°42.3'	6	19	105°18.5'	N22°42.5'	6	20	120°18.4'	N22°42.8'
6	21	135°18.3'	N22°43.0'	6	22	150°18.1'	N22°43.3'	6	23	165°18.0'	N22°43.5'
7	0	180°17.9'	N22°43.8'	7	1	195°17.8'	N22°44.0'	7	2	210°17.7'	N22°44.2'
7	3	225°17.6'	N22°44.5'	7	4	240°17.4'	N22°44.7'	7	5	255°17.3'	N22°45.0'
7	6	270°17.2'	N22°45.2'	7	7	285°17.1'	N22°45.5'	7	8	300°17.0'	N22°45.7'
7	9	315°16.9'	N22°45.9'	7	10	330°16.7'	N22°46.2'	7	11	345°16.6'	N22°46.4'
7	12	0°16.5'	N22°46.6'	7	13	15°16.4'	N22°46.9'	7	14	30°16.3'	N22°47.1'
7	15	45°16.2'	N22°47.3'	7	16	60°16.0'	N22°47.6'	7	17	75°15.9'	N22°47.8'
7	18	90°15.8'	N22°48.0'	7	19	105°15.7'	N22°48.3'	7	20	120°15.6'	N22°48.5'
7	21	135°15.4'	N22°48.7'	7	22	150°15.3'	N22°49.0'	7	23	165°15.2'	N22°49.2'
8	0	180°15.1'	N22°49.4'	8	1	195°15.0'	N22°49.7'	8	2	210°14.8'	N22°49.9'
8	3	225°14.7'	N22°50.1'	8	4	240°14.6'	N22°50.3'	8	5	255°14.5'	N22°50.6'
8	6	270°14.4'	N22°50.8'	8	7	285°14.2'	N22°51.0'	8	8	300°14.1'	N22°51.2'
8	9	315°14.0'	N22°51.4'	8	10	330°13.9'	N22°51.7'	8	11	345°13.8'	N22°51.9'
8	12	0°13.6'	N22°52.1'	8	13	15°13.5'	N22°52.3'	8	14	30°13.4'	N22°52.5'
8	15	45°13.3'	N22°52.8'	8	16	60°13.2'	N22°53.0'	8	17	75°13.0'	N22°53.2'
8	18	90°12.9'	N22°53.4'	8	19	105°12.8'	N22°53.6'	8	20	120°12.7'	N22°53.8'
8	21	135°12.6'	N22°54.1'	8	22	150°12.4'	N22°54.3'	8	23	165°12.3'	N22°54.5'
9	0	180°12.2'	N22°54.7'	9	1	195°12.1'	N22°54.9'	9	2	210°12.0'	N22°55.1'
9	3	225°11.8'	N22°55.3'	9	4	240°11.7'	N22°55.5'	9	5	255°11.6'	N22°55.7'
9	6	270°11.5'	N22°55.9'	9	7	285°11.3'	N22°56.2'	9	8	300°11.2'	N22°56.4'
9	9	315°11.1'	N22°56.6'	9	10	330°11.0'	N22°56.8'	9	11	345°10.9'	N22°57.0'
9	12	0°10.7'	N22°57.2'	9	13	15°10.6'	N22°57.4'	9	14	30°10.5'	N22°57.6'
9	15	45°10.4'	N22°57.8'	9	16	60°10.2'	N22°58.0'	9	17	75°10.1'	N22°58.2'
9	18	90°10.0'	N22°58.4'	9	19	105°09.9'	N22°58.6'	9	20	120°09.7'	N22°58.8'
9	21	135°09.6'	N22°59.0'	9	22	150°09.5'	N22°59.2'	9	23	165°09.4'	N22°59.4'
10	0	180°09.2'	N22°59.6'	10	1	195°09.1'	N22°59.8'	10	2	210°09.0'	N22°59.9'
10	3	225°08.9'	N23°00.1'	10	4	240°08.7'	N23°00.3'	10	5	255°08.6'	N23°00.5'
10	6	270°08.5'	N23°00.7'	10	7	285°08.4'	N23°00.9'	10	8	300°08.2'	N23°01.1'
10	9	315°08.1'	N23°01.3'	10	10	330°08.0'	N23°01.5'	10	11	345°07.9'	N23°01.7'
10	12	0°07.7'	N23°01.8'	10	13	15°07.6'	N23°02.0'	10	14	30°07.5'	N23°02.2'
10	15	45°07.4'	N23°02.4'	10	16	60°07.2'	N23°02.6'	10	17	75°07.1'	N23°02.8'
10	18	90°07.0'	N23°02.9'	10	19	105°06.9'	N23°03.1'	10	20	120°06.7'	N23°03.3'
10	21	135°06.6'	N23°03.5'	10	22	150°06.5'	N23°03.7'	10	23	165°06.4'	N23°03.8'
11	0	180°06.2'	N23°04.0'	11	1	195°06.1'	N23°04.2'	11	2	210°06.0'	N23°04.4'
11	3	225°05.9'	N23°04.5'	11	4	240°05.7'	N23°04.7'	11	5	255°05.6'	N23°04.9'
11	6	270°05.5'	N23°05.1'	11	7	285°05.3'	N23°05.2'	11	8	300°05.2'	N23°05.4'

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Sun — GHA and Declination, June 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
11	9	315°05.1'	N23°05.6'	11	10	330°05.0'	N23°05.8'	11	11	345°04.8'	N23°05.9'
11	12	0°04.7'	N23°06.1'	11	13	15°04.6'	N23°06.3'	11	14	30°04.5'	N23°06.4'
11	15	45°04.3'	N23°06.6'	11	16	60°04.2'	N23°06.8'	11	17	75°04.1'	N23°06.9'
11	18	90°03.9'	N23°07.1'	11	19	105°03.8'	N23°07.3'	11	20	120°03.7'	N23°07.4'
11	21	135°03.6'	N23°07.6'	11	22	150°03.4'	N23°07.7'	11	23	165°03.3'	N23°07.9'
12	0	180°03.2'	N23°08.1'	12	1	195°03.0'	N23°08.2'	12	2	210°02.9'	N23°08.4'
12	3	225°02.8'	N23°08.5'	12	4	240°02.7'	N23°08.7'	12	5	255°02.5'	N23°08.9'
12	6	270°02.4'	N23°09.0'	12	7	285°02.3'	N23°09.2'	12	8	300°02.1'	N23°09.3'
12	9	315°02.0'	N23°09.5'	12	10	330°01.9'	N23°09.6'	12	11	345°01.8'	N23°09.8'
12	12	0°01.6'	N23°09.9'	12	13	15°01.5'	N23°10.1'	12	14	30°01.4'	N23°10.2'
12	15	45°01.2'	N23°10.4'	12	16	60°01.1'	N23°10.5'	12	17	75°01.0'	N23°10.7'
12	18	90°00.9'	N23°10.8'	12	19	105°00.7'	N23°11.0'	12	20	120°00.6'	N23°11.1'
12	21	135°00.5'	N23°11.3'	12	22	150°00.3'	N23°11.4'	12	23	165°00.2'	N23°11.6'
13	0	180°00.1'	N23°11.7'	13	1	194°59.9'	N23°11.9'	13	2	209°59.8'	N23°12.0'
13	3	224°59.7'	N23°12.1'	13	4	239°59.5'	N23°12.3'	13	5	254°59.4'	N23°12.4'
13	6	269°59.3'	N23°12.6'	13	7	284°59.2'	N23°12.7'	13	8	299°59.0'	N23°12.8'
13	9	314°58.9'	N23°13.0'	13	10	329°58.8'	N23°13.1'	13	11	344°58.6'	N23°13.2'
13	12	359°58.5'	N23°13.4'	13	13	14°58.4'	N23°13.5'	13	14	29°58.2'	N23°13.7'
13	15	44°58.1'	N23°13.8'	13	16	59°58.0'	N23°13.9'	13	17	74°57.8'	N23°14.1'
13	18	89°57.7'	N23°14.2'	13	19	104°57.6'	N23°14.3'	13	20	119°57.5'	N23°14.4'
13	21	134°57.3'	N23°14.6'	13	22	149°57.2'	N23°14.7'	13	23	164°57.1'	N23°14.8'
14	0	179°56.9'	N23°15.0'	14	1	194°56.8'	N23°15.1'	14	2	209°56.7'	N23°15.2'
14	3	224°56.5'	N23°15.3'	14	4	239°56.4'	N23°15.5'	14	5	254°56.3'	N23°15.6'
14	6	269°56.1'	N23°15.7'	14	7	284°56.0'	N23°15.8'	14	8	299°55.9'	N23°15.9'
14	9	314°55.7'	N23°16.1'	14	10	329°55.6'	N23°16.2'	14	11	344°55.5'	N23°16.3'
14	12	359°55.3'	N23°16.4'	14	13	14°55.2'	N23°16.5'	14	14	29°55.1'	N23°16.7'
14	15	44°54.9'	N23°16.8'	14	16	59°54.8'	N23°16.9'	14	17	74°54.7'	N23°17.0'
14	18	89°54.5'	N23°17.1'	14	19	104°54.4'	N23°17.2'	14	20	119°54.3'	N23°17.3'
14	21	134°54.1'	N23°17.4'	14	22	149°54.0'	N23°17.6'	14	23	164°53.9'	N23°17.7'
15	0	179°53.7'	N23°17.8'	15	1	194°53.6'	N23°17.9'	15	2	209°53.5'	N23°18.0'
15	3	224°53.3'	N23°18.1'	15	4	239°53.2'	N23°18.2'	15	5	254°53.1'	N23°18.3'
15	6	269°52.9'	N23°18.4'	15	7	284°52.8'	N23°18.5'	15	8	299°52.7'	N23°18.6'
15	9	314°52.5'	N23°18.7'	15	10	329°52.4'	N23°18.8'	15	11	344°52.3'	N23°18.9'
15	12	359°52.1'	N23°19.0'	15	13	14°52.0'	N23°19.1'	15	14	29°51.9'	N23°19.2'
15	15	44°51.7'	N23°19.3'	15	16	59°51.6'	N23°19.4'	15	17	74°51.5'	N23°19.5'
15	18	89°51.3'	N23°19.6'	15	19	104°51.2'	N23°19.7'	15	20	119°51.1'	N23°19.8'
15	21	134°50.9'	N23°19.9'	15	22	149°50.8'	N23°20.0'	15	23	164°50.7'	N23°20.1'
16	0	179°50.5'	N23°20.2'	16	1	194°50.4'	N23°20.3'	16	2	209°50.3'	N23°20.4'
16	3	224°50.1'	N23°20.5'	16	4	239°50.0'	N23°20.6'	16	5	254°49.9'	N23°20.6'
16	6	269°49.7'	N23°20.7'	16	7	284°49.6'	N23°20.8'	16	8	299°49.5'	N23°20.9'
16	9	314°49.3'	N23°21.0'	16	10	329°49.2'	N23°21.1'	16	11	344°49.1'	N23°21.2'
16	12	359°48.9'	N23°21.2'	16	13	14°48.8'	N23°21.3'	16	14	29°48.6'	N23°21.4'
16	15	44°48.5'	N23°21.5'	16	16	59°48.4'	N23°21.6'	16	17	74°48.2'	N23°21.7'
16	18	89°48.1'	N23°21.7'	16	19	104°48.0'	N23°21.8'	16	20	119°47.8'	N23°21.9'
16	21	134°47.7'	N23°22.0'	16	22	149°47.6'	N23°22.0'	16	23	164°47.4'	N23°22.1'
17	0	179°47.3'	N23°22.2'	17	1	194°47.2'	N23°22.3'	17	2	209°47.0'	N23°22.3'
17	3	224°46.9'	N23°22.4'	17	4	239°46.8'	N23°22.5'	17	5	254°46.6'	N23°22.6'
17	6	269°46.5'	N23°22.6'	17	7	284°46.3'	N23°22.7'	17	8	299°46.2'	N23°22.8'
17	9	314°46.1'	N23°22.8'	17	10	329°45.9'	N23°22.9'	17	11	344°45.8'	N23°23.0'
17	12	359°45.7'	N23°23.0'	17	13	14°45.5'	N23°23.1'	17	14	29°45.4'	N23°23.2'
17	15	44°45.3'	N23°23.2'	17	16	59°45.1'	N23°23.3'	17	17	74°45.0'	N23°23.4'
17	18	89°44.9'	N23°23.4'	17	19	104°44.7'	N23°23.5'	17	20	119°44.6'	N23°23.6'
17	21	134°44.5'	N23°23.6'	17	22	149°44.3'	N23°23.7'	17	23	164°44.2'	N23°23.7'
18	0	179°44.0'	N23°23.8'	18	1	194°43.9'	N23°23.8'	18	2	209°43.8'	N23°23.9'
18	3	224°43.6'	N23°24.0'	18	4	239°43.5'	N23°24.0'	18	5	254°43.4'	N23°24.1'
18	6	269°43.2'	N23°24.1'	18	7	284°43.1'	N23°24.2'	18	8	299°43.0'	N23°24.2'
18	9	314°42.8'	N23°24.3'	18	10	329°42.7'	N23°24.3'	18	11	344°42.5'	N23°24.4'
18	12	359°42.4'	N23°24.4'	18	13	14°42.3'	N23°24.5'	18	14	29°42.1'	N23°24.5'
18	15	44°42.0'	N23°24.6'	18	16	59°41.9'	N23°24.6'	18	17	74°41.7'	N23°24.7'
18	18	89°41.6'	N23°24.7'	18	19	104°41.5'	N23°24.8'	18	20	119°41.3'	N23°24.8'
18	21	134°41.2'	N23°24.8'	18	22	149°41.1'	N23°24.9'	18	23	164°40.9'	N23°24.9'
19	0	179°40.8'	N23°25.0'	19	1	194°40.6'	N23°25.0'	19	2	209°40.5'	N23°25.0'
19	3	224°40.4'	N23°25.1'	19	4	239°40.2'	N23°25.1'	19	5	254°40.1'	N23°25.2'
19	6	269°40.0'	N23°25.2'	19	7	284°39.8'	N23°25.2'	19	8	299°39.7'	N23°25.3'

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Sun — GHA and Declination, June 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
19	9	314°39.6'	N23°25.3'	19	10	329°39.4'	N23°25.3'	19	11	344°39.3'	N23°25.4'
19	12	359°39.1'	N23°25.4'	19	13	14°39.0'	N23°25.4'	19	14	29°38.9'	N23°25.5'
19	15	44°38.7'	N23°25.5'	19	16	59°38.6'	N23°25.5'	19	17	74°38.5'	N23°25.6'
19	18	89°38.3'	N23°25.6'	19	19	104°38.2'	N23°25.6'	19	20	119°38.1'	N23°25.6'
19	21	134°37.9'	N23°25.7'	19	22	149°37.8'	N23°25.7'	19	23	164°37.6'	N23°25.7'
20	0	179°37.5'	N23°25.7'	20	1	194°37.4'	N23°25.8'	20	2	209°37.2'	N23°25.8'
20	3	224°37.1'	N23°25.8'	20	4	239°37.0'	N23°25.8'	20	5	254°36.8'	N23°25.8'
20	6	269°36.7'	N23°25.9'	20	7	284°36.6'	N23°25.9'	20	8	299°36.4'	N23°25.9'
20	9	314°36.3'	N23°25.9'	20	10	329°36.1'	N23°25.9'	20	11	344°36.0'	N23°25.9'
20	12	359°35.9'	N23°26.0'	20	13	14°35.7'	N23°26.0'	20	14	29°35.6'	N23°26.0'
20	15	44°35.5'	N23°26.0'	20	16	59°35.3'	N23°26.0'	20	17	74°35.2'	N23°26.0'
20	18	89°35.1'	N23°26.0'	20	19	104°34.9'	N23°26.0'	20	20	119°34.8'	N23°26.1'
20	21	134°34.6'	N23°26.1'	20	22	149°34.5'	N23°26.1'	20	23	164°34.4'	N23°26.1'
21	0	179°34.2'	N23°26.1'	21	1	194°34.1'	N23°26.1'	21	2	209°34.0'	N23°26.1'
21	3	224°33.8'	N23°26.1'	21	4	239°33.7'	N23°26.1'	21	5	254°33.6'	N23°26.1'
21	6	269°33.4'	N23°26.1'	21	7	284°33.3'	N23°26.1'	21	8	299°33.2'	N23°26.1'
21	9	314°33.0'	N23°26.1'	21	10	329°32.9'	N23°26.1'	21	11	344°32.7'	N23°26.1'
21	12	359°32.6'	N23°26.1'	21	13	14°32.5'	N23°26.1'	21	14	29°32.3'	N23°26.1'
21	15	44°32.2'	N23°26.1'	21	16	59°32.1'	N23°26.1'	21	17	74°31.9'	N23°26.1'
21	18	89°31.8'	N23°26.1'	21	19	104°31.7'	N23°26.1'	21	20	119°31.5'	N23°26.1'
21	21	134°31.4'	N23°26.1'	21	22	149°31.2'	N23°26.0'	21	23	164°31.1'	N23°26.0'
22	0	179°31.0'	N23°26.0'	22	1	194°30.8'	N23°26.0'	22	2	209°30.7'	N23°26.0'
22	3	224°30.6'	N23°26.0'	22	4	239°30.4'	N23°26.0'	22	5	254°30.3'	N23°26.0'
22	6	269°30.2'	N23°25.9'	22	7	284°30.0'	N23°25.9'	22	8	299°29.9'	N23°25.9'
22	9	314°29.8'	N23°25.9'	22	10	329°29.6'	N23°25.9'	22	11	344°29.5'	N23°25.9'
22	12	359°29.3'	N23°25.8'	22	13	14°29.2'	N23°25.8'	22	14	29°29.1'	N23°25.8'
22	15	44°28.9'	N23°25.8'	22	16	59°28.8'	N23°25.7'	22	17	74°28.7'	N23°25.7'
22	18	89°28.5'	N23°25.7'	22	19	104°28.4'	N23°25.7'	22	20	119°28.3'	N23°25.7'
22	21	134°28.1'	N23°25.6'	22	22	149°28.0'	N23°25.6'	22	23	164°27.9'	N23°25.6'
23	0	179°27.7'	N23°25.5'	23	1	194°27.6'	N23°25.5'	23	2	209°27.5'	N23°25.5'
23	3	224°27.3'	N23°25.5'	23	4	239°27.2'	N23°25.4'	23	5	254°27.0'	N23°25.4'
23	6	269°26.9'	N23°25.4'	23	7	284°26.8'	N23°25.3'	23	8	299°26.6'	N23°25.3'
23	9	314°26.5'	N23°25.3'	23	10	329°26.4'	N23°25.2'	23	11	344°26.2'	N23°25.2'
23	12	359°26.1'	N23°25.2'	23	13	14°26.0'	N23°25.1'	23	14	29°25.8'	N23°25.1'
23	15	44°25.7'	N23°25.0'	23	16	59°25.6'	N23°25.0'	23	17	74°25.4'	N23°25.0'
23	18	89°25.3'	N23°24.9'	23	19	104°25.2'	N23°24.9'	23	20	119°25.0'	N23°24.8'
23	21	134°24.9'	N23°24.8'	23	22	149°24.8'	N23°24.7'	23	23	164°24.6'	N23°24.7'
24	0	179°24.5'	N23°24.7'	24	1	194°24.4'	N23°24.6'	24	2	209°24.2'	N23°24.6'
24	3	224°24.1'	N23°24.5'	24	4	239°24.0'	N23°24.5'	24	5	254°23.8'	N23°24.4'
24	6	269°23.7'	N23°24.4'	24	7	284°23.6'	N23°24.3'	24	8	299°23.4'	N23°24.3'
24	9	314°23.3'	N23°24.2'	24	10	329°23.2'	N23°24.2'	24	11	344°23.0'	N23°24.1'
24	12	359°22.9'	N23°24.1'	24	13	14°22.7'	N23°24.0'	24	14	29°22.6'	N23°23.9'
24	15	44°22.5'	N23°23.9'	24	16	59°22.3'	N23°23.8'	24	17	74°22.2'	N23°23.8'
24	18	89°22.1'	N23°23.7'	24	19	104°21.9'	N23°23.7'	24	20	119°21.8'	N23°23.6'
24	21	134°21.7'	N23°23.5'	24	22	149°21.5'	N23°23.5'	24	23	164°21.4'	N23°23.4'
25	0	179°21.3'	N23°23.4'	25	1	194°21.1'	N23°23.3'	25	2	209°21.0'	N23°23.2'
25	3	224°20.9'	N23°23.2'	25	4	239°20.7'	N23°23.1'	25	5	254°20.6'	N23°23.0'
25	6	269°20.5'	N23°23.0'	25	7	284°20.4'	N23°22.9'	25	8	299°20.2'	N23°22.8'
25	9	314°20.1'	N23°22.8'	25	10	329°20.0'	N23°22.7'	25	11	344°19.8'	N23°22.6'
25	12	359°19.7'	N23°22.6'	25	13	14°19.6'	N23°22.5'	25	14	29°19.4'	N23°22.4'
25	15	44°19.3'	N23°22.3'	25	16	59°19.2'	N23°22.3'	25	17	74°19.0'	N23°22.2'
25	18	89°18.9'	N23°22.1'	25	19	104°18.8'	N23°22.0'	25	20	119°18.6'	N23°22.0'
25	21	134°18.5'	N23°21.9'	25	22	149°18.4'	N23°21.8'	25	23	164°18.2'	N23°21.7'
26	0	179°18.1'	N23°21.6'	26	1	194°18.0'	N23°21.6'	26	2	209°17.8'	N23°21.5'
26	3	224°17.7'	N23°21.4'	26	4	239°17.6'	N23°21.3'	26	5	254°17.4'	N23°21.2'
26	6	269°17.3'	N23°21.2'	26	7	284°17.2'	N23°21.1'	26	8	299°17.0'	N23°21.0'
26	9	314°16.9'	N23°20.9'	26	10	329°16.8'	N23°20.8'	26	11	344°16.7'	N23°20.7'
26	12	359°16.5'	N23°20.6'	26	13	14°16.4'	N23°20.5'	26	14	29°16.3'	N23°20.5'
26	15	44°16.1'	N23°20.4'	26	16	59°16.0'	N23°20.3'	26	17	74°15.9'	N23°20.2'
26	18	89°15.7'	N23°20.1'	26	19	104°15.6'	N23°20.0'	26	20	119°15.5'	N23°19.9'
26	21	134°15.3'	N23°19.8'	26	22	149°15.2'	N23°19.7'	26	23	164°15.1'	N23°19.6'
27	0	179°15.0'	N23°19.5'	27	1	194°14.8'	N23°19.4'	27	2	209°14.7'	N23°19.3'
27	3	224°14.6'	N23°19.2'	27	4	239°14.4'	N23°19.1'	27	5	254°14.3'	N23°19.0'
27	6	269°14.2'	N23°18.9'	27	7	284°14.0'	N23°18.8'	27	8	299°13.9'	N23°18.7'

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Sun — GHA and Declination, June 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
27	9	314°13.8'	N23°18.6'	27	10	329°13.7'	N23°18.5'	27	11	344°13.5'	N23°18.4'
27	12	359°13.4'	N23°18.3'	27	13	14°13.3'	N23°18.2'	27	14	29°13.1'	N23°18.1'
27	15	44°13.0'	N23°18.0'	27	16	59°12.9'	N23°17.9'	27	17	74°12.8'	N23°17.8'
27	18	89°12.6'	N23°17.7'	27	19	104°12.5'	N23°17.6'	27	20	119°12.4'	N23°17.4'
27	21	134°12.2'	N23°17.3'	27	22	149°12.1'	N23°17.2'	27	23	164°12.0'	N23°17.1'
28	0	179°11.9'	N23°17.0'	28	1	194°11.7'	N23°16.9'	28	2	209°11.6'	N23°16.8'
28	3	224°11.5'	N23°16.6'	28	4	239°11.3'	N23°16.5'	28	5	254°11.2'	N23°16.4'
28	6	269°11.1'	N23°16.3'	28	7	284°11.0'	N23°16.2'	28	8	299°10.8'	N23°16.1'
28	9	314°10.7'	N23°15.9'	28	10	329°10.6'	N23°15.8'	28	11	344°10.4'	N23°15.7'
28	12	359°10.3'	N23°15.6'	28	13	14°10.2'	N23°15.4'	28	14	29°10.1'	N23°15.3'
28	15	44°09.9'	N23°15.2'	28	16	59°09.8'	N23°15.1'	28	17	74°09.7'	N23°14.9'
28	18	89°09.6'	N23°14.8'	28	19	104°09.4'	N23°14.7'	28	20	119°09.3'	N23°14.6'
28	21	134°09.2'	N23°14.4'	28	22	149°09.0'	N23°14.3'	28	23	164°08.9'	N23°14.2'
29	0	179°08.8'	N23°14.0'	29	1	194°08.7'	N23°13.9'	29	2	209°08.5'	N23°13.8'
29	3	224°08.4'	N23°13.7'	29	4	239°08.3'	N23°13.5'	29	5	254°08.2'	N23°13.4'
29	6	269°08.0'	N23°13.2'	29	7	284°07.9'	N23°13.1'	29	8	299°07.8'	N23°13.0'
29	9	314°07.7'	N23°12.8'	29	10	329°07.5'	N23°12.7'	29	11	344°07.4'	N23°12.6'
29	12	359°07.3'	N23°12.4'	29	13	14°07.2'	N23°12.3'	29	14	29°07.0'	N23°12.1'
29	15	44°06.9'	N23°12.0'	29	16	59°06.8'	N23°11.9'	29	17	74°06.7'	N23°11.7'
29	18	89°06.5'	N23°11.6'	29	19	104°06.4'	N23°11.4'	29	20	119°06.3'	N23°11.3'
29	21	134°06.2'	N23°11.1'	29	22	149°06.0'	N23°11.0'	29	23	164°05.9'	N23°10.8'
30	0	179°05.8'	N23°10.7'	30	1	194°05.7'	N23°10.6'	30	2	209°05.5'	N23°10.4'
30	3	224°05.4'	N23°10.3'	30	4	239°05.3'	N23°10.1'	30	5	254°05.2'	N23°10.0'
30	6	269°05.1'	N23°09.8'	30	7	284°04.9'	N23°09.6'	30	8	299°04.8'	N23°09.5'
30	9	314°04.7'	N23°09.3'	30	10	329°04.6'	N23°09.2'	30	11	344°04.4'	N23°09.0'
30	12	359°04.3'	N23°08.9'	30	13	14°04.2'	N23°08.7'	30	14	29°04.1'	N23°08.6'
30	15	44°03.9'	N23°08.4'	30	16	59°03.8'	N23°08.2'	30	17	74°03.7'	N23°08.1'
30	18	89°03.6'	N23°07.9'	30	19	104°03.5'	N23°07.8'	30	20	119°03.3'	N23°07.6'
30	21	134°03.2'	N23°07.4'	30	22	149°03.1'	N23°07.3'	30	23	164°03.0'	N23°07.1'

Table 9.7: Sun — GHA and Declination, July 2026 (UT)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
1	0	179°02.8'	N23°06.9'	1	1	194°02.7'	N23°06.8'	1	2	209°02.6'	N23°06.6'
1	3	224°02.5'	N23°06.4'	1	4	239°02.4'	N23°06.3'	1	5	254°02.2'	N23°06.1'
1	6	269°02.1'	N23°05.9'	1	7	284°02.0'	N23°05.8'	1	8	299°01.9'	N23°05.6'
1	9	314°01.8'	N23°05.4'	1	10	329°01.6'	N23°05.3'	1	11	344°01.5'	N23°05.1'
1	12	359°01.4'	N23°04.9'	1	13	14°01.3'	N23°04.7'	1	14	29°01.2'	N23°04.6'
1	15	44°01.0'	N23°04.4'	1	16	59°00.9'	N23°04.2'	1	17	74°00.8'	N23°04.0'
1	18	89°00.7'	N23°03.9'	1	19	104°00.6'	N23°03.7'	1	20	119°00.4'	N23°03.5'
1	21	134°00.3'	N23°03.3'	1	22	149°00.2'	N23°03.1'	1	23	164°00.1'	N23°03.0'
2	0	179°00.0'	N23°02.8'	2	1	193°59.8'	N23°02.6'	2	2	208°59.7'	N23°02.4'
2	3	223°59.6'	N23°02.2'	2	4	238°59.5'	N23°02.1'	2	5	253°59.4'	N23°01.9'
2	6	268°59.3'	N23°01.7'	2	7	283°59.1'	N23°01.5'	2	8	298°59.0'	N23°01.3'
2	9	313°58.9'	N23°01.1'	2	10	328°58.8'	N23°00.9'	2	11	343°58.7'	N23°00.7'
2	12	358°58.5'	N23°00.6'	2	13	13°58.4'	N23°00.4'	2	14	28°58.3'	N23°00.2'
2	15	43°58.2'	N23°00.0'	2	16	58°58.1'	N22°59.8'	2	17	73°58.0'	N22°59.6'
2	18	88°57.8'	N22°59.4'	2	19	103°57.7'	N22°59.2'	2	20	118°57.6'	N22°59.0'
2	21	133°57.5'	N22°58.8'	2	22	148°57.4'	N22°58.6'	2	23	163°57.3'	N22°58.4'
3	0	178°57.2'	N22°58.2'	3	1	193°57.0'	N22°58.0'	3	2	208°56.9'	N22°57.8'
3	3	223°56.8'	N22°57.6'	3	4	238°56.7'	N22°57.4'	3	5	253°56.6'	N22°57.2'
3	6	268°56.5'	N22°57.0'	3	7	283°56.3'	N22°56.8'	3	8	298°56.2'	N22°56.6'
3	9	313°56.1'	N22°56.4'	3	10	328°56.0'	N22°56.2'	3	11	343°55.9'	N22°56.0'
3	12	358°55.8'	N22°55.8'	3	13	13°55.7'	N22°55.6'	3	14	28°55.5'	N22°55.4'
3	15	43°55.4'	N22°55.2'	3	16	58°55.3'	N22°55.0'	3	17	73°55.2'	N22°54.8'
3	18	88°55.1'	N22°54.5'	3	19	103°55.0'	N22°54.3'	3	20	118°54.9'	N22°54.1'
3	21	133°54.7'	N22°53.9'	3	22	148°54.6'	N22°53.7'	3	23	163°54.5'	N22°53.5'
4	0	178°54.4'	N22°53.3'	4	1	193°54.3'	N22°53.0'	4	2	208°54.2'	N22°52.8'
4	3	223°54.1'	N22°52.6'	4	4	238°54.0'	N22°52.4'	4	5	253°53.9'	N22°52.2'
4	6	268°53.7'	N22°52.0'	4	7	283°53.6'	N22°51.7'	4	8	298°53.5'	N22°51.5'
4	9	313°53.4'	N22°51.3'	4	10	328°53.3'	N22°51.1'	4	11	343°53.2'	N22°50.9'
4	12	358°53.1'	N22°50.6'	4	13	13°53.0'	N22°50.4'	4	14	28°52.9'	N22°50.2'

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Sun — GHA and Declination, July 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
4	15	43°52.7'	N22°50.0'	4	16	58°52.6'	N22°49.7'	4	17	73°52.5'	N22°49.5'
4	18	88°52.4'	N22°49.3'	4	19	103°52.3'	N22°49.1'	4	20	118°52.2'	N22°48.8'
4	21	133°52.1'	N22°48.6'	4	22	148°52.0'	N22°48.4'	4	23	163°51.9'	N22°48.1'
5	0	178°51.8'	N22°47.9'	5	1	193°51.6'	N22°47.7'	5	2	208°51.5'	N22°47.4'
5	3	223°51.4'	N22°47.2'	5	4	238°51.3'	N22°47.0'	5	5	253°51.2'	N22°46.7'
5	6	268°51.1'	N22°46.5'	5	7	283°51.0'	N22°46.3'	5	8	298°50.9'	N22°46.0'
5	9	313°50.8'	N22°45.8'	5	10	328°50.7'	N22°45.6'	5	11	343°50.6'	N22°45.3'
5	12	358°50.5'	N22°45.1'	5	13	13°50.3'	N22°44.8'	5	14	28°50.2'	N22°44.6'
5	15	43°50.1'	N22°44.4'	5	16	58°50.0'	N22°44.1'	5	17	73°49.9'	N22°43.9'
5	18	88°49.8'	N22°43.6'	5	19	103°49.7'	N22°43.4'	5	20	118°49.6'	N22°43.1'
5	21	133°49.5'	N22°42.9'	5	22	148°49.4'	N22°42.6'	5	23	163°49.3'	N22°42.4'
6	0	178°49.2'	N22°42.1'	6	1	193°49.1'	N22°41.9'	6	2	208°49.0'	N22°41.6'
6	3	223°48.9'	N22°41.4'	6	4	238°48.8'	N22°41.1'	6	5	253°48.6'	N22°40.9'
6	6	268°48.5'	N22°40.6'	6	7	283°48.4'	N22°40.4'	6	8	298°48.3'	N22°40.1'
6	9	313°48.2'	N22°39.9'	6	10	328°48.1'	N22°39.6'	6	11	343°48.0'	N22°39.4'
6	12	358°47.9'	N22°39.1'	6	13	13°47.8'	N22°38.9'	6	14	28°47.7'	N22°38.6'
6	15	43°47.6'	N22°38.3'	6	16	58°47.5'	N22°38.1'	6	17	73°47.4'	N22°37.8'
6	18	88°47.3'	N22°37.6'	6	19	103°47.2'	N22°37.3'	6	20	118°47.1'	N22°37.0'
6	21	133°47.0'	N22°36.8'	6	22	148°46.9'	N22°36.5'	6	23	163°46.8'	N22°36.3'
7	0	178°46.7'	N22°36.0'	7	1	193°46.6'	N22°35.7'	7	2	208°46.5'	N22°35.5'
7	3	223°46.4'	N22°35.2'	7	4	238°46.3'	N22°34.9'	7	5	253°46.2'	N22°34.7'
7	6	268°46.1'	N22°34.4'	7	7	283°46.0'	N22°34.1'	7	8	298°45.9'	N22°33.9'
7	9	313°45.8'	N22°33.6'	7	10	328°45.7'	N22°33.3'	7	11	343°45.6'	N22°33.0'
7	12	358°45.5'	N22°32.8'	7	13	13°45.4'	N22°32.5'	7	14	28°45.3'	N22°32.2'
7	15	43°45.2'	N22°32.0'	7	16	58°45.1'	N22°31.7'	7	17	73°45.0'	N22°31.4'
7	18	88°44.9'	N22°31.1'	7	19	103°44.8'	N22°30.8'	7	20	118°44.7'	N22°30.6'
7	21	133°44.6'	N22°30.3'	7	22	148°44.5'	N22°30.0'	7	23	163°44.4'	N22°29.7'
8	0	178°44.3'	N22°29.5'	8	1	193°44.2'	N22°29.2'	8	2	208°44.1'	N22°28.9'
8	3	223°44.0'	N22°28.6'	8	4	238°43.9'	N22°28.3'	8	5	253°43.8'	N22°28.0'
8	6	268°43.7'	N22°27.8'	8	7	283°43.6'	N22°27.5'	8	8	298°43.5'	N22°27.2'
8	9	313°43.4'	N22°26.9'	8	10	328°43.3'	N22°26.6'	8	11	343°43.2'	N22°26.3'
8	12	358°43.1'	N22°26.0'	8	13	13°43.0'	N22°25.7'	8	14	28°42.9'	N22°25.5'
8	15	43°42.8'	N22°25.2'	8	16	58°42.8'	N22°24.9'	8	17	73°42.7'	N22°24.6'
8	18	88°42.6'	N22°24.3'	8	19	103°42.5'	N22°24.0'	8	20	118°42.4'	N22°23.7'
8	21	133°42.3'	N22°23.4'	8	22	148°42.2'	N22°23.1'	8	23	163°42.1'	N22°22.8'
9	0	178°42.0'	N22°22.5'	9	1	193°41.9'	N22°22.2'	9	2	208°41.8'	N22°21.9'
9	3	223°41.7'	N22°21.6'	9	4	238°41.6'	N22°21.3'	9	5	253°41.5'	N22°21.0'
9	6	268°41.4'	N22°20.7'	9	7	283°41.3'	N22°20.4'	9	8	298°41.3'	N22°20.1'
9	9	313°41.2'	N22°19.8'	9	10	328°41.1'	N22°19.5'	9	11	343°41.0'	N22°19.2'
9	12	358°40.9'	N22°18.9'	9	13	13°40.8'	N22°18.6'	9	14	28°40.7'	N22°18.3'
9	15	43°40.6'	N22°18.0'	9	16	58°40.5'	N22°17.7'	9	17	73°40.4'	N22°17.4'
9	18	88°40.3'	N22°17.1'	9	19	103°40.3'	N22°16.8'	9	20	118°40.2'	N22°16.5'
9	21	133°40.1'	N22°16.1'	9	22	148°40.0'	N22°15.8'	9	23	163°39.9'	N22°15.5'
10	0	178°39.8'	N22°15.2'	10	1	193°39.7'	N22°14.9'	10	2	208°39.6'	N22°14.6'
10	3	223°39.5'	N22°14.3'	10	4	238°39.5'	N22°13.9'	10	5	253°39.4'	N22°13.6'
10	6	268°39.3'	N22°13.3'	10	7	283°39.2'	N22°13.0'	10	8	298°39.1'	N22°12.7'
10	9	313°39.0'	N22°12.4'	10	10	328°38.9'	N22°12.0'	10	11	343°38.8'	N22°11.7'
10	12	358°38.8'	N22°11.4'	10	13	13°38.7'	N22°11.1'	10	14	28°38.6'	N22°10.8'
10	15	43°38.5'	N22°10.4'	10	16	58°38.4'	N22°10.1'	10	17	73°38.3'	N22°09.8'
10	18	88°38.2'	N22°09.5'	10	19	103°38.1'	N22°09.1'	10	20	118°38.1'	N22°08.8'
10	21	133°38.0'	N22°08.5'	10	22	148°37.9'	N22°08.2'	10	23	163°37.8'	N22°07.8'
11	0	178°37.7'	N22°07.5'	11	1	193°37.6'	N22°07.2'	11	2	208°37.6'	N22°06.8'
11	3	223°37.5'	N22°06.5'	11	4	238°37.4'	N22°06.2'	11	5	253°37.3'	N22°05.9'
11	6	268°37.2'	N22°05.5'	11	7	283°37.1'	N22°05.2'	11	8	298°37.1'	N22°04.9'
11	9	313°37.0'	N22°04.5'	11	10	328°36.9'	N22°04.2'	11	11	343°36.8'	N22°03.8'
11	12	358°36.7'	N22°03.5'	11	13	13°36.6'	N22°03.2'	11	14	28°36.6'	N22°02.8'
11	15	43°36.5'	N22°02.5'	11	16	58°36.4'	N22°02.2'	11	17	73°36.3'	N22°01.8'
11	18	88°36.2'	N22°01.5'	11	19	103°36.1'	N22°01.1'	11	20	118°36.1'	N22°00.8'
11	21	133°36.0'	N22°00.5'	11	22	148°35.9'	N22°00.1'	11	23	163°35.8'	N21°59.8'
12	0	178°35.7'	N21°59.4'	12	1	193°35.7'	N21°59.1'	12	2	208°35.6'	N21°58.7'
12	3	223°35.5'	N21°58.4'	12	4	238°35.4'	N21°58.0'	12	5	253°35.4'	N21°57.7'
12	6	268°35.3'	N21°57.3'	12	7	283°35.2'	N21°57.0'	12	8	298°35.1'	N21°56.6'
12	9	313°35.0'	N21°56.3'	12	10	328°35.0'	N21°55.9'	12	11	343°34.9'	N21°55.6'
12	12	358°34.8'	N21°55.2'	12	13	13°34.7'	N21°54.9'	12	14	28°34.6'	N21°54.5'

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Sun — GHA and Declination, July 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
12	15	43°34.6'	N21°54.2'	12	16	58°34.5'	N21°53.8'	12	17	73°34.4'	N21°53.5'
12	18	88°34.3'	N21°53.1'	12	19	103°34.3'	N21°52.8'	12	20	118°34.2'	N21°52.4'
12	21	133°34.1'	N21°52.0'	12	22	148°34.0'	N21°51.7'	12	23	163°34.0'	N21°51.3'
13	0	178°33.9'	N21°51.0'	13	1	193°33.8'	N21°50.6'	13	2	208°33.7'	N21°50.2'
13	3	223°33.7'	N21°49.9'	13	4	238°33.6'	N21°49.5'	13	5	253°33.5'	N21°49.2'
13	6	268°33.4'	N21°48.8'	13	7	283°33.4'	N21°48.4'	13	8	298°33.3'	N21°48.1'
13	9	313°33.2'	N21°47.7'	13	10	328°33.1'	N21°47.3'	13	11	343°33.1'	N21°47.0'
13	12	358°33.0'	N21°46.6'	13	13	13°32.9'	N21°46.2'	13	14	28°32.9'	N21°45.9'
13	15	43°32.8'	N21°45.5'	13	16	58°32.7'	N21°45.1'	13	17	73°32.6'	N21°44.8'
13	18	88°32.6'	N21°44.4'	13	19	103°32.5'	N21°44.0'	13	20	118°32.4'	N21°43.6'
13	21	133°32.4'	N21°43.3'	13	22	148°32.3'	N21°42.9'	13	23	163°32.2'	N21°42.5'
14	0	178°32.1'	N21°42.1'	14	1	193°32.1'	N21°41.8'	14	2	208°32.0'	N21°41.4'
14	3	223°31.9'	N21°41.0'	14	4	238°31.9'	N21°40.6'	14	5	253°31.8'	N21°40.2'
14	6	268°31.7'	N21°39.9'	14	7	283°31.7'	N21°39.5'	14	8	298°31.6'	N21°39.1'
14	9	313°31.5'	N21°38.7'	14	10	328°31.5'	N21°38.3'	14	11	343°31.4'	N21°38.0'
14	12	358°31.3'	N21°37.6'	14	13	13°31.3'	N21°37.2'	14	14	28°31.2'	N21°36.8'
14	15	43°31.1'	N21°36.4'	14	16	58°31.1'	N21°36.0'	14	17	73°31.0'	N21°35.7'
14	18	88°30.9'	N21°35.3'	14	19	103°30.9'	N21°34.9'	14	20	118°30.8'	N21°34.5'
14	21	133°30.7'	N21°34.1'	14	22	148°30.7'	N21°33.7'	14	23	163°30.6'	N21°33.3'
15	0	178°30.5'	N21°32.9'	15	1	193°30.5'	N21°32.5'	15	2	208°30.4'	N21°32.2'
15	3	223°30.3'	N21°31.8'	15	4	238°30.3'	N21°31.4'	15	5	253°30.2'	N21°31.0'
15	6	268°30.1'	N21°30.6'	15	7	283°30.1'	N21°30.2'	15	8	298°30.0'	N21°29.8'
15	9	313°30.0'	N21°29.4'	15	10	328°29.9'	N21°29.0'	15	11	343°29.8'	N21°28.6'
15	12	358°29.8'	N21°28.2'	15	13	13°29.7'	N21°27.8'	15	14	28°29.6'	N21°27.4'
15	15	43°29.6'	N21°27.0'	15	16	58°29.5'	N21°26.6'	15	17	73°29.5'	N21°26.2'
15	18	88°29.4'	N21°25.8'	15	19	103°29.3'	N21°25.4'	15	20	118°29.3'	N21°25.0'
15	21	133°29.2'	N21°24.6'	15	22	148°29.2'	N21°24.2'	15	23	163°29.1'	N21°23.8'
16	0	178°29.0'	N21°23.4'	16	1	193°29.0'	N21°23.0'	16	2	208°28.9'	N21°22.6'
16	3	223°28.9'	N21°22.1'	16	4	238°28.8'	N21°21.7'	16	5	253°28.7'	N21°21.3'
16	6	268°28.7'	N21°20.9'	16	7	283°28.6'	N21°20.5'	16	8	298°28.6'	N21°20.1'
16	9	313°28.5'	N21°19.7'	16	10	328°28.4'	N21°19.3'	16	11	343°28.4'	N21°18.9'
16	12	358°28.3'	N21°18.4'	16	13	13°28.3'	N21°18.0'	16	14	28°28.2'	N21°17.6'
16	15	43°28.2'	N21°17.2'	16	16	58°28.1'	N21°16.8'	16	17	73°28.1'	N21°16.4'
16	18	88°28.0'	N21°16.0'	16	19	103°27.9'	N21°15.5'	16	20	118°27.9'	N21°15.1'
16	21	133°27.8'	N21°14.7'	16	22	148°27.8'	N21°14.3'	16	23	163°27.7'	N21°13.9'
17	0	178°27.7'	N21°13.4'	17	1	193°27.6'	N21°13.0'	17	2	208°27.6'	N21°12.6'
17	3	223°27.5'	N21°12.2'	17	4	238°27.5'	N21°11.7'	17	5	253°27.4'	N21°11.3'
17	6	268°27.3'	N21°10.9'	17	7	283°27.3'	N21°10.5'	17	8	298°27.2'	N21°10.0'
17	9	313°27.2'	N21°09.6'	17	10	328°27.1'	N21°09.2'	17	11	343°27.1'	N21°08.8'
17	12	358°27.0'	N21°08.3'	17	13	13°27.0'	N21°07.9'	17	14	28°26.9'	N21°07.5'
17	15	43°26.9'	N21°07.0'	17	16	58°26.8'	N21°06.6'	17	17	73°26.8'	N21°06.2'
17	18	88°26.7'	N21°05.7'	17	19	103°26.7'	N21°05.3'	17	20	118°26.6'	N21°04.9'
17	21	133°26.6'	N21°04.4'	17	22	148°26.5'	N21°04.0'	17	23	163°26.5'	N21°03.6'
18	0	178°26.4'	N21°03.1'	18	1	193°26.4'	N21°02.7'	18	2	208°26.3'	N21°02.3'
18	3	223°26.3'	N21°01.8'	18	4	238°26.2'	N21°01.4'	18	5	253°26.2'	N21°01.0'
18	6	268°26.1'	N21°00.5'	18	7	283°26.1'	N21°00.1'	18	8	298°26.1'	N20°59.6'
18	9	313°26.0'	N20°59.2'	18	10	328°26.0'	N20°58.7'	18	11	343°25.9'	N20°58.3'
18	12	358°25.9'	N20°57.9'	18	13	13°25.8'	N20°57.4'	18	14	28°25.8'	N20°57.0'
18	15	43°25.7'	N20°56.5'	18	16	58°25.7'	N20°56.1'	18	17	73°25.6'	N20°55.6'
18	18	88°25.6'	N20°55.2'	18	19	103°25.6'	N20°54.7'	18	20	118°25.5'	N20°54.3'
18	21	133°25.5'	N20°53.8'	18	22	148°25.4'	N20°53.4'	18	23	163°25.4'	N20°52.9'
19	0	178°25.3'	N20°52.5'	19	1	193°25.3'	N20°52.0'	19	2	208°25.2'	N20°51.6'
19	3	223°25.2'	N20°51.1'	19	4	238°25.2'	N20°50.7'	19	5	253°25.1'	N20°50.2'
19	6	268°25.1'	N20°49.8'	19	7	283°25.0'	N20°49.3'	19	8	298°25.0'	N20°48.9'
19	9	313°25.0'	N20°48.4'	19	10	328°24.9'	N20°48.0'	19	11	343°24.9'	N20°47.5'
19	12	358°24.8'	N20°47.0'	19	13	13°24.8'	N20°46.6'	19	14	28°24.8'	N20°46.1'
19	15	43°24.7'	N20°45.7'	19	16	58°24.7'	N20°45.2'	19	17	73°24.6'	N20°44.7'
19	18	88°24.6'	N20°44.3'	19	19	103°24.6'	N20°43.8'	19	20	118°24.5'	N20°43.4'
19	21	133°24.5'	N20°42.9'	19	22	148°24.4'	N20°42.4'	19	23	163°24.4'	N20°42.0'
20	0	178°24.4'	N20°41.5'	20	1	193°24.3'	N20°41.0'	20	2	208°24.3'	N20°40.6'
20	3	223°24.3'	N20°40.1'	20	4	238°24.2'	N20°39.6'	20	5	253°24.2'	N20°39.2'
20	6	268°24.2'	N20°38.7'	20	7	283°24.1'	N20°38.2'	20	8	298°24.1'	N20°37.8'
20	9	313°24.0'	N20°37.3'	20	10	328°24.0'	N20°36.8'	20	11	343°24.0'	N20°36.3'
20	12	358°23.9'	N20°35.9'	20	13	13°23.9'	N20°35.4'	20	14	28°23.9'	N20°34.9'

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Sun — GHA and Declination, July 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
20	15	43°23.8'	N20°34.4'	20	16	58°23.8'	N20°34.0'	20	17	73°23.8'	N20°33.5'
20	18	88°23.7'	N20°33.0'	20	19	103°23.7'	N20°32.5'	20	20	118°23.7'	N20°32.1'
20	21	133°23.6'	N20°31.6'	20	22	148°23.6'	N20°31.1'	20	23	163°23.6'	N20°30.6'
21	0	178°23.5'	N20°30.1'	21	1	193°23.5'	N20°29.7'	21	2	208°23.5'	N20°29.2'
21	3	223°23.5'	N20°28.7'	21	4	238°23.4'	N20°28.2'	21	5	253°23.4'	N20°27.7'
21	6	268°23.4'	N20°27.3'	21	7	283°23.3'	N20°26.8'	21	8	298°23.3'	N20°26.3'
21	9	313°23.3'	N20°25.8'	21	10	328°23.2'	N20°25.3'	21	11	343°23.2'	N20°24.8'
21	12	358°23.2'	N20°24.3'	21	13	13°23.2'	N20°23.9'	21	14	28°23.1'	N20°23.4'
21	15	43°23.1'	N20°22.9'	21	16	58°23.1'	N20°22.4'	21	17	73°23.1'	N20°21.9'
21	18	88°23.0'	N20°21.4'	21	19	103°23.0'	N20°20.9'	21	20	118°23.0'	N20°20.4'
21	21	133°22.9'	N20°19.9'	21	22	148°22.9'	N20°19.4'	21	23	163°22.9'	N20°19.0'
22	0	178°22.9'	N20°18.5'	22	1	193°22.8'	N20°18.0'	22	2	208°22.8'	N20°17.5'
22	3	223°22.8'	N20°17.0'	22	4	238°22.8'	N20°16.5'	22	5	253°22.7'	N20°16.0'
22	6	268°22.7'	N20°15.5'	22	7	283°22.7'	N20°15.0'	22	8	298°22.7'	N20°14.5'
22	9	313°22.6'	N20°14.0'	22	10	328°22.6'	N20°13.5'	22	11	343°22.6'	N20°13.0'
22	12	358°22.6'	N20°12.5'	22	13	13°22.6'	N20°12.0'	22	14	28°22.5'	N20°11.5'
22	15	43°22.5'	N20°11.0'	22	16	58°22.5'	N20°10.5'	22	17	73°22.5'	N20°10.0'
22	18	88°22.5'	N20°09.5'	22	19	103°22.4'	N20°09.0'	22	20	118°22.4'	N20°08.5'
22	21	133°22.4'	N20°07.9'	22	22	148°22.4'	N20°07.4'	22	23	163°22.3'	N20°06.9'
23	0	178°22.3'	N20°06.4'	23	1	193°22.3'	N20°05.9'	23	2	208°22.3'	N20°05.4'
23	3	223°22.3'	N20°04.9'	23	4	238°22.3'	N20°04.4'	23	5	253°22.2'	N20°03.9'
23	6	268°22.2'	N20°03.4'	23	7	283°22.2'	N20°02.9'	23	8	298°22.2'	N20°02.3'
23	9	313°22.2'	N20°01.8'	23	10	328°22.1'	N20°01.3'	23	11	343°22.1'	N20°00.8'
23	12	358°22.1'	N20°00.3'	23	13	13°22.1'	N19°59.8'	23	14	28°22.1'	N19°59.3'
23	15	43°22.1'	N19°58.7'	23	16	58°22.1'	N19°58.2'	23	17	73°22.0'	N19°57.7'
23	18	88°22.0'	N19°57.2'	23	19	103°22.0'	N19°56.7'	23	20	118°22.0'	N19°56.1'
23	21	133°22.0'	N19°55.6'	23	22	148°22.0'	N19°55.1'	23	23	163°21.9'	N19°54.6'
24	0	178°21.9'	N19°54.1'	24	1	193°21.9'	N19°53.5'	24	2	208°21.9'	N19°53.0'
24	3	223°21.9'	N19°52.5'	24	4	238°21.9'	N19°52.0'	24	5	253°21.9'	N19°51.4'
24	6	268°21.9'	N19°50.9'	24	7	283°21.8'	N19°50.4'	24	8	298°21.8'	N19°49.9'
24	9	313°21.8'	N19°49.3'	24	10	328°21.8'	N19°48.8'	24	11	343°21.8'	N19°48.3'
24	12	358°21.8'	N19°47.8'	24	13	13°21.8'	N19°47.2'	24	14	28°21.8'	N19°46.7'
24	15	43°21.8'	N19°46.2'	24	16	58°21.8'	N19°45.6'	24	17	73°21.7'	N19°45.1'
24	18	88°21.7'	N19°44.6'	24	19	103°21.7'	N19°44.0'	24	20	118°21.7'	N19°43.5'
24	21	133°21.7'	N19°43.0'	24	22	148°21.7'	N19°42.4'	24	23	163°21.7'	N19°41.9'
25	0	178°21.7'	N19°41.4'	25	1	193°21.7'	N19°40.8'	25	2	208°21.7'	N19°40.3'
25	3	223°21.7'	N19°39.8'	25	4	238°21.7'	N19°39.2'	25	5	253°21.7'	N19°38.7'
25	6	268°21.7'	N19°38.1'	25	7	283°21.6'	N19°37.6'	25	8	298°21.6'	N19°37.1'
25	9	313°21.6'	N19°36.5'	25	10	328°21.6'	N19°36.0'	25	11	343°21.6'	N19°35.4'
25	12	358°21.6'	N19°34.9'	25	13	13°21.6'	N19°34.3'	25	14	28°21.6'	N19°33.8'
25	15	43°21.6'	N19°33.3'	25	16	58°21.6'	N19°32.7'	25	17	73°21.6'	N19°32.2'
25	18	88°21.6'	N19°31.6'	25	19	103°21.6'	N19°31.1'	25	20	118°21.6'	N19°30.5'
25	21	133°21.6'	N19°30.0'	25	22	148°21.6'	N19°29.4'	25	23	163°21.6'	N19°28.9'
26	0	178°21.6'	N19°28.3'	26	1	193°21.6'	N19°27.8'	26	2	208°21.6'	N19°27.2'
26	3	223°21.6'	N19°26.7'	26	4	238°21.6'	N19°26.1'	26	5	253°21.6'	N19°25.6'
26	6	268°21.6'	N19°25.0'	26	7	283°21.6'	N19°24.5'	26	8	298°21.6'	N19°23.9'
26	9	313°21.6'	N19°23.4'	26	10	328°21.6'	N19°22.8'	26	11	343°21.6'	N19°22.3'
26	12	358°21.6'	N19°21.7'	26	13	13°21.6'	N19°21.1'	26	14	28°21.6'	N19°20.6'
26	15	43°21.6'	N19°20.0'	26	16	58°21.6'	N19°19.5'	26	17	73°21.6'	N19°18.9'
26	18	88°21.6'	N19°18.4'	26	19	103°21.6'	N19°17.8'	26	20	118°21.6'	N19°17.2'
26	21	133°21.6'	N19°16.7'	26	22	148°21.6'	N19°16.1'	26	23	163°21.6'	N19°15.6'
27	0	178°21.6'	N19°15.0'	27	1	193°21.6'	N19°14.4'	27	2	208°21.7'	N19°13.9'
27	3	223°21.7'	N19°13.3'	27	4	238°21.7'	N19°12.7'	27	5	253°21.7'	N19°12.2'
27	6	268°21.7'	N19°11.6'	27	7	283°21.7'	N19°11.0'	27	8	298°21.7'	N19°10.5'
27	9	313°21.7'	N19°09.9'	27	10	328°21.7'	N19°09.3'	27	11	343°21.7'	N19°08.8'
27	12	358°21.7'	N19°08.2'	27	13	13°21.7'	N19°07.6'	27	14	28°21.7'	N19°07.1'
27	15	43°21.7'	N19°06.5'	27	16	58°21.8'	N19°05.9'	27	17	73°21.8'	N19°05.3'
27	18	88°21.8'	N19°04.8'	27	19	103°21.8'	N19°04.2'	27	20	118°21.8'	N19°03.6'
27	21	133°21.8'	N19°03.1'	27	22	148°21.8'	N19°02.5'	27	23	163°21.8'	N19°01.9'
28	0	178°21.8'	N19°01.3'	28	1	193°21.9'	N19°00.7'	28	2	208°21.9'	N19°00.2'
28	3	223°21.9'	N18°59.6'	28	4	238°21.9'	N18°59.0'	28	5	253°21.9'	N18°58.4'
28	6	268°21.9'	N18°57.9'	28	7	283°21.9'	N18°57.3'	28	8	298°21.9'	N18°56.7'
28	9	313°22.0'	N18°56.1'	28	10	328°22.0'	N18°55.5'	28	11	343°22.0'	N18°55.0'
28	12	358°22.0'	N18°54.4'	28	13	13°22.0'	N18°53.8'	28	14	28°22.0'	N18°53.2'

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Sun — GHA and Declination, July 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
28	15	43°22.0'	N18°52.6'	28	16	58°22.1'	N18°52.0'	28	17	73°22.1'	N18°51.5'
28	18	88°22.1'	N18°50.9'	28	19	103°22.1'	N18°50.3'	28	20	118°22.1'	N18°49.7'
28	21	133°22.1'	N18°49.1'	28	22	148°22.2'	N18°48.5'	28	23	163°22.2'	N18°47.9'
29	0	178°22.2'	N18°47.3'	29	1	193°22.2'	N18°46.8'	29	2	208°22.2'	N18°46.2'
29	3	223°22.2'	N18°45.6'	29	4	238°22.3'	N18°45.0'	29	5	253°22.3'	N18°44.4'
29	6	268°22.3'	N18°43.8'	29	7	283°22.3'	N18°43.2'	29	8	298°22.3'	N18°42.6'
29	9	313°22.4'	N18°42.0'	29	10	328°22.4'	N18°41.4'	29	11	343°22.4'	N18°40.8'
29	12	358°22.4'	N18°40.2'	29	13	13°22.4'	N18°39.6'	29	14	28°22.5'	N18°39.0'
29	15	43°22.5'	N18°38.5'	29	16	58°22.5'	N18°37.9'	29	17	73°22.5'	N18°37.3'
29	18	88°22.6'	N18°36.7'	29	19	103°22.6'	N18°36.1'	29	20	118°22.6'	N18°35.5'
29	21	133°22.6'	N18°34.9'	29	22	148°22.6'	N18°34.3'	29	23	163°22.7'	N18°33.7'
30	0	178°22.7'	N18°33.1'	30	1	193°22.7'	N18°32.5'	30	2	208°22.7'	N18°31.9'
30	3	223°22.8'	N18°31.3'	30	4	238°22.8'	N18°30.6'	30	5	253°22.8'	N18°30.0'
30	6	268°22.8'	N18°29.4'	30	7	283°22.9'	N18°28.8'	30	8	298°22.9'	N18°28.2'
30	9	313°22.9'	N18°27.6'	30	10	328°23.0'	N18°27.0'	30	11	343°23.0'	N18°26.4'
30	12	358°23.0'	N18°25.8'	30	13	13°23.0'	N18°25.2'	30	14	28°23.1'	N18°24.6'
30	15	43°23.1'	N18°24.0'	30	16	58°23.1'	N18°23.4'	30	17	73°23.1'	N18°22.8'
30	18	88°23.2'	N18°22.1'	30	19	103°23.2'	N18°21.5'	30	20	118°23.2'	N18°20.9'
30	21	133°23.3'	N18°20.3'	30	22	148°23.3'	N18°19.7'	30	23	163°23.3'	N18°19.1'
31	0	178°23.4'	N18°18.5'	31	1	193°23.4'	N18°17.9'	31	2	208°23.4'	N18°17.2'
31	3	223°23.4'	N18°16.6'	31	4	238°23.5'	N18°16.0'	31	5	253°23.5'	N18°15.4'
31	6	268°23.5'	N18°14.8'	31	7	283°23.6'	N18°14.2'	31	8	298°23.6'	N18°13.5'
31	9	313°23.6'	N18°12.9'	31	10	328°23.7'	N18°12.3'	31	11	343°23.7'	N18°11.7'
31	12	358°23.7'	N18°11.1'	31	13	13°23.8'	N18°10.4'	31	14	28°23.8'	N18°09.8'
31	15	43°23.8'	N18°09.2'	31	16	58°23.9'	N18°08.6'	31	17	73°23.9'	N18°07.9'
31	18	88°23.9'	N18°07.3'	31	19	103°24.0'	N18°06.7'	31	20	118°24.0'	N18°06.1'
31	21	133°24.1'	N18°05.5'	31	22	148°24.1'	N18°04.8'	31	23	163°24.1'	N18°04.2'

Table 9.8: Sun — GHA and Declination, August 2026 (UT)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
1	0	178°24.2'	N18°03.6'	1	1	193°24.2'	N18°02.9'	1	2	208°24.2'	N18°02.3'
1	3	223°24.3'	N18°01.7'	1	4	238°24.3'	N18°01.1'	1	5	253°24.3'	N18°00.4'
1	6	268°24.4'	N17°59.8'	1	7	283°24.4'	N17°59.2'	1	8	298°24.5'	N17°58.5'
1	9	313°24.5'	N17°57.9'	1	10	328°24.5'	N17°57.3'	1	11	343°24.6'	N17°56.7'
1	12	358°24.6'	N17°56.0'	1	13	13°24.7'	N17°55.4'	1	14	28°24.7'	N17°54.8'
1	15	43°24.7'	N17°54.1'	1	16	58°24.8'	N17°53.5'	1	17	73°24.8'	N17°52.8'
1	18	88°24.9'	N17°52.2'	1	19	103°24.9'	N17°51.6'	1	20	118°24.9'	N17°50.9'
1	21	133°25.0'	N17°50.3'	1	22	148°25.0'	N17°49.7'	1	23	163°25.1'	N17°49.0'
2	0	178°25.1'	N17°48.4'	2	1	193°25.2'	N17°47.8'	2	2	208°25.2'	N17°47.1'
2	3	223°25.2'	N17°46.5'	2	4	238°25.3'	N17°45.8'	2	5	253°25.3'	N17°45.2'
2	6	268°25.4'	N17°44.5'	2	7	283°25.4'	N17°43.9'	2	8	298°25.5'	N17°43.3'
2	9	313°25.5'	N17°42.6'	2	10	328°25.6'	N17°42.0'	2	11	343°25.6'	N17°41.3'
2	12	358°25.7'	N17°40.7'	2	13	13°25.7'	N17°40.0'	2	14	28°25.7'	N17°39.4'
2	15	43°25.8'	N17°38.8'	2	16	58°25.8'	N17°38.1'	2	17	73°25.9'	N17°37.5'
2	18	88°25.9'	N17°36.8'	2	19	103°26.0'	N17°36.2'	2	20	118°26.0'	N17°35.5'
2	21	133°26.1'	N17°34.9'	2	22	148°26.1'	N17°34.2'	2	23	163°26.2'	N17°33.6'
3	0	178°26.2'	N17°32.9'	3	1	193°26.3'	N17°32.3'	3	2	208°26.3'	N17°31.6'
3	3	223°26.4'	N17°31.0'	3	4	238°26.4'	N17°30.3'	3	5	253°26.5'	N17°29.7'
3	6	268°26.5'	N17°29.0'	3	7	283°26.6'	N17°28.3'	3	8	298°26.6'	N17°27.7'
3	9	313°26.7'	N17°27.0'	3	10	328°26.7'	N17°26.4'	3	11	343°26.8'	N17°25.7'
3	12	358°26.8'	N17°25.1'	3	13	13°26.9'	N17°24.4'	3	14	28°26.9'	N17°23.8'
3	15	43°27.0'	N17°23.1'	3	16	58°27.1'	N17°22.4'	3	17	73°27.1'	N17°21.8'
3	18	88°27.2'	N17°21.1'	3	19	103°27.2'	N17°20.5'	3	20	118°27.3'	N17°19.8'
3	21	133°27.3'	N17°19.1'	3	22	148°27.4'	N17°18.5'	3	23	163°27.4'	N17°17.8'
4	0	178°27.5'	N17°17.2'	4	1	193°27.5'	N17°16.5'	4	2	208°27.6'	N17°15.8'
4	3	223°27.7'	N17°15.2'	4	4	238°27.7'	N17°14.5'	4	5	253°27.8'	N17°13.8'
4	6	268°27.8'	N17°13.2'	4	7	283°27.9'	N17°12.5'	4	8	298°27.9'	N17°11.8'
4	9	313°28.0'	N17°11.2'	4	10	328°28.1'	N17°10.5'	4	11	343°28.1'	N17°09.8'
4	12	358°28.2'	N17°09.2'	4	13	13°28.2'	N17°08.5'	4	14	28°28.3'	N17°07.8'
4	15	43°28.4'	N17°07.2'	4	16	58°28.4'	N17°06.5'	4	17	73°28.5'	N17°05.8'
4	18	88°28.5'	N17°05.1'	4	19	103°28.6'	N17°04.5'	4	20	118°28.7'	N17°03.8'

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Sun — GHA and Declination, August 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
4	21	133°28.7'	N17°03.1'	4	22	148°28.8'	N17°02.5'	4	23	163°28.8'	N17°01.8'
5	0	178°28.9'	N17°01.1'	5	1	193°29.0'	N17°00.4'	5	2	208°29.0'	N16°59.8'
5	3	223°29.1'	N16°59.1'	5	4	238°29.2'	N16°58.4'	5	5	253°29.2'	N16°57.7'
5	6	268°29.3'	N16°57.1'	5	7	283°29.3'	N16°56.4'	5	8	298°29.4'	N16°55.7'
5	9	313°29.5'	N16°55.0'	5	10	328°29.5'	N16°54.3'	5	11	343°29.6'	N16°53.7'
5	12	358°29.7'	N16°53.0'	5	13	13°29.7'	N16°52.3'	5	14	28°29.8'	N16°51.6'
5	15	43°29.9'	N16°50.9'	5	16	58°29.9'	N16°50.3'	5	17	73°30.0'	N16°49.6'
5	18	88°30.1'	N16°48.9'	5	19	103°30.1'	N16°48.2'	5	20	118°30.2'	N16°47.5'
5	21	133°30.3'	N16°46.9'	5	22	148°30.3'	N16°46.2'	5	23	163°30.4'	N16°45.5'
6	0	178°30.5'	N16°44.8'	6	1	193°30.5'	N16°44.1'	6	2	208°30.6'	N16°43.4'
6	3	223°30.7'	N16°42.7'	6	4	238°30.7'	N16°42.1'	6	5	253°30.8'	N16°41.4'
6	6	268°30.9'	N16°40.7'	6	7	283°31.0'	N16°40.0'	6	8	298°31.0'	N16°39.3'
6	9	313°31.1'	N16°38.6'	6	10	328°31.2'	N16°37.9'	6	11	343°31.2'	N16°37.2'
6	12	358°31.3'	N16°36.5'	6	13	13°31.4'	N16°35.8'	6	14	28°31.5'	N16°35.2'
6	15	43°31.5'	N16°34.5'	6	16	58°31.6'	N16°33.8'	6	17	73°31.7'	N16°33.1'
6	18	88°31.7'	N16°32.4'	6	19	103°31.8'	N16°31.7'	6	20	118°31.9'	N16°31.0'
6	21	133°32.0'	N16°30.3'	6	22	148°32.0'	N16°29.6'	6	23	163°32.1'	N16°28.9'
7	0	178°32.2'	N16°28.2'	7	1	193°32.3'	N16°27.5'	7	2	208°32.3'	N16°26.8'
7	3	223°32.4'	N16°26.1'	7	4	238°32.5'	N16°25.4'	7	5	253°32.6'	N16°24.7'
7	6	268°32.6'	N16°24.0'	7	7	283°32.7'	N16°23.3'	7	8	298°32.8'	N16°22.6'
7	9	313°32.9'	N16°21.9'	7	10	328°32.9'	N16°21.2'	7	11	343°33.0'	N16°20.5'
7	12	358°33.1'	N16°19.8'	7	13	13°33.2'	N16°19.1'	7	14	28°33.3'	N16°18.4'
7	15	43°33.3'	N16°17.7'	7	16	58°33.4'	N16°17.0'	7	17	73°33.5'	N16°16.3'
7	18	88°33.6'	N16°15.6'	7	19	103°33.6'	N16°14.9'	7	20	118°33.7'	N16°14.2'
7	21	133°33.8'	N16°13.5'	7	22	148°33.9'	N16°12.8'	7	23	163°34.0'	N16°12.1'
8	0	178°34.0'	N16°11.4'	8	1	193°34.1'	N16°10.7'	8	2	208°34.2'	N16°09.9'
8	3	223°34.3'	N16°09.2'	8	4	238°34.4'	N16°08.5'	8	5	253°34.5'	N16°07.8'
8	6	268°34.5'	N16°07.1'	8	7	283°34.6'	N16°06.4'	8	8	298°34.7'	N16°05.7'
8	9	313°34.8'	N16°05.0'	8	10	328°34.9'	N16°04.3'	8	11	343°34.9'	N16°03.6'
8	12	358°35.0'	N16°02.8'	8	13	13°35.1'	N16°02.1'	8	14	28°35.2'	N16°01.4'
8	15	43°35.3'	N16°00.7'	8	16	58°35.4'	N16°00.0'	8	17	73°35.5'	N15°59.3'
8	18	88°35.5'	N15°58.6'	8	19	103°35.6'	N15°57.8'	8	20	118°35.7'	N15°57.1'
8	21	133°35.8'	N15°56.4'	8	22	148°35.9'	N15°55.7'	8	23	163°36.0'	N15°55.0'
9	0	178°36.1'	N15°54.3'	9	1	193°36.1'	N15°53.5'	9	2	208°36.2'	N15°52.8'
9	3	223°36.3'	N15°52.1'	9	4	238°36.4'	N15°51.4'	9	5	253°36.5'	N15°50.7'
9	6	268°36.6'	N15°49.9'	9	7	283°36.7'	N15°49.2'	9	8	298°36.8'	N15°48.5'
9	9	313°36.8'	N15°47.8'	9	10	328°36.9'	N15°47.0'	9	11	343°37.0'	N15°46.3'
9	12	358°37.1'	N15°45.6'	9	13	13°37.2'	N15°44.9'	9	14	28°37.3'	N15°44.2'
9	15	43°37.4'	N15°43.4'	9	16	58°37.5'	N15°42.7'	9	17	73°37.6'	N15°42.0'
9	18	88°37.7'	N15°41.3'	9	19	103°37.7'	N15°40.5'	9	20	118°37.8'	N15°39.8'
9	21	133°37.9'	N15°39.1'	9	22	148°38.0'	N15°38.3'	9	23	163°38.1'	N15°37.6'
10	0	178°38.2'	N15°36.9'	10	1	193°38.3'	N15°36.2'	10	2	208°38.4'	N15°35.4'
10	3	223°38.5'	N15°34.7'	10	4	238°38.6'	N15°34.0'	10	5	253°38.7'	N15°33.2'
10	6	268°38.8'	N15°32.5'	10	7	283°38.9'	N15°31.8'	10	8	298°39.0'	N15°31.0'
10	9	313°39.1'	N15°30.3'	10	10	328°39.1'	N15°29.6'	10	11	343°39.2'	N15°28.8'
10	12	358°39.3'	N15°28.1'	10	13	13°39.4'	N15°27.4'	10	14	28°39.5'	N15°26.6'
10	15	43°39.6'	N15°25.9'	10	16	58°39.7'	N15°25.2'	10	17	73°39.8'	N15°24.4'
10	18	88°39.9'	N15°23.7'	10	19	103°40.0'	N15°23.0'	10	20	118°40.1'	N15°22.2'
10	21	133°40.2'	N15°21.5'	10	22	148°40.3'	N15°20.7'	10	23	163°40.4'	N15°20.0'
11	0	178°40.5'	N15°19.3'	11	1	193°40.6'	N15°18.5'	11	2	208°40.7'	N15°17.8'
11	3	223°40.8'	N15°17.1'	11	4	238°40.9'	N15°16.3'	11	5	253°41.0'	N15°15.6'
11	6	268°41.1'	N15°14.8'	11	7	283°41.2'	N15°14.1'	11	8	298°41.3'	N15°13.3'
11	9	313°41.4'	N15°12.6'	11	10	328°41.5'	N15°11.9'	11	11	343°41.6'	N15°11.1'
11	12	358°41.7'	N15°10.4'	11	13	13°41.8'	N15°09.6'	11	14	28°41.9'	N15°08.9'
11	15	43°42.0'	N15°08.1'	11	16	58°42.1'	N15°07.4'	11	17	73°42.2'	N15°06.6'
11	18	88°42.3'	N15°05.9'	11	19	103°42.4'	N15°05.2'	11	20	118°42.5'	N15°04.4'
11	21	133°42.6'	N15°03.7'	11	22	148°42.7'	N15°02.9'	11	23	163°42.8'	N15°02.2'
12	0	178°42.9'	N15°01.4'	12	1	193°43.0'	N15°00.7'	12	2	208°43.2'	N14°59.9'
12	3	223°43.3'	N14°59.2'	12	4	238°43.4'	N14°58.4'	12	5	253°43.5'	N14°57.7'
12	6	268°43.6'	N14°56.9'	12	7	283°43.7'	N14°56.2'	12	8	298°43.8'	N14°55.4'
12	9	313°43.9'	N14°54.7'	12	10	328°44.0'	N14°53.9'	12	11	343°44.1'	N14°53.1'
12	12	358°44.2'	N14°52.4'	12	13	13°44.3'	N14°51.6'	12	14	28°44.4'	N14°50.9'
12	15	43°44.5'	N14°50.1'	12	16	58°44.6'	N14°49.4'	12	17	73°44.8'	N14°48.6'
12	18	88°44.9'	N14°47.9'	12	19	103°45.0'	N14°47.1'	12	20	118°45.1'	N14°46.3'

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Sun — GHA and Declination, August 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
12	21	133°45.2'	N14°45.6'	12	22	148°45.3'	N14°44.8'	12	23	163°45.4'	N14°44.1'
13	0	178°45.5'	N14°43.3'	13	1	193°45.6'	N14°42.6'	13	2	208°45.7'	N14°41.8'
13	3	223°45.9'	N14°41.0'	13	4	238°46.0'	N14°40.3'	13	5	253°46.1'	N14°39.5'
13	6	268°46.2'	N14°38.8'	13	7	283°46.3'	N14°38.0'	13	8	298°46.4'	N14°37.2'
13	9	313°46.5'	N14°36.5'	13	10	328°46.6'	N14°35.7'	13	11	343°46.8'	N14°34.9'
13	12	358°46.9'	N14°34.2'	13	13	13°47.0'	N14°33.4'	13	14	28°47.1'	N14°32.6'
13	15	43°47.2'	N14°31.9'	13	16	58°47.3'	N14°31.1'	13	17	73°47.4'	N14°30.4'
13	18	88°47.5'	N14°29.6'	13	19	103°47.7'	N14°28.8'	13	20	118°47.8'	N14°28.1'
13	21	133°47.9'	N14°27.3'	13	22	148°48.0'	N14°26.5'	13	23	163°48.1'	N14°25.7'
14	0	178°48.2'	N14°25.0'	14	1	193°48.4'	N14°24.2'	14	2	208°48.5'	N14°23.4'
14	3	223°48.6'	N14°22.7'	14	4	238°48.7'	N14°21.9'	14	5	253°48.8'	N14°21.1'
14	6	268°48.9'	N14°20.4'	14	7	283°49.1'	N14°19.6'	14	8	298°49.2'	N14°18.8'
14	9	313°49.3'	N14°18.0'	14	10	328°49.4'	N14°17.3'	14	11	343°49.5'	N14°16.5'
14	12	358°49.7'	N14°15.7'	14	13	13°49.8'	N14°15.0'	14	14	28°49.9'	N14°14.2'
14	15	43°50.0'	N14°13.4'	14	16	58°50.1'	N14°12.6'	14	17	73°50.2'	N14°11.9'
14	18	88°50.4'	N14°11.1'	14	19	103°50.5'	N14°10.3'	14	20	118°50.6'	N14°09.5'
14	21	133°50.7'	N14°08.7'	14	22	148°50.9'	N14°08.0'	14	23	163°51.0'	N14°07.2'
15	0	178°51.1'	N14°06.4'	15	1	193°51.2'	N14°05.6'	15	2	208°51.3'	N14°04.9'
15	3	223°51.5'	N14°04.1'	15	4	238°51.6'	N14°03.3'	15	5	253°51.7'	N14°02.5'
15	6	268°51.8'	N14°01.7'	15	7	283°52.0'	N14°01.0'	15	8	298°52.1'	N14°00.2'
15	9	313°52.2'	N13°59.4'	15	10	328°52.3'	N13°58.6'	15	11	343°52.4'	N13°57.8'
15	12	358°52.6'	N13°57.1'	15	13	13°52.7'	N13°56.3'	15	14	28°52.8'	N13°55.5'
15	15	43°52.9'	N13°54.7'	15	16	58°53.1'	N13°53.9'	15	17	73°53.2'	N13°53.1'
15	18	88°53.3'	N13°52.3'	15	19	103°53.4'	N13°51.6'	15	20	118°53.6'	N13°50.8'
15	21	133°53.7'	N13°50.0'	15	22	148°53.8'	N13°49.2'	15	23	163°54.0'	N13°48.4'
16	0	178°54.1'	N13°47.6'	16	1	193°54.2'	N13°46.8'	16	2	208°54.3'	N13°46.1'
16	3	223°54.5'	N13°45.3'	16	4	238°54.6'	N13°44.5'	16	5	253°54.7'	N13°43.7'
16	6	268°54.8'	N13°42.9'	16	7	283°55.0'	N13°42.1'	16	8	298°55.1'	N13°41.3'
16	9	313°55.2'	N13°40.5'	16	10	328°55.4'	N13°39.7'	16	11	343°55.5'	N13°38.9'
16	12	358°55.6'	N13°38.2'	16	13	13°55.8'	N13°37.4'	16	14	28°55.9'	N13°36.6'
16	15	43°56.0'	N13°35.8'	16	16	58°56.1'	N13°35.0'	16	17	73°56.3'	N13°34.2'
16	18	88°56.4'	N13°33.4'	16	19	103°56.5'	N13°32.6'	16	20	118°56.7'	N13°31.8'
16	21	133°56.8'	N13°31.0'	16	22	148°56.9'	N13°30.2'	16	23	163°57.1'	N13°29.4'
17	0	178°57.2'	N13°28.6'	17	1	193°57.3'	N13°27.8'	17	2	208°57.5'	N13°27.0'
17	3	223°57.6'	N13°26.2'	17	4	238°57.7'	N13°25.4'	17	5	253°57.9'	N13°24.6'
17	6	268°58.0'	N13°23.8'	17	7	283°58.1'	N13°23.0'	17	8	298°58.3'	N13°22.2'
17	9	313°58.4'	N13°21.4'	17	10	328°58.5'	N13°20.6'	17	11	343°58.7'	N13°19.8'
17	12	358°58.8'	N13°19.0'	17	13	13°58.9'	N13°18.2'	17	14	28°59.1'	N13°17.4'
17	15	43°59.2'	N13°16.6'	17	16	58°59.4'	N13°15.8'	17	17	73°59.5'	N13°15.0'
17	18	88°59.6'	N13°14.2'	17	19	103°59.8'	N13°13.4'	17	20	118°59.9'	N13°12.6'
17	21	134°00.0'	N13°11.8'	17	22	149°00.2'	N13°11.0'	17	23	164°00.3'	N13°10.2'
18	0	179°00.4'	N13°09.4'	18	1	194°00.6'	N13°08.6'	18	2	209°00.7'	N13°07.8'
18	3	224°00.9'	N13°07.0'	18	4	239°01.0'	N13°06.2'	18	5	254°01.1'	N13°05.4'
18	6	269°01.3'	N13°04.6'	18	7	284°01.4'	N13°03.8'	18	8	299°01.6'	N13°03.0'
18	9	314°01.7'	N13°02.1'	18	10	329°01.8'	N13°01.3'	18	11	344°02.0'	N13°00.5'
18	12	359°02.1'	N12°59.7'	18	13	14°02.3'	N12°58.9'	18	14	29°02.4'	N12°58.1'
18	15	44°02.5'	N12°57.3'	18	16	59°02.7'	N12°56.5'	18	17	74°02.8'	N12°55.7'
18	18	89°03.0'	N12°54.9'	18	19	104°03.1'	N12°54.0'	18	20	119°03.2'	N12°53.2'
18	21	134°03.4'	N12°52.4'	18	22	149°03.5'	N12°51.6'	18	23	164°03.7'	N12°50.8'
19	0	179°03.8'	N12°50.0'	19	1	194°04.0'	N12°49.2'	19	2	209°04.1'	N12°48.3'
19	3	224°04.2'	N12°47.5'	19	4	239°04.4'	N12°46.7'	19	5	254°04.5'	N12°45.9'
19	6	269°04.7'	N12°45.1'	19	7	284°04.8'	N12°44.3'	19	8	299°05.0'	N12°43.5'
19	9	314°05.1'	N12°42.6'	19	10	329°05.3'	N12°41.8'	19	11	344°05.4'	N12°41.0'
19	12	359°05.6'	N12°40.2'	19	13	14°05.7'	N12°39.4'	19	14	29°05.8'	N12°38.6'
19	15	44°06.0'	N12°37.7'	19	16	59°06.1'	N12°36.9'	19	17	74°06.3'	N12°36.1'
19	18	89°06.4'	N12°35.3'	19	19	104°06.6'	N12°34.5'	19	20	119°06.7'	N12°33.6'
19	21	134°06.9'	N12°32.8'	19	22	149°07.0'	N12°32.0'	19	23	164°07.2'	N12°31.2'
20	0	179°07.3'	N12°30.3'	20	1	194°07.5'	N12°29.5'	20	2	209°07.6'	N12°28.7'
20	3	224°07.8'	N12°27.9'	20	4	239°07.9'	N12°27.1'	20	5	254°08.1'	N12°26.2'
20	6	269°08.2'	N12°25.4'	20	7	284°08.4'	N12°24.6'	20	8	299°08.5'	N12°23.8'
20	9	314°08.7'	N12°22.9'	20	10	329°08.8'	N12°22.1'	20	11	344°09.0'	N12°21.3'
20	12	359°09.1'	N12°20.5'	20	13	14°09.3'	N12°19.6'	20	14	29°09.4'	N12°18.8'
20	15	44°09.6'	N12°18.0'	20	16	59°09.7'	N12°17.2'	20	17	74°09.9'	N12°16.3'
20	18	89°10.0'	N12°15.5'	20	19	104°10.2'	N12°14.7'	20	20	119°10.3'	N12°13.8'

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Sun — GHA and Declination, August 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
20	21	134°10.5'	N12°13.0'	20	22	149°10.6'	N12°12.2'	20	23	164°10.8'	N12°11.4'
21	0	179°10.9'	N12°10.5'	21	1	194°11.1'	N12°09.7'	21	2	209°11.2'	N12°08.9'
21	3	224°11.4'	N12°08.0'	21	4	239°11.5'	N12°07.2'	21	5	254°11.7'	N12°06.4'
21	6	269°11.9'	N12°05.5'	21	7	284°12.0'	N12°04.7'	21	8	299°12.2'	N12°03.9'
21	9	314°12.3'	N12°03.0'	21	10	329°12.5'	N12°02.2'	21	11	344°12.6'	N12°01.4'
21	12	359°12.8'	N12°00.5'	21	13	14°12.9'	N11°59.7'	21	14	29°13.1'	N11°58.9'
21	15	44°13.2'	N11°58.0'	21	16	59°13.4'	N11°57.2'	21	17	74°13.6'	N11°56.4'
21	18	89°13.7'	N11°55.5'	21	19	104°13.9'	N11°54.7'	21	20	119°14.0'	N11°53.8'
21	21	134°14.2'	N11°53.0'	21	22	149°14.3'	N11°52.2'	21	23	164°14.5'	N11°51.3'
22	0	179°14.7'	N11°50.5'	22	1	194°14.8'	N11°49.7'	22	2	209°15.0'	N11°48.8'
22	3	224°15.1'	N11°48.0'	22	4	239°15.3'	N11°47.1'	22	5	254°15.5'	N11°46.3'
22	6	269°15.6'	N11°45.5'	22	7	284°15.8'	N11°44.6'	22	8	299°15.9'	N11°43.8'
22	9	314°16.1'	N11°42.9'	22	10	329°16.2'	N11°42.1'	22	11	344°16.4'	N11°41.3'
22	12	359°16.6'	N11°40.4'	22	13	14°16.7'	N11°39.6'	22	14	29°16.9'	N11°38.7'
22	15	44°17.0'	N11°37.9'	22	16	59°17.2'	N11°37.0'	22	17	74°17.4'	N11°36.2'
22	18	89°17.5'	N11°35.4'	22	19	104°17.7'	N11°34.5'	22	20	119°17.9'	N11°33.7'
22	21	134°18.0'	N11°32.8'	22	22	149°18.2'	N11°32.0'	22	23	164°18.3'	N11°31.1'
23	0	179°18.5'	N11°30.3'	23	1	194°18.7'	N11°29.4'	23	2	209°18.8'	N11°28.6'
23	3	224°19.0'	N11°27.8'	23	4	239°19.2'	N11°26.9'	23	5	254°19.3'	N11°26.1'
23	6	269°19.5'	N11°25.2'	23	7	284°19.6'	N11°24.4'	23	8	299°19.8'	N11°23.5'
23	9	314°20.0'	N11°22.7'	23	10	329°20.1'	N11°21.8'	23	11	344°20.3'	N11°21.0'
23	12	359°20.5'	N11°20.1'	23	13	14°20.6'	N11°19.3'	23	14	29°20.8'	N11°18.4'
23	15	44°21.0'	N11°17.6'	23	16	59°21.1'	N11°16.7'	23	17	74°21.3'	N11°15.9'
23	18	89°21.5'	N11°15.0'	23	19	104°21.6'	N11°14.2'	23	20	119°21.8'	N11°13.3'
23	21	134°22.0'	N11°12.5'	23	22	149°22.1'	N11°11.6'	23	23	164°22.3'	N11°10.8'
24	0	179°22.5'	N11°09.9'	24	1	194°22.6'	N11°09.0'	24	2	209°22.8'	N11°08.2'
24	3	224°23.0'	N11°07.3'	24	4	239°23.1'	N11°06.5'	24	5	254°23.3'	N11°05.6'
24	6	269°23.5'	N11°04.8'	24	7	284°23.6'	N11°03.9'	24	8	299°23.8'	N11°03.1'
24	9	314°24.0'	N11°02.2'	24	10	329°24.1'	N11°01.4'	24	11	344°24.3'	N11°00.5'
24	12	359°24.5'	N10°59.6'	24	13	14°24.6'	N10°58.8'	24	14	29°24.8'	N10°57.9'
24	15	44°25.0'	N10°57.1'	24	16	59°25.2'	N10°56.2'	24	17	74°25.3'	N10°55.4'
24	18	89°25.5'	N10°54.5'	24	19	104°25.7'	N10°53.6'	24	20	119°25.8'	N10°52.8'
24	21	134°26.0'	N10°51.9'	24	22	149°26.2'	N10°51.1'	24	23	164°26.3'	N10°50.2'
25	0	179°26.5'	N10°49.3'	25	1	194°26.7'	N10°48.5'	25	2	209°26.9'	N10°47.6'
25	3	224°27.0'	N10°46.8'	25	4	239°27.2'	N10°45.9'	25	5	254°27.4'	N10°45.0'
25	6	269°27.6'	N10°44.2'	25	7	284°27.7'	N10°43.3'	25	8	299°27.9'	N10°42.4'
25	9	314°28.1'	N10°41.6'	25	10	329°28.2'	N10°40.7'	25	11	344°28.4'	N10°39.9'
25	12	359°28.6'	N10°39.0'	25	13	14°28.8'	N10°38.1'	25	14	29°28.9'	N10°37.3'
25	15	44°29.1'	N10°36.4'	25	16	59°29.3'	N10°35.5'	25	17	74°29.5'	N10°34.7'
25	18	89°29.6'	N10°33.8'	25	19	104°29.8'	N10°32.9'	25	20	119°30.0'	N10°32.1'
25	21	134°30.2'	N10°31.2'	25	22	149°30.3'	N10°30.3'	25	23	164°30.5'	N10°29.5'
26	0	179°30.7'	N10°28.6'	26	1	194°30.9'	N10°27.7'	26	2	209°31.0'	N10°26.9'
26	3	224°31.2'	N10°26.0'	26	4	239°31.4'	N10°25.1'	26	5	254°31.6'	N10°24.3'
26	6	269°31.7'	N10°23.4'	26	7	284°31.9'	N10°22.5'	26	8	299°32.1'	N10°21.7'
26	9	314°32.3'	N10°20.8'	26	10	329°32.4'	N10°19.9'	26	11	344°32.6'	N10°19.0'
26	12	359°32.8'	N10°18.2'	26	13	14°33.0'	N10°17.3'	26	14	29°33.2'	N10°16.4'
26	15	44°33.3'	N10°15.6'	26	16	59°33.5'	N10°14.7'	26	17	74°33.7'	N10°13.8'
26	18	89°33.9'	N10°12.9'	26	19	104°34.1'	N10°12.1'	26	20	119°34.2'	N10°11.2'
26	21	134°34.4'	N10°10.3'	26	22	149°34.6'	N10°09.4'	26	23	164°34.8'	N10°08.6'
27	0	179°34.9'	N10°07.7'	27	1	194°35.1'	N10°06.8'	27	2	209°35.3'	N10°05.9'
27	3	224°35.5'	N10°05.1'	27	4	239°35.7'	N10°04.2'	27	5	254°35.8'	N10°03.3'
27	6	269°36.0'	N10°02.4'	27	7	284°36.2'	N10°01.6'	27	8	299°36.4'	N10°00.7'
27	9	314°36.6'	N9°59.8'	27	10	329°36.8'	N9°58.9'	27	11	344°36.9'	N9°58.1'
27	12	359°37.1'	N9°57.2'	27	13	14°37.3'	N9°56.3'	27	14	29°37.5'	N9°55.4'
27	15	44°37.7'	N9°54.6'	27	16	59°37.8'	N9°53.7'	27	17	74°38.0'	N9°52.8'
27	18	89°38.2'	N9°51.9'	27	19	104°38.4'	N9°51.0'	27	20	119°38.6'	N9°50.2'
27	21	134°38.8'	N9°49.3'	27	22	149°38.9'	N9°48.4'	27	23	164°39.1'	N9°47.5'
28	0	179°39.3'	N9°46.6'	28	1	194°39.5'	N9°45.8'	28	2	209°39.7'	N9°44.9'
28	3	224°39.9'	N9°44.0'	28	4	239°40.0'	N9°43.1'	28	5	254°40.2'	N9°42.2'
28	6	269°40.4'	N9°41.3'	28	7	284°40.6'	N9°40.5'	28	8	299°40.8'	N9°39.6'
28	9	314°41.0'	N9°38.7'	28	10	329°41.2'	N9°37.8'	28	11	344°41.3'	N9°36.9'
28	12	359°41.5'	N9°36.0'	28	13	14°41.7'	N9°35.2'	28	14	29°41.9'	N9°34.3'
28	15	44°42.1'	N9°33.4'	28	16	59°42.3'	N9°32.5'	28	17	74°42.5'	N9°31.6'
28	18	89°42.6'	N9°30.7'	28	19	104°42.8'	N9°29.8'	28	20	119°43.0'	N9°29.0'

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Sun — GHA and Declination, August 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
28	21	134°43.2'	N9°28.1'	28	22	149°43.4'	N9°27.2'	28	23	164°43.6'	N9°26.3'
29	0	179°43.8'	N9°25.4'	29	1	194°43.9'	N9°24.5'	29	2	209°44.1'	N9°23.6'
29	3	224°44.3'	N9°22.8'	29	4	239°44.5'	N9°21.9'	29	5	254°44.7'	N9°21.0'
29	6	269°44.9'	N9°20.1'	29	7	284°45.1'	N9°19.2'	29	8	299°45.3'	N9°18.3'
29	9	314°45.4'	N9°17.4'	29	10	329°45.6'	N9°16.5'	29	11	344°45.8'	N9°15.6'
29	12	359°46.0'	N9°14.7'	29	13	14°46.2'	N9°13.9'	29	14	29°46.4'	N9°13.0'
29	15	44°46.6'	N9°12.1'	29	16	59°46.8'	N9°11.2'	29	17	74°47.0'	N9°10.3'
29	18	89°47.2'	N9°09.4'	29	19	104°47.3'	N9°08.5'	29	20	119°47.5'	N9°07.6'
29	21	134°47.7'	N9°06.7'	29	22	149°47.9'	N9°05.8'	29	23	164°48.1'	N9°04.9'
30	0	179°48.3'	N9°04.0'	30	1	194°48.5'	N9°03.2'	30	2	209°48.7'	N9°02.3'
30	3	224°48.9'	N9°01.4'	30	4	239°49.1'	N9°00.5'	30	5	254°49.3'	N8°59.6'
30	6	269°49.4'	N8°58.7'	30	7	284°49.6'	N8°57.8'	30	8	299°49.8'	N8°56.9'
30	9	314°50.0'	N8°56.0'	30	10	329°50.2'	N8°55.1'	30	11	344°50.4'	N8°54.2'
30	12	359°50.6'	N8°53.3'	30	13	14°50.8'	N8°52.4'	30	14	29°51.0'	N8°51.5'
30	15	44°51.2'	N8°50.6'	30	16	59°51.4'	N8°49.7'	30	17	74°51.6'	N8°48.8'
30	18	89°51.8'	N8°47.9'	30	19	104°51.9'	N8°47.0'	30	20	119°52.1'	N8°46.1'
30	21	134°52.3'	N8°45.2'	30	22	149°52.5'	N8°44.3'	30	23	164°52.7'	N8°43.4'
31	0	179°52.9'	N8°42.5'	31	1	194°53.1'	N8°41.6'	31	2	209°53.3'	N8°40.7'
31	3	224°53.5'	N8°39.8'	31	4	239°53.7'	N8°38.9'	31	5	254°53.9'	N8°38.0'
31	6	269°54.1'	N8°37.1'	31	7	284°54.3'	N8°36.2'	31	8	299°54.5'	N8°35.3'
31	9	314°54.7'	N8°34.4'	31	10	329°54.9'	N8°33.5'	31	11	344°55.1'	N8°32.6'
31	12	359°55.3'	N8°31.7'	31	13	14°55.5'	N8°30.8'	31	14	29°55.7'	N8°29.9'
31	15	44°55.8'	N8°29.0'	31	16	59°56.0'	N8°28.1'	31	17	74°56.2'	N8°27.2'
31	18	89°56.4'	N8°26.3'	31	19	104°56.6'	N8°25.4'	31	20	119°56.8'	N8°24.5'
31	21	134°57.0'	N8°23.6'	31	22	149°57.2'	N8°22.7'	31	23	164°57.4'	N8°21.8'

Table 9.9: Sun — GHA and Declination, September 2026 (UT)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
1	0	179°57.6'	N8°20.9'	1	1	194°57.8'	N8°20.0'	1	2	209°58.0'	N8°19.1'
1	3	224°58.2'	N8°18.2'	1	4	239°58.4'	N8°17.3'	1	5	254°58.6'	N8°16.4'
1	6	269°58.8'	N8°15.4'	1	7	284°59.0'	N8°14.5'	1	8	299°59.2'	N8°13.6'
1	9	314°59.4'	N8°12.7'	1	10	329°59.6'	N8°11.8'	1	11	344°59.8'	N8°10.9'
1	12	0°00.0'	N8°10.0'	1	13	15°00.2'	N8°09.1'	1	14	30°00.4'	N8°08.2'
1	15	45°00.6'	N8°07.3'	1	16	60°00.8'	N8°06.4'	1	17	75°01.0'	N8°05.5'
1	18	90°01.2'	N8°04.6'	1	19	105°01.4'	N8°03.6'	1	20	120°01.6'	N8°02.7'
1	21	135°01.8'	N8°01.8'	1	22	150°02.0'	N8°00.9'	1	23	165°02.2'	N8°00.0'
2	0	180°02.4'	N7°59.1'	2	1	195°02.6'	N7°58.2'	2	2	210°02.8'	N7°57.3'
2	3	225°03.0'	N7°56.4'	2	4	240°03.2'	N7°55.5'	2	5	255°03.4'	N7°54.5'
2	6	270°03.6'	N7°53.6'	2	7	285°03.8'	N7°52.7'	2	8	300°04.0'	N7°51.8'
2	9	315°04.2'	N7°50.9'	2	10	330°04.4'	N7°50.0'	2	11	345°04.6'	N7°49.1'
2	12	0°04.8'	N7°48.2'	2	13	15°05.0'	N7°47.2'	2	14	30°05.2'	N7°46.3'
2	15	45°05.4'	N7°45.4'	2	16	60°05.6'	N7°44.5'	2	17	75°05.8'	N7°43.6'
2	18	90°06.0'	N7°42.7'	2	19	105°06.2'	N7°41.8'	2	20	120°06.4'	N7°40.8'
2	21	135°06.6'	N7°39.9'	2	22	150°06.8'	N7°39.0'	2	23	165°07.0'	N7°38.1'
3	0	180°07.2'	N7°37.2'	3	1	195°07.4'	N7°36.3'	3	2	210°07.7'	N7°35.3'
3	3	225°07.9'	N7°34.4'	3	4	240°08.1'	N7°33.5'	3	5	255°08.3'	N7°32.6'
3	6	270°08.5'	N7°31.7'	3	7	285°08.7'	N7°30.8'	3	8	300°08.9'	N7°29.8'
3	9	315°09.1'	N7°28.9'	3	10	330°09.3'	N7°28.0'	3	11	345°09.5'	N7°27.1'
3	12	0°09.7'	N7°26.2'	3	13	15°09.9'	N7°25.3'	3	14	30°10.1'	N7°24.3'
3	15	45°10.3'	N7°23.4'	3	16	60°10.5'	N7°22.5'	3	17	75°10.7'	N7°21.6'
3	18	90°10.9'	N7°20.7'	3	19	105°11.1'	N7°19.7'	3	20	120°11.3'	N7°18.8'
3	21	135°11.5'	N7°17.9'	3	22	150°11.7'	N7°17.0'	3	23	165°12.0'	N7°16.1'
4	0	180°12.2'	N7°15.1'	4	1	195°12.4'	N7°14.2'	4	2	210°12.6'	N7°13.3'
4	3	225°12.8'	N7°12.4'	4	4	240°13.0'	N7°11.5'	4	5	255°13.2'	N7°10.5'
4	6	270°13.4'	N7°09.6'	4	7	285°13.6'	N7°08.7'	4	8	300°13.8'	N7°07.8'
4	9	315°14.0'	N7°06.8'	4	10	330°14.2'	N7°05.9'	4	11	345°14.4'	N7°05.0'
4	12	0°14.6'	N7°04.1'	4	13	15°14.8'	N7°03.2'	4	14	30°15.1'	N7°02.2'
4	15	45°15.3'	N7°01.3'	4	16	60°15.5'	N7°00.4'	4	17	75°15.7'	N6°59.5'
4	18	90°15.9'	N6°58.5'	4	19	105°16.1'	N6°57.6'	4	20	120°16.3'	N6°56.7'
4	21	135°16.5'	N6°55.8'	4	22	150°16.7'	N6°54.8'	4	23	165°16.9'	N6°53.9'
5	0	180°17.1'	N6°53.0'	5	1	195°17.3'	N6°52.1'	5	2	210°17.6'	N6°51.1'

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Sun — GHA and Declination, September 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
5	3	225°17.8'	N6°50.2'	5	4	240°18.0'	N6°49.3'	5	5	255°18.2'	N6°48.4'
5	6	270°18.4'	N6°47.4'	5	7	285°18.6'	N6°46.5'	5	8	300°18.8'	N6°45.6'
5	9	315°19.0'	N6°44.7'	5	10	330°19.2'	N6°43.7'	5	11	345°19.4'	N6°42.8'
5	12	0°19.6'	N6°41.9'	5	13	15°19.9'	N6°40.9'	5	14	30°20.1'	N6°40.0'
5	15	45°20.3'	N6°39.1'	5	16	60°20.5'	N6°38.2'	5	17	75°20.7'	N6°37.2'
5	18	90°20.9'	N6°36.3'	5	19	105°21.1'	N6°35.4'	5	20	120°21.3'	N6°34.4'
5	21	135°21.5'	N6°33.5'	5	22	150°21.8'	N6°32.6'	5	23	165°22.0'	N6°31.7'
6	0	180°22.2'	N6°30.7'	6	1	195°22.4'	N6°29.8'	6	2	210°22.6'	N6°28.9'
6	3	225°22.8'	N6°27.9'	6	4	240°23.0'	N6°27.0'	6	5	255°23.2'	N6°26.1'
6	6	270°23.4'	N6°25.1'	6	7	285°23.7'	N6°24.2'	6	8	300°23.9'	N6°23.3'
6	9	315°24.1'	N6°22.3'	6	10	330°24.3'	N6°21.4'	6	11	345°24.5'	N6°20.5'
6	12	0°24.7'	N6°19.5'	6	13	15°24.9'	N6°18.6'	6	14	30°25.1'	N6°17.7'
6	15	45°25.3'	N6°16.7'	6	16	60°25.6'	N6°15.8'	6	17	75°25.8'	N6°14.9'
6	18	90°26.0'	N6°13.9'	6	19	105°26.2'	N6°13.0'	6	20	120°26.4'	N6°12.1'
6	21	135°26.6'	N6°11.1'	6	22	150°26.8'	N6°10.2'	6	23	165°27.1'	N6°09.3'
7	0	180°27.3'	N6°08.3'	7	1	195°27.5'	N6°07.4'	7	2	210°27.7'	N6°06.5'
7	3	225°27.9'	N6°05.5'	7	4	240°28.1'	N6°04.6'	7	5	255°28.3'	N6°03.7'
7	6	270°28.5'	N6°02.7'	7	7	285°28.8'	N6°01.8'	7	8	300°29.0'	N6°00.9'
7	9	315°29.2'	N5°59.9'	7	10	330°29.4'	N5°59.0'	7	11	345°29.6'	N5°58.1'
7	12	0°29.8'	N5°57.1'	7	13	15°30.0'	N5°56.2'	7	14	30°30.3'	N5°55.2'
7	15	45°30.5'	N5°54.3'	7	16	60°30.7'	N5°53.4'	7	17	75°30.9'	N5°52.4'
7	18	90°31.1'	N5°51.5'	7	19	105°31.3'	N5°50.6'	7	20	120°31.5'	N5°49.6'
7	21	135°31.8'	N5°48.7'	7	22	150°32.0'	N5°47.7'	7	23	165°32.2'	N5°46.8'
8	0	180°32.4'	N5°45.9'	8	1	195°32.6'	N5°44.9'	8	2	210°32.8'	N5°44.0'
8	3	225°33.0'	N5°43.1'	8	4	240°33.3'	N5°42.1'	8	5	255°33.5'	N5°41.2'
8	6	270°33.7'	N5°40.2'	8	7	285°33.9'	N5°39.3'	8	8	300°34.1'	N5°38.4'
8	9	315°34.3'	N5°37.4'	8	10	330°34.6'	N5°36.5'	8	11	345°34.8'	N5°35.5'
8	12	0°35.0'	N5°34.6'	8	13	15°35.2'	N5°33.7'	8	14	30°35.4'	N5°32.7'
8	15	45°35.6'	N5°31.8'	8	16	60°35.9'	N5°30.8'	8	17	75°36.1'	N5°29.9'
8	18	90°36.3'	N5°29.0'	8	19	105°36.5'	N5°28.0'	8	20	120°36.7'	N5°27.1'
8	21	135°36.9'	N5°26.1'	8	22	150°37.1'	N5°25.2'	8	23	165°37.4'	N5°24.2'
9	0	180°37.6'	N5°23.3'	9	1	195°37.8'	N5°22.4'	9	2	210°38.0'	N5°21.4'
9	3	225°38.2'	N5°20.5'	9	4	240°38.5'	N5°19.5'	9	5	255°38.7'	N5°18.6'
9	6	270°38.9'	N5°17.6'	9	7	285°39.1'	N5°16.7'	9	8	300°39.3'	N5°15.8'
9	9	315°39.5'	N5°14.8'	9	10	330°39.8'	N5°13.9'	9	11	345°40.0'	N5°12.9'
9	12	0°40.2'	N5°12.0'	9	13	15°40.4'	N5°11.0'	9	14	30°40.6'	N5°10.1'
9	15	45°40.8'	N5°09.1'	9	16	60°41.1'	N5°08.2'	9	17	75°41.3'	N5°07.3'
9	18	90°41.5'	N5°06.3'	9	19	105°41.7'	N5°05.4'	9	20	120°41.9'	N5°04.4'
9	21	135°42.2'	N5°03.5'	9	22	150°42.4'	N5°02.5'	9	23	165°42.6'	N5°01.6'
10	0	180°42.8'	N5°00.6'	10	1	195°43.0'	N4°59.7'	10	2	210°43.2'	N4°58.7'
10	3	225°43.5'	N4°57.8'	10	4	240°43.7'	N4°56.9'	10	5	255°43.9'	N4°55.9'
10	6	270°44.1'	N4°55.0'	10	7	285°44.3'	N4°54.0'	10	8	300°44.6'	N4°53.1'
10	9	315°44.8'	N4°52.1'	10	10	330°45.0'	N4°51.2'	10	11	345°45.2'	N4°50.2'
10	12	0°45.4'	N4°49.3'	10	13	15°45.6'	N4°48.3'	10	14	30°45.9'	N4°47.4'
10	15	45°46.1'	N4°46.4'	10	16	60°46.3'	N4°45.5'	10	17	75°46.5'	N4°44.5'
10	18	90°46.7'	N4°43.6'	10	19	105°47.0'	N4°42.6'	10	20	120°47.2'	N4°41.7'
10	21	135°47.4'	N4°40.7'	10	22	150°47.6'	N4°39.8'	10	23	165°47.8'	N4°38.8'
11	0	180°48.1'	N4°37.9'	11	1	195°48.3'	N4°36.9'	11	2	210°48.5'	N4°36.0'
11	3	225°48.7'	N4°35.0'	11	4	240°48.9'	N4°34.1'	11	5	255°49.2'	N4°33.1'
11	6	270°49.4'	N4°32.2'	11	7	285°49.6'	N4°31.2'	11	8	300°49.8'	N4°30.3'
11	9	315°50.0'	N4°29.3'	11	10	330°50.3'	N4°28.4'	11	11	345°50.5'	N4°27.4'
11	12	0°50.7'	N4°26.5'	11	13	15°50.9'	N4°25.5'	11	14	30°51.1'	N4°24.6'
11	15	45°51.4'	N4°23.6'	11	16	60°51.6'	N4°22.7'	11	17	75°51.8'	N4°21.7'
11	18	90°52.0'	N4°20.8'	11	19	105°52.2'	N4°19.8'	11	20	120°52.5'	N4°18.9'
11	21	135°52.7'	N4°17.9'	11	22	150°52.9'	N4°17.0'	11	23	165°53.1'	N4°16.0'
12	0	180°53.3'	N4°15.1'	12	1	195°53.6'	N4°14.1'	12	2	210°53.8'	N4°13.2'
12	3	225°54.0'	N4°12.2'	12	4	240°54.2'	N4°11.2'	12	5	255°54.5'	N4°10.3'
12	6	270°54.7'	N4°09.3'	12	7	285°54.9'	N4°08.4'	12	8	300°55.1'	N4°07.4'
12	9	315°55.3'	N4°06.5'	12	10	330°55.6'	N4°05.5'	12	11	345°55.8'	N4°04.6'
12	12	0°56.0'	N4°03.6'	12	13	15°56.2'	N4°02.7'	12	14	30°56.4'	N4°01.7'
12	15	45°56.7'	N4°00.8'	12	16	60°56.9'	N3°59.8'	12	17	75°57.1'	N3°58.8'
12	18	90°57.3'	N3°57.9'	12	19	105°57.6'	N3°56.9'	12	20	120°57.8'	N3°56.0'
12	21	135°58.0'	N3°55.0'	12	22	150°58.2'	N3°54.1'	12	23	165°58.4'	N3°53.1'
13	0	180°58.7'	N3°52.2'	13	1	195°58.9'	N3°51.2'	13	2	210°59.1'	N3°50.2'

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Sun — GHA and Declination, September 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
13	3	225°59.3'	N3°49.3'	13	4	240°59.5'	N3°48.3'	13	5	255°59.8'	N3°47.4'
13	6	271°00.0'	N3°46.4'	13	7	286°00.2'	N3°45.5'	13	8	301°00.4'	N3°44.5'
13	9	316°00.7'	N3°43.6'	13	10	331°00.9'	N3°42.6'	13	11	346°01.1'	N3°41.6'
13	12	1°01.3'	N3°40.7'	13	13	16°01.5'	N3°39.7'	13	14	31°01.8'	N3°38.8'
13	15	46°02.0'	N3°37.8'	13	16	61°02.2'	N3°36.9'	13	17	76°02.4'	N3°35.9'
13	18	91°02.7'	N3°34.9'	13	19	106°02.9'	N3°34.0'	13	20	121°03.1'	N3°33.0'
13	21	136°03.3'	N3°32.1'	13	22	151°03.5'	N3°31.1'	13	23	166°03.8'	N3°30.1'
14	0	181°04.0'	N3°29.2'	14	1	196°04.2'	N3°28.2'	14	2	211°04.4'	N3°27.3'
14	3	226°04.7'	N3°26.3'	14	4	241°04.9'	N3°25.4'	14	5	256°05.1'	N3°24.4'
14	6	271°05.3'	N3°23.4'	14	7	286°05.5'	N3°22.5'	14	8	301°05.8'	N3°21.5'
14	9	316°06.0'	N3°20.6'	14	10	331°06.2'	N3°19.6'	14	11	346°06.4'	N3°18.6'
14	12	1°06.7'	N3°17.7'	14	13	16°06.9'	N3°16.7'	14	14	31°07.1'	N3°15.8'
14	15	46°07.3'	N3°14.8'	14	16	61°07.6'	N3°13.8'	14	17	76°07.8'	N3°12.9'
14	18	91°08.0'	N3°11.9'	14	19	106°08.2'	N3°11.0'	14	20	121°08.4'	N3°10.0'
14	21	136°08.7'	N3°09.0'	14	22	151°08.9'	N3°08.1'	14	23	166°09.1'	N3°07.1'
15	0	181°09.3'	N3°06.2'	15	1	196°09.6'	N3°05.2'	15	2	211°09.8'	N3°04.2'
15	3	226°10.0'	N3°03.3'	15	4	241°10.2'	N3°02.3'	15	5	256°10.5'	N3°01.3'
15	6	271°10.7'	N3°00.4'	15	7	286°10.9'	N2°59.4'	15	8	301°11.1'	N2°58.5'
15	9	316°11.3'	N2°57.5'	15	10	331°11.6'	N2°56.5'	15	11	346°11.8'	N2°55.6'
15	12	1°12.0'	N2°54.6'	15	13	16°12.2'	N2°53.7'	15	14	31°12.5'	N2°52.7'
15	15	46°12.7'	N2°51.7'	15	16	61°12.9'	N2°50.8'	15	17	76°13.1'	N2°49.8'
15	18	91°13.4'	N2°48.8'	15	19	106°13.6'	N2°47.9'	15	20	121°13.8'	N2°46.9'
15	21	136°14.0'	N2°46.0'	15	22	151°14.2'	N2°45.0'	15	23	166°14.5'	N2°44.0'
16	0	181°14.7'	N2°43.1'	16	1	196°14.9'	N2°42.1'	16	2	211°15.1'	N2°41.1'
16	3	226°15.4'	N2°40.2'	16	4	241°15.6'	N2°39.2'	16	5	256°15.8'	N2°38.2'
16	6	271°16.0'	N2°37.3'	16	7	286°16.3'	N2°36.3'	16	8	301°16.5'	N2°35.4'
16	9	316°16.7'	N2°34.4'	16	10	331°16.9'	N2°33.4'	16	11	346°17.2'	N2°32.5'
16	12	1°17.4'	N2°31.5'	16	13	16°17.6'	N2°30.5'	16	14	31°17.8'	N2°29.6'
16	15	46°18.0'	N2°28.6'	16	16	61°18.3'	N2°27.6'	16	17	76°18.5'	N2°26.7'
16	18	91°18.7'	N2°25.7'	16	19	106°18.9'	N2°24.7'	16	20	121°19.2'	N2°23.8'
16	21	136°19.4'	N2°22.8'	16	22	151°19.6'	N2°21.9'	16	23	166°19.8'	N2°20.9'
17	0	181°20.1'	N2°19.9'	17	1	196°20.3'	N2°19.0'	17	2	211°20.5'	N2°18.0'
17	3	226°20.7'	N2°17.0'	17	4	241°21.0'	N2°16.1'	17	5	256°21.2'	N2°15.1'
17	6	271°21.4'	N2°14.1'	17	7	286°21.6'	N2°13.2'	17	8	301°21.8'	N2°12.2'
17	9	316°22.1'	N2°11.2'	17	10	331°22.3'	N2°10.3'	17	11	346°22.5'	N2°09.3'
17	12	1°22.7'	N2°08.3'	17	13	16°23.0'	N2°07.4'	17	14	31°23.2'	N2°06.4'
17	15	46°23.4'	N2°05.4'	17	16	61°23.6'	N2°04.5'	17	17	76°23.9'	N2°03.5'
17	18	91°24.1'	N2°02.5'	17	19	106°24.3'	N2°01.6'	17	20	121°24.5'	N2°00.6'
17	21	136°24.8'	N1°59.6'	17	22	151°25.0'	N1°58.7'	17	23	166°25.2'	N1°57.7'
18	0	181°25.4'	N1°56.7'	18	1	196°25.6'	N1°55.8'	18	2	211°25.9'	N1°54.8'
18	3	226°26.1'	N1°53.8'	18	4	241°26.3'	N1°52.9'	18	5	256°26.5'	N1°51.9'
18	6	271°26.8'	N1°50.9'	18	7	286°27.0'	N1°50.0'	18	8	301°27.2'	N1°49.0'
18	9	316°27.4'	N1°48.0'	18	10	331°27.7'	N1°47.1'	18	11	346°27.9'	N1°46.1'
18	12	1°28.1'	N1°45.1'	18	13	16°28.3'	N1°44.2'	18	14	31°28.5'	N1°43.2'
18	15	46°28.8'	N1°42.2'	18	16	61°29.0'	N1°41.2'	18	17	76°29.2'	N1°40.3'
18	18	91°29.4'	N1°39.3'	18	19	106°29.7'	N1°38.3'	18	20	121°29.9'	N1°37.4'
18	21	136°30.1'	N1°36.4'	18	22	151°30.3'	N1°35.4'	18	23	166°30.6'	N1°34.5'
19	0	181°30.8'	N1°33.5'	19	1	196°31.0'	N1°32.5'	19	2	211°31.2'	N1°31.6'
19	3	226°31.5'	N1°30.6'	19	4	241°31.7'	N1°29.6'	19	5	256°31.9'	N1°28.7'
19	6	271°32.1'	N1°27.7'	19	7	286°32.3'	N1°26.7'	19	8	301°32.6'	N1°25.7'
19	9	316°32.8'	N1°24.8'	19	10	331°33.0'	N1°23.8'	19	11	346°33.2'	N1°22.8'
19	12	1°33.5'	N1°21.9'	19	13	16°33.7'	N1°20.9'	19	14	31°33.9'	N1°19.9'
19	15	46°34.1'	N1°19.0'	19	16	61°34.3'	N1°18.0'	19	17	76°34.6'	N1°17.0'
19	18	91°34.8'	N1°16.1'	19	19	106°35.0'	N1°15.1'	19	20	121°35.2'	N1°14.1'
19	21	136°35.5'	N1°13.1'	19	22	151°35.7'	N1°12.2'	19	23	166°35.9'	N1°11.2'
20	0	181°36.1'	N1°10.2'	20	1	196°36.4'	N1°09.3'	20	2	211°36.6'	N1°08.3'
20	3	226°36.8'	N1°07.3'	20	4	241°37.0'	N1°06.4'	20	5	256°37.2'	N1°05.4'
20	6	271°37.5'	N1°04.4'	20	7	286°37.7'	N1°03.4'	20	8	301°37.9'	N1°02.5'
20	9	316°38.1'	N1°01.5'	20	10	331°38.4'	N1°00.5'	20	11	346°38.6'	N0°59.6'
20	12	1°38.8'	N0°58.6'	20	13	16°39.0'	N0°57.6'	20	14	31°39.2'	N0°56.6'
20	15	46°39.5'	N0°55.7'	20	16	61°39.7'	N0°54.7'	20	17	76°39.9'	N0°53.7'
20	18	91°40.1'	N0°52.8'	20	19	106°40.4'	N0°51.8'	20	20	121°40.6'	N0°50.8'
20	21	136°40.8'	N0°49.8'	20	22	151°41.0'	N0°48.9'	20	23	166°41.2'	N0°47.9'
21	0	181°41.5'	N0°46.9'	21	1	196°41.7'	N0°46.0'	21	2	211°41.9'	N0°45.0'

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Sun — GHA and Declination, September 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
21	3	226°42.1'	N0°44.0'	21	4	241°42.4'	N0°43.0'	21	5	256°42.6'	N0°42.1'
21	6	271°42.8'	N0°41.1'	21	7	286°43.0'	N0°40.1'	21	8	301°43.2'	N0°39.2'
21	9	316°43.5'	N0°38.2'	21	10	331°43.7'	N0°37.2'	21	11	346°43.9'	N0°36.2'
21	12	1°44.1'	N0°35.3'	21	13	16°44.3'	N0°34.3'	21	14	31°44.6'	N0°33.3'
21	15	46°44.8'	N0°32.4'	21	16	61°45.0'	N0°31.4'	21	17	76°45.2'	N0°30.4'
21	18	91°45.5'	N0°29.4'	21	19	106°45.7'	N0°28.5'	21	20	121°45.9'	N0°27.5'
21	21	136°46.1'	N0°26.5'	21	22	151°46.3'	N0°25.6'	21	23	166°46.6'	N0°24.6'
22	0	181°46.8'	N0°23.6'	22	1	196°47.0'	N0°22.6'	22	2	211°47.2'	N0°21.7'
22	3	226°47.4'	N0°20.7'	22	4	241°47.7'	N0°19.7'	22	5	256°47.9'	N0°18.8'
22	6	271°48.1'	N0°17.8'	22	7	286°48.3'	N0°16.8'	22	8	301°48.5'	N0°15.8'
22	9	316°48.8'	N0°14.9'	22	10	331°49.0'	N0°13.9'	22	11	346°49.2'	N0°12.9'
22	12	1°49.4'	N0°11.9'	22	13	16°49.6'	N0°11.0'	22	14	31°49.9'	N0°10.0'
22	15	46°50.1'	N0°09.0'	22	16	61°50.3'	N0°08.1'	22	17	76°50.5'	N0°07.1'
22	18	91°50.7'	N0°06.1'	22	19	106°51.0'	N0°05.1'	22	20	121°51.2'	N0°04.2'
22	21	136°51.4'	N0°03.2'	22	22	151°51.6'	N0°02.2'	22	23	166°51.8'	N0°01.2'
23	0	181°52.1'	N0°00.3'	23	1	196°52.3'	S0°00.7'	23	2	211°52.5'	S0°01.7'
23	3	226°52.7'	S0°02.6'	23	4	241°52.9'	S0°03.6'	23	5	256°53.2'	S0°04.6'
23	6	271°53.4'	S0°05.6'	23	7	286°53.6'	S0°06.5'	23	8	301°53.8'	S0°07.5'
23	9	316°54.0'	S0°08.5'	23	10	331°54.3'	S0°09.5'	23	11	346°54.5'	S0°10.4'
23	12	1°54.7'	S0°11.4'	23	13	16°54.9'	S0°12.4'	23	14	31°55.1'	S0°13.3'
23	15	46°55.4'	S0°14.3'	23	16	61°55.6'	S0°15.3'	23	17	76°55.8'	S0°16.3'
23	18	91°56.0'	S0°17.2'	23	19	106°56.2'	S0°18.2'	23	20	121°56.5'	S0°19.2'
23	21	136°56.7'	S0°20.2'	23	22	151°56.9'	S0°21.1'	23	23	166°57.1'	S0°22.1'
24	0	181°57.3'	S0°23.1'	24	1	196°57.5'	S0°24.1'	24	2	211°57.8'	S0°25.0'
24	3	226°58.0'	S0°26.0'	24	4	241°58.2'	S0°27.0'	24	5	256°58.4'	S0°27.9'
24	6	271°58.6'	S0°28.9'	24	7	286°58.9'	S0°29.9'	24	8	301°59.1'	S0°30.9'
24	9	316°59.3'	S0°31.8'	24	10	331°59.5'	S0°32.8'	24	11	346°59.7'	S0°33.8'
24	12	1°59.9'	S0°34.8'	24	13	17°00.2'	S0°35.7'	24	14	32°00.4'	S0°36.7'
24	15	47°00.6'	S0°37.7'	24	16	62°00.8'	S0°38.7'	24	17	77°01.0'	S0°39.6'
24	18	92°01.3'	S0°40.6'	24	19	107°01.5'	S0°41.6'	24	20	122°01.7'	S0°42.5'
24	21	137°01.9'	S0°43.5'	24	22	152°02.1'	S0°44.5'	24	23	167°02.3'	S0°45.5'
25	0	182°02.6'	S0°46.4'	25	1	197°02.8'	S0°47.4'	25	2	212°03.0'	S0°48.4'
25	3	227°03.2'	S0°49.4'	25	4	242°03.4'	S0°50.3'	25	5	257°03.6'	S0°51.3'
25	6	272°03.9'	S0°52.3'	25	7	287°04.1'	S0°53.3'	25	8	302°04.3'	S0°54.2'
25	9	317°04.5'	S0°55.2'	25	10	332°04.7'	S0°56.2'	25	11	347°04.9'	S0°57.2'
25	12	2°05.2'	S0°58.1'	25	13	17°05.4'	S0°59.1'	25	14	32°05.6'	S1°00.1'
25	15	47°05.8'	S1°01.0'	25	16	62°06.0'	S1°02.0'	25	17	77°06.2'	S1°03.0'
25	18	92°06.4'	S1°04.0'	25	19	107°06.7'	S1°04.9'	25	20	122°06.9'	S1°05.9'
25	21	137°07.1'	S1°06.9'	25	22	152°07.3'	S1°07.9'	25	23	167°07.5'	S1°08.8'
26	0	182°07.7'	S1°09.8'	26	1	197°08.0'	S1°10.8'	26	2	212°08.2'	S1°11.8'
26	3	227°08.4'	S1°12.7'	26	4	242°08.6'	S1°13.7'	26	5	257°08.8'	S1°14.7'
26	6	272°09.0'	S1°15.6'	26	7	287°09.2'	S1°16.6'	26	8	302°09.5'	S1°17.6'
26	9	317°09.7'	S1°18.6'	26	10	332°09.9'	S1°19.5'	26	11	347°10.1'	S1°20.5'
26	12	2°10.3'	S1°21.5'	26	13	17°10.5'	S1°22.5'	26	14	32°10.7'	S1°23.4'
26	15	47°11.0'	S1°24.4'	26	16	62°11.2'	S1°25.4'	26	17	77°11.4'	S1°26.4'
26	18	92°11.6'	S1°27.3'	26	19	107°11.8'	S1°28.3'	26	20	122°12.0'	S1°29.3'
26	21	137°12.2'	S1°30.2'	26	22	152°12.5'	S1°31.2'	26	23	167°12.7'	S1°32.2'
27	0	182°12.9'	S1°33.2'	27	1	197°13.1'	S1°34.1'	27	2	212°13.3'	S1°35.1'
27	3	227°13.5'	S1°36.1'	27	4	242°13.7'	S1°37.1'	27	5	257°13.9'	S1°38.0'
27	6	272°14.2'	S1°39.0'	27	7	287°14.4'	S1°40.0'	27	8	302°14.6'	S1°40.9'
27	9	317°14.8'	S1°41.9'	27	10	332°15.0'	S1°42.9'	27	11	347°15.2'	S1°43.9'
27	12	2°15.4'	S1°44.8'	27	13	17°15.6'	S1°45.8'	27	14	32°15.9'	S1°46.8'
27	15	47°16.1'	S1°47.8'	27	16	62°16.3'	S1°48.7'	27	17	77°16.5'	S1°49.7'
27	18	92°16.7'	S1°50.7'	27	19	107°16.9'	S1°51.6'	27	20	122°17.1'	S1°52.6'
27	21	137°17.3'	S1°53.6'	27	22	152°17.5'	S1°54.6'	27	23	167°17.8'	S1°55.5'
28	0	182°18.0'	S1°56.5'	28	1	197°18.2'	S1°57.5'	28	2	212°18.4'	S1°58.5'
28	3	227°18.6'	S1°59.4'	28	4	242°18.8'	S2°00.4'	28	5	257°19.0'	S2°01.4'
28	6	272°19.2'	S2°02.3'	28	7	287°19.4'	S2°03.3'	28	8	302°19.7'	S2°04.3'
28	9	317°19.9'	S2°05.3'	28	10	332°20.1'	S2°06.2'	28	11	347°20.3'	S2°07.2'
28	12	2°20.5'	S2°08.2'	28	13	17°20.7'	S2°09.2'	28	14	32°20.9'	S2°10.1'
28	15	47°21.1'	S2°11.1'	28	16	62°21.3'	S2°12.1'	28	17	77°21.5'	S2°13.0'
28	18	92°21.7'	S2°14.0'	28	19	107°22.0'	S2°15.0'	28	20	122°22.2'	S2°16.0'
28	21	137°22.4'	S2°16.9'	28	22	152°22.6'	S2°17.9'	28	23	167°22.8'	S2°18.9'
29	0	182°23.0'	S2°19.8'	29	1	197°23.2'	S2°20.8'	29	2	212°23.4'	S2°21.8'

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Sun — GHA and Declination, September 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
29	3	227°23.6'	S2°22.8'	29	4	242°23.8'	S2°23.7'	29	5	257°24.0'	S2°24.7'
29	6	272°24.2'	S2°25.7'	29	7	287°24.5'	S2°26.6'	29	8	302°24.7'	S2°27.6'
29	9	317°24.9'	S2°28.6'	29	10	332°25.1'	S2°29.6'	29	11	347°25.3'	S2°30.5'
29	12	2°25.5'	S2°31.5'	29	13	17°25.7'	S2°32.5'	29	14	32°25.9'	S2°33.4'
29	15	47°26.1'	S2°34.4'	29	16	62°26.3'	S2°35.4'	29	17	77°26.5'	S2°36.4'
29	18	92°26.7'	S2°37.3'	29	19	107°26.9'	S2°38.3'	29	20	122°27.1'	S2°39.3'
29	21	137°27.4'	S2°40.2'	29	22	152°27.6'	S2°41.2'	29	23	167°27.8'	S2°42.2'
30	0	182°28.0'	S2°43.2'	30	1	197°28.2'	S2°44.1'	30	2	212°28.4'	S2°45.1'
30	3	227°28.6'	S2°46.1'	30	4	242°28.8'	S2°47.0'	30	5	257°29.0'	S2°48.0'
30	6	272°29.2'	S2°49.0'	30	7	287°29.4'	S2°50.0'	30	8	302°29.6'	S2°50.9'
30	9	317°29.8'	S2°51.9'	30	10	332°30.0'	S2°52.9'	30	11	347°30.2'	S2°53.8'
30	12	2°30.4'	S2°54.8'	30	13	17°30.6'	S2°55.8'	30	14	32°30.8'	S2°56.7'
30	15	47°31.0'	S2°57.7'	30	16	62°31.3'	S2°58.7'	30	17	77°31.5'	S2°59.7'
30	18	92°31.7'	S3°00.6'	30	19	107°31.9'	S3°01.6'	30	20	122°32.1'	S3°02.6'
30	21	137°32.3'	S3°03.5'	30	22	152°32.5'	S3°04.5'	30	23	167°32.7'	S3°05.5'

Table 9.10: Sun — GHA and Declination, October 2026 (UT)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
1	0	182°32.9'	S3°06.4'	1	1	197°33.1'	S3°07.4'	1	2	212°33.3'	S3°08.4'
1	3	227°33.5'	S3°09.4'	1	4	242°33.7'	S3°10.3'	1	5	257°33.9'	S3°11.3'
1	6	272°34.1'	S3°12.3'	1	7	287°34.3'	S3°13.2'	1	8	302°34.5'	S3°14.2'
1	9	317°34.7'	S3°15.2'	1	10	332°34.9'	S3°16.1'	1	11	347°35.1'	S3°17.1'
1	12	2°35.3'	S3°18.1'	1	13	17°35.5'	S3°19.0'	1	14	32°35.7'	S3°20.0'
1	15	47°35.9'	S3°21.0'	1	16	62°36.1'	S3°21.9'	1	17	77°36.3'	S3°22.9'
1	18	92°36.5'	S3°23.9'	1	19	107°36.7'	S3°24.9'	1	20	122°36.9'	S3°25.8'
1	21	137°37.1'	S3°26.8'	1	22	152°37.3'	S3°27.8'	1	23	167°37.5'	S3°28.7'
2	0	182°37.7'	S3°29.7'	2	1	197°37.9'	S3°30.7'	2	2	212°38.1'	S3°31.6'
2	3	227°38.3'	S3°32.6'	2	4	242°38.5'	S3°33.6'	2	5	257°38.7'	S3°34.5'
2	6	272°38.9'	S3°35.5'	2	7	287°39.1'	S3°36.5'	2	8	302°39.3'	S3°37.4'
2	9	317°39.5'	S3°38.4'	2	10	332°39.7'	S3°39.4'	2	11	347°39.9'	S3°40.3'
2	12	2°40.1'	S3°41.3'	2	13	17°40.3'	S3°42.3'	2	14	32°40.5'	S3°43.2'
2	15	47°40.7'	S3°44.2'	2	16	62°40.9'	S3°45.2'	2	17	77°41.1'	S3°46.1'
2	18	92°41.3'	S3°47.1'	2	19	107°41.5'	S3°48.1'	2	20	122°41.7'	S3°49.0'
2	21	137°41.9'	S3°50.0'	2	22	152°42.1'	S3°51.0'	2	23	167°42.3'	S3°51.9'
3	0	182°42.5'	S3°52.9'	3	1	197°42.7'	S3°53.9'	3	2	212°42.9'	S3°54.8'
3	3	227°43.1'	S3°55.8'	3	4	242°43.3'	S3°56.8'	3	5	257°43.5'	S3°57.7'
3	6	272°43.7'	S3°58.7'	3	7	287°43.8'	S3°59.7'	3	8	302°44.0'	S4°00.6'
3	9	317°44.2'	S4°01.6'	3	10	332°44.4'	S4°02.6'	3	11	347°44.6'	S4°03.5'
3	12	2°44.8'	S4°04.5'	3	13	17°45.0'	S4°05.5'	3	14	32°45.2'	S4°06.4'
3	15	47°45.4'	S4°07.4'	3	16	62°45.6'	S4°08.4'	3	17	77°45.8'	S4°09.3'
3	18	92°46.0'	S4°10.3'	3	19	107°46.2'	S4°11.3'	3	20	122°46.4'	S4°12.2'
3	21	137°46.6'	S4°13.2'	3	22	152°46.8'	S4°14.1'	3	23	167°47.0'	S4°15.1'
4	0	182°47.2'	S4°16.1'	4	1	197°47.3'	S4°17.0'	4	2	212°47.5'	S4°18.0'
4	3	227°47.7'	S4°19.0'	4	4	242°47.9'	S4°19.9'	4	5	257°48.1'	S4°20.9'
4	6	272°48.3'	S4°21.9'	4	7	287°48.5'	S4°22.8'	4	8	302°48.7'	S4°23.8'
4	9	317°48.9'	S4°24.7'	4	10	332°49.1'	S4°25.7'	4	11	347°49.3'	S4°26.7'
4	12	2°49.5'	S4°27.6'	4	13	17°49.7'	S4°28.6'	4	14	32°49.8'	S4°29.6'
4	15	47°50.0'	S4°30.5'	4	16	62°50.2'	S4°31.5'	4	17	77°50.4'	S4°32.5'
4	18	92°50.6'	S4°33.4'	4	19	107°50.8'	S4°34.4'	4	20	122°51.0'	S4°35.3'
4	21	137°51.2'	S4°36.3'	4	22	152°51.4'	S4°37.3'	4	23	167°51.6'	S4°38.2'
5	0	182°51.7'	S4°39.2'	5	1	197°51.9'	S4°40.2'	5	2	212°52.1'	S4°41.1'
5	3	227°52.3'	S4°42.1'	5	4	242°52.5'	S4°43.0'	5	5	257°52.7'	S4°44.0'
5	6	272°52.9'	S4°45.0'	5	7	287°53.1'	S4°45.9'	5	8	302°53.3'	S4°46.9'
5	9	317°53.4'	S4°47.8'	5	10	332°53.6'	S4°48.8'	5	11	347°53.8'	S4°49.8'
5	12	2°54.0'	S4°50.7'	5	13	17°54.2'	S4°51.7'	5	14	32°54.4'	S4°52.6'
5	15	47°54.6'	S4°53.6'	5	16	62°54.8'	S4°54.6'	5	17	77°54.9'	S4°55.5'
5	18	92°55.1'	S4°56.5'	5	19	107°55.3'	S4°57.5'	5	20	122°55.5'	S4°58.4'
5	21	137°55.7'	S4°59.4'	5	22	152°55.9'	S5°00.3'	5	23	167°56.1'	S5°01.3'
6	0	182°56.2'	S5°02.3'	6	1	197°56.4'	S5°03.2'	6	2	212°56.6'	S5°04.2'
6	3	227°56.8'	S5°05.1'	6	4	242°57.0'	S5°06.1'	6	5	257°57.2'	S5°07.0'
6	6	272°57.4'	S5°08.0'	6	7	287°57.5'	S5°09.0'	6	8	302°57.7'	S5°09.9'

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Sun — GHA and Declination, October 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
6	9	317°57.9'	S5°10.9'	6	10	332°58.1'	S5°11.8'	6	11	347°58.3'	S5°12.8'
6	12	2°58.5'	S5°13.8'	6	13	17°58.6'	S5°14.7'	6	14	32°58.8'	S5°15.7'
6	15	47°59.0'	S5°16.6'	6	16	62°59.2'	S5°17.6'	6	17	77°59.4'	S5°18.5'
6	18	92°59.6'	S5°19.5'	6	19	107°59.7'	S5°20.5'	6	20	122°59.9'	S5°21.4'
6	21	138°00.1'	S5°22.4'	6	22	153°00.3'	S5°23.3'	6	23	168°00.5'	S5°24.3'
7	0	183°00.7'	S5°25.2'	7	1	198°00.8'	S5°26.2'	7	2	213°01.0'	S5°27.2'
7	3	228°01.2'	S5°28.1'	7	4	243°01.4'	S5°29.1'	7	5	258°01.6'	S5°30.0'
7	6	273°01.7'	S5°31.0'	7	7	288°01.9'	S5°31.9'	7	8	303°02.1'	S5°32.9'
7	9	318°02.3'	S5°33.9'	7	10	333°02.5'	S5°34.8'	7	11	348°02.6'	S5°35.8'
7	12	3°02.8'	S5°36.7'	7	13	18°03.0'	S5°37.7'	7	14	33°03.2'	S5°38.6'
7	15	48°03.4'	S5°39.6'	7	16	63°03.5'	S5°40.5'	7	17	78°03.7'	S5°41.5'
7	18	93°03.9'	S5°42.5'	7	19	108°04.1'	S5°43.4'	7	20	123°04.2'	S5°44.4'
7	21	138°04.4'	S5°45.3'	7	22	153°04.6'	S5°46.3'	7	23	168°04.8'	S5°47.2'
8	0	183°05.0'	S5°48.2'	8	1	198°05.1'	S5°49.1'	8	2	213°05.3'	S5°50.1'
8	3	228°05.5'	S5°51.0'	8	4	243°05.7'	S5°52.0'	8	5	258°05.8'	S5°52.9'
8	6	273°06.0'	S5°53.9'	8	7	288°06.2'	S5°54.9'	8	8	303°06.4'	S5°55.8'
8	9	318°06.5'	S5°56.8'	8	10	333°06.7'	S5°57.7'	8	11	348°06.9'	S5°58.7'
8	12	3°07.1'	S5°59.6'	8	13	18°07.2'	S6°00.6'	8	14	33°07.4'	S6°01.5'
8	15	48°07.6'	S6°02.5'	8	16	63°07.8'	S6°03.4'	8	17	78°07.9'	S6°04.4'
8	18	93°08.1'	S6°05.3'	8	19	108°08.3'	S6°06.3'	8	20	123°08.5'	S6°07.2'
8	21	138°08.6'	S6°08.2'	8	22	153°08.8'	S6°09.1'	8	23	168°09.0'	S6°10.1'
9	0	183°09.2'	S6°11.0'	9	1	198°09.3'	S6°12.0'	9	2	213°09.5'	S6°12.9'
9	3	228°09.7'	S6°13.9'	9	4	243°09.8'	S6°14.8'	9	5	258°10.0'	S6°15.8'
9	6	273°10.2'	S6°16.7'	9	7	288°10.4'	S6°17.7'	9	8	303°10.5'	S6°18.6'
9	9	318°10.7'	S6°19.6'	9	10	333°10.9'	S6°20.5'	9	11	348°11.0'	S6°21.5'
9	12	3°11.2'	S6°22.4'	9	13	18°11.4'	S6°23.4'	9	14	33°11.6'	S6°24.3'
9	15	48°11.7'	S6°25.3'	9	16	63°11.9'	S6°26.2'	9	17	78°12.1'	S6°27.2'
9	18	93°12.2'	S6°28.1'	9	19	108°12.4'	S6°29.1'	9	20	123°12.6'	S6°30.0'
9	21	138°12.7'	S6°31.0'	9	22	153°12.9'	S6°31.9'	9	23	168°13.1'	S6°32.9'
10	0	183°13.2'	S6°33.8'	10	1	198°13.4'	S6°34.7'	10	2	213°13.6'	S6°35.7'
10	3	228°13.7'	S6°36.6'	10	4	243°13.9'	S6°37.6'	10	5	258°14.1'	S6°38.5'
10	6	273°14.2'	S6°39.5'	10	7	288°14.4'	S6°40.4'	10	8	303°14.6'	S6°41.4'
10	9	318°14.7'	S6°42.3'	10	10	333°14.9'	S6°43.3'	10	11	348°15.1'	S6°44.2'
10	12	3°15.2'	S6°45.2'	10	13	18°15.4'	S6°46.1'	10	14	33°15.6'	S6°47.0'
10	15	48°15.7'	S6°48.0'	10	16	63°15.9'	S6°48.9'	10	17	78°16.1'	S6°49.9'
10	18	93°16.2'	S6°50.8'	10	19	108°16.4'	S6°51.8'	10	20	123°16.6'	S6°52.7'
10	21	138°16.7'	S6°53.7'	10	22	153°16.9'	S6°54.6'	10	23	168°17.0'	S6°55.5'
11	0	183°17.2'	S6°56.5'	11	1	198°17.4'	S6°57.4'	11	2	213°17.5'	S6°58.4'
11	3	228°17.7'	S6°59.3'	11	4	243°17.9'	S7°00.3'	11	5	258°18.0'	S7°01.2'
11	6	273°18.2'	S7°02.1'	11	7	288°18.3'	S7°03.1'	11	8	303°18.5'	S7°04.0'
11	9	318°18.7'	S7°05.0'	11	10	333°18.8'	S7°05.9'	11	11	348°19.0'	S7°06.9'
11	12	3°19.2'	S7°07.8'	11	13	18°19.3'	S7°08.7'	11	14	33°19.5'	S7°09.7'
11	15	48°19.6'	S7°10.6'	11	16	63°19.8'	S7°11.6'	11	17	78°20.0'	S7°12.5'
11	18	93°20.1'	S7°13.4'	11	19	108°20.3'	S7°14.4'	11	20	123°20.4'	S7°15.3'
11	21	138°20.6'	S7°16.3'	11	22	153°20.7'	S7°17.2'	11	23	168°20.9'	S7°18.1'
12	0	183°21.1'	S7°19.1'	12	1	198°21.2'	S7°20.0'	12	2	213°21.4'	S7°21.0'
12	3	228°21.5'	S7°21.9'	12	4	243°21.7'	S7°22.8'	12	5	258°21.9'	S7°23.8'
12	6	273°22.0'	S7°24.7'	12	7	288°22.2'	S7°25.7'	12	8	303°22.3'	S7°26.6'
12	9	318°22.5'	S7°27.5'	12	10	333°22.6'	S7°28.5'	12	11	348°22.8'	S7°29.4'
12	12	3°22.9'	S7°30.3'	12	13	18°23.1'	S7°31.3'	12	14	33°23.3'	S7°32.2'
12	15	48°23.4'	S7°33.1'	12	16	63°23.6'	S7°34.1'	12	17	78°23.7'	S7°35.0'
12	18	93°23.9'	S7°36.0'	12	19	108°24.0'	S7°36.9'	12	20	123°24.2'	S7°37.8'
12	21	138°24.3'	S7°38.8'	12	22	153°24.5'	S7°39.7'	12	23	168°24.6'	S7°40.6'
13	0	183°24.8'	S7°41.6'	13	1	198°24.9'	S7°42.5'	13	2	213°25.1'	S7°43.4'
13	3	228°25.2'	S7°44.4'	13	4	243°25.4'	S7°45.3'	13	5	258°25.6'	S7°46.2'
13	6	273°25.7'	S7°47.2'	13	7	288°25.9'	S7°48.1'	13	8	303°26.0'	S7°49.0'
13	9	318°26.2'	S7°50.0'	13	10	333°26.3'	S7°50.9'	13	11	348°26.5'	S7°51.8'
13	12	3°26.6'	S7°52.8'	13	13	18°26.8'	S7°53.7'	13	14	33°26.9'	S7°54.6'
13	15	48°27.1'	S7°55.6'	13	16	63°27.2'	S7°56.5'	13	17	78°27.4'	S7°57.4'
13	18	93°27.5'	S7°58.4'	13	19	108°27.6'	S7°59.3'	13	20	123°27.8'	S8°00.2'
13	21	138°27.9'	S8°01.2'	13	22	153°28.1'	S8°02.1'	13	23	168°28.2'	S8°03.0'
14	0	183°28.4'	S8°04.0'	14	1	198°28.5'	S8°04.9'	14	2	213°28.7'	S8°05.8'
14	3	228°28.8'	S8°06.7'	14	4	243°29.0'	S8°07.7'	14	5	258°29.1'	S8°08.6'
14	6	273°29.3'	S8°09.5'	14	7	288°29.4'	S8°10.5'	14	8	303°29.6'	S8°11.4'

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Sun — GHA and Declination, October 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
14	9	318°29.7'	S8°12.3'	14	10	333°29.8'	S8°13.3'	14	11	348°30.0'	S8°14.2'
14	12	3°30.1'	S8°15.1'	14	13	18°30.3'	S8°16.0'	14	14	33°30.4'	S8°17.0'
14	15	48°30.6'	S8°17.9'	14	16	63°30.7'	S8°18.8'	14	17	78°30.9'	S8°19.7'
14	18	93°31.0'	S8°20.7'	14	19	108°31.1'	S8°21.6'	14	20	123°31.3'	S8°22.5'
14	21	138°31.4'	S8°23.5'	14	22	153°31.6'	S8°24.4'	14	23	168°31.7'	S8°25.3'
15	0	183°31.9'	S8°26.2'	15	1	198°32.0'	S8°27.2'	15	2	213°32.1'	S8°28.1'
15	3	228°32.3'	S8°29.0'	15	4	243°32.4'	S8°29.9'	15	5	258°32.6'	S8°30.9'
15	6	273°32.7'	S8°31.8'	15	7	288°32.8'	S8°32.7'	15	8	303°33.0'	S8°33.6'
15	9	318°33.1'	S8°34.6'	15	10	333°33.3'	S8°35.5'	15	11	348°33.4'	S8°36.4'
15	12	3°33.5'	S8°37.3'	15	13	18°33.7'	S8°38.2'	15	14	33°33.8'	S8°39.2'
15	15	48°33.9'	S8°40.1'	15	16	63°34.1'	S8°41.0'	15	17	78°34.2'	S8°41.9'
15	18	93°34.4'	S8°42.9'	15	19	108°34.5'	S8°43.8'	15	20	123°34.6'	S8°44.7'
15	21	138°34.8'	S8°45.6'	15	22	153°34.9'	S8°46.5'	15	23	168°35.0'	S8°47.5'
16	0	183°35.2'	S8°48.4'	16	1	198°35.3'	S8°49.3'	16	2	213°35.4'	S8°50.2'
16	3	228°35.6'	S8°51.1'	16	4	243°35.7'	S8°52.1'	16	5	258°35.8'	S8°53.0'
16	6	273°36.0'	S8°53.9'	16	7	288°36.1'	S8°54.8'	16	8	303°36.3'	S8°55.7'
16	9	318°36.4'	S8°56.7'	16	10	333°36.5'	S8°57.6'	16	11	348°36.7'	S8°58.5'
16	12	3°36.8'	S8°59.4'	16	13	18°36.9'	S9°00.3'	16	14	33°37.0'	S9°01.3'
16	15	48°37.2'	S9°02.2'	16	16	63°37.3'	S9°03.1'	16	17	78°37.4'	S9°04.0'
16	18	93°37.6'	S9°04.9'	16	19	108°37.7'	S9°05.8'	16	20	123°37.8'	S9°06.8'
16	21	138°38.0'	S9°07.7'	16	22	153°38.1'	S9°08.6'	16	23	168°38.2'	S9°09.5'
17	0	183°38.4'	S9°10.4'	17	1	198°38.5'	S9°11.3'	17	2	213°38.6'	S9°12.2'
17	3	228°38.7'	S9°13.2'	17	4	243°38.9'	S9°14.1'	17	5	258°39.0'	S9°15.0'
17	6	273°39.1'	S9°15.9'	17	7	288°39.3'	S9°16.8'	17	8	303°39.4'	S9°17.7'
17	9	318°39.5'	S9°18.6'	17	10	333°39.6'	S9°19.6'	17	11	348°39.8'	S9°20.5'
17	12	3°39.9'	S9°21.4'	17	13	18°40.0'	S9°22.3'	17	14	33°40.1'	S9°23.2'
17	15	48°40.3'	S9°24.1'	17	16	63°40.4'	S9°25.0'	17	17	78°40.5'	S9°25.9'
17	18	93°40.6'	S9°26.9'	17	19	108°40.8'	S9°27.8'	17	20	123°40.9'	S9°28.7'
17	21	138°41.0'	S9°29.6'	17	22	153°41.1'	S9°30.5'	17	23	168°41.3'	S9°31.4'
18	0	183°41.4'	S9°32.3'	18	1	198°41.5'	S9°33.2'	18	2	213°41.6'	S9°34.1'
18	3	228°41.8'	S9°35.0'	18	4	243°41.9'	S9°36.0'	18	5	258°42.0'	S9°36.9'
18	6	273°42.1'	S9°37.8'	18	7	288°42.2'	S9°38.7'	18	8	303°42.4'	S9°39.6'
18	9	318°42.5'	S9°40.5'	18	10	333°42.6'	S9°41.4'	18	11	348°42.7'	S9°42.3'
18	12	3°42.8'	S9°43.2'	18	13	18°43.0'	S9°44.1'	18	14	33°43.1'	S9°45.0'
18	15	48°43.2'	S9°45.9'	18	16	63°43.3'	S9°46.8'	18	17	78°43.4'	S9°47.7'
18	18	93°43.6'	S9°48.7'	18	19	108°43.7'	S9°49.6'	18	20	123°43.8'	S9°50.5'
18	21	138°43.9'	S9°51.4'	18	22	153°44.0'	S9°52.3'	18	23	168°44.2'	S9°53.2'
19	0	183°44.3'	S9°54.1'	19	1	198°44.4'	S9°55.0'	19	2	213°44.5'	S9°55.9'
19	3	228°44.6'	S9°56.8'	19	4	243°44.7'	S9°57.7'	19	5	258°44.9'	S9°58.6'
19	6	273°45.0'	S9°59.5'	19	7	288°45.1'	S10°00.4'	19	8	303°45.2'	S10°01.3'
19	9	318°45.3'	S10°02.2'	19	10	333°45.4'	S10°03.1'	19	11	348°45.5'	S10°04.0'
19	12	3°45.7'	S10°04.9'	19	13	18°45.8'	S10°05.8'	19	14	33°45.9'	S10°06.7'
19	15	48°46.0'	S10°07.6'	19	16	63°46.1'	S10°08.5'	19	17	78°46.2'	S10°09.4'
19	18	93°46.3'	S10°10.3'	19	19	108°46.4'	S10°11.2'	19	20	123°46.6'	S10°12.1'
19	21	138°46.7'	S10°13.0'	19	22	153°46.8'	S10°13.9'	19	23	168°46.9'	S10°14.8'
20	0	183°47.0'	S10°15.7'	20	1	198°47.1'	S10°16.6'	20	2	213°47.2'	S10°17.5'
20	3	228°47.3'	S10°18.4'	20	4	243°47.4'	S10°19.3'	20	5	258°47.5'	S10°20.2'
20	6	273°47.7'	S10°21.1'	20	7	288°47.8'	S10°22.0'	20	8	303°47.9'	S10°22.9'
20	9	318°48.0'	S10°23.8'	20	10	333°48.1'	S10°24.7'	20	11	348°48.2'	S10°25.6'
20	12	3°48.3'	S10°26.4'	20	13	18°48.4'	S10°27.3'	20	14	33°48.5'	S10°28.2'
20	15	48°48.6'	S10°29.1'	20	16	63°48.7'	S10°30.0'	20	17	78°48.8'	S10°30.9'
20	18	93°48.9'	S10°31.8'	20	19	108°49.0'	S10°32.7'	20	20	123°49.1'	S10°33.6'
20	21	138°49.3'	S10°34.5'	20	22	153°49.4'	S10°35.4'	20	23	168°49.5'	S10°36.3'
21	0	183°49.6'	S10°37.2'	21	1	198°49.7'	S10°38.1'	21	2	213°49.8'	S10°38.9'
21	3	228°49.9'	S10°39.8'	21	4	243°50.0'	S10°40.7'	21	5	258°50.1'	S10°41.6'
21	6	273°50.2'	S10°42.5'	21	7	288°50.3'	S10°43.4'	21	8	303°50.4'	S10°44.3'
21	9	318°50.5'	S10°45.2'	21	10	333°50.6'	S10°46.1'	21	11	348°50.7'	S10°47.0'
21	12	3°50.8'	S10°47.8'	21	13	18°50.9'	S10°48.7'	21	14	33°51.0'	S10°49.6'
21	15	48°51.1'	S10°50.5'	21	16	63°51.2'	S10°51.4'	21	17	78°51.3'	S10°52.3'
21	18	93°51.4'	S10°53.2'	21	19	108°51.5'	S10°54.0'	21	20	123°51.6'	S10°54.9'
21	21	138°51.7'	S10°55.8'	21	22	153°51.8'	S10°56.7'	21	23	168°51.9'	S10°57.6'
22	0	183°52.0'	S10°58.5'	22	1	198°52.1'	S10°59.4'	22	2	213°52.2'	S11°00.2'
22	3	228°52.3'	S11°01.1'	22	4	243°52.3'	S11°02.0'	22	5	258°52.4'	S11°02.9'
22	6	273°52.5'	S11°03.8'	22	7	288°52.6'	S11°04.7'	22	8	303°52.7'	S11°05.5'

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Sun — GHA and Declination, October 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
22	9	318°52.8'	S11°06.4'	22	10	333°52.9'	S11°07.3'	22	11	348°53.0'	S11°08.2'
22	12	3°53.1'	S11°09.1'	22	13	18°53.2'	S11°09.9'	22	14	33°53.3'	S11°10.8'
22	15	48°53.4'	S11°11.7'	22	16	63°53.5'	S11°12.6'	22	17	78°53.6'	S11°13.5'
22	18	93°53.7'	S11°14.3'	22	19	108°53.7'	S11°15.2'	22	20	123°53.8'	S11°16.1'
22	21	138°53.9'	S11°17.0'	22	22	153°54.0'	S11°17.9'	22	23	168°54.1'	S11°18.7'
23	0	183°54.2'	S11°19.6'	23	1	198°54.3'	S11°20.5'	23	2	213°54.4'	S11°21.4'
23	3	228°54.5'	S11°22.3'	23	4	243°54.6'	S11°23.1'	23	5	258°54.6'	S11°24.0'
23	6	273°54.7'	S11°24.9'	23	7	288°54.8'	S11°25.8'	23	8	303°54.9'	S11°26.6'
23	9	318°55.0'	S11°27.5'	23	10	333°55.1'	S11°28.4'	23	11	348°55.2'	S11°29.3'
23	12	3°55.3'	S11°30.1'	23	13	18°55.3'	S11°31.0'	23	14	33°55.4'	S11°31.9'
23	15	48°55.5'	S11°32.8'	23	16	63°55.6'	S11°33.6'	23	17	78°55.7'	S11°34.5'
23	18	93°55.8'	S11°35.4'	23	19	108°55.8'	S11°36.2'	23	20	123°55.9'	S11°37.1'
23	21	138°56.0'	S11°38.0'	23	22	153°56.1'	S11°38.9'	23	23	168°56.2'	S11°39.7'
24	0	183°56.3'	S11°40.6'	24	1	198°56.3'	S11°41.5'	24	2	213°56.4'	S11°42.3'
24	3	228°56.5'	S11°43.2'	24	4	243°56.6'	S11°44.1'	24	5	258°56.7'	S11°44.9'
24	6	273°56.8'	S11°45.8'	24	7	288°56.8'	S11°46.7'	24	8	303°56.9'	S11°47.5'
24	9	318°57.0'	S11°48.4'	24	10	333°57.1'	S11°49.3'	24	11	348°57.1'	S11°50.2'
24	12	3°57.2'	S11°51.0'	24	13	18°57.3'	S11°51.9'	24	14	33°57.4'	S11°52.8'
24	15	48°57.5'	S11°53.6'	24	16	63°57.5'	S11°54.5'	24	17	78°57.6'	S11°55.3'
24	18	93°57.7'	S11°56.2'	24	19	108°57.8'	S11°57.1'	24	20	123°57.8'	S11°57.9'
24	21	138°57.9'	S11°58.8'	24	22	153°58.0'	S11°59.7'	24	23	168°58.1'	S12°00.5'
25	0	183°58.1'	S12°01.4'	25	1	198°58.2'	S12°02.3'	25	2	213°58.3'	S12°03.1'
25	3	228°58.4'	S12°04.0'	25	4	243°58.4'	S12°04.8'	25	5	258°58.5'	S12°05.7'
25	6	273°58.6'	S12°06.6'	25	7	288°58.7'	S12°07.4'	25	8	303°58.7'	S12°08.3'
25	9	318°58.8'	S12°09.1'	25	10	333°58.9'	S12°10.0'	25	11	348°59.0'	S12°10.9'
25	12	3°59.0'	S12°11.7'	25	13	18°59.1'	S12°12.6'	25	14	33°59.2'	S12°13.4'
25	15	48°59.2'	S12°14.3'	25	16	63°59.3'	S12°15.2'	25	17	78°59.4'	S12°16.0'
25	18	93°59.4'	S12°16.9'	25	19	108°59.5'	S12°17.7'	25	20	123°59.6'	S12°18.6'
25	21	138°59.7'	S12°19.4'	25	22	153°59.7'	S12°20.3'	25	23	168°59.8'	S12°21.2'
26	0	183°59.9'	S12°22.0'	26	1	198°59.9'	S12°22.9'	26	2	214°00.0'	S12°23.7'
26	3	229°00.1'	S12°24.6'	26	4	244°00.1'	S12°25.4'	26	5	259°00.2'	S12°26.3'
26	6	274°00.3'	S12°27.1'	26	7	289°00.3'	S12°28.0'	26	8	304°00.4'	S12°28.8'
26	9	319°00.5'	S12°29.7'	26	10	334°00.5'	S12°30.5'	26	11	349°00.6'	S12°31.4'
26	12	4°00.6'	S12°32.2'	26	13	19°00.7'	S12°33.1'	26	14	34°00.8'	S12°33.9'
26	15	49°00.8'	S12°34.8'	26	16	64°00.9'	S12°35.6'	26	17	79°01.0'	S12°36.5'
26	18	94°01.0'	S12°37.3'	26	19	109°01.1'	S12°38.2'	26	20	124°01.1'	S12°39.0'
26	21	139°01.2'	S12°39.9'	26	22	154°01.3'	S12°40.7'	26	23	169°01.3'	S12°41.6'
27	0	184°01.4'	S12°42.4'	27	1	199°01.4'	S12°43.3'	27	2	214°01.5'	S12°44.1'
27	3	229°01.6'	S12°45.0'	27	4	244°01.6'	S12°45.8'	27	5	259°01.7'	S12°46.7'
27	6	274°01.7'	S12°47.5'	27	7	289°01.8'	S12°48.4'	27	8	304°01.9'	S12°49.2'
27	9	319°01.9'	S12°50.0'	27	10	334°02.0'	S12°50.9'	27	11	349°02.0'	S12°51.7'
27	12	4°02.1'	S12°52.6'	27	13	19°02.1'	S12°53.4'	27	14	34°02.2'	S12°54.3'
27	15	49°02.2'	S12°55.1'	27	16	64°02.3'	S12°55.9'	27	17	79°02.4'	S12°56.8'
27	18	94°02.4'	S12°57.6'	27	19	109°02.5'	S12°58.5'	27	20	124°02.5'	S12°59.3'
27	21	139°02.6'	S13°00.1'	27	22	154°02.6'	S13°01.0'	27	23	169°02.7'	S13°01.8'
28	0	184°02.7'	S13°02.7'	28	1	199°02.8'	S13°03.5'	28	2	214°02.8'	S13°04.3'
28	3	229°02.9'	S13°05.2'	28	4	244°02.9'	S13°06.0'	28	5	259°03.0'	S13°06.9'
28	6	274°03.0'	S13°07.7'	28	7	289°03.1'	S13°08.5'	28	8	304°03.1'	S13°09.4'
28	9	319°03.2'	S13°10.2'	28	10	334°03.2'	S13°11.0'	28	11	349°03.3'	S13°11.9'
28	12	4°03.3'	S13°12.7'	28	13	19°03.4'	S13°13.5'	28	14	34°03.4'	S13°14.4'
28	15	49°03.5'	S13°15.2'	28	16	64°03.5'	S13°16.0'	28	17	79°03.6'	S13°16.9'
28	18	94°03.6'	S13°17.7'	28	19	109°03.7'	S13°18.5'	28	20	124°03.7'	S13°19.4'
28	21	139°03.7'	S13°20.2'	28	22	154°03.8'	S13°21.0'	28	23	169°03.8'	S13°21.9'
29	0	184°03.9'	S13°22.7'	29	1	199°03.9'	S13°23.5'	29	2	214°04.0'	S13°24.3'
29	3	229°04.0'	S13°25.2'	29	4	244°04.1'	S13°26.0'	29	5	259°04.1'	S13°26.8'
29	6	274°04.1'	S13°27.7'	29	7	289°04.2'	S13°28.5'	29	8	304°04.2'	S13°29.3'
29	9	319°04.3'	S13°30.1'	29	10	334°04.3'	S13°31.0'	29	11	349°04.3'	S13°31.8'
29	12	4°04.4'	S13°32.6'	29	13	19°04.4'	S13°33.4'	29	14	34°04.5'	S13°34.3'
29	15	49°04.5'	S13°35.1'	29	16	64°04.5'	S13°35.9'	29	17	79°04.6'	S13°36.7'
29	18	94°04.6'	S13°37.6'	29	19	109°04.7'	S13°38.4'	29	20	124°04.7'	S13°39.2'
29	21	139°04.7'	S13°40.0'	29	22	154°04.8'	S13°40.9'	29	23	169°04.8'	S13°41.7'
30	0	184°04.8'	S13°42.5'	30	1	199°04.9'	S13°43.3'	30	2	214°04.9'	S13°44.1'
30	3	229°04.9'	S13°45.0'	30	4	244°05.0'	S13°45.8'	30	5	259°05.0'	S13°46.6'
30	6	274°05.1'	S13°47.4'	30	7	289°05.1'	S13°48.2'	30	8	304°05.1'	S13°49.1'

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Sun — GHA and Declination, October 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
30	9	319°05.2'	S13°49.9'	30	10	334°05.2'	S13°50.7'	30	11	349°05.2'	S13°51.5'
30	12	4°05.3'	S13°52.3'	30	13	19°05.3'	S13°53.1'	30	14	34°05.3'	S13°54.0'
30	15	49°05.3'	S13°54.8'	30	16	64°05.4'	S13°55.6'	30	17	79°05.4'	S13°56.4'
30	18	94°05.4'	S13°57.2'	30	19	109°05.5'	S13°58.0'	30	20	124°05.5'	S13°58.9'
30	21	139°05.5'	S13°59.7'	30	22	154°05.6'	S14°00.5'	30	23	169°05.6'	S14°01.3'
31	0	184°05.6'	S14°02.1'	31	1	199°05.6'	S14°02.9'	31	2	214°05.7'	S14°03.7'
31	3	229°05.7'	S14°04.5'	31	4	244°05.7'	S14°05.3'	31	5	259°05.7'	S14°06.2'
31	6	274°05.8'	S14°07.0'	31	7	289°05.8'	S14°07.8'	31	8	304°05.8'	S14°08.6'
31	9	319°05.8'	S14°09.4'	31	10	334°05.9'	S14°10.2'	31	11	349°05.9'	S14°11.0'
31	12	4°05.9'	S14°11.8'	31	13	19°05.9'	S14°12.6'	31	14	34°06.0'	S14°13.4'
31	15	49°06.0'	S14°14.2'	31	16	64°06.0'	S14°15.0'	31	17	79°06.0'	S14°15.8'
31	18	94°06.1'	S14°16.7'	31	19	109°06.1'	S14°17.5'	31	20	124°06.1'	S14°18.3'
31	21	139°06.1'	S14°19.1'	31	22	154°06.1'	S14°19.9'	31	23	169°06.2'	S14°20.7'

Table 9.11: Sun — GHA and Declination, November 2026 (UT)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
1	0	184°06.2'	S14°21.5'	1	1	199°06.2'	S14°22.3'	1	2	214°06.2'	S14°23.1'
1	3	229°06.2'	S14°23.9'	1	4	244°06.3'	S14°24.7'	1	5	259°06.3'	S14°25.5'
1	6	274°06.3'	S14°26.3'	1	7	289°06.3'	S14°27.1'	1	8	304°06.3'	S14°27.9'
1	9	319°06.3'	S14°28.7'	1	10	334°06.4'	S14°29.5'	1	11	349°06.4'	S14°30.3'
1	12	4°06.4'	S14°31.1'	1	13	19°06.4'	S14°31.9'	1	14	34°06.4'	S14°32.7'
1	15	49°06.4'	S14°33.5'	1	16	64°06.5'	S14°34.3'	1	17	79°06.5'	S14°35.1'
1	18	94°06.5'	S14°35.9'	1	19	109°06.5'	S14°36.7'	1	20	124°06.5'	S14°37.4'
1	21	139°06.5'	S14°38.2'	1	22	154°06.5'	S14°39.0'	1	23	169°06.5'	S14°39.8'
2	0	184°06.6'	S14°40.6'	2	1	199°06.6'	S14°41.4'	2	2	214°06.6'	S14°42.2'
2	3	229°06.6'	S14°43.0'	2	4	244°06.6'	S14°43.8'	2	5	259°06.6'	S14°44.6'
2	6	274°06.6'	S14°45.4'	2	7	289°06.6'	S14°46.2'	2	8	304°06.6'	S14°47.0'
2	9	319°06.6'	S14°47.7'	2	10	334°06.7'	S14°48.5'	2	11	349°06.7'	S14°49.3'
2	12	4°06.7'	S14°50.1'	2	13	19°06.7'	S14°50.9'	2	14	34°06.7'	S14°51.7'
2	15	49°06.7'	S14°52.5'	2	16	64°06.7'	S14°53.3'	2	17	79°06.7'	S14°54.0'
2	18	94°06.7'	S14°54.8'	2	19	109°06.7'	S14°55.6'	2	20	124°06.7'	S14°56.4'
2	21	139°06.7'	S14°57.2'	2	22	154°06.7'	S14°58.0'	2	23	169°06.7'	S14°58.7'
3	0	184°06.7'	S14°59.5'	3	1	199°06.7'	S15°00.3'	3	2	214°06.7'	S15°01.1'
3	3	229°06.7'	S15°01.9'	3	4	244°06.7'	S15°02.7'	3	5	259°06.7'	S15°03.4'
3	6	274°06.7'	S15°04.2'	3	7	289°06.7'	S15°05.0'	3	8	304°06.7'	S15°05.8'
3	9	319°06.7'	S15°06.6'	3	10	334°06.7'	S15°07.3'	3	11	349°06.7'	S15°08.1'
3	12	4°06.7'	S15°08.9'	3	13	19°06.7'	S15°09.7'	3	14	34°06.7'	S15°10.4'
3	15	49°06.7'	S15°11.2'	3	16	64°06.7'	S15°12.0'	3	17	79°06.7'	S15°12.8'
3	18	94°06.7'	S15°13.5'	3	19	109°06.7'	S15°14.3'	3	20	124°06.7'	S15°15.1'
3	21	139°06.7'	S15°15.9'	3	22	154°06.7'	S15°16.6'	3	23	169°06.7'	S15°17.4'
4	0	184°06.7'	S15°18.2'	4	1	199°06.7'	S15°19.0'	4	2	214°06.7'	S15°19.7'
4	3	229°06.7'	S15°20.5'	4	4	244°06.7'	S15°21.3'	4	5	259°06.7'	S15°22.0'
4	6	274°06.7'	S15°22.8'	4	7	289°06.6'	S15°23.6'	4	8	304°06.6'	S15°24.4'
4	9	319°06.6'	S15°25.1'	4	10	334°06.6'	S15°25.9'	4	11	349°06.6'	S15°26.7'
4	12	4°06.6'	S15°27.4'	4	13	19°06.6'	S15°28.2'	4	14	34°06.6'	S15°29.0'
4	15	49°06.6'	S15°29.7'	4	16	64°06.6'	S15°30.5'	4	17	79°06.5'	S15°31.3'
4	18	94°06.5'	S15°32.0'	4	19	109°06.5'	S15°32.8'	4	20	124°06.5'	S15°33.6'
4	21	139°06.5'	S15°34.3'	4	22	154°06.5'	S15°35.1'	4	23	169°06.5'	S15°35.8'
5	0	184°06.5'	S15°36.6'	5	1	199°06.4'	S15°37.4'	5	2	214°06.4'	S15°38.1'
5	3	229°06.4'	S15°38.9'	5	4	244°06.4'	S15°39.6'	5	5	259°06.4'	S15°40.4'
5	6	274°06.4'	S15°41.2'	5	7	289°06.3'	S15°41.9'	5	8	304°06.3'	S15°42.7'
5	9	319°06.3'	S15°43.4'	5	10	334°06.3'	S15°44.2'	5	11	349°06.3'	S15°45.0'
5	12	4°06.3'	S15°45.7'	5	13	19°06.2'	S15°46.5'	5	14	34°06.2'	S15°47.2'
5	15	49°06.2'	S15°48.0'	5	16	64°06.2'	S15°48.7'	5	17	79°06.2'	S15°49.5'
5	18	94°06.1'	S15°50.2'	5	19	109°06.1'	S15°51.0'	5	20	124°06.1'	S15°51.8'
5	21	139°06.1'	S15°52.5'	5	22	154°06.1'	S15°53.3'	5	23	169°06.0'	S15°54.0'
6	0	184°06.0'	S15°54.8'	6	1	199°06.0'	S15°55.5'	6	2	214°06.0'	S15°56.3'
6	3	229°05.9'	S15°57.0'	6	4	244°05.9'	S15°57.8'	6	5	259°05.9'	S15°58.5'
6	6	274°05.9'	S15°59.3'	6	7	289°05.8'	S16°00.0'	6	8	304°05.8'	S16°00.8'
6	9	319°05.8'	S16°01.5'	6	10	334°05.8'	S16°02.2'	6	11	349°05.7'	S16°03.0'
6	12	4°05.7'	S16°03.7'	6	13	19°05.7'	S16°04.5'	6	14	34°05.7'	S16°05.2'

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Sun — GHA and Declination, November 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
6	15	49°05.6'	S16°06.0'	6	16	64°05.6'	S16°06.7'	6	17	79°05.6'	S16°07.5'
6	18	94°05.5'	S16°08.2'	6	19	109°05.5'	S16°08.9'	6	20	124°05.5'	S16°09.7'
6	21	139°05.5'	S16°10.4'	6	22	154°05.4'	S16°11.2'	6	23	169°05.4'	S16°11.9'
7	0	184°05.4'	S16°12.6'	7	1	199°05.3'	S16°13.4'	7	2	214°05.3'	S16°14.1'
7	3	229°05.3'	S16°14.9'	7	4	244°05.2'	S16°15.6'	7	5	259°05.2'	S16°16.3'
7	6	274°05.2'	S16°17.1'	7	7	289°05.1'	S16°17.8'	7	8	304°05.1'	S16°18.6'
7	9	319°05.1'	S16°19.3'	7	10	334°05.0'	S16°20.0'	7	11	349°05.0'	S16°20.8'
7	12	4°05.0'	S16°21.5'	7	13	19°04.9'	S16°22.2'	7	14	34°04.9'	S16°23.0'
7	15	49°04.8'	S16°23.7'	7	16	64°04.8'	S16°24.4'	7	17	79°04.8'	S16°25.2'
7	18	94°04.7'	S16°25.9'	7	19	109°04.7'	S16°26.6'	7	20	124°04.7'	S16°27.4'
7	21	139°04.6'	S16°28.1'	7	22	154°04.6'	S16°28.8'	7	23	169°04.5'	S16°29.5'
8	0	184°04.5'	S16°30.3'	8	1	199°04.5'	S16°31.0'	8	2	214°04.4'	S16°31.7'
8	3	229°04.4'	S16°32.5'	8	4	244°04.3'	S16°33.2'	8	5	259°04.3'	S16°33.9'
8	6	274°04.2'	S16°34.6'	8	7	289°04.2'	S16°35.4'	8	8	304°04.2'	S16°36.1'
8	9	319°04.1'	S16°36.8'	8	10	334°04.1'	S16°37.5'	8	11	349°04.0'	S16°38.3'
8	12	4°04.0'	S16°39.0'	8	13	19°03.9'	S16°39.7'	8	14	34°03.9'	S16°40.4'
8	15	49°03.9'	S16°41.1'	8	16	64°03.8'	S16°41.9'	8	17	79°03.8'	S16°42.6'
8	18	94°03.7'	S16°43.3'	8	19	109°03.7'	S16°44.0'	8	20	124°03.6'	S16°44.7'
8	21	139°03.6'	S16°45.5'	8	22	154°03.5'	S16°46.2'	8	23	169°03.5'	S16°46.9'
9	0	184°03.4'	S16°47.6'	9	1	199°03.4'	S16°48.3'	9	2	214°03.3'	S16°49.0'
9	3	229°03.3'	S16°49.8'	9	4	244°03.2'	S16°50.5'	9	5	259°03.2'	S16°51.2'
9	6	274°03.1'	S16°51.9'	9	7	289°03.1'	S16°52.6'	9	8	304°03.0'	S16°53.3'
9	9	319°03.0'	S16°54.0'	9	10	334°02.9'	S16°54.8'	9	11	349°02.9'	S16°55.5'
9	12	4°02.8'	S16°56.2'	9	13	19°02.8'	S16°56.9'	9	14	34°02.7'	S16°57.6'
9	15	49°02.6'	S16°58.3'	9	16	64°02.6'	S16°59.0'	9	17	79°02.5'	S16°59.7'
9	18	94°02.5'	S17°00.4'	9	19	109°02.4'	S17°01.1'	9	20	124°02.4'	S17°01.8'
9	21	139°02.3'	S17°02.6'	9	22	154°02.3'	S17°03.3'	9	23	169°02.2'	S17°04.0'
10	0	184°02.1'	S17°04.7'	10	1	199°02.1'	S17°05.4'	10	2	214°02.0'	S17°06.1'
10	3	229°02.0'	S17°06.8'	10	4	244°01.9'	S17°07.5'	10	5	259°01.8'	S17°08.2'
10	6	274°01.8'	S17°08.9'	10	7	289°01.7'	S17°09.6'	10	8	304°01.7'	S17°10.3'
10	9	319°01.6'	S17°11.0'	10	10	334°01.5'	S17°11.7'	10	11	349°01.5'	S17°12.4'
10	12	4°01.4'	S17°13.1'	10	13	19°01.4'	S17°13.8'	10	14	34°01.3'	S17°14.5'
10	15	49°01.2'	S17°15.2'	10	16	64°01.2'	S17°15.9'	10	17	79°01.1'	S17°16.6'
10	18	94°01.0'	S17°17.3'	10	19	109°01.0'	S17°18.0'	10	20	124°00.9'	S17°18.7'
10	21	139°00.8'	S17°19.4'	10	22	154°00.8'	S17°20.0'	10	23	169°00.7'	S17°20.7'
11	0	184°00.6'	S17°21.4'	11	1	199°00.6'	S17°22.1'	11	2	214°00.5'	S17°22.8'
11	3	229°00.4'	S17°23.5'	11	4	244°00.4'	S17°24.2'	11	5	259°00.3'	S17°24.9'
11	6	274°00.2'	S17°25.6'	11	7	289°00.2'	S17°26.3'	11	8	304°00.1'	S17°27.0'
11	9	319°00.0'	S17°27.6'	11	10	334°00.0'	S17°28.3'	11	11	348°59.9'	S17°29.0'
11	12	3°59.8'	S17°29.7'	11	13	18°59.7'	S17°30.4'	11	14	33°59.7'	S17°31.1'
11	15	48°59.6'	S17°31.8'	11	16	63°59.5'	S17°32.4'	11	17	78°59.5'	S17°33.1'
11	18	93°59.4'	S17°33.8'	11	19	108°59.3'	S17°34.5'	11	20	123°59.2'	S17°35.2'
11	21	138°59.2'	S17°35.9'	11	22	153°59.1'	S17°36.5'	11	23	168°59.0'	S17°37.2'
12	0	183°58.9'	S17°37.9'	12	1	198°58.9'	S17°38.6'	12	2	213°58.8'	S17°39.3'
12	3	228°58.7'	S17°39.9'	12	4	243°58.6'	S17°40.6'	12	5	258°58.6'	S17°41.3'
12	6	273°58.5'	S17°42.0'	12	7	288°58.4'	S17°42.6'	12	8	303°58.3'	S17°43.3'
12	9	318°58.2'	S17°44.0'	12	10	333°58.2'	S17°44.7'	12	11	348°58.1'	S17°45.3'
12	12	3°58.0'	S17°46.0'	12	13	18°57.9'	S17°46.7'	12	14	33°57.8'	S17°47.4'
12	15	48°57.8'	S17°48.0'	12	16	63°57.7'	S17°48.7'	12	17	78°57.6'	S17°49.4'
12	18	93°57.5'	S17°50.1'	12	19	108°57.4'	S17°50.7'	12	20	123°57.4'	S17°51.4'
12	21	138°57.3'	S17°52.1'	12	22	153°57.2'	S17°52.7'	12	23	168°57.1'	S17°53.4'
13	0	183°57.0'	S17°54.1'	13	1	198°56.9'	S17°54.7'	13	2	213°56.8'	S17°55.4'
13	3	228°56.8'	S17°56.1'	13	4	243°56.7'	S17°56.7'	13	5	258°56.6'	S17°57.4'
13	6	273°56.5'	S17°58.1'	13	7	288°56.4'	S17°58.7'	13	8	303°56.3'	S17°59.4'
13	9	318°56.2'	S18°00.0'	13	10	333°56.2'	S18°00.7'	13	11	348°56.1'	S18°01.4'
13	12	3°56.0'	S18°02.0'	13	13	18°55.9'	S18°02.7'	13	14	33°55.8'	S18°03.4'
13	15	48°55.7'	S18°04.0'	13	16	63°55.6'	S18°04.7'	13	17	78°55.5'	S18°05.3'
13	18	93°55.4'	S18°06.0'	13	19	108°55.3'	S18°06.6'	13	20	123°55.3'	S18°07.3'
13	21	138°55.2'	S18°08.0'	13	22	153°55.1'	S18°08.6'	13	23	168°55.0'	S18°09.3'
14	0	183°54.9'	S18°09.9'	14	1	198°54.8'	S18°10.6'	14	2	213°54.7'	S18°11.2'
14	3	228°54.6'	S18°11.9'	14	4	243°54.5'	S18°12.5'	14	5	258°54.4'	S18°13.2'
14	6	273°54.3'	S18°13.8'	14	7	288°54.2'	S18°14.5'	14	8	303°54.1'	S18°15.1'
14	9	318°54.0'	S18°15.8'	14	10	333°53.9'	S18°16.4'	14	11	348°53.8'	S18°17.1'
14	12	3°53.7'	S18°17.7'	14	13	18°53.6'	S18°18.4'	14	14	33°53.5'	S18°19.0'

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Sun — GHA and Declination, November 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
14	15	48°53.4'	S18°19.7'	14	16	63°53.3'	S18°20.3'	14	17	78°53.2'	S18°21.0'
14	18	93°53.1'	S18°21.6'	14	19	108°53.0'	S18°22.2'	14	20	123°52.9'	S18°22.9'
14	21	138°52.8'	S18°23.5'	14	22	153°52.7'	S18°24.2'	14	23	168°52.6'	S18°24.8'
15	0	183°52.5'	S18°25.5'	15	1	198°52.4'	S18°26.1'	15	2	213°52.3'	S18°26.7'
15	3	228°52.2'	S18°27.4'	15	4	243°52.1'	S18°28.0'	15	5	258°52.0'	S18°28.7'
15	6	273°51.9'	S18°29.3'	15	7	288°51.8'	S18°29.9'	15	8	303°51.7'	S18°30.6'
15	9	318°51.6'	S18°31.2'	15	10	333°51.5'	S18°31.8'	15	11	348°51.4'	S18°32.5'
15	12	3°51.3'	S18°33.1'	15	13	18°51.2'	S18°33.7'	15	14	33°51.1'	S18°34.4'
15	15	48°51.0'	S18°35.0'	15	16	63°50.9'	S18°35.6'	15	17	78°50.7'	S18°36.3'
15	18	93°50.6'	S18°36.9'	15	19	108°50.5'	S18°37.5'	15	20	123°50.4'	S18°38.2'
15	21	138°50.3'	S18°38.8'	15	22	153°50.2'	S18°39.4'	15	23	168°50.1'	S18°40.0'
16	0	183°50.0'	S18°40.7'	16	1	198°49.9'	S18°41.3'	16	2	213°49.8'	S18°41.9'
16	3	228°49.6'	S18°42.5'	16	4	243°49.5'	S18°43.2'	16	5	258°49.4'	S18°43.8'
16	6	273°49.3'	S18°44.4'	16	7	288°49.2'	S18°45.0'	16	8	303°49.1'	S18°45.7'
16	9	318°49.0'	S18°46.3'	16	10	333°48.9'	S18°46.9'	16	11	348°48.7'	S18°47.5'
16	12	3°48.6'	S18°48.2'	16	13	18°48.5'	S18°48.8'	16	14	33°48.4'	S18°49.4'
16	15	48°48.3'	S18°50.0'	16	16	63°48.2'	S18°50.6'	16	17	78°48.0'	S18°51.2'
16	18	93°47.9'	S18°51.9'	16	19	108°47.8'	S18°52.5'	16	20	123°47.7'	S18°53.1'
16	21	138°47.6'	S18°53.7'	16	22	153°47.5'	S18°54.3'	16	23	168°47.3'	S18°54.9'
17	0	183°47.2'	S18°55.6'	17	1	198°47.1'	S18°56.2'	17	2	213°47.0'	S18°56.8'
17	3	228°46.9'	S18°57.4'	17	4	243°46.7'	S18°58.0'	17	5	258°46.6'	S18°58.6'
17	6	273°46.5'	S18°59.2'	17	7	288°46.4'	S18°59.8'	17	8	303°46.2'	S19°00.4'
17	9	318°46.1'	S19°01.0'	17	10	333°46.0'	S19°01.7'	17	11	348°45.9'	S19°02.3'
17	12	3°45.8'	S19°02.9'	17	13	18°45.6'	S19°03.5'	17	14	33°45.5'	S19°04.1'
17	15	48°45.4'	S19°04.7'	17	16	63°45.3'	S19°05.3'	17	17	78°45.1'	S19°05.9'
17	18	93°45.0'	S19°06.5'	17	19	108°44.9'	S19°07.1'	17	20	123°44.7'	S19°07.7'
17	21	138°44.6'	S19°08.3'	17	22	153°44.5'	S19°08.9'	17	23	168°44.4'	S19°09.5'
18	0	183°44.2'	S19°10.1'	18	1	198°44.1'	S19°10.7'	18	2	213°44.0'	S19°11.3'
18	3	228°43.9'	S19°11.9'	18	4	243°43.7'	S19°12.5'	18	5	258°43.6'	S19°13.1'
18	6	273°43.5'	S19°13.7'	18	7	288°43.3'	S19°14.3'	18	8	303°43.2'	S19°14.9'
18	9	318°43.1'	S19°15.5'	18	10	333°42.9'	S19°16.1'	18	11	348°42.8'	S19°16.6'
18	12	3°42.7'	S19°17.2'	18	13	18°42.5'	S19°17.8'	18	14	33°42.4'	S19°18.4'
18	15	48°42.3'	S19°19.0'	18	16	63°42.1'	S19°19.6'	18	17	78°42.0'	S19°20.2'
18	18	93°41.9'	S19°20.8'	18	19	108°41.7'	S19°21.4'	18	20	123°41.6'	S19°22.0'
18	21	138°41.5'	S19°22.5'	18	22	153°41.3'	S19°23.1'	18	23	168°41.2'	S19°23.7'
19	0	183°41.1'	S19°24.3'	19	1	198°40.9'	S19°24.9'	19	2	213°40.8'	S19°25.5'
19	3	228°40.6'	S19°26.0'	19	4	243°40.5'	S19°26.6'	19	5	258°40.4'	S19°27.2'
19	6	273°40.2'	S19°27.8'	19	7	288°40.1'	S19°28.4'	19	8	303°39.9'	S19°29.0'
19	9	318°39.8'	S19°29.5'	19	10	333°39.7'	S19°30.1'	19	11	348°39.5'	S19°30.7'
19	12	3°39.4'	S19°31.3'	19	13	18°39.2'	S19°31.8'	19	14	33°39.1'	S19°32.4'
19	15	48°39.0'	S19°33.0'	19	16	63°38.8'	S19°33.6'	19	17	78°38.7'	S19°34.1'
19	18	93°38.5'	S19°34.7'	19	19	108°38.4'	S19°35.3'	19	20	123°38.2'	S19°35.9'
19	21	138°38.1'	S19°36.4'	19	22	153°38.0'	S19°37.0'	19	23	168°37.8'	S19°37.6'
20	0	183°37.7'	S19°38.2'	20	1	198°37.5'	S19°38.7'	20	2	213°37.4'	S19°39.3'
20	3	228°37.2'	S19°39.9'	20	4	243°37.1'	S19°40.4'	20	5	258°36.9'	S19°41.0'
20	6	273°36.8'	S19°41.6'	20	7	288°36.6'	S19°42.1'	20	8	303°36.5'	S19°42.7'
20	9	318°36.3'	S19°43.3'	20	10	333°36.2'	S19°43.8'	20	11	348°36.0'	S19°44.4'
20	12	3°35.9'	S19°44.9'	20	13	18°35.7'	S19°45.5'	20	14	33°35.6'	S19°46.1'
20	15	48°35.4'	S19°46.6'	20	16	63°35.3'	S19°47.2'	20	17	78°35.1'	S19°47.7'
20	18	93°35.0'	S19°48.3'	20	19	108°34.8'	S19°48.9'	20	20	123°34.7'	S19°49.4'
20	21	138°34.5'	S19°50.0'	20	22	153°34.4'	S19°50.5'	20	23	168°34.2'	S19°51.1'
21	0	183°34.1'	S19°51.6'	21	1	198°33.9'	S19°52.2'	21	2	213°33.8'	S19°52.8'
21	3	228°33.6'	S19°53.3'	21	4	243°33.5'	S19°53.9'	21	5	258°33.3'	S19°54.4'
21	6	273°33.1'	S19°55.0'	21	7	288°33.0'	S19°55.5'	21	8	303°32.8'	S19°56.1'
21	9	318°32.7'	S19°56.6'	21	10	333°32.5'	S19°57.2'	21	11	348°32.4'	S19°57.7'
21	12	3°32.2'	S19°58.3'	21	13	18°32.0'	S19°58.8'	21	14	33°31.9'	S19°59.4'
21	15	48°31.7'	S19°59.9'	21	16	63°31.6'	S20°00.4'	21	17	78°31.4'	S20°01.0'
21	18	93°31.2'	S20°01.5'	21	19	108°31.1'	S20°02.1'	21	20	123°30.9'	S20°02.6'
21	21	138°30.8'	S20°03.2'	21	22	153°30.6'	S20°03.7'	21	23	168°30.4'	S20°04.2'
22	0	183°30.3'	S20°04.8'	22	1	198°30.1'	S20°05.3'	22	2	213°30.0'	S20°05.9'
22	3	228°29.8'	S20°06.4'	22	4	243°29.6'	S20°06.9'	22	5	258°29.5'	S20°07.5'
22	6	273°29.3'	S20°08.0'	22	7	288°29.1'	S20°08.5'	22	8	303°29.0'	S20°09.1'
22	9	318°28.8'	S20°09.6'	22	10	333°28.6'	S20°10.1'	22	11	348°28.5'	S20°10.7'
22	12	3°28.3'	S20°11.2'	22	13	18°28.1'	S20°11.7'	22	14	33°28.0'	S20°12.3'

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Sun — GHA and Declination, November 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
22	15	48°27.8'	S20°12.8'	22	16	63°27.6'	S20°13.3'	22	17	78°27.5'	S20°13.9'
22	18	93°27.3'	S20°14.4'	22	19	108°27.1'	S20°14.9'	22	20	123°27.0'	S20°15.4'
22	21	138°26.8'	S20°16.0'	22	22	153°26.6'	S20°16.5'	22	23	168°26.5'	S20°17.0'
23	0	183°26.3'	S20°17.6'	23	1	198°26.1'	S20°18.1'	23	2	213°25.9'	S20°18.6'
23	3	228°25.8'	S20°19.1'	23	4	243°25.6'	S20°19.6'	23	5	258°25.4'	S20°20.2'
23	6	273°25.3'	S20°20.7'	23	7	288°25.1'	S20°21.2'	23	8	303°24.9'	S20°21.7'
23	9	318°24.7'	S20°22.2'	23	10	333°24.6'	S20°22.8'	23	11	348°24.4'	S20°23.3'
23	12	3°24.2'	S20°23.8'	23	13	18°24.0'	S20°24.3'	23	14	33°23.9'	S20°24.8'
23	15	48°23.7'	S20°25.3'	23	16	63°23.5'	S20°25.9'	23	17	78°23.3'	S20°26.4'
23	18	93°23.2'	S20°26.9'	23	19	108°23.0'	S20°27.4'	23	20	123°22.8'	S20°27.9'
23	21	138°22.6'	S20°28.4'	23	22	153°22.5'	S20°28.9'	23	23	168°22.3'	S20°29.4'
24	0	183°22.1'	S20°29.9'	24	1	198°21.9'	S20°30.5'	24	2	213°21.7'	S20°31.0'
24	3	228°21.6'	S20°31.5'	24	4	243°21.4'	S20°32.0'	24	5	258°21.2'	S20°32.5'
24	6	273°21.0'	S20°33.0'	24	7	288°20.8'	S20°33.5'	24	8	303°20.7'	S20°34.0'
24	9	318°20.5'	S20°34.5'	24	10	333°20.3'	S20°35.0'	24	11	348°20.1'	S20°35.5'
24	12	3°19.9'	S20°36.0'	24	13	18°19.7'	S20°36.5'	24	14	33°19.6'	S20°37.0'
24	15	48°19.4'	S20°37.5'	24	16	63°19.2'	S20°38.0'	24	17	78°19.0'	S20°38.5'
24	18	93°18.8'	S20°39.0'	24	19	108°18.6'	S20°39.5'	24	20	123°18.5'	S20°40.0'
24	21	138°18.3'	S20°40.5'	24	22	153°18.1'	S20°41.0'	24	23	168°17.9'	S20°41.5'
25	0	183°17.7'	S20°42.0'	25	1	198°17.5'	S20°42.5'	25	2	213°17.3'	S20°42.9'
25	3	228°17.2'	S20°43.4'	25	4	243°17.0'	S20°43.9'	25	5	258°16.8'	S20°44.4'
25	6	273°16.6'	S20°44.9'	25	7	288°16.4'	S20°45.4'	25	8	303°16.2'	S20°45.9'
25	9	318°16.0'	S20°46.4'	25	10	333°15.8'	S20°46.9'	25	11	348°15.6'	S20°47.3'
25	12	3°15.5'	S20°47.8'	25	13	18°15.3'	S20°48.3'	25	14	33°15.1'	S20°48.8'
25	15	48°14.9'	S20°49.3'	25	16	63°14.7'	S20°49.8'	25	17	78°14.5'	S20°50.2'
25	18	93°14.3'	S20°50.7'	25	19	108°14.1'	S20°51.2'	25	20	123°13.9'	S20°51.7'
25	21	138°13.7'	S20°52.2'	25	22	153°13.5'	S20°52.6'	25	23	168°13.3'	S20°53.1'
26	0	183°13.1'	S20°53.6'	26	1	198°12.9'	S20°54.1'	26	2	213°12.8'	S20°54.5'
26	3	228°12.6'	S20°55.0'	26	4	243°12.4'	S20°55.5'	26	5	258°12.2'	S20°56.0'
26	6	273°12.0'	S20°56.4'	26	7	288°11.8'	S20°56.9'	26	8	303°11.6'	S20°57.4'
26	9	318°11.4'	S20°57.9'	26	10	333°11.2'	S20°58.3'	26	11	348°11.0'	S20°58.8'
26	12	3°10.8'	S20°59.3'	26	13	18°10.6'	S20°59.7'	26	14	33°10.4'	S21°00.2'
26	15	48°10.2'	S21°00.7'	26	16	63°10.0'	S21°01.1'	26	17	78°09.8'	S21°01.6'
26	18	93°09.6'	S21°02.1'	26	19	108°09.4'	S21°02.5'	26	20	123°09.2'	S21°03.0'
26	21	138°09.0'	S21°03.5'	26	22	153°08.8'	S21°03.9'	26	23	168°08.6'	S21°04.4'
27	0	183°08.4'	S21°04.8'	27	1	198°08.2'	S21°05.3'	27	2	213°08.0'	S21°05.8'
27	3	228°07.8'	S21°06.2'	27	4	243°07.6'	S21°06.7'	27	5	258°07.4'	S21°07.1'
27	6	273°07.2'	S21°07.6'	27	7	288°07.0'	S21°08.0'	27	8	303°06.8'	S21°08.5'
27	9	318°06.6'	S21°08.9'	27	10	333°06.3'	S21°09.4'	27	11	348°06.1'	S21°09.9'
27	12	3°05.9'	S21°10.3'	27	13	18°05.7'	S21°10.8'	27	14	33°05.5'	S21°11.2'
27	15	48°05.3'	S21°11.7'	27	16	63°05.1'	S21°12.1'	27	17	78°04.9'	S21°12.6'
27	18	93°04.7'	S21°13.0'	27	19	108°04.5'	S21°13.5'	27	20	123°04.3'	S21°13.9'
27	21	138°04.1'	S21°14.3'	27	22	153°03.9'	S21°14.8'	27	23	168°03.7'	S21°15.2'
28	0	183°03.4'	S21°15.7'	28	1	198°03.2'	S21°16.1'	28	2	213°03.0'	S21°16.6'
28	3	228°02.8'	S21°17.0'	28	4	243°02.6'	S21°17.4'	28	5	258°02.4'	S21°17.9'
28	6	273°02.2'	S21°18.3'	28	7	288°02.0'	S21°18.8'	28	8	303°01.8'	S21°19.2'
28	9	318°01.5'	S21°19.6'	28	10	333°01.3'	S21°20.1'	28	11	348°01.1'	S21°20.5'
28	12	3°00.9'	S21°21.0'	28	13	18°00.7'	S21°21.4'	28	14	33°00.5'	S21°21.8'
28	15	48°00.3'	S21°22.3'	28	16	63°00.1'	S21°22.7'	28	17	77°59.8'	S21°23.1'
28	18	92°59.6'	S21°23.6'	28	19	107°59.4'	S21°24.0'	28	20	122°59.2'	S21°24.4'
28	21	137°59.0'	S21°24.8'	28	22	152°58.8'	S21°25.3'	28	23	167°58.5'	S21°25.7'
29	0	182°58.3'	S21°26.1'	29	1	197°58.1'	S21°26.5'	29	2	212°57.9'	S21°27.0'
29	3	227°57.7'	S21°27.4'	29	4	242°57.5'	S21°27.8'	29	5	257°57.2'	S21°28.2'
29	6	272°57.0'	S21°28.7'	29	7	287°56.8'	S21°29.1'	29	8	302°56.6'	S21°29.5'
29	9	317°56.4'	S21°29.9'	29	10	332°56.1'	S21°30.4'	29	11	347°55.9'	S21°30.8'
29	12	2°55.7'	S21°31.2'	29	13	17°55.5'	S21°31.6'	29	14	32°55.3'	S21°32.0'
29	15	47°55.0'	S21°32.4'	29	16	62°54.8'	S21°32.9'	29	17	77°54.6'	S21°33.3'
29	18	92°54.4'	S21°33.7'	29	19	107°54.2'	S21°34.1'	29	20	122°53.9'	S21°34.5'
29	21	137°53.7'	S21°34.9'	29	22	152°53.5'	S21°35.3'	29	23	167°53.3'	S21°35.8'
30	0	182°53.0'	S21°36.2'	30	1	197°52.8'	S21°36.6'	30	2	212°52.6'	S21°37.0'
30	3	227°52.4'	S21°37.4'	30	4	242°52.1'	S21°37.8'	30	5	257°51.9'	S21°38.2'
30	6	272°51.7'	S21°38.6'	30	7	287°51.5'	S21°39.0'	30	8	302°51.2'	S21°39.4'
30	9	317°51.0'	S21°39.8'	30	10	332°50.8'	S21°40.2'	30	11	347°50.6'	S21°40.6'
30	12	2°50.3'	S21°41.0'	30	13	17°50.1'	S21°41.4'	30	14	32°49.9'	S21°41.8'

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Sun — GHA and Declination, November 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
30	15	47°49.6'	S21°42.2'	30	16	62°49.4'	S21°42.6'	30	17	77°49.2'	S21°43.0'
30	18	92°49.0'	S21°43.4'	30	19	107°48.7'	S21°43.8'	30	20	122°48.5'	S21°44.2'
30	21	137°48.3'	S21°44.6'	30	22	152°48.0'	S21°45.0'	30	23	167°47.8'	S21°45.4'

Table 9.12: Sun — GHA and Declination, December 2026 (UT)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
1	0	182°47.6'	S21°45.8'	1	1	197°47.4'	S21°46.2'	1	2	212°47.1'	S21°46.6'
1	3	227°46.9'	S21°47.0'	1	4	242°46.7'	S21°47.4'	1	5	257°46.4'	S21°47.7'
1	6	272°46.2'	S21°48.1'	1	7	287°46.0'	S21°48.5'	1	8	302°45.7'	S21°48.9'
1	9	317°45.5'	S21°49.3'	1	10	332°45.3'	S21°49.7'	1	11	347°45.0'	S21°50.1'
1	12	2°44.8'	S21°50.4'	1	13	17°44.6'	S21°50.8'	1	14	32°44.3'	S21°51.2'
1	15	47°44.1'	S21°51.6'	1	16	62°43.9'	S21°52.0'	1	17	77°43.6'	S21°52.4'
1	18	92°43.4'	S21°52.7'	1	19	107°43.1'	S21°53.1'	1	20	122°42.9'	S21°53.5'
1	21	137°42.7'	S21°53.9'	1	22	152°42.4'	S21°54.2'	1	23	167°42.2'	S21°54.6'
2	0	182°42.0'	S21°55.0'	2	1	197°41.7'	S21°55.4'	2	2	212°41.5'	S21°55.7'
2	3	227°41.2'	S21°56.1'	2	4	242°41.0'	S21°56.5'	2	5	257°40.8'	S21°56.9'
2	6	272°40.5'	S21°57.2'	2	7	287°40.3'	S21°57.6'	2	8	302°40.1'	S21°58.0'
2	9	317°39.8'	S21°58.3'	2	10	332°39.6'	S21°58.7'	2	11	347°39.3'	S21°59.1'
2	12	2°39.1'	S21°59.4'	2	13	17°38.9'	S21°59.8'	2	14	32°38.6'	S22°00.2'
2	15	47°38.4'	S22°00.5'	2	16	62°38.1'	S22°00.9'	2	17	77°37.9'	S22°01.3'
2	18	92°37.6'	S22°01.6'	2	19	107°37.4'	S22°02.0'	2	20	122°37.2'	S22°02.4'
2	21	137°36.9'	S22°02.7'	2	22	152°36.7'	S22°03.1'	2	23	167°36.4'	S22°03.4'
3	0	182°36.2'	S22°03.8'	3	1	197°35.9'	S22°04.1'	3	2	212°35.7'	S22°04.5'
3	3	227°35.5'	S22°04.9'	3	4	242°35.2'	S22°05.2'	3	5	257°35.0'	S22°05.6'
3	6	272°34.7'	S22°05.9'	3	7	287°34.5'	S22°06.3'	3	8	302°34.2'	S22°06.6'
3	9	317°34.0'	S22°07.0'	3	10	332°33.7'	S22°07.3'	3	11	347°33.5'	S22°07.7'
3	12	2°33.2'	S22°08.0'	3	13	17°33.0'	S22°08.4'	3	14	32°32.7'	S22°08.7'
3	15	47°32.5'	S22°09.1'	3	16	62°32.2'	S22°09.4'	3	17	77°32.0'	S22°09.8'
3	18	92°31.8'	S22°10.1'	3	19	107°31.5'	S22°10.4'	3	20	122°31.3'	S22°10.8'
3	21	137°31.0'	S22°11.1'	3	22	152°30.8'	S22°11.5'	3	23	167°30.5'	S22°11.8'
4	0	182°30.3'	S22°12.2'	4	1	197°30.0'	S22°12.5'	4	2	212°29.8'	S22°12.8'
4	3	227°29.5'	S22°13.2'	4	4	242°29.3'	S22°13.5'	4	5	257°29.0'	S22°13.8'
4	6	272°28.8'	S22°14.2'	4	7	287°28.5'	S22°14.5'	4	8	302°28.2'	S22°14.8'
4	9	317°28.0'	S22°15.2'	4	10	332°27.7'	S22°15.5'	4	11	347°27.5'	S22°15.8'
4	12	2°27.2'	S22°16.2'	4	13	17°27.0'	S22°16.5'	4	14	32°26.7'	S22°16.8'
4	15	47°26.5'	S22°17.2'	4	16	62°26.2'	S22°17.5'	4	17	77°26.0'	S22°17.8'
4	18	92°25.7'	S22°18.1'	4	19	107°25.5'	S22°18.5'	4	20	122°25.2'	S22°18.8'
4	21	137°24.9'	S22°19.1'	4	22	152°24.7'	S22°19.4'	4	23	167°24.4'	S22°19.8'
5	0	182°24.2'	S22°20.1'	5	1	197°23.9'	S22°20.4'	5	2	212°23.7'	S22°20.7'
5	3	227°23.4'	S22°21.1'	5	4	242°23.2'	S22°21.4'	5	5	257°22.9'	S22°21.7'
5	6	272°22.6'	S22°22.0'	5	7	287°22.4'	S22°22.3'	5	8	302°22.1'	S22°22.6'
5	9	317°21.9'	S22°23.0'	5	10	332°21.6'	S22°23.3'	5	11	347°21.3'	S22°23.6'
5	12	2°21.1'	S22°23.9'	5	13	17°20.8'	S22°24.2'	5	14	32°20.6'	S22°24.5'
5	15	47°20.3'	S22°24.8'	5	16	62°20.1'	S22°25.1'	5	17	77°19.8'	S22°25.5'
5	18	92°19.5'	S22°25.8'	5	19	107°19.3'	S22°26.1'	5	20	122°19.0'	S22°26.4'
5	21	137°18.8'	S22°26.7'	5	22	152°18.5'	S22°27.0'	5	23	167°18.2'	S22°27.3'
6	0	182°18.0'	S22°27.6'	6	1	197°17.7'	S22°27.9'	6	2	212°17.4'	S22°28.2'
6	3	227°17.2'	S22°28.5'	6	4	242°16.9'	S22°28.8'	6	5	257°16.7'	S22°29.1'
6	6	272°16.4'	S22°29.4'	6	7	287°16.1'	S22°29.7'	6	8	302°15.9'	S22°30.0'
6	9	317°15.6'	S22°30.3'	6	10	332°15.3'	S22°30.6'	6	11	347°15.1'	S22°30.9'
6	12	2°14.8'	S22°31.2'	6	13	17°14.5'	S22°31.5'	6	14	32°14.3'	S22°31.8'
6	15	47°14.0'	S22°32.1'	6	16	62°13.7'	S22°32.4'	6	17	77°13.5'	S22°32.6'
6	18	92°13.2'	S22°32.9'	6	19	107°13.0'	S22°33.2'	6	20	122°12.7'	S22°33.5'
6	21	137°12.4'	S22°33.8'	6	22	152°12.2'	S22°34.1'	6	23	167°11.9'	S22°34.4'
7	0	182°11.6'	S22°34.7'	7	1	197°11.4'	S22°34.9'	7	2	212°11.1'	S22°35.2'
7	3	227°10.8'	S22°35.5'	7	4	242°10.5'	S22°35.8'	7	5	257°10.3'	S22°36.1'
7	6	272°10.0'	S22°36.4'	7	7	287°09.7'	S22°36.6'	7	8	302°09.5'	S22°36.9'
7	9	317°09.2'	S22°37.2'	7	10	332°08.9'	S22°37.5'	7	11	347°08.7'	S22°37.8'
7	12	2°08.4'	S22°38.0'	7	13	17°08.1'	S22°38.3'	7	14	32°07.9'	S22°38.6'
7	15	47°07.6'	S22°38.9'	7	16	62°07.3'	S22°39.1'	7	17	77°07.0'	S22°39.4'
7	18	92°06.8'	S22°39.7'	7	19	107°06.5'	S22°39.9'	7	20	122°06.2'	S22°40.2'

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Sun — GHA and Declination, December 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
7	21	137°06.0'	S22°40.5'	7	22	152°05.7'	S22°40.7'	7	23	167°05.4'	S22°41.0'
8	0	182°05.1'	S22°41.3'	8	1	197°04.9'	S22°41.5'	8	2	212°04.6'	S22°41.8'
8	3	227°04.3'	S22°42.1'	8	4	242°04.1'	S22°42.3'	8	5	257°03.8'	S22°42.6'
8	6	272°03.5'	S22°42.9'	8	7	287°03.2'	S22°43.1'	8	8	302°03.0'	S22°43.4'
8	9	317°02.7'	S22°43.7'	8	10	332°02.4'	S22°43.9'	8	11	347°02.1'	S22°44.2'
8	12	2°01.9'	S22°44.4'	8	13	17°01.6'	S22°44.7'	8	14	32°01.3'	S22°44.9'
8	15	47°01.0'	S22°45.2'	8	16	62°00.8'	S22°45.5'	8	17	77°00.5'	S22°45.7'
8	18	92°00.2'	S22°46.0'	8	19	106°59.9'	S22°46.2'	8	20	121°59.7'	S22°46.5'
8	21	136°59.4'	S22°46.7'	8	22	151°59.1'	S22°47.0'	8	23	166°58.8'	S22°47.2'
9	0	181°58.5'	S22°47.5'	9	1	196°58.3'	S22°47.7'	9	2	211°58.0'	S22°48.0'
9	3	226°57.7'	S22°48.2'	9	4	241°57.4'	S22°48.4'	9	5	256°57.2'	S22°48.7'
9	6	271°56.9'	S22°48.9'	9	7	286°56.6'	S22°49.2'	9	8	301°56.3'	S22°49.4'
9	9	316°56.0'	S22°49.7'	9	10	331°55.8'	S22°49.9'	9	11	346°55.5'	S22°50.1'
9	12	1°55.2'	S22°50.4'	9	13	16°54.9'	S22°50.6'	9	14	31°54.6'	S22°50.9'
9	15	46°54.4'	S22°51.1'	9	16	61°54.1'	S22°51.3'	9	17	76°53.8'	S22°51.6'
9	18	91°53.5'	S22°51.8'	9	19	106°53.2'	S22°52.0'	9	20	121°53.0'	S22°52.3'
9	21	136°52.7'	S22°52.5'	9	22	151°52.4'	S22°52.7'	9	23	166°52.1'	S22°53.0'
10	0	181°51.8'	S22°53.2'	10	1	196°51.6'	S22°53.4'	10	2	211°51.3'	S22°53.7'
10	3	226°51.0'	S22°53.9'	10	4	241°50.7'	S22°54.1'	10	5	256°50.4'	S22°54.3'
10	6	271°50.1'	S22°54.6'	10	7	286°49.9'	S22°54.8'	10	8	301°49.6'	S22°55.0'
10	9	316°49.3'	S22°55.2'	10	10	331°49.0'	S22°55.4'	10	11	346°48.7'	S22°55.7'
10	12	1°48.4'	S22°55.9'	10	13	16°48.2'	S22°56.1'	10	14	31°47.9'	S22°56.3'
10	15	46°47.6'	S22°56.5'	10	16	61°47.3'	S22°56.8'	10	17	76°47.0'	S22°57.0'
10	18	91°46.7'	S22°57.2'	10	19	106°46.5'	S22°57.4'	10	20	121°46.2'	S22°57.6'
10	21	136°45.9'	S22°57.8'	10	22	151°45.6'	S22°58.1'	10	23	166°45.3'	S22°58.3'
11	0	181°45.0'	S22°58.5'	11	1	196°44.7'	S22°58.7'	11	2	211°44.5'	S22°58.9'
11	3	226°44.2'	S22°59.1'	11	4	241°43.9'	S22°59.3'	11	5	256°43.6'	S22°59.5'
11	6	271°43.3'	S22°59.7'	11	7	286°43.0'	S22°59.9'	11	8	301°42.7'	S23°00.1'
11	9	316°42.4'	S23°00.3'	11	10	331°42.2'	S23°00.5'	11	11	346°41.9'	S23°00.7'
11	12	1°41.6'	S23°00.9'	11	13	16°41.3'	S23°01.1'	11	14	31°41.0'	S23°01.3'
11	15	46°40.7'	S23°01.5'	11	16	61°40.4'	S23°01.7'	11	17	76°40.1'	S23°01.9'
11	18	91°39.9'	S23°02.1'	11	19	106°39.6'	S23°02.3'	11	20	121°39.3'	S23°02.5'
11	21	136°39.0'	S23°02.7'	11	22	151°38.7'	S23°02.9'	11	23	166°38.4'	S23°03.1'
12	0	181°38.1'	S23°03.3'	12	1	196°37.8'	S23°03.5'	12	2	211°37.5'	S23°03.7'
12	3	226°37.2'	S23°03.9'	12	4	241°37.0'	S23°04.1'	12	5	256°36.7'	S23°04.2'
12	6	271°36.4'	S23°04.4'	12	7	286°36.1'	S23°04.6'	12	8	301°35.8'	S23°04.8'
12	9	316°35.5'	S23°05.0'	12	10	331°35.2'	S23°05.2'	12	11	346°34.9'	S23°05.4'
12	12	1°34.6'	S23°05.5'	12	13	16°34.3'	S23°05.7'	12	14	31°34.0'	S23°05.9'
12	15	46°33.7'	S23°06.1'	12	16	61°33.5'	S23°06.3'	12	17	76°33.2'	S23°06.4'
12	18	91°32.9'	S23°06.6'	12	19	106°32.6'	S23°06.8'	12	20	121°32.3'	S23°07.0'
12	21	136°32.0'	S23°07.1'	12	22	151°31.7'	S23°07.3'	12	23	166°31.4'	S23°07.5'
13	0	181°31.1'	S23°07.7'	13	1	196°30.8'	S23°07.8'	13	2	211°30.5'	S23°08.0'
13	3	226°30.2'	S23°08.2'	13	4	241°29.9'	S23°08.4'	13	5	256°29.6'	S23°08.5'
13	6	271°29.3'	S23°08.7'	13	7	286°29.1'	S23°08.9'	13	8	301°28.8'	S23°09.0'
13	9	316°28.5'	S23°09.2'	13	10	331°28.2'	S23°09.4'	13	11	346°27.9'	S23°09.5'
13	12	1°27.6'	S23°09.7'	13	13	16°27.3'	S23°09.8'	13	14	31°27.0'	S23°10.0'
13	15	46°26.7'	S23°10.2'	13	16	61°26.4'	S23°10.3'	13	17	76°26.1'	S23°10.5'
13	18	91°25.8'	S23°10.6'	13	19	106°25.5'	S23°10.8'	13	20	121°25.2'	S23°11.0'
13	21	136°24.9'	S23°11.1'	13	22	151°24.6'	S23°11.3'	13	23	166°24.3'	S23°11.4'
14	0	181°24.0'	S23°11.6'	14	1	196°23.7'	S23°11.7'	14	2	211°23.4'	S23°11.9'
14	3	226°23.1'	S23°12.0'	14	4	241°22.8'	S23°12.2'	14	5	256°22.5'	S23°12.3'
14	6	271°22.2'	S23°12.5'	14	7	286°21.9'	S23°12.6'	14	8	301°21.6'	S23°12.8'
14	9	316°21.4'	S23°12.9'	14	10	331°21.1'	S23°13.1'	14	11	346°20.8'	S23°13.2'
14	12	1°20.5'	S23°13.4'	14	13	16°20.2'	S23°13.5'	14	14	31°19.9'	S23°13.6'
14	15	46°19.6'	S23°13.8'	14	16	61°19.3'	S23°13.9'	14	17	76°19.0'	S23°14.1'
14	18	91°18.7'	S23°14.2'	14	19	106°18.4'	S23°14.3'	14	20	121°18.1'	S23°14.5'
14	21	136°17.8'	S23°14.6'	14	22	151°17.5'	S23°14.8'	14	23	166°17.2'	S23°14.9'
15	0	181°16.9'	S23°15.0'	15	1	196°16.6'	S23°15.2'	15	2	211°16.3'	S23°15.3'
15	3	226°16.0'	S23°15.4'	15	4	241°15.7'	S23°15.6'	15	5	256°15.4'	S23°15.7'
15	6	271°15.1'	S23°15.8'	15	7	286°14.8'	S23°15.9'	15	8	301°14.5'	S23°16.1'
15	9	316°14.2'	S23°16.2'	15	10	331°13.9'	S23°16.3'	15	11	346°13.6'	S23°16.5'
15	12	1°13.3'	S23°16.6'	15	13	16°13.0'	S23°16.7'	15	14	31°12.7'	S23°16.8'
15	15	46°12.4'	S23°16.9'	15	16	61°12.1'	S23°17.1'	15	17	76°11.8'	S23°17.2'
15	18	91°11.5'	S23°17.3'	15	19	106°11.2'	S23°17.4'	15	20	121°10.8'	S23°17.5'

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Sun — GHA and Declination, December 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
15	21	136°10.5'	S23°17.7'	15	22	151°10.2'	S23°17.8'	15	23	166°09.9'	S23°17.9'
16	0	181°09.6'	S23°18.0'	16	1	196°09.3'	S23°18.1'	16	2	211°09.0'	S23°18.2'
16	3	226°08.7'	S23°18.3'	16	4	241°08.4'	S23°18.5'	16	5	256°08.1'	S23°18.6'
16	6	271°07.8'	S23°18.7'	16	7	286°07.5'	S23°18.8'	16	8	301°07.2'	S23°18.9'
16	9	316°06.9'	S23°19.0'	16	10	331°06.6'	S23°19.1'	16	11	346°06.3'	S23°19.2'
16	12	1°06.0'	S23°19.3'	16	13	16°05.7'	S23°19.4'	16	14	31°05.4'	S23°19.5'
16	15	46°05.1'	S23°19.6'	16	16	61°04.8'	S23°19.7'	16	17	76°04.5'	S23°19.8'
16	18	91°04.2'	S23°19.9'	16	19	106°03.9'	S23°20.0'	16	20	121°03.6'	S23°20.1'
16	21	136°03.3'	S23°20.2'	16	22	151°03.0'	S23°20.3'	16	23	166°02.7'	S23°20.4'
17	0	181°02.4'	S23°20.5'	17	1	196°02.1'	S23°20.6'	17	2	211°01.7'	S23°20.7'
17	3	226°01.4'	S23°20.8'	17	4	241°01.1'	S23°20.9'	17	5	256°00.8'	S23°21.0'
17	6	271°00.5'	S23°21.1'	17	7	286°00.2'	S23°21.2'	17	8	300°59.9'	S23°21.3'
17	9	315°59.6'	S23°21.4'	17	10	330°59.3'	S23°21.4'	17	11	345°59.0'	S23°21.5'
17	12	0°58.7'	S23°21.6'	17	13	15°58.4'	S23°21.7'	17	14	30°58.1'	S23°21.8'
17	15	45°57.8'	S23°21.9'	17	16	60°57.5'	S23°21.9'	17	17	75°57.2'	S23°22.0'
17	18	90°56.9'	S23°22.1'	17	19	105°56.6'	S23°22.2'	17	20	120°56.2'	S23°22.3'
17	21	135°55.9'	S23°22.3'	17	22	150°55.6'	S23°22.4'	17	23	165°55.3'	S23°22.5'
18	0	180°55.0'	S23°22.6'	18	1	195°54.7'	S23°22.7'	18	2	210°54.4'	S23°22.7'
18	3	225°54.1'	S23°22.8'	18	4	240°53.8'	S23°22.9'	18	5	255°53.5'	S23°22.9'
18	6	270°53.2'	S23°23.0'	18	7	285°52.9'	S23°23.1'	18	8	300°52.6'	S23°23.2'
18	9	315°52.3'	S23°23.2'	18	10	330°52.0'	S23°23.3'	18	11	345°51.6'	S23°23.4'
18	12	0°51.3'	S23°23.4'	18	13	15°51.0'	S23°23.5'	18	14	30°50.7'	S23°23.6'
18	15	45°50.4'	S23°23.6'	18	16	60°50.1'	S23°23.7'	18	17	75°49.8'	S23°23.8'
18	18	90°49.5'	S23°23.8'	18	19	105°49.2'	S23°23.9'	18	20	120°48.9'	S23°23.9'
18	21	135°48.6'	S23°24.0'	18	22	150°48.3'	S23°24.1'	18	23	165°48.0'	S23°24.1'
19	0	180°47.6'	S23°24.2'	19	1	195°47.3'	S23°24.2'	19	2	210°47.0'	S23°24.3'
19	3	225°46.7'	S23°24.3'	19	4	240°46.4'	S23°24.4'	19	5	255°46.1'	S23°24.4'
19	6	270°45.8'	S23°24.5'	19	7	285°45.5'	S23°24.5'	19	8	300°45.2'	S23°24.6'
19	9	315°44.9'	S23°24.6'	19	10	330°44.6'	S23°24.7'	19	11	345°44.2'	S23°24.7'
19	12	0°43.9'	S23°24.8'	19	13	15°43.6'	S23°24.8'	19	14	30°43.3'	S23°24.9'
19	15	45°43.0'	S23°24.9'	19	16	60°42.7'	S23°25.0'	19	17	75°42.4'	S23°25.0'
19	18	90°42.1'	S23°25.0'	19	19	105°41.8'	S23°25.1'	19	20	120°41.5'	S23°25.1'
19	21	135°41.2'	S23°25.2'	19	22	150°40.8'	S23°25.2'	19	23	165°40.5'	S23°25.2'
20	0	180°40.2'	S23°25.3'	20	1	195°39.9'	S23°25.3'	20	2	210°39.6'	S23°25.4'
20	3	225°39.3'	S23°25.4'	20	4	240°39.0'	S23°25.4'	20	5	255°38.7'	S23°25.5'
20	6	270°38.4'	S23°25.5'	20	7	285°38.1'	S23°25.5'	20	8	300°37.8'	S23°25.5'
20	9	315°37.4'	S23°25.6'	20	10	330°37.1'	S23°25.6'	20	11	345°36.8'	S23°25.6'
20	12	0°36.5'	S23°25.7'	20	13	15°36.2'	S23°25.7'	20	14	30°35.9'	S23°25.7'
20	15	45°35.6'	S23°25.7'	20	16	60°35.3'	S23°25.8'	20	17	75°35.0'	S23°25.8'
20	18	90°34.7'	S23°25.8'	20	19	105°34.3'	S23°25.8'	20	20	120°34.0'	S23°25.9'
20	21	135°33.7'	S23°25.9'	20	22	150°33.4'	S23°25.9'	20	23	165°33.1'	S23°25.9'
21	0	180°32.8'	S23°25.9'	21	1	195°32.5'	S23°25.9'	21	2	210°32.2'	S23°26.0'
21	3	225°31.9'	S23°26.0'	21	4	240°31.6'	S23°26.0'	21	5	255°31.2'	S23°26.0'
21	6	270°30.9'	S23°26.0'	21	7	285°30.6'	S23°26.0'	21	8	300°30.3'	S23°26.0'
21	9	315°30.0'	S23°26.0'	21	10	330°29.7'	S23°26.1'	21	11	345°29.4'	S23°26.1'
21	12	0°29.1'	S23°26.1'	21	13	15°28.8'	S23°26.1'	21	14	30°28.4'	S23°26.1'
21	15	45°28.1'	S23°26.1'	21	16	60°27.8'	S23°26.1'	21	17	75°27.5'	S23°26.1'
21	18	90°27.2'	S23°26.1'	21	19	105°26.9'	S23°26.1'	21	20	120°26.6'	S23°26.1'
21	21	135°26.3'	S23°26.1'	21	22	150°26.0'	S23°26.1'	21	23	165°25.7'	S23°26.1'
22	0	180°25.3'	S23°26.1'	22	1	195°25.0'	S23°26.1'	22	2	210°24.7'	S23°26.1'
22	3	225°24.4'	S23°26.1'	22	4	240°24.1'	S23°26.1'	22	5	255°23.8'	S23°26.1'
22	6	270°23.5'	S23°26.1'	22	7	285°23.2'	S23°26.1'	22	8	300°22.9'	S23°26.1'
22	9	315°22.5'	S23°26.0'	22	10	330°22.2'	S23°26.0'	22	11	345°21.9'	S23°26.0'
22	12	0°21.6'	S23°26.0'	22	13	15°21.3'	S23°26.0'	22	14	30°21.0'	S23°26.0'
22	15	45°20.7'	S23°26.0'	22	16	60°20.4'	S23°26.0'	22	17	75°20.1'	S23°25.9'
22	18	90°19.7'	S23°25.9'	22	19	105°19.4'	S23°25.9'	22	20	120°19.1'	S23°25.9'
22	21	135°18.8'	S23°25.9'	22	22	150°18.5'	S23°25.8'	22	23	165°18.2'	S23°25.8'
23	0	180°17.9'	S23°25.8'	23	1	195°17.6'	S23°25.8'	23	2	210°17.3'	S23°25.8'
23	3	225°16.9'	S23°25.7'	23	4	240°16.6'	S23°25.7'	23	5	255°16.3'	S23°25.7'
23	6	270°16.0'	S23°25.7'	23	7	285°15.7'	S23°25.6'	23	8	300°15.4'	S23°25.6'
23	9	315°15.1'	S23°25.6'	23	10	330°14.8'	S23°25.5'	23	11	345°14.5'	S23°25.5'
23	12	0°14.2'	S23°25.5'	23	13	15°13.8'	S23°25.4'	23	14	30°13.5'	S23°25.4'
23	15	45°13.2'	S23°25.4'	23	16	60°12.9'	S23°25.3'	23	17	75°12.6'	S23°25.3'
23	18	90°12.3'	S23°25.3'	23	19	105°12.0'	S23°25.2'	23	20	120°11.7'	S23°25.2'

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Sun — GHA and Declination, December 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
23	21	135°11.4'	S23°25.2'	23	22	150°11.0'	S23°25.1'	23	23	165°10.7'	S23°25.1'
24	0	180°10.4'	S23°25.0'	24	1	195°10.1'	S23°25.0'	24	2	210°09.8'	S23°25.0'
24	3	225°09.5'	S23°24.9'	24	4	240°09.2'	S23°24.9'	24	5	255°08.9'	S23°24.8'
24	6	270°08.6'	S23°24.8'	24	7	285°08.2'	S23°24.7'	24	8	300°07.9'	S23°24.7'
24	9	315°07.6'	S23°24.6'	24	10	330°07.3'	S23°24.6'	24	11	345°07.0'	S23°24.5'
24	12	0°06.7'	S23°24.5'	24	13	15°06.4'	S23°24.4'	24	14	30°06.1'	S23°24.4'
24	15	45°05.8'	S23°24.3'	24	16	60°05.5'	S23°24.3'	24	17	75°05.1'	S23°24.2'
24	18	90°04.8'	S23°24.2'	24	19	105°04.5'	S23°24.1'	24	20	120°04.2'	S23°24.0'
24	21	135°03.9'	S23°24.0'	24	22	150°03.6'	S23°23.9'	24	23	165°03.3'	S23°23.9'
25	0	180°03.0'	S23°23.8'	25	1	195°02.7'	S23°23.7'	25	2	210°02.4'	S23°23.7'
25	3	225°02.0'	S23°23.6'	25	4	240°01.7'	S23°23.5'	25	5	255°01.4'	S23°23.5'
25	6	270°01.1'	S23°23.4'	25	7	285°00.8'	S23°23.3'	25	8	300°00.5'	S23°23.3'
25	9	315°00.2'	S23°23.2'	25	10	329°59.9'	S23°23.1'	25	11	344°59.6'	S23°23.1'
25	12	359°59.3'	S23°23.0'	25	13	14°58.9'	S23°22.9'	25	14	29°58.6'	S23°22.9'
25	15	44°58.3'	S23°22.8'	25	16	59°58.0'	S23°22.7'	25	17	74°57.7'	S23°22.6'
25	18	89°57.4'	S23°22.6'	25	19	104°57.1'	S23°22.5'	25	20	119°56.8'	S23°22.4'
25	21	134°56.5'	S23°22.3'	25	22	149°56.2'	S23°22.3'	25	23	164°55.8'	S23°22.2'
26	0	179°55.5'	S23°22.1'	26	1	194°55.2'	S23°22.0'	26	2	209°54.9'	S23°21.9'
26	3	224°54.6'	S23°21.8'	26	4	239°54.3'	S23°21.8'	26	5	254°54.0'	S23°21.7'
26	6	269°53.7'	S23°21.6'	26	7	284°53.4'	S23°21.5'	26	8	299°53.1'	S23°21.4'
26	9	314°52.8'	S23°21.3'	26	10	329°52.5'	S23°21.2'	26	11	344°52.1'	S23°21.2'
26	12	359°51.8'	S23°21.1'	26	13	14°51.5'	S23°21.0'	26	14	29°51.2'	S23°20.9'
26	15	44°50.9'	S23°20.8'	26	16	59°50.6'	S23°20.7'	26	17	74°50.3'	S23°20.6'
26	18	89°50.0'	S23°20.5'	26	19	104°49.7'	S23°20.4'	26	20	119°49.4'	S23°20.3'
26	21	134°49.1'	S23°20.2'	26	22	149°48.7'	S23°20.1'	26	23	164°48.4'	S23°20.0'
27	0	179°48.1'	S23°19.9'	27	1	194°47.8'	S23°19.8'	27	2	209°47.5'	S23°19.7'
27	3	224°47.2'	S23°19.6'	27	4	239°46.9'	S23°19.5'	27	5	254°46.6'	S23°19.4'
27	6	269°46.3'	S23°19.3'	27	7	284°46.0'	S23°19.2'	27	8	299°45.7'	S23°19.1'
27	9	314°45.4'	S23°19.0'	27	10	329°45.1'	S23°18.9'	27	11	344°44.8'	S23°18.8'
27	12	359°44.4'	S23°18.7'	27	13	14°44.1'	S23°18.5'	27	14	29°43.8'	S23°18.4'
27	15	44°43.5'	S23°18.3'	27	16	59°43.2'	S23°18.2'	27	17	74°42.9'	S23°18.1'
27	18	89°42.6'	S23°18.0'	27	19	104°42.3'	S23°17.9'	27	20	119°42.0'	S23°17.7'
27	21	134°41.7'	S23°17.6'	27	22	149°41.4'	S23°17.5'	27	23	164°41.1'	S23°17.4'
28	0	179°40.8'	S23°17.3'	28	1	194°40.5'	S23°17.1'	28	2	209°40.2'	S23°17.0'
28	3	224°39.8'	S23°16.9'	28	4	239°39.5'	S23°16.8'	28	5	254°39.2'	S23°16.7'
28	6	269°38.9'	S23°16.5'	28	7	284°38.6'	S23°16.4'	28	8	299°38.3'	S23°16.3'
28	9	314°38.0'	S23°16.2'	28	10	329°37.7'	S23°16.0'	28	11	344°37.4'	S23°15.9'
28	12	359°37.1'	S23°15.8'	28	13	14°36.8'	S23°15.6'	28	14	29°36.5'	S23°15.5'
28	15	44°36.2'	S23°15.4'	28	16	59°35.9'	S23°15.2'	28	17	74°35.6'	S23°15.1'
28	18	89°35.3'	S23°15.0'	28	19	104°35.0'	S23°14.8'	28	20	119°34.7'	S23°14.7'
28	21	134°34.3'	S23°14.6'	28	22	149°34.0'	S23°14.4'	28	23	164°33.7'	S23°14.3'
29	0	179°33.4'	S23°14.2'	29	1	194°33.1'	S23°14.0'	29	2	209°32.8'	S23°13.9'
29	3	224°32.5'	S23°13.7'	29	4	239°32.2'	S23°13.6'	29	5	254°31.9'	S23°13.4'
29	6	269°31.6'	S23°13.3'	29	7	284°31.3'	S23°13.2'	29	8	299°31.0'	S23°13.0'
29	9	314°30.7'	S23°12.9'	29	10	329°30.4'	S23°12.7'	29	11	344°30.1'	S23°12.6'
29	12	359°29.8'	S23°12.4'	29	13	14°29.5'	S23°12.3'	29	14	29°29.2'	S23°12.1'
29	15	44°28.9'	S23°12.0'	29	16	59°28.6'	S23°11.8'	29	17	74°28.3'	S23°11.7'
29	18	89°28.0'	S23°11.5'	29	19	104°27.7'	S23°11.4'	29	20	119°27.4'	S23°11.2'
29	21	134°27.1'	S23°11.1'	29	22	149°26.8'	S23°10.9'	29	23	164°26.5'	S23°10.7'
30	0	179°26.2'	S23°10.6'	30	1	194°25.9'	S23°10.4'	30	2	209°25.6'	S23°10.3'
30	3	224°25.3'	S23°10.1'	30	4	239°24.9'	S23°09.9'	30	5	254°24.6'	S23°09.8'
30	6	269°24.3'	S23°09.6'	30	7	284°24.0'	S23°09.4'	30	8	299°23.7'	S23°09.3'
30	9	314°23.4'	S23°09.1'	30	10	329°23.1'	S23°08.9'	30	11	344°22.8'	S23°08.8'
30	12	359°22.5'	S23°08.6'	30	13	14°22.2'	S23°08.4'	30	14	29°21.9'	S23°08.3'
30	15	44°21.6'	S23°08.1'	30	16	59°21.3'	S23°07.9'	30	17	74°21.0'	S23°07.8'
30	18	89°20.7'	S23°07.6'	30	19	104°20.4'	S23°07.4'	30	20	119°20.1'	S23°07.2'
30	21	134°19.8'	S23°07.1'	30	22	149°19.5'	S23°06.9'	30	23	164°19.2'	S23°06.7'
31	0	179°18.9'	S23°06.5'	31	1	194°18.6'	S23°06.4'	31	2	209°18.3'	S23°06.2'
31	3	224°18.0'	S23°06.0'	31	4	239°17.7'	S23°05.8'	31	5	254°17.4'	S23°05.6'
31	6	269°17.1'	S23°05.5'	31	7	284°16.8'	S23°05.3'	31	8	299°16.5'	S23°05.1'
31	9	314°16.3'	S23°04.9'	31	10	329°16.0'	S23°04.7'	31	11	344°15.7'	S23°04.5'
31	12	359°15.4'	S23°04.3'	31	13	14°15.1'	S23°04.2'	31	14	29°14.8'	S23°04.0'
31	15	44°14.5'	S23°03.8'	31	16	59°14.2'	S23°03.6'	31	17	74°13.9'	S23°03.4'
31	18	89°13.6'	S23°03.2'	31	19	104°13.3'	S23°03.0'	31	20	119°13.0'	S23°02.8'

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Sun — GHA and Declination, December 2026 (UT) (continued)

d	h	GHA	Dec	d	h	GHA	Dec	d	h	GHA	Dec
31	21	134°12.7'	S23°02.6'	31	22	149°12.4'	S23°02.4'	31	23	164°12.1'	S23°02.2'

Part III

GHA Aries & the Navigational Stars, 2026

Chapter 10

The 58 navigational stars

For each star the table gives the **Sidereal Hour Angle** (SHA, the star's angular distance west of the First Point of Aries) and **Declination** at epoch 2026.5, together with the visual magnitude. These are *mean places* for epoch 2026.5; precession and proper motion shift them by up to about half an arcminute toward the ends of the year — within this almanac's $\sim 1'$ budget, but re-generate the tables from the source engine if you need another year. To find a star's GHA at your moment of observation, add its SHA to the GHA of Aries from the next chapter:

$$\text{GHA}_* = \text{GHA Aries} + \text{SHA} \quad (\text{reduce modulo } 360^\circ).$$

Table 10.1: The 58 navigational stars — SHA, Declination (epoch 2026.5), and magnitude

Star	SHA	Dec	Mag	Star	SHA	Dec	Mag
Acamar	315°10.9'	S40°12.0'	2.9	Fomalhaut	15°13.4'	S29°28.9'	1.2
Achernar	335°19.5'	S57°06.2'	0.5	Gacrux	171°50.2'	S57°15.7'	1.6
Acrux	172°58.6'	S63°14.7'	0.8	Gienah	175°42.4'	S17°41.3'	2.6
Adhara	255°05.0'	S29°00.6'	1.5	Hadar	148°34.2'	S60°30.0'	0.6
Aldebaran	290°38.3'	N16°33.6'	0.9	Hamal	327°49.9'	N23°35.2'	2.0
Alioth	166°12.2'	N55°49.0'	1.8	Kaus Australis	83°31.1'	S34°22.2'	1.8
Alkaid	152°51.3'	N49°10.9'	1.9	Kochab	137°20.2'	N74°02.8'	2.1
Al Na'ir	27°31.7'	S46°49.9'	1.7	Markab	13°28.7'	N15°20.9'	2.5
Alnilam	275°36.6'	S1°11.2'	1.7	Menkar	314°05.0'	N4°11.5'	2.5
Alphard	217°46.7'	S8°46.5'	2.0	Menkent	147°56.2'	S36°29.9'	2.1
Alphecca	126°02.8'	N26°37.6'	2.2	Miaplacidus	221°37.7'	S69°49.6'	1.7
Alpheratz	357°33.5'	N29°14.2'	2.1	Mirfak	308°26.6'	N49°57.2'	1.8
Altair	61°58.9'	N8°56.4'	0.8	Nunki	75°46.4'	S26°15.7'	2.0
Ankaa	353°06.2'	S42°09.7'	2.4	Peacock	53°04.1'	S56°38.9'	1.9
Antares	112°14.4'	S26°29.3'	1.1	Polaris	313°19.3'	N89°22.4'	2.0
Arcturus	145°46.9'	N19°02.7'	-0.1	Pollux	243°16.0'	N27°57.6'	1.2
Atria	107°07.6'	S69°04.3'	1.9	Procyon	244°49.7'	N5°09.3'	0.4
Avior	234°14.2'	S59°35.7'	1.9	Rasalhague	95°57.5'	N12°32.6'	2.1
Bellatrix	278°21.7'	N6°22.3'	1.6	Regulus	207°33.3'	N11°50.2'	1.4
Betelgeuse	270°50.9'	N7°24.6'	0.5	Rigel	281°02.8'	S8°10.4'	0.1
Canopus	263°51.9'	S52°42.7'	-0.7	Rigil Kentaurus	139°38.6'	S60°56.6'	-0.3
Capella	280°20.2'	N46°01.3'	0.1	Sabik	102°01.5'	S15°45.3'	2.4
Deneb	49°25.0'	N45°22.6'	1.2	Schedar	349°29.6'	N56°40.9'	2.2
Denebola	182°23.9'	N14°25.4'	2.1	Shaula	96°08.8'	S37°07.2'	1.6
Diphda	348°46.2'	S17°50.5'	2.0	Sirius	258°25.3'	S16°45.3'	-1.5
Dubhe	193°39.9'	N61°36.5'	1.8	Spica	158°21.1'	S11°17.9'	1.0
Elnath	278°00.5'	N28°37.6'	1.7	Suhail	222°45.4'	S43°32.4'	2.2
Eltanin	90°41.7'	N51°29.2'	2.2	Vega	80°32.5'	N38°48.6'	0.0
Enif	33°37.7'	N9°59.9'	2.4	Zubenelgenubi	136°54.8'	S16°09.0'	2.8

Chapter 11

Hourly GHA of Aries

The Greenwich Hour Angle of the First Point of Aries (γ) for each day and whole hour of UT. Add the star's SHA (previous chapter) for its GHA, and the Aries increment from Part IV for the odd minutes and seconds. Entries run left to right, then top to bottom, in four columns of day / hour / GHA γ .

Table 11.1: Aries — GHA of the First Point of Aries, January 2026 (UT)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
1	0	100°39.6'	1	1	115°42.1'	1	2	130°44.6'	1	3	145°47.0'
1	4	160°49.5'	1	5	175°52.0'	1	6	190°54.4'	1	7	205°56.9'
1	8	220°59.4'	1	9	236°01.8'	1	10	251°04.3'	1	11	266°06.8'
1	12	281°09.2'	1	13	296°11.7'	1	14	311°14.1'	1	15	326°16.6'
1	16	341°19.1'	1	17	356°21.5'	1	18	11°24.0'	1	19	26°26.5'
1	20	41°28.9'	1	21	56°31.4'	1	22	71°33.9'	1	23	86°36.3'
2	0	101°38.8'	2	1	116°41.3'	2	2	131°43.7'	2	3	146°46.2'
2	4	161°48.6'	2	5	176°51.1'	2	6	191°53.6'	2	7	206°56.0'
2	8	221°58.5'	2	9	237°01.0'	2	10	252°03.4'	2	11	267°05.9'
2	12	282°08.4'	2	13	297°10.8'	2	14	312°13.3'	2	15	327°15.7'
2	16	342°18.2'	2	17	357°20.7'	2	18	12°23.1'	2	19	27°25.6'
2	20	42°28.1'	2	21	57°30.5'	2	22	72°33.0'	2	23	87°35.5'
3	0	102°37.9'	3	1	117°40.4'	3	2	132°42.9'	3	3	147°45.3'
3	4	162°47.8'	3	5	177°50.2'	3	6	192°52.7'	3	7	207°55.2'
3	8	222°57.6'	3	9	238°00.1'	3	10	253°02.6'	3	11	268°05.0'
3	12	283°07.5'	3	13	298°10.0'	3	14	313°12.4'	3	15	328°14.9'
3	16	343°17.4'	3	17	358°19.8'	3	18	13°22.3'	3	19	28°24.7'
3	20	43°27.2'	3	21	58°29.7'	3	22	73°32.1'	3	23	88°34.6'
4	0	103°37.1'	4	1	118°39.5'	4	2	133°42.0'	4	3	148°44.5'
4	4	163°46.9'	4	5	178°49.4'	4	6	193°51.9'	4	7	208°54.3'
4	8	223°56.8'	4	9	238°59.2'	4	10	254°01.7'	4	11	269°04.2'
4	12	284°06.6'	4	13	299°09.1'	4	14	314°11.6'	4	15	329°14.0'
4	16	344°16.5'	4	17	359°19.0'	4	18	14°21.4'	4	19	29°23.9'
4	20	44°26.3'	4	21	59°28.8'	4	22	74°31.3'	4	23	89°33.7'
5	0	104°36.2'	5	1	119°38.7'	5	2	134°41.1'	5	3	149°43.6'
5	4	164°46.1'	5	5	179°48.5'	5	6	194°51.0'	5	7	209°53.5'
5	8	224°55.9'	5	9	239°58.4'	5	10	255°00.8'	5	11	270°03.3'
5	12	285°05.8'	5	13	300°08.2'	5	14	315°10.7'	5	15	330°13.2'
5	16	345°15.6'	5	17	0°18.1'	5	18	15°20.6'	5	19	30°23.0'
5	20	45°25.5'	5	21	60°28.0'	5	22	75°30.4'	5	23	90°32.9'
6	0	105°35.3'	6	1	120°37.8'	6	2	135°40.3'	6	3	150°42.7'
6	4	165°45.2'	6	5	180°47.7'	6	6	195°50.1'	6	7	210°52.6'
6	8	225°55.1'	6	9	240°57.5'	6	10	256°00.0'	6	11	271°02.4'
6	12	286°04.9'	6	13	301°07.4'	6	14	316°09.8'	6	15	331°12.3'
6	16	346°14.8'	6	17	1°17.2'	6	18	16°19.7'	6	19	31°22.2'
6	20	46°24.6'	6	21	61°27.1'	6	22	76°29.6'	6	23	91°32.0'
7	0	106°34.5'	7	1	121°36.9'	7	2	136°39.4'	7	3	151°41.9'
7	4	166°44.3'	7	5	181°46.8'	7	6	196°49.3'	7	7	211°51.7'
7	8	226°54.2'	7	9	241°56.7'	7	10	256°59.1'	7	11	272°01.6'
7	12	287°04.1'	7	13	302°06.5'	7	14	317°09.0'	7	15	332°11.4'
7	16	347°13.9'	7	17	2°16.4'	7	18	17°18.8'	7	19	32°21.3'

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Aries — GHA of the First Point of Aries, January 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
7	20	47°23.8'	7	21	62°26.2'	7	22	77°28.7'	7	23	92°31.2'
8	0	107°33.6'	8	1	122°36.1'	8	2	137°38.5'	8	3	152°41.0'
8	4	167°43.5'	8	5	182°45.9'	8	6	197°48.4'	8	7	212°50.9'
8	8	227°53.3'	8	9	242°55.8'	8	10	257°58.3'	8	11	273°00.7'
8	12	288°03.2'	8	13	303°05.7'	8	14	318°08.1'	8	15	333°10.6'
8	16	348°13.0'	8	17	3°15.5'	8	18	18°18.0'	8	19	33°20.4'
8	20	48°22.9'	8	21	63°25.4'	8	22	78°27.8'	8	23	93°30.3'
9	0	108°32.8'	9	1	123°35.2'	9	2	138°37.7'	9	3	153°40.2'
9	4	168°42.6'	9	5	183°45.1'	9	6	198°47.5'	9	7	213°50.0'
9	8	228°52.5'	9	9	243°54.9'	9	10	258°57.4'	9	11	273°59.9'
9	12	289°02.3'	9	13	304°04.8'	9	14	319°07.3'	9	15	334°09.7'
9	16	349°12.2'	9	17	4°14.6'	9	18	19°17.1'	9	19	34°19.6'
9	20	49°22.0'	9	21	64°24.5'	9	22	79°27.0'	9	23	94°29.4'
10	0	109°31.9'	10	1	124°34.4'	10	2	139°36.8'	10	3	154°39.3'
10	4	169°41.8'	10	5	184°44.2'	10	6	199°46.7'	10	7	214°49.1'
10	8	229°51.6'	10	9	244°54.1'	10	10	259°56.5'	10	11	274°59.0'
10	12	290°01.5'	10	13	305°03.9'	10	14	320°06.4'	10	15	335°08.9'
10	16	350°11.3'	10	17	5°13.8'	10	18	20°16.3'	10	19	35°18.7'
10	20	50°21.2'	10	21	65°23.6'	10	22	80°26.1'	10	23	95°28.6'
11	0	110°31.0'	11	1	125°33.5'	11	2	140°36.0'	11	3	155°38.4'
11	4	170°40.9'	11	5	185°43.4'	11	6	200°45.8'	11	7	215°48.3'
11	8	230°50.8'	11	9	245°53.2'	11	10	260°55.7'	11	11	275°58.1'
11	12	291°00.6'	11	13	306°03.1'	11	14	321°05.5'	11	15	336°08.0'
11	16	351°10.5'	11	17	6°12.9'	11	18	21°15.4'	11	19	36°17.9'
11	20	51°20.3'	11	21	66°22.8'	11	22	81°25.2'	11	23	96°27.7'
12	0	111°30.2'	12	1	126°32.6'	12	2	141°35.1'	12	3	156°37.6'
12	4	171°40.0'	12	5	186°42.5'	12	6	201°45.0'	12	7	216°47.4'
12	8	231°49.9'	12	9	246°52.4'	12	10	261°54.8'	12	11	276°57.3'
12	12	291°59.7'	12	13	307°02.2'	12	14	322°04.7'	12	15	337°07.1'
12	16	352°09.6'	12	17	7°12.1'	12	18	22°14.5'	12	19	37°17.0'
12	20	52°19.5'	12	21	67°21.9'	12	22	82°24.4'	12	23	97°26.9'
13	0	112°29.3'	13	1	127°31.8'	13	2	142°34.2'	13	3	157°36.7'
13	4	172°39.2'	13	5	187°41.6'	13	6	202°44.1'	13	7	217°46.6'
13	8	232°49.0'	13	9	247°51.5'	13	10	262°54.0'	13	11	277°56.4'
13	12	292°58.9'	13	13	308°01.3'	13	14	323°03.8'	13	15	338°06.3'
13	16	353°08.7'	13	17	8°11.2'	13	18	23°13.7'	13	19	38°16.1'
13	20	53°18.6'	13	21	68°21.1'	13	22	83°23.5'	13	23	98°26.0'
14	0	113°28.5'	14	1	128°30.9'	14	2	143°33.4'	14	3	158°35.8'
14	4	173°38.3'	14	5	188°40.8'	14	6	203°43.2'	14	7	218°45.7'
14	8	233°48.2'	14	9	248°50.6'	14	10	263°53.1'	14	11	278°55.6'
14	12	293°58.0'	14	13	309°00.5'	14	14	324°03.0'	14	15	339°05.4'
14	16	354°07.9'	14	17	9°10.3'	14	18	24°12.8'	14	19	39°15.3'
14	20	54°17.7'	14	21	69°20.2'	14	22	84°22.7'	14	23	99°25.1'
15	0	114°27.6'	15	1	129°30.1'	15	2	144°32.5'	15	3	159°35.0'
15	4	174°37.4'	15	5	189°39.9'	15	6	204°42.4'	15	7	219°44.8'
15	8	234°47.3'	15	9	249°49.8'	15	10	264°52.2'	15	11	279°54.7'
15	12	294°57.2'	15	13	309°59.6'	15	14	325°02.1'	15	15	340°04.6'
15	16	355°07.0'	15	17	10°09.5'	15	18	25°11.9'	15	19	40°14.4'
15	20	55°16.9'	15	21	70°19.3'	15	22	85°21.8'	15	23	100°24.3'
16	0	115°26.7'	16	1	130°29.2'	16	2	145°31.7'	16	3	160°34.1'
16	4	175°36.6'	16	5	190°39.1'	16	6	205°41.5'	16	7	220°44.0'
16	8	235°46.4'	16	9	250°48.9'	16	10	265°51.4'	16	11	280°53.8'
16	12	295°56.3'	16	13	310°58.8'	16	14	326°01.2'	16	15	341°03.7'
16	16	356°06.2'	16	17	11°08.6'	16	18	26°11.1'	16	19	41°13.6'
16	20	56°16.0'	16	21	71°18.5'	16	22	86°20.9'	16	23	101°23.4'
17	0	116°25.9'	17	1	131°28.3'	17	2	146°30.8'	17	3	161°33.3'
17	4	176°35.7'	17	5	191°38.2'	17	6	206°40.7'	17	7	221°43.1'
17	8	236°45.6'	17	9	251°48.0'	17	10	266°50.5'	17	11	281°53.0'
17	12	296°55.4'	17	13	311°57.9'	17	14	327°00.4'	17	15	342°02.8'
17	16	357°05.3'	17	17	12°07.8'	17	18	27°10.2'	17	19	42°12.7'
17	20	57°15.2'	17	21	72°17.6'	17	22	87°20.1'	17	23	102°22.5'
18	0	117°25.0'	18	1	132°27.5'	18	2	147°29.9'	18	3	162°32.4'
18	4	177°34.9'	18	5	192°37.3'	18	6	207°39.8'	18	7	222°42.3'
18	8	237°44.7'	18	9	252°47.2'	18	10	267°49.7'	18	11	282°52.1'

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Aries — GHA of the First Point of Aries, January 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
18	12	297°54.6'	18	13	312°57.0'	18	14	327°59.5'	18	15	343°02.0'
18	16	358°04.4'	18	17	13°06.9'	18	18	28°09.4'	18	19	43°11.8'
18	20	58°14.3'	18	21	73°16.8'	18	22	88°19.2'	18	23	103°21.7'
19	0	118°24.1'	19	1	133°26.6'	19	2	148°29.1'	19	3	163°31.5'
19	4	178°34.0'	19	5	193°36.5'	19	6	208°38.9'	19	7	223°41.4'
19	8	238°43.9'	19	9	253°46.3'	19	10	268°48.8'	19	11	283°51.3'
19	12	298°53.7'	19	13	313°56.2'	19	14	328°58.6'	19	15	344°01.1'
19	16	359°03.6'	19	17	14°06.0'	19	18	29°08.5'	19	19	44°11.0'
19	20	59°13.4'	19	21	74°15.9'	19	22	89°18.4'	19	23	104°20.8'
20	0	119°23.3'	20	1	134°25.8'	20	2	149°28.2'	20	3	164°30.7'
20	4	179°33.1'	20	5	194°35.6'	20	6	209°38.1'	20	7	224°40.5'
20	8	239°43.0'	20	9	254°45.5'	20	10	269°47.9'	20	11	284°50.4'
20	12	299°52.9'	20	13	314°55.3'	20	14	329°57.8'	20	15	345°00.2'
20	16	0°02.7'	20	17	15°05.2'	20	18	30°07.6'	20	19	45°10.1'
20	20	60°12.6'	20	21	75°15.0'	20	22	90°17.5'	20	23	105°20.0'
21	0	120°22.4'	21	1	135°24.9'	21	2	150°27.4'	21	3	165°29.8'
21	4	180°32.3'	21	5	195°34.7'	21	6	210°37.2'	21	7	225°39.7'
21	8	240°42.1'	21	9	255°44.6'	21	10	270°47.1'	21	11	285°49.5'
21	12	300°52.0'	21	13	315°54.5'	21	14	330°56.9'	21	15	345°59.4'
21	16	1°01.9'	21	17	16°04.3'	21	18	31°06.8'	21	19	46°09.2'
21	20	61°11.7'	21	21	76°14.2'	21	22	91°16.6'	21	23	106°19.1'
22	0	121°21.6'	22	1	136°24.0'	22	2	151°26.5'	22	3	166°29.0'
22	4	181°31.4'	22	5	196°33.9'	22	6	211°36.3'	22	7	226°38.8'
22	8	241°41.3'	22	9	256°43.7'	22	10	271°46.2'	22	11	286°48.7'
22	12	301°51.1'	22	13	316°53.6'	22	14	331°56.1'	22	15	346°58.5'
22	16	2°01.0'	22	17	17°03.5'	22	18	32°05.9'	22	19	47°08.4'
22	20	62°10.8'	22	21	77°13.3'	22	22	92°15.8'	22	23	107°18.2'
23	0	122°20.7'	23	1	137°23.2'	23	2	152°25.6'	23	3	167°28.1'
23	4	182°30.6'	23	5	197°33.0'	23	6	212°35.5'	23	7	227°38.0'
23	8	242°40.4'	23	9	257°42.9'	23	10	272°45.3'	23	11	287°47.8'
23	12	302°50.3'	23	13	317°52.7'	23	14	332°55.2'	23	15	347°57.7'
23	16	3°00.1'	23	17	18°02.6'	23	18	33°05.1'	23	19	48°07.5'
23	20	63°10.0'	23	21	78°12.5'	23	22	93°14.9'	23	23	108°17.4'
24	0	123°19.8'	24	1	138°22.3'	24	2	153°24.8'	24	3	168°27.2'
24	4	183°29.7'	24	5	198°32.2'	24	6	213°34.6'	24	7	228°37.1'
24	8	243°39.6'	24	9	258°42.0'	24	10	273°44.5'	24	11	288°46.9'
24	12	303°49.4'	24	13	318°51.9'	24	14	333°54.3'	24	15	348°56.8'
24	16	3°59.3'	24	17	19°01.7'	24	18	34°04.2'	24	19	49°06.7'
24	20	64°09.1'	24	21	79°11.6'	24	22	94°14.1'	24	23	109°16.5'
25	0	124°19.0'	25	1	139°21.4'	25	2	154°23.9'	25	3	169°26.4'
25	4	184°28.8'	25	5	199°31.3'	25	6	214°33.8'	25	7	229°36.2'
25	8	244°38.7'	25	9	259°41.2'	25	10	274°43.6'	25	11	289°46.1'
25	12	304°48.6'	25	13	319°51.0'	25	14	334°53.5'	25	15	349°55.9'
25	16	4°58.4'	25	17	20°00.9'	25	18	35°03.3'	25	19	50°05.8'
25	20	65°08.3'	25	21	80°10.7'	25	22	95°13.2'	25	23	110°15.7'
26	0	125°18.1'	26	1	140°20.6'	26	2	155°23.0'	26	3	170°25.5'
26	4	185°28.0'	26	5	200°30.4'	26	6	215°32.9'	26	7	230°35.4'
26	8	245°37.8'	26	9	260°40.3'	26	10	275°42.8'	26	11	290°45.2'
26	12	305°47.7'	26	13	320°50.2'	26	14	335°52.6'	26	15	350°55.1'
26	16	5°57.5'	26	17	21°00.0'	26	18	36°02.5'	26	19	51°04.9'
26	20	66°07.4'	26	21	81°09.9'	26	22	96°12.3'	26	23	111°14.8'
27	0	126°17.3'	27	1	141°19.7'	27	2	156°22.2'	27	3	171°24.7'
27	4	186°27.1'	27	5	201°29.6'	27	6	216°32.0'	27	7	231°34.5'
27	8	246°37.0'	27	9	261°39.4'	27	10	276°41.9'	27	11	291°44.4'
27	12	306°46.8'	27	13	321°49.3'	27	14	336°51.8'	27	15	351°54.2'
27	16	6°56.7'	27	17	21°59.1'	27	18	37°01.6'	27	19	52°04.1'
27	20	67°06.5'	27	21	82°09.0'	27	22	97°11.5'	27	23	112°13.9'
28	0	127°16.4'	28	1	142°18.9'	28	2	157°21.3'	28	3	172°23.8'
28	4	187°26.3'	28	5	202°28.7'	28	6	217°31.2'	28	7	232°33.6'
28	8	247°36.1'	28	9	262°38.6'	28	10	277°41.0'	28	11	292°43.5'
28	12	307°46.0'	28	13	322°48.4'	28	14	337°50.9'	28	15	352°53.4'
28	16	7°55.8'	28	17	22°58.3'	28	18	38°00.8'	28	19	53°03.2'
28	20	68°05.7'	28	21	83°08.1'	28	22	98°10.6'	28	23	113°13.1'
29	0	128°15.5'	29	1	143°18.0'	29	2	158°20.5'	29	3	173°22.9'

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Aries — GHA of the First Point of Aries, January 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
29	4	188°25.4'	29	5	203°27.9'	29	6	218°30.3'	29	7	233°32.8'
29	8	248°35.2'	29	9	263°37.7'	29	10	278°40.2'	29	11	293°42.6'
29	12	308°45.1'	29	13	323°47.6'	29	14	338°50.0'	29	15	353°52.5'
29	16	8°55.0'	29	17	23°57.4'	29	18	38°59.9'	29	19	54°02.4'
29	20	69°04.8'	29	21	84°07.3'	29	22	99°09.7'	29	23	114°12.2'
30	0	129°14.7'	30	1	144°17.1'	30	2	159°19.6'	30	3	174°22.1'
30	4	189°24.5'	30	5	204°27.0'	30	6	219°29.5'	30	7	234°31.9'
30	8	249°34.4'	30	9	264°36.9'	30	10	279°39.3'	30	11	294°41.8'
30	12	309°44.2'	30	13	324°46.7'	30	14	339°49.2'	30	15	354°51.6'
30	16	9°54.1'	30	17	24°56.6'	30	18	39°59.0'	30	19	55°01.5'
30	20	70°04.0'	30	21	85°06.4'	30	22	100°08.9'	30	23	115°11.4'
31	0	130°13.8'	31	1	145°16.3'	31	2	160°18.7'	31	3	175°21.2'
31	4	190°23.7'	31	5	205°26.1'	31	6	220°28.6'	31	7	235°31.1'
31	8	250°33.5'	31	9	265°36.0'	31	10	280°38.5'	31	11	295°40.9'
31	12	310°43.4'	31	13	325°45.8'	31	14	340°48.3'	31	15	355°50.8'
31	16	10°53.2'	31	17	25°55.7'	31	18	40°58.2'	31	19	56°00.6'
31	20	71°03.1'	31	21	86°05.6'	31	22	101°08.0'	31	23	116°10.5'

Table 11.2: Aries — GHA of the First Point of Aries, February 2026 (UT)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
1	0	131°13.0'	1	1	146°15.4'	1	2	161°17.9'	1	3	176°20.3'
1	4	191°22.8'	1	5	206°25.3'	1	6	221°27.7'	1	7	236°30.2'
1	8	251°32.7'	1	9	266°35.1'	1	10	281°37.6'	1	11	296°40.1'
1	12	311°42.5'	1	13	326°45.0'	1	14	341°47.5'	1	15	356°49.9'
1	16	11°52.4'	1	17	26°54.8'	1	18	41°57.3'	1	19	56°59.8'
1	20	72°02.2'	1	21	87°04.7'	1	22	102°07.2'	1	23	117°09.6'
2	0	132°12.1'	2	1	147°14.6'	2	2	162°17.0'	2	3	177°19.5'
2	4	192°21.9'	2	5	207°24.4'	2	6	222°26.9'	2	7	237°29.3'
2	8	252°31.8'	2	9	267°34.3'	2	10	282°36.7'	2	11	297°39.2'
2	12	312°41.7'	2	13	327°44.1'	2	14	342°46.6'	2	15	357°49.1'
2	16	12°51.5'	2	17	27°54.0'	2	18	42°56.4'	2	19	57°58.9'
2	20	73°01.4'	2	21	88°03.8'	2	22	103°06.3'	2	23	118°08.8'
3	0	133°11.2'	3	1	148°13.7'	3	2	163°16.2'	3	3	178°18.6'
3	4	193°21.1'	3	5	208°23.6'	3	6	223°26.0'	3	7	238°28.5'
3	8	253°30.9'	3	9	268°33.4'	3	10	283°35.9'	3	11	298°38.3'
3	12	313°40.8'	3	13	328°43.3'	3	14	343°45.7'	3	15	358°48.2'
3	16	13°50.7'	3	17	28°53.1'	3	18	43°55.6'	3	19	58°58.0'
3	20	74°00.5'	3	21	89°03.0'	3	22	104°05.4'	3	23	119°07.9'
4	0	134°10.4'	4	1	149°12.8'	4	2	164°15.3'	4	3	179°17.8'
4	4	194°20.2'	4	5	209°22.7'	4	6	224°25.2'	4	7	239°27.6'
4	8	254°30.1'	4	9	269°32.5'	4	10	284°35.0'	4	11	299°37.5'
4	12	314°39.9'	4	13	329°42.4'	4	14	344°44.9'	4	15	359°47.3'
4	16	14°49.8'	4	17	29°52.3'	4	18	44°54.7'	4	19	59°57.2'
4	20	74°59.7'	4	21	90°02.1'	4	22	105°04.6'	4	23	120°07.0'
5	0	135°09.5'	5	1	150°12.0'	5	2	165°14.4'	5	3	180°16.9'
5	4	195°19.4'	5	5	210°21.8'	5	6	225°24.3'	5	7	240°26.8'
5	8	255°29.2'	5	9	270°31.7'	5	10	285°34.1'	5	11	300°36.6'
5	12	315°39.1'	5	13	330°41.5'	5	14	345°44.0'	5	15	0°46.5'
5	16	15°48.9'	5	17	30°51.4'	5	18	45°53.9'	5	19	60°56.3'
5	20	75°58.8'	5	21	91°01.3'	5	22	106°03.7'	5	23	121°06.2'
6	0	136°08.6'	6	1	151°11.1'	6	2	166°13.6'	6	3	181°16.0'
6	4	196°18.5'	6	5	211°21.0'	6	6	226°23.4'	6	7	241°25.9'
6	8	256°28.4'	6	9	271°30.8'	6	10	286°33.3'	6	11	301°35.8'
6	12	316°38.2'	6	13	331°40.7'	6	14	346°43.1'	6	15	1°45.6'
6	16	16°48.1'	6	17	31°50.5'	6	18	46°53.0'	6	19	61°55.5'
6	20	76°57.9'	6	21	92°00.4'	6	22	107°02.9'	6	23	122°05.3'
7	0	137°07.8'	7	1	152°10.3'	7	2	167°12.7'	7	3	182°15.2'
7	4	197°17.6'	7	5	212°20.1'	7	6	227°22.6'	7	7	242°25.0'
7	8	257°27.5'	7	9	272°30.0'	7	10	287°32.4'	7	11	302°34.9'
7	12	317°37.4'	7	13	332°39.8'	7	14	347°42.3'	7	15	2°44.7'
7	16	17°47.2'	7	17	32°49.7'	7	18	47°52.1'	7	19	62°54.6'

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Aries — GHA of the First Point of Aries, February 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
7	20	77°57.1'	7	21	92°59.5'	7	22	108°02.0'	7	23	123°04.5'
8	0	138°06.9'	8	1	153°09.4'	8	2	168°11.9'	8	3	183°14.3'
8	4	198°16.8'	8	5	213°19.2'	8	6	228°21.7'	8	7	243°24.2'
8	8	258°26.6'	8	9	273°29.1'	8	10	288°31.6'	8	11	303°34.0'
8	12	318°36.5'	8	13	333°39.0'	8	14	348°41.4'	8	15	3°43.9'
8	16	18°46.4'	8	17	33°48.8'	8	18	48°51.3'	8	19	63°53.7'
8	20	78°56.2'	8	21	93°58.7'	8	22	109°01.1'	8	23	124°03.6'
9	0	139°06.1'	9	1	154°08.5'	9	2	169°11.0'	9	3	184°13.5'
9	4	199°15.9'	9	5	214°18.4'	9	6	229°20.8'	9	7	244°23.3'
9	8	259°25.8'	9	9	274°28.2'	9	10	289°30.7'	9	11	304°33.2'
9	12	319°35.6'	9	13	334°38.1'	9	14	349°40.6'	9	15	4°43.0'
9	16	19°45.5'	9	17	34°48.0'	9	18	49°50.4'	9	19	64°52.9'
9	20	79°55.3'	9	21	94°57.8'	9	22	110°00.3'	9	23	125°02.7'
10	0	140°05.2'	10	1	155°07.7'	10	2	170°10.1'	10	3	185°12.6'
10	4	200°15.1'	10	5	215°17.5'	10	6	230°20.0'	10	7	245°22.5'
10	8	260°24.9'	10	9	275°27.4'	10	10	290°29.8'	10	11	305°32.3'
10	12	320°34.8'	10	13	335°37.2'	10	14	350°39.7'	10	15	5°42.2'
10	16	20°44.6'	10	17	35°47.1'	10	18	50°49.6'	10	19	65°52.0'
10	20	80°54.5'	10	21	95°56.9'	10	22	110°59.4'	10	23	126°01.9'
11	0	141°04.3'	11	1	156°06.8'	11	2	171°09.3'	11	3	186°11.7'
11	4	201°14.2'	11	5	216°16.7'	11	6	231°19.1'	11	7	246°21.6'
11	8	261°24.1'	11	9	276°26.5'	11	10	291°29.0'	11	11	306°31.4'
11	12	321°33.9'	11	13	336°36.4'	11	14	351°38.8'	11	15	6°41.3'
11	16	21°43.8'	11	17	36°46.2'	11	18	51°48.7'	11	19	66°51.2'
11	20	81°53.6'	11	21	96°56.1'	11	22	111°58.6'	11	23	127°01.0'
12	0	142°03.5'	12	1	157°05.9'	12	2	172°08.4'	12	3	187°10.9'
12	4	202°13.3'	12	5	217°15.8'	12	6	232°18.3'	12	7	247°20.7'
12	8	262°23.2'	12	9	277°25.7'	12	10	292°28.1'	12	11	307°30.6'
12	12	322°33.0'	12	13	337°35.5'	12	14	352°38.0'	12	15	7°40.4'
12	16	22°42.9'	12	17	37°45.4'	12	18	52°47.8'	12	19	67°50.3'
12	20	82°52.8'	12	21	97°55.2'	12	22	112°57.7'	12	23	128°00.2'
13	0	143°02.6'	13	1	158°05.1'	13	2	173°07.5'	13	3	188°10.0'
13	4	203°12.5'	13	5	218°14.9'	13	6	233°17.4'	13	7	248°19.9'
13	8	263°22.3'	13	9	278°24.8'	13	10	293°27.3'	13	11	308°29.7'
13	12	323°32.2'	13	13	338°34.7'	13	14	353°37.1'	13	15	8°39.6'
13	16	23°42.0'	13	17	38°44.5'	13	18	53°47.0'	13	19	68°49.4'
13	20	83°51.9'	13	21	98°54.4'	13	22	113°56.8'	13	23	128°59.3'
14	0	144°01.8'	14	1	159°04.2'	14	2	174°06.7'	14	3	189°09.2'
14	4	204°11.6'	14	5	219°14.1'	14	6	234°16.5'	14	7	249°19.0'
14	8	264°21.5'	14	9	279°23.9'	14	10	294°26.4'	14	11	309°28.9'
14	12	324°31.3'	14	13	339°33.8'	14	14	354°36.3'	14	15	9°38.7'
14	16	24°41.2'	14	17	39°43.6'	14	18	54°46.1'	14	19	69°48.6'
14	20	84°51.0'	14	21	99°53.5'	14	22	114°56.0'	14	23	129°58.4'
15	0	145°00.9'	15	1	160°03.4'	15	2	175°05.8'	15	3	190°08.3'
15	4	205°10.8'	15	5	220°13.2'	15	6	235°15.7'	15	7	250°18.1'
15	8	265°20.6'	15	9	280°23.1'	15	10	295°25.5'	15	11	310°28.0'
15	12	325°30.5'	15	13	340°32.9'	15	14	355°35.4'	15	15	10°37.9'
15	16	25°40.3'	15	17	40°42.8'	15	18	55°45.3'	15	19	70°47.7'
15	20	85°50.2'	15	21	100°52.6'	15	22	115°55.1'	15	23	130°57.6'
16	0	146°00.0'	16	1	161°02.5'	16	2	176°05.0'	16	3	191°07.4'
16	4	206°09.9'	16	5	221°12.4'	16	6	236°14.8'	16	7	251°17.3'
16	8	266°19.7'	16	9	281°22.2'	16	10	296°24.7'	16	11	311°27.1'
16	12	326°29.6'	16	13	341°32.1'	16	14	356°34.5'	16	15	11°37.0'
16	16	26°39.5'	16	17	41°41.9'	16	18	56°44.4'	16	19	71°46.9'
16	20	86°49.3'	16	21	101°51.8'	16	22	116°54.2'	16	23	131°56.7'
17	0	146°59.2'	17	1	162°01.6'	17	2	177°04.1'	17	3	192°06.6'
17	4	207°09.0'	17	5	222°11.5'	17	6	237°14.0'	17	7	252°16.4'
17	8	267°18.9'	17	9	282°21.4'	17	10	297°23.8'	17	11	312°26.3'
17	12	327°28.7'	17	13	342°31.2'	17	14	357°33.7'	17	15	12°36.1'
17	16	27°38.6'	17	17	42°41.1'	17	18	57°43.5'	17	19	72°46.0'
17	20	87°48.5'	17	21	102°50.9'	17	22	117°53.4'	17	23	132°55.8'
18	0	147°58.3'	18	1	163°00.8'	18	2	178°03.2'	18	3	193°05.7'
18	4	208°08.2'	18	5	223°10.6'	18	6	238°13.1'	18	7	253°15.6'
18	8	268°18.0'	18	9	283°20.5'	18	10	298°23.0'	18	11	313°25.4'

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Aries — GHA of the First Point of Aries, February 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
18	12	328°27.9'	18	13	343°30.3'	18	14	358°32.8'	18	15	13°35.3'
18	16	28°37.7'	18	17	43°40.2'	18	18	58°42.7'	18	19	73°45.1'
18	20	88°47.6'	18	21	103°50.1'	18	22	118°52.5'	18	23	133°55.0'
19	0	148°57.5'	19	1	163°59.9'	19	2	179°02.4'	19	3	194°04.8'
19	4	209°07.3'	19	5	224°09.8'	19	6	239°12.2'	19	7	254°14.7'
19	8	269°17.2'	19	9	284°19.6'	19	10	299°22.1'	19	11	314°24.6'
19	12	329°27.0'	19	13	344°29.5'	19	14	359°32.0'	19	15	14°34.4'
19	16	29°36.9'	19	17	44°39.3'	19	18	59°41.8'	19	19	74°44.3'
19	20	89°46.7'	19	21	104°49.2'	19	22	119°51.7'	19	23	134°54.1'
20	0	149°56.6'	20	1	164°59.1'	20	2	180°01.5'	20	3	195°04.0'
20	4	210°06.4'	20	5	225°08.9'	20	6	240°11.4'	20	7	255°13.8'
20	8	270°16.3'	20	9	285°18.8'	20	10	300°21.2'	20	11	315°23.7'
20	12	330°26.2'	20	13	345°28.6'	20	14	0°31.1'	20	15	15°33.6'
20	16	30°36.0'	20	17	45°38.5'	20	18	60°40.9'	20	19	75°43.4'
20	20	90°45.9'	20	21	105°48.3'	20	22	120°50.8'	20	23	135°53.3'
21	0	150°55.7'	21	1	165°58.2'	21	2	181°00.7'	21	3	196°03.1'
21	4	211°05.6'	21	5	226°08.1'	21	6	241°10.5'	21	7	256°13.0'
21	8	271°15.4'	21	9	286°17.9'	21	10	301°20.4'	21	11	316°22.8'
21	12	331°25.3'	21	13	346°27.8'	21	14	1°30.2'	21	15	16°32.7'
21	16	31°35.2'	21	17	46°37.6'	21	18	61°40.1'	21	19	76°42.5'
21	20	91°45.0'	21	21	106°47.5'	21	22	121°49.9'	21	23	136°52.4'
22	0	151°54.9'	22	1	166°57.3'	22	2	181°59.8'	22	3	197°02.3'
22	4	212°04.7'	22	5	227°07.2'	22	6	242°09.7'	22	7	257°12.1'
22	8	272°14.6'	22	9	287°17.0'	22	10	302°19.5'	22	11	317°22.0'
22	12	332°24.4'	22	13	347°26.9'	22	14	2°29.4'	22	15	17°31.8'
22	16	32°34.3'	22	17	47°36.8'	22	18	62°39.2'	22	19	77°41.7'
22	20	92°44.2'	22	21	107°46.6'	22	22	122°49.1'	22	23	137°51.5'
23	0	152°54.0'	23	1	167°56.5'	23	2	182°58.9'	23	3	198°01.4'
23	4	213°03.9'	23	5	228°06.3'	23	6	243°08.8'	23	7	258°11.3'
23	8	273°13.7'	23	9	288°16.2'	23	10	303°18.6'	23	11	318°21.1'
23	12	333°23.6'	23	13	348°26.0'	23	14	3°28.5'	23	15	18°31.0'
23	16	33°33.4'	23	17	48°35.9'	23	18	63°38.4'	23	19	78°40.8'
23	20	93°43.3'	23	21	108°45.8'	23	22	123°48.2'	23	23	138°50.7'
24	0	153°53.1'	24	1	168°55.6'	24	2	183°58.1'	24	3	199°00.5'
24	4	214°03.0'	24	5	229°05.5'	24	6	244°07.9'	24	7	259°10.4'
24	8	274°12.9'	24	9	289°15.3'	24	10	304°17.8'	24	11	319°20.3'
24	12	334°22.7'	24	13	349°25.2'	24	14	4°27.6'	24	15	19°30.1'
24	16	34°32.6'	24	17	49°35.0'	24	18	64°37.5'	24	19	79°40.0'
24	20	94°42.4'	24	21	109°44.9'	24	22	124°47.4'	24	23	139°49.8'
25	0	154°52.3'	25	1	169°54.7'	25	2	184°57.2'	25	3	199°59.7'
25	4	215°02.1'	25	5	230°04.6'	25	6	245°07.1'	25	7	260°09.5'
25	8	275°12.0'	25	9	290°14.5'	25	10	305°16.9'	25	11	320°19.4'
25	12	335°21.9'	25	13	350°24.3'	25	14	5°26.8'	25	15	20°29.2'
25	16	35°31.7'	25	17	50°34.2'	25	18	65°36.6'	25	19	80°39.1'
25	20	95°41.6'	25	21	110°44.0'	25	22	125°46.5'	25	23	140°49.0'
26	0	155°51.4'	26	1	170°53.9'	26	2	185°56.4'	26	3	200°58.8'
26	4	216°01.3'	26	5	231°03.7'	26	6	246°06.2'	26	7	261°08.7'
26	8	276°11.1'	26	9	291°13.6'	26	10	306°16.1'	26	11	321°18.5'
26	12	336°21.0'	26	13	351°23.5'	26	14	6°25.9'	26	15	21°28.4'
26	16	36°30.9'	26	17	51°33.3'	26	18	66°35.8'	26	19	81°38.2'
26	20	96°40.7'	26	21	111°43.2'	26	22	126°45.6'	26	23	141°48.1'
27	0	156°50.6'	27	1	171°53.0'	27	2	186°55.5'	27	3	201°58.0'
27	4	217°00.4'	27	5	232°02.9'	27	6	247°05.3'	27	7	262°07.8'
27	8	277°10.3'	27	9	292°12.7'	27	10	307°15.2'	27	11	322°17.7'
27	12	337°20.1'	27	13	352°22.6'	27	14	7°25.1'	27	15	22°27.5'
27	16	37°30.0'	27	17	52°32.5'	27	18	67°34.9'	27	19	82°37.4'
27	20	97°39.8'	27	21	112°42.3'	27	22	127°44.8'	27	23	142°47.2'
28	0	157°49.7'	28	1	172°52.2'	28	2	187°54.6'	28	3	202°57.1'
28	4	217°59.6'	28	5	233°02.0'	28	6	248°04.5'	28	7	263°07.0'
28	8	278°09.4'	28	9	293°11.9'	28	10	308°14.3'	28	11	323°16.8'
28	12	338°19.3'	28	13	353°21.7'	28	14	8°24.2'	28	15	23°26.7'
28	16	38°29.1'	28	17	53°31.6'	28	18	68°34.1'	28	19	83°36.5'
28	20	98°39.0'	28	21	113°41.4'	28	22	128°43.9'	28	23	143°46.4'

Table 11.3: Aries — GHA of the First Point of Aries, March 2026 (UT)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
1	0	158°48.8'	1	1	173°51.3'	1	2	188°53.8'	1	3	203°56.2'
1	4	218°58.7'	1	5	234°01.2'	1	6	249°03.6'	1	7	264°06.1'
1	8	279°08.6'	1	9	294°11.0'	1	10	309°13.5'	1	11	324°15.9'
1	12	339°18.4'	1	13	354°20.9'	1	14	9°23.3'	1	15	24°25.8'
1	16	39°28.3'	1	17	54°30.7'	1	18	69°33.2'	1	19	84°35.7'
1	20	99°38.1'	1	21	114°40.6'	1	22	129°43.1'	1	23	144°45.5'
2	0	159°48.0'	2	1	174°50.4'	2	2	189°52.9'	2	3	204°55.4'
2	4	219°57.8'	2	5	235°00.3'	2	6	250°02.8'	2	7	265°05.2'
2	8	280°07.7'	2	9	295°10.2'	2	10	310°12.6'	2	11	325°15.1'
2	12	340°17.5'	2	13	355°20.0'	2	14	10°22.5'	2	15	25°24.9'
2	16	40°27.4'	2	17	55°29.9'	2	18	70°32.3'	2	19	85°34.8'
2	20	100°37.3'	2	21	115°39.7'	2	22	130°42.2'	2	23	145°44.7'
3	0	160°47.1'	3	1	175°49.6'	3	2	190°52.0'	3	3	205°54.5'
3	4	220°57.0'	3	5	235°59.4'	3	6	251°01.9'	3	7	266°04.4'
3	8	281°06.8'	3	9	296°09.3'	3	10	311°11.8'	3	11	326°14.2'
3	12	341°16.7'	3	13	356°19.2'	3	14	11°21.6'	3	15	26°24.1'
3	16	41°26.5'	3	17	56°29.0'	3	18	71°31.5'	3	19	86°33.9'
3	20	101°36.4'	3	21	116°38.9'	3	22	131°41.3'	3	23	146°43.8'
4	0	161°46.3'	4	1	176°48.7'	4	2	191°51.2'	4	3	206°53.6'
4	4	221°56.1'	4	5	236°58.6'	4	6	252°01.0'	4	7	267°03.5'
4	8	282°06.0'	4	9	297°08.4'	4	10	312°10.9'	4	11	327°13.4'
4	12	342°15.8'	4	13	357°18.3'	4	14	12°20.8'	4	15	27°23.2'
4	16	42°25.7'	4	17	57°28.1'	4	18	72°30.6'	4	19	87°33.1'
4	20	102°35.5'	4	21	117°38.0'	4	22	132°40.5'	4	23	147°42.9'
5	0	162°45.4'	5	1	177°47.9'	5	2	192°50.3'	5	3	207°52.8'
5	4	222°55.3'	5	5	237°57.7'	5	6	253°00.2'	5	7	268°02.6'
5	8	283°05.1'	5	9	298°07.6'	5	10	313°10.0'	5	11	328°12.5'
5	12	343°15.0'	5	13	358°17.4'	5	14	13°19.9'	5	15	28°22.4'
5	16	43°24.8'	5	17	58°27.3'	5	18	73°29.8'	5	19	88°32.2'
5	20	103°34.7'	5	21	118°37.1'	5	22	133°39.6'	5	23	148°42.1'
6	0	163°44.5'	6	1	178°47.0'	6	2	193°49.5'	6	3	208°51.9'
6	4	223°54.4'	6	5	238°56.9'	6	6	253°59.3'	6	7	269°01.8'
6	8	284°04.2'	6	9	299°06.7'	6	10	314°09.2'	6	11	329°11.6'
6	12	344°14.1'	6	13	359°16.6'	6	14	14°19.0'	6	15	29°21.5'
6	16	44°24.0'	6	17	59°26.4'	6	18	74°28.9'	6	19	89°31.4'
6	20	104°33.8'	6	21	119°36.3'	6	22	134°38.7'	6	23	149°41.2'
7	0	164°43.7'	7	1	179°46.1'	7	2	194°48.6'	7	3	209°51.1'
7	4	224°53.5'	7	5	239°56.0'	7	6	254°58.5'	7	7	270°00.9'
7	8	285°03.4'	7	9	300°05.9'	7	10	315°08.3'	7	11	330°10.8'
7	12	345°13.2'	7	13	0°15.7'	7	14	15°18.2'	7	15	30°20.6'
7	16	45°23.1'	7	17	60°25.6'	7	18	75°28.0'	7	19	90°30.5'
7	20	105°33.0'	7	21	120°35.4'	7	22	135°37.9'	7	23	150°40.3'
8	0	165°42.8'	8	1	180°45.3'	8	2	195°47.7'	8	3	210°50.2'
8	4	225°52.7'	8	5	240°55.1'	8	6	255°57.6'	8	7	271°00.1'
8	8	286°02.5'	8	9	301°05.0'	8	10	316°07.5'	8	11	331°09.9'
8	12	346°12.4'	8	13	1°14.8'	8	14	16°17.3'	8	15	31°19.8'
8	16	46°22.2'	8	17	61°24.7'	8	18	76°27.2'	8	19	91°29.6'
8	20	106°32.1'	8	21	121°34.6'	8	22	136°37.0'	8	23	151°39.5'
9	0	166°42.0'	9	1	181°44.4'	9	2	196°46.9'	9	3	211°49.3'
9	4	226°51.8'	9	5	241°54.3'	9	6	256°56.7'	9	7	271°59.2'
9	8	287°01.7'	9	9	302°04.1'	9	10	317°06.6'	9	11	332°09.1'
9	12	347°11.5'	9	13	2°14.0'	9	14	17°16.4'	9	15	32°18.9'
9	16	47°21.4'	9	17	62°23.8'	9	18	77°26.3'	9	19	92°28.8'
9	20	107°31.2'	9	21	122°33.7'	9	22	137°36.2'	9	23	152°38.6'
10	0	167°41.1'	10	1	182°43.6'	10	2	197°46.0'	10	3	212°48.5'
10	4	227°50.9'	10	5	242°53.4'	10	6	257°55.9'	10	7	272°58.3'
10	8	288°00.8'	10	9	303°03.3'	10	10	318°05.7'	10	11	333°08.2'
10	12	348°10.7'	10	13	3°13.1'	10	14	18°15.6'	10	15	33°18.1'
10	16	48°20.5'	10	17	63°23.0'	10	18	78°25.4'	10	19	93°27.9'
10	20	108°30.4'	10	21	123°32.8'	10	22	138°35.3'	10	23	153°37.8'
11	0	168°40.2'	11	1	183°42.7'	11	2	198°45.2'	11	3	213°47.6'
11	4	228°50.1'	11	5	243°52.5'	11	6	258°55.0'	11	7	273°57.5'
11	8	288°59.9'	11	9	304°02.4'	11	10	319°04.9'	11	11	334°07.3'

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Aries — GHA of the First Point of Aries, March 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
11	12	349°09.8'	11	13	4°12.3'	11	14	19°14.7'	11	15	34°17.2'
11	16	49°19.7'	11	17	64°22.1'	11	18	79°24.6'	11	19	94°27.0'
11	20	109°29.5'	11	21	124°32.0'	11	22	139°34.4'	11	23	154°36.9'
12	0	169°39.4'	12	1	184°41.8'	12	2	199°44.3'	12	3	214°46.8'
12	4	229°49.2'	12	5	244°51.7'	12	6	259°54.2'	12	7	274°56.6'
12	8	289°59.1'	12	9	305°01.5'	12	10	320°04.0'	12	11	335°06.5'
12	12	350°08.9'	12	13	5°11.4'	12	14	20°13.9'	12	15	35°16.3'
12	16	50°18.8'	12	17	65°21.3'	12	18	80°23.7'	12	19	95°26.2'
12	20	110°28.7'	12	21	125°31.1'	12	22	140°33.6'	12	23	155°36.0'
13	0	170°38.5'	13	1	185°41.0'	13	2	200°43.4'	13	3	215°45.9'
13	4	230°48.4'	13	5	245°50.8'	13	6	260°53.3'	13	7	275°55.8'
13	8	290°58.2'	13	9	306°00.7'	13	10	321°03.1'	13	11	336°05.6'
13	12	351°08.1'	13	13	6°10.5'	13	14	21°13.0'	13	15	36°15.5'
13	16	51°17.9'	13	17	66°20.4'	13	18	81°22.9'	13	19	96°25.3'
13	20	111°27.8'	13	21	126°30.3'	13	22	141°32.7'	13	23	156°35.2'
14	0	171°37.6'	14	1	186°40.1'	14	2	201°42.6'	14	3	216°45.0'
14	4	231°47.5'	14	5	246°50.0'	14	6	261°52.4'	14	7	276°54.9'
14	8	291°57.4'	14	9	306°59.8'	14	10	322°02.3'	14	11	337°04.8'
14	12	352°07.2'	14	13	7°09.7'	14	14	22°12.1'	14	15	37°14.6'
14	16	52°17.1'	14	17	67°19.5'	14	18	82°22.0'	14	19	97°24.5'
14	20	112°26.9'	14	21	127°29.4'	14	22	142°31.9'	14	23	157°34.3'
15	0	172°36.8'	15	1	187°39.2'	15	2	202°41.7'	15	3	217°44.2'
15	4	232°46.6'	15	5	247°49.1'	15	6	262°51.6'	15	7	277°54.0'
15	8	292°56.5'	15	9	307°59.0'	15	10	323°01.4'	15	11	338°03.9'
15	12	353°06.4'	15	13	8°08.8'	15	14	23°11.3'	15	15	38°13.7'
15	16	53°16.2'	15	17	68°18.7'	15	18	83°21.1'	15	19	98°23.6'
15	20	113°26.1'	15	21	128°28.5'	15	22	143°31.0'	15	23	158°33.5'
16	0	173°35.9'	16	1	188°38.4'	16	2	203°40.9'	16	3	218°43.3'
16	4	233°45.8'	16	5	248°48.2'	16	6	263°50.7'	16	7	278°53.2'
16	8	293°55.6'	16	9	308°58.1'	16	10	324°00.6'	16	11	339°03.0'
16	12	354°05.5'	16	13	9°08.0'	16	14	24°10.4'	16	15	39°12.9'
16	16	54°15.3'	16	17	69°17.8'	16	18	84°20.3'	16	19	99°22.7'
16	20	114°25.2'	16	21	129°27.7'	16	22	144°30.1'	16	23	159°32.6'
17	0	174°35.1'	17	1	189°37.5'	17	2	204°40.0'	17	3	219°42.5'
17	4	234°44.9'	17	5	249°47.4'	17	6	264°49.8'	17	7	279°52.3'
17	8	294°54.8'	17	9	309°57.2'	17	10	324°59.7'	17	11	340°02.2'
17	12	355°04.6'	17	13	10°07.1'	17	14	25°09.6'	17	15	40°12.0'
17	16	55°14.5'	17	17	70°17.0'	17	18	85°19.4'	17	19	100°21.9'
17	20	115°24.3'	17	21	130°26.8'	17	22	145°29.3'	17	23	160°31.7'
18	0	175°34.2'	18	1	190°36.7'	18	2	205°39.1'	18	3	220°41.6'
18	4	235°44.1'	18	5	250°46.5'	18	6	265°49.0'	18	7	280°51.4'
18	8	295°53.9'	18	9	310°56.4'	18	10	325°58.8'	18	11	341°01.3'
18	12	356°03.8'	18	13	11°06.2'	18	14	26°08.7'	18	15	41°11.2'
18	16	56°13.6'	18	17	71°16.1'	18	18	86°18.6'	18	19	101°21.0'
18	20	116°23.5'	18	21	131°25.9'	18	22	146°28.4'	18	23	161°30.9'
19	0	176°33.3'	19	1	191°35.8'	19	2	206°38.3'	19	3	221°40.7'
19	4	236°43.2'	19	5	251°45.7'	19	6	266°48.1'	19	7	281°50.6'
19	8	296°53.1'	19	9	311°55.5'	19	10	326°58.0'	19	11	342°00.4'
19	12	357°02.9'	19	13	12°05.4'	19	14	27°07.8'	19	15	42°10.3'
19	16	57°12.8'	19	17	72°15.2'	19	18	87°17.7'	19	19	102°20.2'
19	20	117°22.6'	19	21	132°25.1'	19	22	147°27.6'	19	23	162°30.0'
20	0	177°32.5'	20	1	192°34.9'	20	2	207°37.4'	20	3	222°39.9'
20	4	237°42.3'	20	5	252°44.8'	20	6	267°47.3'	20	7	282°49.7'
20	8	297°52.2'	20	9	312°54.7'	20	10	327°57.1'	20	11	342°59.6'
20	12	358°02.0'	20	13	13°04.5'	20	14	28°07.0'	20	15	43°09.4'
20	16	58°11.9'	20	17	73°14.4'	20	18	88°16.8'	20	19	103°19.3'
20	20	118°21.8'	20	21	133°24.2'	20	22	148°26.7'	20	23	163°29.2'
21	0	178°31.6'	21	1	193°34.1'	21	2	208°36.5'	21	3	223°39.0'
21	4	238°41.5'	21	5	253°43.9'	21	6	268°46.4'	21	7	283°48.9'
21	8	298°51.3'	21	9	313°53.8'	21	10	328°56.3'	21	11	343°58.7'
21	12	359°01.2'	21	13	14°03.7'	21	14	29°06.1'	21	15	44°08.6'
21	16	59°11.0'	21	17	74°13.5'	21	18	89°16.0'	21	19	104°18.4'
21	20	119°20.9'	21	21	134°23.4'	21	22	149°25.8'	21	23	164°28.3'
22	0	179°30.8'	22	1	194°33.2'	22	2	209°35.7'	22	3	224°38.1'

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Aries — GHA of the First Point of Aries, March 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
22	4	239°40.6'	22	5	254°43.1'	22	6	269°45.5'	22	7	284°48.0'
22	8	299°50.5'	22	9	314°52.9'	22	10	329°55.4'	22	11	344°57.9'
22	12	0°00.3'	22	13	15°02.8'	22	14	30°05.3'	22	15	45°07.7'
22	16	60°10.2'	22	17	75°12.6'	22	18	90°15.1'	22	19	105°17.6'
22	20	120°20.0'	22	21	135°22.5'	22	22	150°25.0'	22	23	165°27.4'
23	0	180°29.9'	23	1	195°32.4'	23	2	210°34.8'	23	3	225°37.3'
23	4	240°39.8'	23	5	255°42.2'	23	6	270°44.7'	23	7	285°47.1'
23	8	300°49.6'	23	9	315°52.1'	23	10	330°54.5'	23	11	345°57.0'
23	12	0°59.5'	23	13	16°01.9'	23	14	31°04.4'	23	15	46°06.9'
23	16	61°09.3'	23	17	76°11.8'	23	18	91°14.2'	23	19	106°16.7'
23	20	121°19.2'	23	21	136°21.6'	23	22	151°24.1'	23	23	166°26.6'
24	0	181°29.0'	24	1	196°31.5'	24	2	211°34.0'	24	3	226°36.4'
24	4	241°38.9'	24	5	256°41.4'	24	6	271°43.8'	24	7	286°46.3'
24	8	301°48.7'	24	9	316°51.2'	24	10	331°53.7'	24	11	346°56.1'
24	12	1°58.6'	24	13	17°01.1'	24	14	32°03.5'	24	15	47°06.0'
24	16	62°08.5'	24	17	77°10.9'	24	18	92°13.4'	24	19	107°15.9'
24	20	122°18.3'	24	21	137°20.8'	24	22	152°23.2'	24	23	167°25.7'
25	0	182°28.2'	25	1	197°30.6'	25	2	212°33.1'	25	3	227°35.6'
25	4	242°38.0'	25	5	257°40.5'	25	6	272°43.0'	25	7	287°45.4'
25	8	302°47.9'	25	9	317°50.4'	25	10	332°52.8'	25	11	347°55.3'
25	12	2°57.7'	25	13	18°00.2'	25	14	33°02.7'	25	15	48°05.1'
25	16	63°07.6'	25	17	78°10.1'	25	18	93°12.5'	25	19	108°15.0'
25	20	123°17.5'	25	21	138°19.9'	25	22	153°22.4'	25	23	168°24.8'
26	0	183°27.3'	26	1	198°29.8'	26	2	213°32.2'	26	3	228°34.7'
26	4	243°37.2'	26	5	258°39.6'	26	6	273°42.1'	26	7	288°44.6'
26	8	303°47.0'	26	9	318°49.5'	26	10	333°52.0'	26	11	348°54.4'
26	12	3°56.9'	26	13	18°59.3'	26	14	34°01.8'	26	15	49°04.3'
26	16	64°06.7'	26	17	79°09.2'	26	18	94°11.7'	26	19	109°14.1'
26	20	124°16.6'	26	21	139°19.1'	26	22	154°21.5'	26	23	169°24.0'
27	0	184°26.5'	27	1	199°28.9'	27	2	214°31.4'	27	3	229°33.8'
27	4	244°36.3'	27	5	259°38.8'	27	6	274°41.2'	27	7	289°43.7'
27	8	304°46.2'	27	9	319°48.6'	27	10	334°51.1'	27	11	349°53.6'
27	12	4°56.0'	27	13	19°58.5'	27	14	35°00.9'	27	15	50°03.4'
27	16	65°05.9'	27	17	80°08.3'	27	18	95°10.8'	27	19	110°13.3'
27	20	125°15.7'	27	21	140°18.2'	27	22	155°20.7'	27	23	170°23.1'
28	0	185°25.6'	28	1	200°28.1'	28	2	215°30.5'	28	3	230°33.0'
28	4	245°35.4'	28	5	260°37.9'	28	6	275°40.4'	28	7	290°42.8'
28	8	305°45.3'	28	9	320°47.8'	28	10	335°50.2'	28	11	350°52.7'
28	12	5°55.2'	28	13	20°57.6'	28	14	36°00.1'	28	15	51°02.6'
28	16	66°05.0'	28	17	81°07.5'	28	18	96°09.9'	28	19	111°12.4'
28	20	126°14.9'	28	21	141°17.3'	28	22	156°19.8'	28	23	171°22.3'
29	0	186°24.7'	29	1	201°27.2'	29	2	216°29.7'	29	3	231°32.1'
29	4	246°34.6'	29	5	261°37.0'	29	6	276°39.5'	29	7	291°42.0'
29	8	306°44.4'	29	9	321°46.9'	29	10	336°49.4'	29	11	351°51.8'
29	12	6°54.3'	29	13	21°56.8'	29	14	36°59.2'	29	15	52°01.7'
29	16	67°04.2'	29	17	82°06.6'	29	18	97°09.1'	29	19	112°11.5'
29	20	127°14.0'	29	21	142°16.5'	29	22	157°18.9'	29	23	172°21.4'
30	0	187°23.9'	30	1	202°26.3'	30	2	217°28.8'	30	3	232°31.3'
30	4	247°33.7'	30	5	262°36.2'	30	6	277°38.7'	30	7	292°41.1'
30	8	307°43.6'	30	9	322°46.0'	30	10	337°48.5'	30	11	352°51.0'
30	12	7°53.4'	30	13	22°55.9'	30	14	37°58.4'	30	15	53°00.8'
30	16	68°03.3'	30	17	83°05.8'	30	18	98°08.2'	30	19	113°10.7'
30	20	128°13.1'	30	21	143°15.6'	30	22	158°18.1'	30	23	173°20.5'
31	0	188°23.0'	31	1	203°25.5'	31	2	218°27.9'	31	3	233°30.4'
31	4	248°32.9'	31	5	263°35.3'	31	6	278°37.8'	31	7	293°40.3'
31	8	308°42.7'	31	9	323°45.2'	31	10	338°47.6'	31	11	353°50.1'
31	12	8°52.6'	31	13	23°55.0'	31	14	38°57.5'	31	15	54°00.0'
31	16	69°02.4'	31	17	84°04.9'	31	18	99°07.4'	31	19	114°09.8'
31	20	129°12.3'	31	21	144°14.8'	31	22	159°17.2'	31	23	174°19.7'

Table 11.4: Aries — GHA of the First Point of Aries, April 2026 (UT)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
1	0	189°22.1'	1	1	204°24.6'	1	2	219°27.1'	1	3	234°29.5'
1	4	249°32.0'	1	5	264°34.5'	1	6	279°36.9'	1	7	294°39.4'
1	8	309°41.9'	1	9	324°44.3'	1	10	339°46.8'	1	11	354°49.3'
1	12	9°51.7'	1	13	24°54.2'	1	14	39°56.6'	1	15	54°59.1'
1	16	70°01.6'	1	17	85°04.0'	1	18	100°06.5'	1	19	115°09.0'
1	20	130°11.4'	1	21	145°13.9'	1	22	160°16.4'	1	23	175°18.8'
2	0	190°21.3'	2	1	205°23.7'	2	2	220°26.2'	2	3	235°28.7'
2	4	250°31.1'	2	5	265°33.6'	2	6	280°36.1'	2	7	295°38.5'
2	8	310°41.0'	2	9	325°43.5'	2	10	340°45.9'	2	11	355°48.4'
2	12	10°50.9'	2	13	25°53.3'	2	14	40°55.8'	2	15	55°58.2'
2	16	71°00.7'	2	17	86°03.2'	2	18	101°05.6'	2	19	116°08.1'
2	20	131°10.6'	2	21	146°13.0'	2	22	161°15.5'	2	23	176°18.0'
3	0	191°20.4'	3	1	206°22.9'	3	2	221°25.4'	3	3	236°27.8'
3	4	251°30.3'	3	5	266°32.7'	3	6	281°35.2'	3	7	296°37.7'
3	8	311°40.1'	3	9	326°42.6'	3	10	341°45.1'	3	11	356°47.5'
3	12	11°50.0'	3	13	26°52.5'	3	14	41°54.9'	3	15	56°57.4'
3	16	71°59.8'	3	17	87°02.3'	3	18	102°04.8'	3	19	117°07.2'
3	20	132°09.7'	3	21	147°12.2'	3	22	162°14.6'	3	23	177°17.1'
4	0	192°19.6'	4	1	207°22.0'	4	2	222°24.5'	4	3	237°27.0'
4	4	252°29.4'	4	5	267°31.9'	4	6	282°34.3'	4	7	297°36.8'
4	8	312°39.3'	4	9	327°41.7'	4	10	342°44.2'	4	11	357°46.7'
4	12	12°49.1'	4	13	27°51.6'	4	14	42°54.1'	4	15	57°56.5'
4	16	72°59.0'	4	17	88°01.5'	4	18	103°03.9'	4	19	118°06.4'
4	20	133°08.8'	4	21	148°11.3'	4	22	163°13.8'	4	23	178°16.2'
5	0	193°18.7'	5	1	208°21.2'	5	2	223°23.6'	5	3	238°26.1'
5	4	253°28.6'	5	5	268°31.0'	5	6	283°33.5'	5	7	298°35.9'
5	8	313°38.4'	5	9	328°40.9'	5	10	343°43.3'	5	11	358°45.8'
5	12	13°48.3'	5	13	28°50.7'	5	14	43°53.2'	5	15	58°55.7'
5	16	73°58.1'	5	17	89°00.6'	5	18	104°03.1'	5	19	119°05.5'
5	20	134°08.0'	5	21	149°10.4'	5	22	164°12.9'	5	23	179°15.4'
6	0	194°17.8'	6	1	209°20.3'	6	2	224°22.8'	6	3	239°25.2'
6	4	254°27.7'	6	5	269°30.2'	6	6	284°32.6'	6	7	299°35.1'
6	8	314°37.6'	6	9	329°40.0'	6	10	344°42.5'	6	11	359°44.9'
6	12	14°47.4'	6	13	29°49.9'	6	14	44°52.3'	6	15	59°54.8'
6	16	74°57.3'	6	17	89°59.7'	6	18	105°02.2'	6	19	120°04.7'
6	20	135°07.1'	6	21	150°09.6'	6	22	165°12.0'	6	23	180°14.5'
7	0	195°17.0'	7	1	210°19.4'	7	2	225°21.9'	7	3	240°24.4'
7	4	255°26.8'	7	5	270°29.3'	7	6	285°31.8'	7	7	300°34.2'
7	8	315°36.7'	7	9	330°39.2'	7	10	345°41.6'	7	11	0°44.1'
7	12	15°46.5'	7	13	30°49.0'	7	14	45°51.5'	7	15	60°53.9'
7	16	75°56.4'	7	17	90°58.9'	7	18	106°01.3'	7	19	121°03.8'
7	20	136°06.3'	7	21	151°08.7'	7	22	166°11.2'	7	23	181°13.7'
8	0	196°16.1'	8	1	211°18.6'	8	2	226°21.0'	8	3	241°23.5'
8	4	256°26.0'	8	5	271°28.4'	8	6	286°30.9'	8	7	301°33.4'
8	8	316°35.8'	8	9	331°38.3'	8	10	346°40.8'	8	11	1°43.2'
8	12	16°45.7'	8	13	31°48.2'	8	14	46°50.6'	8	15	61°53.1'
8	16	76°55.5'	8	17	91°58.0'	8	18	107°00.5'	8	19	122°02.9'
8	20	137°05.4'	8	21	152°07.9'	8	22	167°10.3'	8	23	182°12.8'
9	0	197°15.3'	9	1	212°17.7'	9	2	227°20.2'	9	3	242°22.6'
9	4	257°25.1'	9	5	272°27.6'	9	6	287°30.0'	9	7	302°32.5'
9	8	317°35.0'	9	9	332°37.4'	9	10	347°39.9'	9	11	2°42.4'
9	12	17°44.8'	9	13	32°47.3'	9	14	47°49.8'	9	15	62°52.2'
9	16	77°54.7'	9	17	92°57.1'	9	18	107°59.6'	9	19	123°02.1'
9	20	138°04.5'	9	21	153°07.0'	9	22	168°09.5'	9	23	183°11.9'
10	0	198°14.4'	10	1	213°16.9'	10	2	228°19.3'	10	3	243°21.8'
10	4	258°24.3'	10	5	273°26.7'	10	6	288°29.2'	10	7	303°31.6'
10	8	318°34.1'	10	9	333°36.6'	10	10	348°39.0'	10	11	3°41.5'
10	12	18°44.0'	10	13	33°46.4'	10	14	48°48.9'	10	15	63°51.4'
10	16	78°53.8'	10	17	93°56.3'	10	18	108°58.7'	10	19	124°01.2'
10	20	139°03.7'	10	21	154°06.1'	10	22	169°08.6'	10	23	184°11.1'
11	0	199°13.5'	11	1	214°16.0'	11	2	229°18.5'	11	3	244°20.9'
11	4	259°23.4'	11	5	274°25.9'	11	6	289°28.3'	11	7	304°30.8'
11	8	319°33.2'	11	9	334°35.7'	11	10	349°38.2'	11	11	4°40.6'

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Aries — GHA of the First Point of Aries, April 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
11	12	19°43.1'	11	13	34°45.6'	11	14	49°48.0'	11	15	64°50.5'
11	16	79°53.0'	11	17	94°55.4'	11	18	109°57.9'	11	19	125°00.4'
11	20	140°02.8'	11	21	155°05.3'	11	22	170°07.7'	11	23	185°10.2'
12	0	200°12.7'	12	1	215°15.1'	12	2	230°17.6'	12	3	245°20.1'
12	4	260°22.5'	12	5	275°25.0'	12	6	290°27.5'	12	7	305°29.9'
12	8	320°32.4'	12	9	335°34.8'	12	10	350°37.3'	12	11	5°39.8'
12	12	20°42.2'	12	13	35°44.7'	12	14	50°47.2'	12	15	65°49.6'
12	16	80°52.1'	12	17	95°54.6'	12	18	110°57.0'	12	19	125°59.5'
12	20	141°02.0'	12	21	156°04.4'	12	22	171°06.9'	12	23	186°09.3'
13	0	201°11.8'	13	1	216°14.3'	13	2	231°16.7'	13	3	246°19.2'
13	4	261°21.7'	13	5	276°24.1'	13	6	291°26.6'	13	7	306°29.1'
13	8	321°31.5'	13	9	336°34.0'	13	10	351°36.5'	13	11	6°38.9'
13	12	21°41.4'	13	13	36°43.8'	13	14	51°46.3'	13	15	66°48.8'
13	16	81°51.2'	13	17	96°53.7'	13	18	111°56.2'	13	19	126°58.6'
13	20	142°01.1'	13	21	157°03.6'	13	22	172°06.0'	13	23	187°08.5'
14	0	202°10.9'	14	1	217°13.4'	14	2	232°15.9'	14	3	247°18.3'
14	4	262°20.8'	14	5	277°23.3'	14	6	292°25.7'	14	7	307°28.2'
14	8	322°30.7'	14	9	337°33.1'	14	10	352°35.6'	14	11	7°38.1'
14	12	22°40.5'	14	13	37°43.0'	14	14	52°45.4'	14	15	67°47.9'
14	16	82°50.4'	14	17	97°52.8'	14	18	112°55.3'	14	19	127°57.8'
14	20	143°00.2'	14	21	158°02.7'	14	22	173°05.2'	14	23	188°07.6'
15	0	203°10.1'	15	1	218°12.6'	15	2	233°15.0'	15	3	248°17.5'
15	4	263°19.9'	15	5	278°22.4'	15	6	293°24.9'	15	7	308°27.3'
15	8	323°29.8'	15	9	338°32.3'	15	10	353°34.7'	15	11	8°37.2'
15	12	23°39.7'	15	13	38°42.1'	15	14	53°44.6'	15	15	68°47.1'
15	16	83°49.5'	15	17	98°52.0'	15	18	113°54.4'	15	19	128°56.9'
15	20	143°59.4'	15	21	159°01.8'	15	22	174°04.3'	15	23	189°06.8'
16	0	204°09.2'	16	1	219°11.7'	16	2	234°14.2'	16	3	249°16.6'
16	4	264°19.1'	16	5	279°21.5'	16	6	294°24.0'	16	7	309°26.5'
16	8	324°28.9'	16	9	339°31.4'	16	10	354°33.9'	16	11	9°36.3'
16	12	24°38.8'	16	13	39°41.3'	16	14	54°43.7'	16	15	69°46.2'
16	16	84°48.7'	16	17	99°51.1'	16	18	114°53.6'	16	19	129°56.0'
16	20	144°58.5'	16	21	160°01.0'	16	22	175°03.4'	16	23	190°05.9'
17	0	205°08.4'	17	1	220°10.8'	17	2	235°13.3'	17	3	250°15.8'
17	4	265°18.2'	17	5	280°20.7'	17	6	295°23.2'	17	7	310°25.6'
17	8	325°28.1'	17	9	340°30.5'	17	10	355°33.0'	17	11	10°35.5'
17	12	25°37.9'	17	13	40°40.4'	17	14	55°42.9'	17	15	70°45.3'
17	16	85°47.8'	17	17	100°50.3'	17	18	115°52.7'	17	19	130°55.2'
17	20	145°57.6'	17	21	161°00.1'	17	22	176°02.6'	17	23	191°05.0'
18	0	206°07.5'	18	1	221°10.0'	18	2	236°12.4'	18	3	251°14.9'
18	4	266°17.4'	18	5	281°19.8'	18	6	296°22.3'	18	7	311°24.8'
18	8	326°27.2'	18	9	341°29.7'	18	10	356°32.1'	18	11	11°34.6'
18	12	26°37.1'	18	13	41°39.5'	18	14	56°42.0'	18	15	71°44.5'
18	16	86°46.9'	18	17	101°49.4'	18	18	116°51.9'	18	19	131°54.3'
18	20	146°56.8'	18	21	161°59.3'	18	22	177°01.7'	18	23	192°04.2'
19	0	207°06.6'	19	1	222°09.1'	19	2	237°11.6'	19	3	252°14.0'
19	4	267°16.5'	19	5	282°19.0'	19	6	297°21.4'	19	7	312°23.9'
19	8	327°26.4'	19	9	342°28.8'	19	10	357°31.3'	19	11	12°33.7'
19	12	27°36.2'	19	13	42°38.7'	19	14	57°41.1'	19	15	72°43.6'
19	16	87°46.1'	19	17	102°48.5'	19	18	117°51.0'	19	19	132°53.5'
19	20	147°55.9'	19	21	162°58.4'	19	22	178°00.9'	19	23	193°03.3'
20	0	208°05.8'	20	1	223°08.2'	20	2	238°10.7'	20	3	253°13.2'
20	4	268°15.6'	20	5	283°18.1'	20	6	298°20.6'	20	7	313°23.0'
20	8	328°25.5'	20	9	343°28.0'	20	10	358°30.4'	20	11	13°32.9'
20	12	28°35.4'	20	13	43°37.8'	20	14	58°40.3'	20	15	73°42.7'
20	16	88°45.2'	20	17	103°47.7'	20	18	118°50.1'	20	19	133°52.6'
20	20	148°55.1'	20	21	163°57.5'	20	22	179°00.0'	20	23	194°02.5'
21	0	209°04.9'	21	1	224°07.4'	21	2	239°09.8'	21	3	254°12.3'
21	4	269°14.8'	21	5	284°17.2'	21	6	299°19.7'	21	7	314°22.2'
21	8	329°24.6'	21	9	344°27.1'	21	10	359°29.6'	21	11	14°32.0'
21	12	29°34.5'	21	13	44°37.0'	21	14	59°39.4'	21	15	74°41.9'
21	16	89°44.3'	21	17	104°46.8'	21	18	119°49.3'	21	19	134°51.7'
21	20	149°54.2'	21	21	164°56.7'	21	22	179°59.1'	21	23	195°01.6'
22	0	210°04.1'	22	1	225°06.5'	22	2	240°09.0'	22	3	255°11.5'

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Aries — GHA of the First Point of Aries, April 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
22	4	270°13.9'	22	5	285°16.4'	22	6	300°18.8'	22	7	315°21.3'
22	8	330°23.8'	22	9	345°26.2'	22	10	0°28.7'	22	11	15°31.2'
22	12	30°33.6'	22	13	45°36.1'	22	14	60°38.6'	22	15	75°41.0'
22	16	90°43.5'	22	17	105°46.0'	22	18	120°48.4'	22	19	135°50.9'
22	20	150°53.3'	22	21	165°55.8'	22	22	180°58.3'	22	23	196°00.7'
23	0	211°03.2'	23	1	226°05.7'	23	2	241°08.1'	23	3	256°10.6'
23	4	271°13.1'	23	5	286°15.5'	23	6	301°18.0'	23	7	316°20.4'
23	8	331°22.9'	23	9	346°25.4'	23	10	1°27.8'	23	11	16°30.3'
23	12	31°32.8'	23	13	46°35.2'	23	14	61°37.7'	23	15	76°40.2'
23	16	91°42.6'	23	17	106°45.1'	23	18	121°47.6'	23	19	136°50.0'
23	20	151°52.5'	23	21	166°54.9'	23	22	181°57.4'	23	23	196°59.9'
24	0	212°02.3'	24	1	227°04.8'	24	2	242°07.3'	24	3	257°09.7'
24	4	272°12.2'	24	5	287°14.7'	24	6	302°17.1'	24	7	317°19.6'
24	8	332°22.1'	24	9	347°24.5'	24	10	2°27.0'	24	11	17°29.4'
24	12	32°31.9'	24	13	47°34.4'	24	14	62°36.8'	24	15	77°39.3'
24	16	92°41.8'	24	17	107°44.2'	24	18	122°46.7'	24	19	137°49.2'
24	20	152°51.6'	24	21	167°54.1'	24	22	182°56.5'	24	23	197°59.0'
25	0	213°01.5'	25	1	228°03.9'	25	2	243°06.4'	25	3	258°08.9'
25	4	273°11.3'	25	5	288°13.8'	25	6	303°16.3'	25	7	318°18.7'
25	8	333°21.2'	25	9	348°23.7'	25	10	3°26.1'	25	11	18°28.6'
25	12	33°31.0'	25	13	48°33.5'	25	14	63°36.0'	25	15	78°38.4'
25	16	93°40.9'	25	17	108°43.4'	25	18	123°45.8'	25	19	138°48.3'
25	20	153°50.8'	25	21	168°53.2'	25	22	183°55.7'	25	23	198°58.2'
26	0	214°00.6'	26	1	229°03.1'	26	2	244°05.5'	26	3	259°08.0'
26	4	274°10.5'	26	5	289°12.9'	26	6	304°15.4'	26	7	319°17.9'
26	8	334°20.3'	26	9	349°22.8'	26	10	4°25.3'	26	11	19°27.7'
26	12	34°30.2'	26	13	49°32.6'	26	14	64°35.1'	26	15	79°37.6'
26	16	94°40.0'	26	17	109°42.5'	26	18	124°45.0'	26	19	139°47.4'
26	20	154°49.9'	26	21	169°52.4'	26	22	184°54.8'	26	23	199°57.3'
27	0	214°59.8'	27	1	230°02.2'	27	2	245°04.7'	27	3	260°07.1'
27	4	275°09.6'	27	5	290°12.1'	27	6	305°14.5'	27	7	320°17.0'
27	8	335°19.5'	27	9	350°21.9'	27	10	5°24.4'	27	11	20°26.9'
27	12	35°29.3'	27	13	50°31.8'	27	14	65°34.3'	27	15	80°36.7'
27	16	95°39.2'	27	17	110°41.6'	27	18	125°44.1'	27	19	140°46.6'
27	20	155°49.0'	27	21	170°51.5'	27	22	185°54.0'	27	23	200°56.4'
28	0	215°58.9'	28	1	231°01.4'	28	2	246°03.8'	28	3	261°06.3'
28	4	276°08.8'	28	5	291°11.2'	28	6	306°13.7'	28	7	321°16.1'
28	8	336°18.6'	28	9	351°21.1'	28	10	6°23.5'	28	11	21°26.0'
28	12	36°28.5'	28	13	51°30.9'	28	14	66°33.4'	28	15	81°35.9'
28	16	96°38.3'	28	17	111°40.8'	28	18	126°43.2'	28	19	141°45.7'
28	20	156°48.2'	28	21	171°50.6'	28	22	186°53.1'	28	23	201°55.6'
29	0	216°58.0'	29	1	232°00.5'	29	2	247°03.0'	29	3	262°05.4'
29	4	277°07.9'	29	5	292°10.4'	29	6	307°12.8'	29	7	322°15.3'
29	8	337°17.7'	29	9	352°20.2'	29	10	7°22.7'	29	11	22°25.1'
29	12	37°27.6'	29	13	52°30.1'	29	14	67°32.5'	29	15	82°35.0'
29	16	97°37.5'	29	17	112°39.9'	29	18	127°42.4'	29	19	142°44.9'
29	20	157°47.3'	29	21	172°49.8'	29	22	187°52.2'	29	23	202°54.7'
30	0	217°57.2'	30	1	232°59.6'	30	2	248°02.1'	30	3	263°04.6'
30	4	278°07.0'	30	5	293°09.5'	30	6	308°12.0'	30	7	323°14.4'
30	8	338°16.9'	30	9	353°19.3'	30	10	8°21.8'	30	11	23°24.3'
30	12	38°26.7'	30	13	53°29.2'	30	14	68°31.7'	30	15	83°34.1'
30	16	98°36.6'	30	17	113°39.1'	30	18	128°41.5'	30	19	143°44.0'
30	20	158°46.5'	30	21	173°48.9'	30	22	188°51.4'	30	23	203°53.8'

Table 11.5: Aries — GHA of the First Point of Aries, May 2026 (UT)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
1	0	218°56.3'	1	1	233°58.8'	1	2	249°01.2'	1	3	264°03.7'
1	4	279°06.2'	1	5	294°08.6'	1	6	309°11.1'	1	7	324°13.6'
1	8	339°16.0'	1	9	354°18.5'	1	10	9°21.0'	1	11	24°23.4'
1	12	39°25.9'	1	13	54°28.3'	1	14	69°30.8'	1	15	84°33.3'
1	16	99°35.7'	1	17	114°38.2'	1	18	129°40.7'	1	19	144°43.1'

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Aries — GHA of the First Point of Aries, May 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
1	20	159°45.6'	1	21	174°48.1'	1	22	189°50.5'	1	23	204°53.0'
2	0	219°55.4'	2	1	234°57.9'	2	2	250°00.4'	2	3	265°02.8'
2	4	280°05.3'	2	5	295°07.8'	2	6	310°10.2'	2	7	325°12.7'
2	8	340°15.2'	2	9	355°17.6'	2	10	10°20.1'	2	11	25°22.6'
2	12	40°25.0'	2	13	55°27.5'	2	14	70°29.9'	2	15	85°32.4'
2	16	100°34.9'	2	17	115°37.3'	2	18	130°39.8'	2	19	145°42.3'
2	20	160°44.7'	2	21	175°47.2'	2	22	190°49.7'	2	23	205°52.1'
3	0	220°54.6'	3	1	235°57.1'	3	2	250°59.5'	3	3	266°02.0'
3	4	281°04.4'	3	5	296°06.9'	3	6	311°09.4'	3	7	326°11.8'
3	8	341°14.3'	3	9	356°16.8'	3	10	11°19.2'	3	11	26°21.7'
3	12	41°24.2'	3	13	56°26.6'	3	14	71°29.1'	3	15	86°31.5'
3	16	101°34.0'	3	17	116°36.5'	3	18	131°38.9'	3	19	146°41.4'
3	20	161°43.9'	3	21	176°46.3'	3	22	191°48.8'	3	23	206°51.3'
4	0	221°53.7'	4	1	236°56.2'	4	2	251°58.7'	4	3	267°01.1'
4	4	282°03.6'	4	5	297°06.0'	4	6	312°08.5'	4	7	327°11.0'
4	8	342°13.4'	4	9	357°15.9'	4	10	12°18.4'	4	11	27°20.8'
4	12	42°23.3'	4	13	57°25.8'	4	14	72°28.2'	4	15	87°30.7'
4	16	102°33.2'	4	17	117°35.6'	4	18	132°38.1'	4	19	147°40.5'
4	20	162°43.0'	4	21	177°45.5'	4	22	192°47.9'	4	23	207°50.4'
5	0	222°52.9'	5	1	237°55.3'	5	2	252°57.8'	5	3	268°00.3'
5	4	283°02.7'	5	5	298°05.2'	5	6	313°07.7'	5	7	328°10.1'
5	8	343°12.6'	5	9	358°15.0'	5	10	13°17.5'	5	11	28°20.0'
5	12	43°22.4'	5	13	58°24.9'	5	14	73°27.4'	5	15	88°29.8'
5	16	103°32.3'	5	17	118°34.8'	5	18	133°37.2'	5	19	148°39.7'
5	20	163°42.1'	5	21	178°44.6'	5	22	193°47.1'	5	23	208°49.5'
6	0	223°52.0'	6	1	238°54.5'	6	2	253°56.9'	6	3	268°59.4'
6	4	284°01.9'	6	5	299°04.3'	6	6	314°06.8'	6	7	329°09.3'
6	8	344°11.7'	6	9	359°14.2'	6	10	14°16.6'	6	11	29°19.1'
6	12	44°21.6'	6	13	59°24.0'	6	14	74°26.5'	6	15	89°29.0'
6	16	104°31.4'	6	17	119°33.9'	6	18	134°36.4'	6	19	149°38.8'
6	20	164°41.3'	6	21	179°43.8'	6	22	194°46.2'	6	23	209°48.7'
7	0	224°51.1'	7	1	239°53.6'	7	2	254°56.1'	7	3	269°58.5'
7	4	285°01.0'	7	5	300°03.5'	7	6	315°05.9'	7	7	330°08.4'
7	8	345°10.9'	7	9	0°13.3'	7	10	15°15.8'	7	11	30°18.2'
7	12	45°20.7'	7	13	60°23.2'	7	14	75°25.6'	7	15	90°28.1'
7	16	105°30.6'	7	17	120°33.0'	7	18	135°35.5'	7	19	150°38.0'
7	20	165°40.4'	7	21	180°42.9'	7	22	195°45.4'	7	23	210°47.8'
8	0	225°50.3'	8	1	240°52.7'	8	2	255°55.2'	8	3	270°57.7'
8	4	286°00.1'	8	5	301°02.6'	8	6	316°05.1'	8	7	331°07.5'
8	8	346°10.0'	8	9	1°12.5'	8	10	16°14.9'	8	11	31°17.4'
8	12	46°19.9'	8	13	61°22.3'	8	14	76°24.8'	8	15	91°27.2'
8	16	106°29.7'	8	17	121°32.2'	8	18	136°34.6'	8	19	151°37.1'
8	20	166°39.6'	8	21	181°42.0'	8	22	196°44.5'	8	23	211°47.0'
9	0	226°49.4'	9	1	241°51.9'	9	2	256°54.3'	9	3	271°56.8'
9	4	286°59.3'	9	5	302°01.7'	9	6	317°04.2'	9	7	332°06.7'
9	8	347°09.1'	9	9	2°11.6'	9	10	17°14.1'	9	11	32°16.5'
9	12	47°19.0'	9	13	62°21.5'	9	14	77°23.9'	9	15	92°26.4'
9	16	107°28.8'	9	17	122°31.3'	9	18	137°33.8'	9	19	152°36.2'
9	20	167°38.7'	9	21	182°41.2'	9	22	197°43.6'	9	23	212°46.1'
10	0	227°48.6'	10	1	242°51.0'	10	2	257°53.5'	10	3	272°56.0'
10	4	287°58.4'	10	5	303°00.9'	10	6	318°03.3'	10	7	333°05.8'
10	8	348°08.3'	10	9	3°10.7'	10	10	18°13.2'	10	11	33°15.7'
10	12	48°18.1'	10	13	63°20.6'	10	14	78°23.1'	10	15	93°25.5'
10	16	108°28.0'	10	17	123°30.4'	10	18	138°32.9'	10	19	153°35.4'
10	20	168°37.8'	10	21	183°40.3'	10	22	198°42.8'	10	23	213°45.2'
11	0	228°47.7'	11	1	243°50.2'	11	2	258°52.6'	11	3	273°55.1'
11	4	288°57.6'	11	5	304°00.0'	11	6	319°02.5'	11	7	334°04.9'
11	8	349°07.4'	11	9	4°09.9'	11	10	19°12.3'	11	11	34°14.8'
11	12	49°17.3'	11	13	64°19.7'	11	14	79°22.2'	11	15	94°24.7'
11	16	109°27.1'	11	17	124°29.6'	11	18	139°32.1'	11	19	154°34.5'
11	20	169°37.0'	11	21	184°39.4'	11	22	199°41.9'	11	23	214°44.4'
12	0	229°46.8'	12	1	244°49.3'	12	2	259°51.8'	12	3	274°54.2'
12	4	289°56.7'	12	5	304°59.2'	12	6	320°01.6'	12	7	335°04.1'
12	8	350°06.6'	12	9	5°09.0'	12	10	20°11.5'	12	11	35°13.9'

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Aries — GHA of the First Point of Aries, May 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
12	12	50°16.4'	12	13	65°18.9'	12	14	80°21.3'	12	15	95°23.8'
12	16	110°26.3'	12	17	125°28.7'	12	18	140°31.2'	12	19	155°33.7'
12	20	170°36.1'	12	21	185°38.6'	12	22	200°41.0'	12	23	215°43.5'
13	0	230°46.0'	13	1	245°48.4'	13	2	260°50.9'	13	3	275°53.4'
13	4	290°55.8'	13	5	305°58.3'	13	6	321°00.8'	13	7	336°03.2'
13	8	351°05.7'	13	9	6°08.2'	13	10	21°10.6'	13	11	36°13.1'
13	12	51°15.5'	13	13	66°18.0'	13	14	81°20.5'	13	15	96°22.9'
13	16	111°25.4'	13	17	126°27.9'	13	18	141°30.3'	13	19	156°32.8'
13	20	171°35.3'	13	21	186°37.7'	13	22	201°40.2'	13	23	216°42.7'
14	0	231°45.1'	14	1	246°47.6'	14	2	261°50.0'	14	3	276°52.5'
14	4	291°55.0'	14	5	306°57.4'	14	6	321°59.9'	14	7	337°02.4'
14	8	352°04.8'	14	9	7°07.3'	14	10	22°09.8'	14	11	37°12.2'
14	12	52°14.7'	14	13	67°17.1'	14	14	82°19.6'	14	15	97°22.1'
14	16	112°24.5'	14	17	127°27.0'	14	18	142°29.5'	14	19	157°31.9'
14	20	172°34.4'	14	21	187°36.9'	14	22	202°39.3'	14	23	217°41.8'
15	0	232°44.3'	15	1	247°46.7'	15	2	262°49.2'	15	3	277°51.6'
15	4	292°54.1'	15	5	307°56.6'	15	6	322°59.0'	15	7	338°01.5'
15	8	353°04.0'	15	9	8°06.4'	15	10	23°08.9'	15	11	38°11.4'
15	12	53°13.8'	15	13	68°16.3'	15	14	83°18.8'	15	15	98°21.2'
15	16	113°23.7'	15	17	128°26.1'	15	18	143°28.6'	15	19	158°31.1'
15	20	173°33.5'	15	21	188°36.0'	15	22	203°38.5'	15	23	218°40.9'
16	0	233°43.4'	16	1	248°45.9'	16	2	263°48.3'	16	3	278°50.8'
16	4	293°53.2'	16	5	308°55.7'	16	6	323°58.2'	16	7	339°00.6'
16	8	354°03.1'	16	9	9°05.6'	16	10	24°08.0'	16	11	39°10.5'
16	12	54°13.0'	16	13	69°15.4'	16	14	84°17.9'	16	15	99°20.4'
16	16	114°22.8'	16	17	129°25.3'	16	18	144°27.7'	16	19	159°30.2'
16	20	174°32.7'	16	21	189°35.1'	16	22	204°37.6'	16	23	219°40.1'
17	0	234°42.5'	17	1	249°45.0'	17	2	264°47.5'	17	3	279°49.9'
17	4	294°52.4'	17	5	309°54.9'	17	6	324°57.3'	17	7	339°59.8'
17	8	355°02.2'	17	9	10°04.7'	17	10	25°07.2'	17	11	40°09.6'
17	12	55°12.1'	17	13	70°14.6'	17	14	85°17.0'	17	15	100°19.5'
17	16	115°22.0'	17	17	130°24.4'	17	18	145°26.9'	17	19	160°29.3'
17	20	175°31.8'	17	21	190°34.3'	17	22	205°36.7'	17	23	220°39.2'
18	0	235°41.7'	18	1	250°44.1'	18	2	265°46.6'	18	3	280°49.1'
18	4	295°51.5'	18	5	310°54.0'	18	6	325°56.5'	18	7	340°58.9'
18	8	356°01.4'	18	9	11°03.8'	18	10	26°06.3'	18	11	41°08.8'
18	12	56°11.2'	18	13	71°13.7'	18	14	86°16.2'	18	15	101°18.6'
18	16	116°21.1'	18	17	131°23.6'	18	18	146°26.0'	18	19	161°28.5'
18	20	176°31.0'	18	21	191°33.4'	18	22	206°35.9'	18	23	221°38.3'
19	0	236°40.8'	19	1	251°43.3'	19	2	266°45.7'	19	3	281°48.2'
19	4	296°50.7'	19	5	311°53.1'	19	6	326°55.6'	19	7	341°58.1'
19	8	357°00.5'	19	9	12°03.0'	19	10	27°05.5'	19	11	42°07.9'
19	12	57°10.4'	19	13	72°12.8'	19	14	87°15.3'	19	15	102°17.8'
19	16	117°20.2'	19	17	132°22.7'	19	18	147°25.2'	19	19	162°27.6'
19	20	177°30.1'	19	21	192°32.6'	19	22	207°35.0'	19	23	222°37.5'
20	0	237°39.9'	20	1	252°42.4'	20	2	267°44.9'	20	3	282°47.3'
20	4	297°49.8'	20	5	312°52.3'	20	6	327°54.7'	20	7	342°57.2'
20	8	357°59.7'	20	9	13°02.1'	20	10	28°04.6'	20	11	43°07.1'
20	12	58°09.5'	20	13	73°12.0'	20	14	88°14.4'	20	15	103°16.9'
20	16	118°19.4'	20	17	133°21.8'	20	18	148°24.3'	20	19	163°26.8'
20	20	178°29.2'	20	21	193°31.7'	20	22	208°34.2'	20	23	223°36.6'
21	0	238°39.1'	21	1	253°41.6'	21	2	268°44.0'	21	3	283°46.5'
21	4	298°48.9'	21	5	313°51.4'	21	6	328°53.9'	21	7	343°56.3'
21	8	358°58.8'	21	9	14°01.3'	21	10	29°03.7'	21	11	44°06.2'
21	12	59°08.7'	21	13	74°11.1'	21	14	89°13.6'	21	15	104°16.0'
21	16	119°18.5'	21	17	134°21.0'	21	18	149°23.4'	21	19	164°25.9'
21	20	179°28.4'	21	21	194°30.8'	21	22	209°33.3'	21	23	224°35.8'
22	0	239°38.2'	22	1	254°40.7'	22	2	269°43.2'	22	3	284°45.6'
22	4	299°48.1'	22	5	314°50.5'	22	6	329°53.0'	22	7	344°55.5'
22	8	359°57.9'	22	9	15°00.4'	22	10	30°02.9'	22	11	45°05.3'
22	12	60°07.8'	22	13	75°10.3'	22	14	90°12.7'	22	15	105°15.2'
22	16	120°17.7'	22	17	135°20.1'	22	18	150°22.6'	22	19	165°25.0'
22	20	180°27.5'	22	21	195°30.0'	22	22	210°32.4'	22	23	225°34.9'
23	0	240°37.4'	23	1	255°39.8'	23	2	270°42.3'	23	3	285°44.8'

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Aries — GHA of the First Point of Aries, May 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
23	4	300°47.2'	23	5	315°49.7'	23	6	330°52.1'	23	7	345°54.6'
23	8	0°57.1'	23	9	15°59.5'	23	10	31°02.0'	23	11	46°04.5'
23	12	61°06.9'	23	13	76°09.4'	23	14	91°11.9'	23	15	106°14.3'
23	16	121°16.8'	23	17	136°19.3'	23	18	151°21.7'	23	19	166°24.2'
23	20	181°26.6'	23	21	196°29.1'	23	22	211°31.6'	23	23	226°34.0'
24	0	241°36.5'	24	1	256°39.0'	24	2	271°41.4'	24	3	286°43.9'
24	4	301°46.4'	24	5	316°48.8'	24	6	331°51.3'	24	7	346°53.8'
24	8	1°56.2'	24	9	16°58.7'	24	10	32°01.1'	24	11	47°03.6'
24	12	62°06.1'	24	13	77°08.5'	24	14	92°11.0'	24	15	107°13.5'
24	16	122°15.9'	24	17	137°18.4'	24	18	152°20.9'	24	19	167°23.3'
24	20	182°25.8'	24	21	197°28.3'	24	22	212°30.7'	24	23	227°33.2'
25	0	242°35.6'	25	1	257°38.1'	25	2	272°40.6'	25	3	287°43.0'
25	4	302°45.5'	25	5	317°48.0'	25	6	332°50.4'	25	7	347°52.9'
25	8	2°55.4'	25	9	17°57.8'	25	10	33°00.3'	25	11	48°02.7'
25	12	63°05.2'	25	13	78°07.7'	25	14	93°10.1'	25	15	108°12.6'
25	16	123°15.1'	25	17	138°17.5'	25	18	153°20.0'	25	19	168°22.5'
25	20	183°24.9'	25	21	198°27.4'	25	22	213°29.9'	25	23	228°32.3'
26	0	243°34.8'	26	1	258°37.2'	26	2	273°39.7'	26	3	288°42.2'
26	4	303°44.6'	26	5	318°47.1'	26	6	333°49.6'	26	7	348°52.0'
26	8	3°54.5'	26	9	18°57.0'	26	10	33°59.4'	26	11	49°01.9'
26	12	64°04.4'	26	13	79°06.8'	26	14	94°09.3'	26	15	109°11.7'
26	16	124°14.2'	26	17	139°16.7'	26	18	154°19.1'	26	19	169°21.6'
26	20	184°24.1'	26	21	199°26.5'	26	22	214°29.0'	26	23	229°31.5'
27	0	244°33.9'	27	1	259°36.4'	27	2	274°38.8'	27	3	289°41.3'
27	4	304°43.8'	27	5	319°46.2'	27	6	334°48.7'	27	7	349°51.2'
27	8	4°53.6'	27	9	19°56.1'	27	10	34°58.6'	27	11	50°01.0'
27	12	65°03.5'	27	13	80°06.0'	27	14	95°08.4'	27	15	110°10.9'
27	16	125°13.3'	27	17	140°15.8'	27	18	155°18.3'	27	19	170°20.7'
27	20	185°23.2'	27	21	200°25.7'	27	22	215°28.1'	27	23	230°30.6'
28	0	245°33.1'	28	1	260°35.5'	28	2	275°38.0'	28	3	290°40.5'
28	4	305°42.9'	28	5	320°45.4'	28	6	335°47.8'	28	7	350°50.3'
28	8	5°52.8'	28	9	20°55.2'	28	10	35°57.7'	28	11	51°00.2'
28	12	66°02.6'	28	13	81°05.1'	28	14	96°07.6'	28	15	111°10.0'
28	16	126°12.5'	28	17	141°14.9'	28	18	156°17.4'	28	19	171°19.9'
28	20	186°22.3'	28	21	201°24.8'	28	22	216°27.3'	28	23	231°29.7'
29	0	246°32.2'	29	1	261°34.7'	29	2	276°37.1'	29	3	291°39.6'
29	4	306°42.1'	29	5	321°44.5'	29	6	336°47.0'	29	7	351°49.4'
29	8	6°51.9'	29	9	21°54.4'	29	10	36°56.8'	29	11	51°59.3'
29	12	67°01.8'	29	13	82°04.2'	29	14	97°06.7'	29	15	112°09.2'
29	16	127°11.6'	29	17	142°14.1'	29	18	157°16.6'	29	19	172°19.0'
29	20	187°21.5'	29	21	202°23.9'	29	22	217°26.4'	29	23	232°28.9'
30	0	247°31.3'	30	1	262°33.8'	30	2	277°36.3'	30	3	292°38.7'
30	4	307°41.2'	30	5	322°43.7'	30	6	337°46.1'	30	7	352°48.6'
30	8	7°51.0'	30	9	22°53.5'	30	10	37°56.0'	30	11	52°58.4'
30	12	68°00.9'	30	13	83°03.4'	30	14	98°05.8'	30	15	113°08.3'
30	16	128°10.8'	30	17	143°13.2'	30	18	158°15.7'	30	19	173°18.2'
30	20	188°20.6'	30	21	203°23.1'	30	22	218°25.5'	30	23	233°28.0'
31	0	248°30.5'	31	1	263°32.9'	31	2	278°35.4'	31	3	293°37.9'
31	4	308°40.3'	31	5	323°42.8'	31	6	338°45.3'	31	7	353°47.7'
31	8	8°50.2'	31	9	23°52.7'	31	10	38°55.1'	31	11	53°57.6'
31	12	69°00.0'	31	13	84°02.5'	31	14	99°05.0'	31	15	114°07.4'
31	16	129°09.9'	31	17	144°12.4'	31	18	159°14.8'	31	19	174°17.3'
31	20	189°19.8'	31	21	204°22.2'	31	22	219°24.7'	31	23	234°27.2'

Table 11.6: Aries — GHA of the First Point of Aries, June 2026 (UT)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
1	0	249°29.6'	1	1	264°32.1'	1	2	279°34.5'	1	3	294°37.0'
1	4	309°39.5'	1	5	324°41.9'	1	6	339°44.4'	1	7	354°46.9'
1	8	9°49.3'	1	9	24°51.8'	1	10	39°54.3'	1	11	54°56.7'
1	12	69°59.2'	1	13	85°01.6'	1	14	100°04.1'	1	15	115°06.6'
1	16	130°09.0'	1	17	145°11.5'	1	18	160°14.0'	1	19	175°16.4'

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Aries — GHA of the First Point of Aries, June 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
1	20	190°18.9'	1	21	205°21.4'	1	22	220°23.8'	1	23	235°26.3'
2	0	250°28.8'	2	1	265°31.2'	2	2	280°33.7'	2	3	295°36.1'
2	4	310°38.6'	2	5	325°41.1'	2	6	340°43.5'	2	7	355°46.0'
2	8	10°48.5'	2	9	25°50.9'	2	10	40°53.4'	2	11	55°55.9'
2	12	70°58.3'	2	13	86°00.8'	2	14	101°03.3'	2	15	116°05.7'
2	16	131°08.2'	2	17	146°10.6'	2	18	161°13.1'	2	19	176°15.6'
2	20	191°18.0'	2	21	206°20.5'	2	22	221°23.0'	2	23	236°25.4'
3	0	251°27.9'	3	1	266°30.4'	3	2	281°32.8'	3	3	296°35.3'
3	4	311°37.7'	3	5	326°40.2'	3	6	341°42.7'	3	7	356°45.1'
3	8	11°47.6'	3	9	26°50.1'	3	10	41°52.5'	3	11	56°55.0'
3	12	71°57.5'	3	13	86°59.9'	3	14	102°02.4'	3	15	117°04.9'
3	16	132°07.3'	3	17	147°09.8'	3	18	162°12.2'	3	19	177°14.7'
3	20	192°17.2'	3	21	207°19.6'	3	22	222°22.1'	3	23	237°24.6'
4	0	252°27.0'	4	1	267°29.5'	4	2	282°32.0'	4	3	297°34.4'
4	4	312°36.9'	4	5	327°39.4'	4	6	342°41.8'	4	7	357°44.3'
4	8	12°46.7'	4	9	27°49.2'	4	10	42°51.7'	4	11	57°54.1'
4	12	72°56.6'	4	13	87°59.1'	4	14	103°01.5'	4	15	118°04.0'
4	16	133°06.5'	4	17	148°08.9'	4	18	163°11.4'	4	19	178°13.8'
4	20	193°16.3'	4	21	208°18.8'	4	22	223°21.2'	4	23	238°23.7'
5	0	253°26.2'	5	1	268°28.6'	5	2	283°31.1'	5	3	298°33.6'
5	4	313°36.0'	5	5	328°38.5'	5	6	343°41.0'	5	7	358°43.4'
5	8	13°45.9'	5	9	28°48.3'	5	10	43°50.8'	5	11	58°53.3'
5	12	73°55.7'	5	13	88°58.2'	5	14	104°00.7'	5	15	119°03.1'
5	16	134°05.6'	5	17	149°08.1'	5	18	164°10.5'	5	19	179°13.0'
5	20	194°15.5'	5	21	209°17.9'	5	22	224°20.4'	5	23	239°22.8'
6	0	254°25.3'	6	1	269°27.8'	6	2	284°30.2'	6	3	299°32.7'
6	4	314°35.2'	6	5	329°37.6'	6	6	344°40.1'	6	7	359°42.6'
6	8	14°45.0'	6	9	29°47.5'	6	10	44°49.9'	6	11	59°52.4'
6	12	74°54.9'	6	13	89°57.3'	6	14	104°59.8'	6	15	120°02.3'
6	16	135°04.7'	6	17	150°07.2'	6	18	165°09.7'	6	19	180°12.1'
6	20	195°14.6'	6	21	210°17.1'	6	22	225°19.5'	6	23	240°22.0'
7	0	255°24.4'	7	1	270°26.9'	7	2	285°29.4'	7	3	300°31.8'
7	4	315°34.3'	7	5	330°36.8'	7	6	345°39.2'	7	7	0°41.7'
7	8	15°44.2'	7	9	30°46.6'	7	10	45°49.1'	7	11	60°51.6'
7	12	75°54.0'	7	13	90°56.5'	7	14	105°58.9'	7	15	121°01.4'
7	16	136°03.9'	7	17	151°06.3'	7	18	166°08.8'	7	19	181°11.3'
7	20	196°13.7'	7	21	211°16.2'	7	22	226°18.7'	7	23	241°21.1'
8	0	256°23.6'	8	1	271°26.1'	8	2	286°28.5'	8	3	301°31.0'
8	4	316°33.4'	8	5	331°35.9'	8	6	346°38.4'	8	7	1°40.8'
8	8	16°43.3'	8	9	31°45.8'	8	10	46°48.2'	8	11	61°50.7'
8	12	76°53.2'	8	13	91°55.6'	8	14	106°58.1'	8	15	122°00.5'
8	16	137°03.0'	8	17	152°05.5'	8	18	167°07.9'	8	19	182°10.4'
8	20	197°12.9'	8	21	212°15.3'	8	22	227°17.8'	8	23	242°20.3'
9	0	257°22.7'	9	1	272°25.2'	9	2	287°27.7'	9	3	302°30.1'
9	4	317°32.6'	9	5	332°35.0'	9	6	347°37.5'	9	7	2°40.0'
9	8	17°42.4'	9	9	32°44.9'	9	10	47°47.4'	9	11	62°49.8'
9	12	77°52.3'	9	13	92°54.8'	9	14	107°57.2'	9	15	122°59.7'
9	16	138°02.2'	9	17	153°04.6'	9	18	168°07.1'	9	19	183°09.5'
9	20	198°12.0'	9	21	213°14.5'	9	22	228°16.9'	9	23	243°19.4'
10	0	258°21.9'	10	1	273°24.3'	10	2	288°26.8'	10	3	303°29.3'
10	4	318°31.7'	10	5	333°34.2'	10	6	348°36.6'	10	7	3°39.1'
10	8	18°41.6'	10	9	33°44.0'	10	10	48°46.5'	10	11	63°49.0'
10	12	78°51.4'	10	13	93°53.9'	10	14	108°56.4'	10	15	123°58.8'
10	16	139°01.3'	10	17	154°03.8'	10	18	169°06.2'	10	19	184°08.7'
10	20	199°11.1'	10	21	214°13.6'	10	22	229°16.1'	10	23	244°18.5'
11	0	259°21.0'	11	1	274°23.5'	11	2	289°25.9'	11	3	304°28.4'
11	4	319°30.9'	11	5	334°33.3'	11	6	349°35.8'	11	7	4°38.3'
11	8	19°40.7'	11	9	34°43.2'	11	10	49°45.6'	11	11	64°48.1'
11	12	79°50.6'	11	13	94°53.0'	11	14	109°55.5'	11	15	124°58.0'
11	16	140°00.4'	11	17	155°02.9'	11	18	170°05.4'	11	19	185°07.8'
11	20	200°10.3'	11	21	215°12.7'	11	22	230°15.2'	11	23	245°17.7'
12	0	260°20.1'	12	1	275°22.6'	12	2	290°25.1'	12	3	305°27.5'
12	4	320°30.0'	12	5	335°32.5'	12	6	350°34.9'	12	7	5°37.4'
12	8	20°39.9'	12	9	35°42.3'	12	10	50°44.8'	12	11	65°47.2'

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Aries — GHA of the First Point of Aries, June 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
12	12	80°49.7'	12	13	95°52.2'	12	14	110°54.6'	12	15	125°57.1'
12	16	140°59.6'	12	17	156°02.0'	12	18	171°04.5'	12	19	186°07.0'
12	20	201°09.4'	12	21	216°11.9'	12	22	231°14.4'	12	23	246°16.8'
13	0	261°19.3'	13	1	276°21.7'	13	2	291°24.2'	13	3	306°26.7'
13	4	321°29.1'	13	5	336°31.6'	13	6	351°34.1'	13	7	6°36.5'
13	8	21°39.0'	13	9	36°41.5'	13	10	51°43.9'	13	11	66°46.4'
13	12	81°48.8'	13	13	96°51.3'	13	14	111°53.8'	13	15	126°56.2'
13	16	141°58.7'	13	17	157°01.2'	13	18	172°03.6'	13	19	187°06.1'
13	20	202°08.6'	13	21	217°11.0'	13	22	232°13.5'	13	23	247°16.0'
14	0	262°18.4'	14	1	277°20.9'	14	2	292°23.3'	14	3	307°25.8'
14	4	322°28.3'	14	5	337°30.7'	14	6	352°33.2'	14	7	7°35.7'
14	8	22°38.1'	14	9	37°40.6'	14	10	52°43.1'	14	11	67°45.5'
14	12	82°48.0'	14	13	97°50.5'	14	14	112°52.9'	14	15	127°55.4'
14	16	142°57.8'	14	17	158°00.3'	14	18	173°02.8'	14	19	188°05.2'
14	20	203°07.7'	14	21	218°10.2'	14	22	233°12.6'	14	23	248°15.1'
15	0	263°17.6'	15	1	278°20.0'	15	2	293°22.5'	15	3	308°25.0'
15	4	323°27.4'	15	5	338°29.9'	15	6	353°32.3'	15	7	8°34.8'
15	8	23°37.3'	15	9	38°39.7'	15	10	53°42.2'	15	11	68°44.7'
15	12	83°47.1'	15	13	98°49.6'	15	14	113°52.1'	15	15	128°54.5'
15	16	143°57.0'	15	17	158°59.4'	15	18	174°01.9'	15	19	189°04.4'
15	20	204°06.8'	15	21	219°09.3'	15	22	234°11.8'	15	23	249°14.2'
16	0	264°16.7'	16	1	279°19.2'	16	2	294°21.6'	16	3	309°24.1'
16	4	324°26.6'	16	5	339°29.0'	16	6	354°31.5'	16	7	9°33.9'
16	8	24°36.4'	16	9	39°38.9'	16	10	54°41.3'	16	11	69°43.8'
16	12	84°46.3'	16	13	99°48.7'	16	14	114°51.2'	16	15	129°53.7'
16	16	144°56.1'	16	17	159°58.6'	16	18	175°01.1'	16	19	190°03.5'
16	20	205°06.0'	16	21	220°08.4'	16	22	235°10.9'	16	23	250°13.4'
17	0	265°15.8'	17	1	280°18.3'	17	2	295°20.8'	17	3	310°23.2'
17	4	325°25.7'	17	5	340°28.2'	17	6	355°30.6'	17	7	10°33.1'
17	8	25°35.5'	17	9	40°38.0'	17	10	55°40.5'	17	11	70°42.9'
17	12	85°45.4'	17	13	100°47.9'	17	14	115°50.3'	17	15	130°52.8'
17	16	145°55.3'	17	17	160°57.7'	17	18	176°00.2'	17	19	191°02.7'
17	20	206°05.1'	17	21	221°07.6'	17	22	236°10.0'	17	23	251°12.5'
18	0	266°15.0'	18	1	281°17.4'	18	2	296°19.9'	18	3	311°22.4'
18	4	326°24.8'	18	5	341°27.3'	18	6	356°29.8'	18	7	11°32.2'
18	8	26°34.7'	18	9	41°37.2'	18	10	56°39.6'	18	11	71°42.1'
18	12	86°44.5'	18	13	101°47.0'	18	14	116°49.5'	18	15	131°51.9'
18	16	146°54.4'	18	17	161°56.9'	18	18	176°59.3'	18	19	192°01.8'
18	20	207°04.3'	18	21	222°06.7'	18	22	237°09.2'	18	23	252°11.6'
19	0	267°14.1'	19	1	282°16.6'	19	2	297°19.0'	19	3	312°21.5'
19	4	327°24.0'	19	5	342°26.4'	19	6	357°28.9'	19	7	12°31.4'
19	8	27°33.8'	19	9	42°36.3'	19	10	57°38.8'	19	11	72°41.2'
19	12	87°43.7'	19	13	102°46.1'	19	14	117°48.6'	19	15	132°51.1'
19	16	147°53.5'	19	17	162°56.0'	19	18	177°58.5'	19	19	193°00.9'
19	20	208°03.4'	19	21	223°05.9'	19	22	238°08.3'	19	23	253°10.8'
20	0	268°13.3'	20	1	283°15.7'	20	2	298°18.2'	20	3	313°20.6'
20	4	328°23.1'	20	5	343°25.6'	20	6	358°28.0'	20	7	13°30.5'
20	8	28°33.0'	20	9	43°35.4'	20	10	58°37.9'	20	11	73°40.4'
20	12	88°42.8'	20	13	103°45.3'	20	14	118°47.7'	20	15	133°50.2'
20	16	148°52.7'	20	17	163°55.1'	20	18	178°57.6'	20	19	194°00.1'
20	20	209°02.5'	20	21	224°05.0'	20	22	239°07.5'	20	23	254°09.9'
21	0	269°12.4'	21	1	284°14.9'	21	2	299°17.3'	21	3	314°19.8'
21	4	329°22.2'	21	5	344°24.7'	21	6	359°27.2'	21	7	14°29.6'
21	8	29°32.1'	21	9	44°34.6'	21	10	59°37.0'	21	11	74°39.5'
21	12	89°42.0'	21	13	104°44.4'	21	14	119°46.9'	21	15	134°49.4'
21	16	149°51.8'	21	17	164°54.3'	21	18	179°56.7'	21	19	194°59.2'
21	20	210°01.7'	21	21	225°04.1'	21	22	240°06.6'	21	23	255°09.1'
22	0	270°11.5'	22	1	285°14.0'	22	2	300°16.5'	22	3	315°18.9'
22	4	330°21.4'	22	5	345°23.9'	22	6	0°26.3'	22	7	15°28.8'
22	8	30°31.2'	22	9	45°33.7'	22	10	60°36.2'	22	11	75°38.6'
22	12	90°41.1'	22	13	105°43.6'	22	14	120°46.0'	22	15	135°48.5'
22	16	150°51.0'	22	17	165°53.4'	22	18	180°55.9'	22	19	195°58.3'
22	20	211°00.8'	22	21	226°03.3'	22	22	241°05.7'	22	23	256°08.2'
23	0	271°10.7'	23	1	286°13.1'	23	2	301°15.6'	23	3	316°18.1'

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Aries — GHA of the First Point of Aries, June 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
23	4	331°20.5'	23	5	346°23.0'	23	6	1°25.5'	23	7	16°27.9'
23	8	31°30.4'	23	9	46°32.8'	23	10	61°35.3'	23	11	76°37.8'
23	12	91°40.2'	23	13	106°42.7'	23	14	121°45.2'	23	15	136°47.6'
23	16	151°50.1'	23	17	166°52.6'	23	18	181°55.0'	23	19	196°57.5'
23	20	212°00.0'	23	21	227°02.4'	23	22	242°04.9'	23	23	257°07.3'
24	0	272°09.8'	24	1	287°12.3'	24	2	302°14.7'	24	3	317°17.2'
24	4	332°19.7'	24	5	347°22.1'	24	6	2°24.6'	24	7	17°27.1'
24	8	32°29.5'	24	9	47°32.0'	24	10	62°34.4'	24	11	77°36.9'
24	12	92°39.4'	24	13	107°41.8'	24	14	122°44.3'	24	15	137°46.8'
24	16	152°49.2'	24	17	167°51.7'	24	18	182°54.2'	24	19	197°56.6'
24	20	212°59.1'	24	21	228°01.6'	24	22	243°04.0'	24	23	258°06.5'
25	0	273°08.9'	25	1	288°11.4'	25	2	303°13.9'	25	3	318°16.3'
25	4	333°18.8'	25	5	348°21.3'	25	6	3°23.7'	25	7	18°26.2'
25	8	33°28.7'	25	9	48°31.1'	25	10	63°33.6'	25	11	78°36.1'
25	12	93°38.5'	25	13	108°41.0'	25	14	123°43.4'	25	15	138°45.9'
25	16	153°48.4'	25	17	168°50.8'	25	18	183°53.3'	25	19	198°55.8'
25	20	213°58.2'	25	21	229°00.7'	25	22	244°03.2'	25	23	259°05.6'
26	0	274°08.1'	26	1	289°10.5'	26	2	304°13.0'	26	3	319°15.5'
26	4	334°17.9'	26	5	349°20.4'	26	6	4°22.9'	26	7	19°25.3'
26	8	34°27.8'	26	9	49°30.3'	26	10	64°32.7'	26	11	79°35.2'
26	12	94°37.7'	26	13	109°40.1'	26	14	124°42.6'	26	15	139°45.0'
26	16	154°47.5'	26	17	169°50.0'	26	18	184°52.4'	26	19	199°54.9'
26	20	214°57.4'	26	21	229°59.8'	26	22	245°02.3'	26	23	260°04.8'
27	0	275°07.2'	27	1	290°09.7'	27	2	305°12.2'	27	3	320°14.6'
27	4	335°17.1'	27	5	350°19.5'	27	6	5°22.0'	27	7	20°24.5'
27	8	35°26.9'	27	9	50°29.4'	27	10	65°31.9'	27	11	80°34.3'
27	12	95°36.8'	27	13	110°39.3'	27	14	125°41.7'	27	15	140°44.2'
27	16	155°46.7'	27	17	170°49.1'	27	18	185°51.6'	27	19	200°54.0'
27	20	215°56.5'	27	21	230°59.0'	27	22	246°01.4'	27	23	261°03.9'
28	0	276°06.4'	28	1	291°08.8'	28	2	306°11.3'	28	3	321°13.8'
28	4	336°16.2'	28	5	351°18.7'	28	6	6°21.1'	28	7	21°23.6'
28	8	36°26.1'	28	9	51°28.5'	28	10	66°31.0'	28	11	81°33.5'
28	12	96°35.9'	28	13	111°38.4'	28	14	126°40.9'	28	15	141°43.3'
28	16	156°45.8'	28	17	171°48.3'	28	18	186°50.7'	28	19	201°53.2'
28	20	216°55.6'	28	21	231°58.1'	28	22	247°00.6'	28	23	262°03.0'
29	0	277°05.5'	29	1	292°08.0'	29	2	307°10.4'	29	3	322°12.9'
29	4	337°15.4'	29	5	352°17.8'	29	6	7°20.3'	29	7	22°22.8'
29	8	37°25.2'	29	9	52°27.7'	29	10	67°30.1'	29	11	82°32.6'
29	12	97°35.1'	29	13	112°37.5'	29	14	127°40.0'	29	15	142°42.5'
29	16	157°44.9'	29	17	172°47.4'	29	18	187°49.9'	29	19	202°52.3'
29	20	217°54.8'	29	21	232°57.2'	29	22	247°59.7'	29	23	263°02.2'
30	0	278°04.6'	30	1	293°07.1'	30	2	308°09.6'	30	3	323°12.0'
30	4	338°14.5'	30	5	353°17.0'	30	6	8°19.4'	30	7	23°21.9'
30	8	38°24.4'	30	9	53°26.8'	30	10	68°29.3'	30	11	83°31.7'
30	12	98°34.2'	30	13	113°36.7'	30	14	128°39.1'	30	15	143°41.6'
30	16	158°44.1'	30	17	173°46.5'	30	18	188°49.0'	30	19	203°51.5'
30	20	218°53.9'	30	21	233°56.4'	30	22	248°58.9'	30	23	264°01.3'

Table 11.7: Aries — GHA of the First Point of Aries, July 2026 (UT)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
1	0	279°03.8'	1	1	294°06.2'	1	2	309°08.7'	1	3	324°11.2'
1	4	339°13.6'	1	5	354°16.1'	1	6	9°18.6'	1	7	24°21.0'
1	8	39°23.5'	1	9	54°26.0'	1	10	69°28.4'	1	11	84°30.9'
1	12	99°33.3'	1	13	114°35.8'	1	14	129°38.3'	1	15	144°40.7'
1	16	159°43.2'	1	17	174°45.7'	1	18	189°48.1'	1	19	204°50.6'
1	20	219°53.1'	1	21	234°55.5'	1	22	249°58.0'	1	23	265°00.5'
2	0	280°02.9'	2	1	295°05.4'	2	2	310°07.8'	2	3	325°10.3'
2	4	340°12.8'	2	5	355°15.2'	2	6	10°17.7'	2	7	25°20.2'
2	8	40°22.6'	2	9	55°25.1'	2	10	70°27.6'	2	11	85°30.0'
2	12	100°32.5'	2	13	115°35.0'	2	14	130°37.4'	2	15	145°39.9'
2	16	160°42.3'	2	17	175°44.8'	2	18	190°47.3'	2	19	205°49.7'

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Aries — GHA of the First Point of Aries, July 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
2	20	220°52.2'	2	21	235°54.7'	2	22	250°57.1'	2	23	265°59.6'
3	0	281°02.1'	3	1	296°04.5'	3	2	311°07.0'	3	3	326°09.4'
3	4	341°11.9'	3	5	356°14.4'	3	6	11°16.8'	3	7	26°19.3'
3	8	41°21.8'	3	9	56°24.2'	3	10	71°26.7'	3	11	86°29.2'
3	12	101°31.6'	3	13	116°34.1'	3	14	131°36.6'	3	15	146°39.0'
3	16	161°41.5'	3	17	176°43.9'	3	18	191°46.4'	3	19	206°48.9'
3	20	221°51.3'	3	21	236°53.8'	3	22	251°56.3'	3	23	266°58.7'
4	0	282°01.2'	4	1	297°03.7'	4	2	312°06.1'	4	3	327°08.6'
4	4	342°11.1'	4	5	357°13.5'	4	6	12°16.0'	4	7	27°18.4'
4	8	42°20.9'	4	9	57°23.4'	4	10	72°25.8'	4	11	87°28.3'
4	12	102°30.8'	4	13	117°33.2'	4	14	132°35.7'	4	15	147°38.2'
4	16	162°40.6'	4	17	177°43.1'	4	18	192°45.6'	4	19	207°48.0'
4	20	222°50.5'	4	21	237°52.9'	4	22	252°55.4'	4	23	267°57.9'
5	0	283°00.3'	5	1	298°02.8'	5	2	313°05.3'	5	3	328°07.7'
5	4	343°10.2'	5	5	358°12.7'	5	6	13°15.1'	5	7	28°17.6'
5	8	43°20.0'	5	9	58°22.5'	5	10	73°25.0'	5	11	88°27.4'
5	12	103°29.9'	5	13	118°32.4'	5	14	133°34.8'	5	15	148°37.3'
5	16	163°39.8'	5	17	178°42.2'	5	18	193°44.7'	5	19	208°47.2'
5	20	223°49.6'	5	21	238°52.1'	5	22	253°54.5'	5	23	268°57.0'
6	0	283°59.5'	6	1	299°01.9'	6	2	314°04.4'	6	3	329°06.9'
6	4	344°09.3'	6	5	359°11.8'	6	6	14°14.3'	6	7	29°16.7'
6	8	44°19.2'	6	9	59°21.7'	6	10	74°24.1'	6	11	89°26.6'
6	12	104°29.0'	6	13	119°31.5'	6	14	134°34.0'	6	15	149°36.4'
6	16	164°38.9'	6	17	179°41.4'	6	18	194°43.8'	6	19	209°46.3'
6	20	224°48.8'	6	21	239°51.2'	6	22	254°53.7'	6	23	269°56.1'
7	0	284°58.6'	7	1	300°01.1'	7	2	315°03.5'	7	3	330°06.0'
7	4	345°08.5'	7	5	0°10.9'	7	6	15°13.4'	7	7	30°15.9'
7	8	45°18.3'	7	9	60°20.8'	7	10	75°23.3'	7	11	90°25.7'
7	12	105°28.2'	7	13	120°30.6'	7	14	135°33.1'	7	15	150°35.6'
7	16	165°38.0'	7	17	180°40.5'	7	18	195°43.0'	7	19	210°45.4'
7	20	225°47.9'	7	21	240°50.4'	7	22	255°52.8'	7	23	270°55.3'
8	0	285°57.8'	8	1	301°00.2'	8	2	316°02.7'	8	3	331°05.1'
8	4	346°07.6'	8	5	1°10.1'	8	6	16°12.5'	8	7	31°15.0'
8	8	46°17.5'	8	9	61°19.9'	8	10	76°22.4'	8	11	91°24.9'
8	12	106°27.3'	8	13	121°29.8'	8	14	136°32.2'	8	15	151°34.7'
8	16	166°37.2'	8	17	181°39.6'	8	18	196°42.1'	8	19	211°44.6'
8	20	226°47.0'	8	21	241°49.5'	8	22	256°52.0'	8	23	271°54.4'
9	0	286°56.9'	9	1	301°59.4'	9	2	317°01.8'	9	3	332°04.3'
9	4	347°06.7'	9	5	2°09.2'	9	6	17°11.7'	9	7	32°14.1'
9	8	47°16.6'	9	9	62°19.1'	9	10	77°21.5'	9	11	92°24.0'
9	12	107°26.5'	9	13	122°28.9'	9	14	137°31.4'	9	15	152°33.9'
9	16	167°36.3'	9	17	182°38.8'	9	18	197°41.2'	9	19	212°43.7'
9	20	227°46.2'	9	21	242°48.6'	9	22	257°51.1'	9	23	272°53.6'
10	0	287°56.0'	10	1	302°58.5'	10	2	318°01.0'	10	3	333°03.4'
10	4	348°05.9'	10	5	3°08.3'	10	6	18°10.8'	10	7	33°13.3'
10	8	48°15.7'	10	9	63°18.2'	10	10	78°20.7'	10	11	93°23.1'
10	12	108°25.6'	10	13	123°28.1'	10	14	138°30.5'	10	15	153°33.0'
10	16	168°35.5'	10	17	183°37.9'	10	18	198°40.4'	10	19	213°42.8'
10	20	228°45.3'	10	21	243°47.8'	10	22	258°50.2'	10	23	273°52.7'
11	0	288°55.2'	11	1	303°57.6'	11	2	319°00.1'	11	3	334°02.6'
11	4	349°05.0'	11	5	4°07.5'	11	6	19°10.0'	11	7	34°12.4'
11	8	49°14.9'	11	9	64°17.3'	11	10	79°19.8'	11	11	94°22.3'
11	12	109°24.7'	11	13	124°27.2'	11	14	139°29.7'	11	15	154°32.1'
11	16	169°34.6'	11	17	184°37.1'	11	18	199°39.5'	11	19	214°42.0'
11	20	229°44.5'	11	21	244°46.9'	11	22	259°49.4'	11	23	274°51.8'
12	0	289°54.3'	12	1	304°56.8'	12	2	319°59.2'	12	3	335°01.7'
12	4	350°04.2'	12	5	5°06.6'	12	6	20°09.1'	12	7	35°11.6'
12	8	50°14.0'	12	9	65°16.5'	12	10	80°18.9'	12	11	95°21.4'
12	12	110°23.9'	12	13	125°26.3'	12	14	140°28.8'	12	15	155°31.3'
12	16	170°33.7'	12	17	185°36.2'	12	18	200°38.7'	12	19	215°41.1'
12	20	230°43.6'	12	21	245°46.1'	12	22	260°48.5'	12	23	275°51.0'
13	0	290°53.4'	13	1	305°55.9'	13	2	320°58.4'	13	3	336°00.8'
13	4	351°03.3'	13	5	6°05.8'	13	6	21°08.2'	13	7	36°10.7'
13	8	51°13.2'	13	9	66°15.6'	13	10	81°18.1'	13	11	96°20.6'

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Aries — GHA of the First Point of Aries, July 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
13	12	111°23.0'	13	13	126°25.5'	13	14	141°27.9'	13	15	156°30.4'
13	16	171°32.9'	13	17	186°35.3'	13	18	201°37.8'	13	19	216°40.3'
13	20	231°42.7'	13	21	246°45.2'	13	22	261°47.7'	13	23	276°50.1'
14	0	291°52.6'	14	1	306°55.0'	14	2	321°57.5'	14	3	337°00.0'
14	4	352°02.4'	14	5	7°04.9'	14	6	22°07.4'	14	7	37°09.8'
14	8	52°12.3'	14	9	67°14.8'	14	10	82°17.2'	14	11	97°19.7'
14	12	112°22.2'	14	13	127°24.6'	14	14	142°27.1'	14	15	157°29.5'
14	16	172°32.0'	14	17	187°34.5'	14	18	202°36.9'	14	19	217°39.4'
14	20	232°41.9'	14	21	247°44.3'	14	22	262°46.8'	14	23	277°49.3'
15	0	292°51.7'	15	1	307°54.2'	15	2	322°56.7'	15	3	337°59.1'
15	4	353°01.6'	15	5	8°04.0'	15	6	23°06.5'	15	7	38°09.0'
15	8	53°11.4'	15	9	68°13.9'	15	10	83°16.4'	15	11	98°18.8'
15	12	113°21.3'	15	13	128°23.8'	15	14	143°26.2'	15	15	158°28.7'
15	16	173°31.1'	15	17	188°33.6'	15	18	203°36.1'	15	19	218°38.5'
15	20	233°41.0'	15	21	248°43.5'	15	22	263°45.9'	15	23	278°48.4'
16	0	293°50.9'	16	1	308°53.3'	16	2	323°55.8'	16	3	338°58.3'
16	4	354°00.7'	16	5	9°03.2'	16	6	24°05.6'	16	7	39°08.1'
16	8	54°10.6'	16	9	69°13.0'	16	10	84°15.5'	16	11	99°18.0'
16	12	114°20.4'	16	13	129°22.9'	16	14	144°25.4'	16	15	159°27.8'
16	16	174°30.3'	16	17	189°32.8'	16	18	204°35.2'	16	19	219°37.7'
16	20	234°40.1'	16	21	249°42.6'	16	22	264°45.1'	16	23	279°47.5'
17	0	294°50.0'	17	1	309°52.5'	17	2	324°54.9'	17	3	339°57.4'
17	4	354°59.9'	17	5	10°02.3'	17	6	25°04.8'	17	7	40°07.2'
17	8	55°09.7'	17	9	70°12.2'	17	10	85°14.6'	17	11	100°17.1'
17	12	115°19.6'	17	13	130°22.0'	17	14	145°24.5'	17	15	160°27.0'
17	16	175°29.4'	17	17	190°31.9'	17	18	205°34.4'	17	19	220°36.8'
17	20	235°39.3'	17	21	250°41.7'	17	22	265°44.2'	17	23	280°46.7'
18	0	295°49.1'	18	1	310°51.6'	18	2	325°54.1'	18	3	340°56.5'
18	4	355°59.0'	18	5	11°01.5'	18	6	26°03.9'	18	7	41°06.4'
18	8	56°08.9'	18	9	71°11.3'	18	10	86°13.8'	18	11	101°16.2'
18	12	116°18.7'	18	13	131°21.2'	18	14	146°23.6'	18	15	161°26.1'
18	16	176°28.6'	18	17	191°31.0'	18	18	206°33.5'	18	19	221°36.0'
18	20	236°38.4'	18	21	251°40.9'	18	22	266°43.4'	18	23	281°45.8'
19	0	296°48.3'	19	1	311°50.7'	19	2	326°53.2'	19	3	341°55.7'
19	4	356°58.1'	19	5	12°00.6'	19	6	27°03.1'	19	7	42°05.5'
19	8	57°08.0'	19	9	72°10.5'	19	10	87°12.9'	19	11	102°15.4'
19	12	117°17.8'	19	13	132°20.3'	19	14	147°22.8'	19	15	162°25.2'
19	16	177°27.7'	19	17	192°30.2'	19	18	207°32.6'	19	19	222°35.1'
19	20	237°37.6'	19	21	252°40.0'	19	22	267°42.5'	19	23	282°45.0'
20	0	297°47.4'	20	1	312°49.9'	20	2	327°52.3'	20	3	342°54.8'
20	4	357°57.3'	20	5	12°59.7'	20	6	28°02.2'	20	7	43°04.7'
20	8	58°07.1'	20	9	73°09.6'	20	10	88°12.1'	20	11	103°14.5'
20	12	118°17.0'	20	13	133°19.5'	20	14	148°21.9'	20	15	163°24.4'
20	16	178°26.8'	20	17	193°29.3'	20	18	208°31.8'	20	19	223°34.2'
20	20	238°36.7'	20	21	253°39.2'	20	22	268°41.6'	20	23	283°44.1'
21	0	298°46.6'	21	1	313°49.0'	21	2	328°51.5'	21	3	343°53.9'
21	4	358°56.4'	21	5	13°58.9'	21	6	29°01.3'	21	7	44°03.8'
21	8	59°06.3'	21	9	74°08.7'	21	10	89°11.2'	21	11	104°13.7'
21	12	119°16.1'	21	13	134°18.6'	21	14	149°21.1'	21	15	164°23.5'
21	16	179°26.0'	21	17	194°28.4'	21	18	209°30.9'	21	19	224°33.4'
21	20	239°35.8'	21	21	254°38.3'	21	22	269°40.8'	21	23	284°43.2'
22	0	299°45.7'	22	1	314°48.2'	22	2	329°50.6'	22	3	344°53.1'
22	4	359°55.6'	22	5	14°58.0'	22	6	30°00.5'	22	7	45°02.9'
22	8	60°05.4'	22	9	75°07.9'	22	10	90°10.3'	22	11	105°12.8'
22	12	120°15.3'	22	13	135°17.7'	22	14	150°20.2'	22	15	165°22.7'
22	16	180°25.1'	22	17	195°27.6'	22	18	210°30.0'	22	19	225°32.5'
22	20	240°35.0'	22	21	255°37.4'	22	22	270°39.9'	22	23	285°42.4'
23	0	300°44.8'	23	1	315°47.3'	23	2	330°49.8'	23	3	345°52.2'
23	4	0°54.7'	23	5	15°57.2'	23	6	30°59.6'	23	7	46°02.1'
23	8	61°04.5'	23	9	76°07.0'	23	10	91°09.5'	23	11	106°11.9'
23	12	121°14.4'	23	13	136°16.9'	23	14	151°19.3'	23	15	166°21.8'
23	16	181°24.3'	23	17	196°26.7'	23	18	211°29.2'	23	19	226°31.7'
23	20	241°34.1'	23	21	256°36.6'	23	22	271°39.0'	23	23	286°41.5'
24	0	301°44.0'	24	1	316°46.4'	24	2	331°48.9'	24	3	346°51.4'

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Aries — GHA of the First Point of Aries, July 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
24	4	1°53.8'	24	5	16°56.3'	24	6	31°58.8'	24	7	47°01.2'
24	8	62°03.7'	24	9	77°06.1'	24	10	92°08.6'	24	11	107°11.1'
24	12	122°13.5'	24	13	137°16.0'	24	14	152°18.5'	24	15	167°20.9'
24	16	182°23.4'	24	17	197°25.9'	24	18	212°28.3'	24	19	227°30.8'
24	20	242°33.3'	24	21	257°35.7'	24	22	272°38.2'	24	23	287°40.6'
25	0	302°43.1'	25	1	317°45.6'	25	2	332°48.0'	25	3	347°50.5'
25	4	2°53.0'	25	5	17°55.4'	25	6	32°57.9'	25	7	48°00.4'
25	8	63°02.8'	25	9	78°05.3'	25	10	93°07.8'	25	11	108°10.2'
25	12	123°12.7'	25	13	138°15.1'	25	14	153°17.6'	25	15	168°20.1'
25	16	183°22.5'	25	17	198°25.0'	25	18	213°27.5'	25	19	228°29.9'
25	20	243°32.4'	25	21	258°34.9'	25	22	273°37.3'	25	23	288°39.8'
26	0	303°42.3'	26	1	318°44.7'	26	2	333°47.2'	26	3	348°49.6'
26	4	3°52.1'	26	5	18°54.6'	26	6	33°57.0'	26	7	48°59.5'
26	8	64°02.0'	26	9	79°04.4'	26	10	94°06.9'	26	11	109°09.4'
26	12	124°11.8'	26	13	139°14.3'	26	14	154°16.7'	26	15	169°19.2'
26	16	184°21.7'	26	17	199°24.1'	26	18	214°26.6'	26	19	229°29.1'
26	20	244°31.5'	26	21	259°34.0'	26	22	274°36.5'	26	23	289°38.9'
27	0	304°41.4'	27	1	319°43.9'	27	2	334°46.3'	27	3	349°48.8'
27	4	4°51.2'	27	5	19°53.7'	27	6	34°56.2'	27	7	49°58.6'
27	8	65°01.1'	27	9	80°03.6'	27	10	95°06.0'	27	11	110°08.5'
27	12	125°11.0'	27	13	140°13.4'	27	14	155°15.9'	27	15	170°18.4'
27	16	185°20.8'	27	17	200°23.3'	27	18	215°25.7'	27	19	230°28.2'
27	20	245°30.7'	27	21	260°33.1'	27	22	275°35.6'	27	23	290°38.1'
28	0	305°40.5'	28	1	320°43.0'	28	2	335°45.5'	28	3	350°47.9'
28	4	5°50.4'	28	5	20°52.8'	28	6	35°55.3'	28	7	50°57.8'
28	8	66°00.2'	28	9	81°02.7'	28	10	96°05.2'	28	11	111°07.6'
28	12	126°10.1'	28	13	141°12.6'	28	14	156°15.0'	28	15	171°17.5'
28	16	186°20.0'	28	17	201°22.4'	28	18	216°24.9'	28	19	231°27.3'
28	20	246°29.8'	28	21	261°32.3'	28	22	276°34.7'	28	23	291°37.2'
29	0	306°39.7'	29	1	321°42.1'	29	2	336°44.6'	29	3	351°47.1'
29	4	6°49.5'	29	5	21°52.0'	29	6	36°54.5'	29	7	51°56.9'
29	8	66°59.4'	29	9	82°01.8'	29	10	97°04.3'	29	11	112°06.8'
29	12	127°09.2'	29	13	142°11.7'	29	14	157°14.2'	29	15	172°16.6'
29	16	187°19.1'	29	17	202°21.6'	29	18	217°24.0'	29	19	232°26.5'
29	20	247°28.9'	29	21	262°31.4'	29	22	277°33.9'	29	23	292°36.3'
30	0	307°38.8'	30	1	322°41.3'	30	2	337°43.7'	30	3	352°46.2'
30	4	7°48.7'	30	5	22°51.1'	30	6	37°53.6'	30	7	52°56.1'
30	8	67°58.5'	30	9	83°01.0'	30	10	98°03.4'	30	11	113°05.9'
30	12	128°08.4'	30	13	143°10.8'	30	14	158°13.3'	30	15	173°15.8'
30	16	188°18.2'	30	17	203°20.7'	30	18	218°23.2'	30	19	233°25.6'
30	20	248°28.1'	30	21	263°30.6'	30	22	278°33.0'	30	23	293°35.5'
31	0	308°37.9'	31	1	323°40.4'	31	2	338°42.9'	31	3	353°45.3'
31	4	8°47.8'	31	5	23°50.3'	31	6	38°52.7'	31	7	53°55.2'
31	8	68°57.7'	31	9	84°00.1'	31	10	99°02.6'	31	11	114°05.1'
31	12	129°07.5'	31	13	144°10.0'	31	14	159°12.4'	31	15	174°14.9'
31	16	189°17.4'	31	17	204°19.8'	31	18	219°22.3'	31	19	234°24.8'
31	20	249°27.2'	31	21	264°29.7'	31	22	279°32.2'	31	23	294°34.6'

Table 11.8: Aries — GHA of the First Point of Aries, August 2026 (UT)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
1	0	309°37.1'	1	1	324°39.5'	1	2	339°42.0'	1	3	354°44.5'
1	4	9°46.9'	1	5	24°49.4'	1	6	39°51.9'	1	7	54°54.3'
1	8	69°56.8'	1	9	84°59.3'	1	10	100°01.7'	1	11	115°04.2'
1	12	130°06.7'	1	13	145°09.1'	1	14	160°11.6'	1	15	175°14.0'
1	16	190°16.5'	1	17	205°19.0'	1	18	220°21.4'	1	19	235°23.9'
1	20	250°26.4'	1	21	265°28.8'	1	22	280°31.3'	1	23	295°33.8'
2	0	310°36.2'	2	1	325°38.7'	2	2	340°41.2'	2	3	355°43.6'
2	4	10°46.1'	2	5	25°48.5'	2	6	40°51.0'	2	7	55°53.5'
2	8	70°55.9'	2	9	85°58.4'	2	10	101°00.9'	2	11	116°03.3'
2	12	131°05.8'	2	13	146°08.3'	2	14	161°10.7'	2	15	176°13.2'
2	16	191°15.6'	2	17	206°18.1'	2	18	221°20.6'	2	19	236°23.0'

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Aries — GHA of the First Point of Aries, August 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
2	20	251°25.5'	2	21	266°28.0'	2	22	281°30.4'	2	23	296°32.9'
3	0	311°35.4'	3	1	326°37.8'	3	2	341°40.3'	3	3	356°42.8'
3	4	11°45.2'	3	5	26°47.7'	3	6	41°50.1'	3	7	56°52.6'
3	8	71°55.1'	3	9	86°57.5'	3	10	102°00.0'	3	11	117°02.5'
3	12	132°04.9'	3	13	147°07.4'	3	14	162°09.9'	3	15	177°12.3'
3	16	192°14.8'	3	17	207°17.3'	3	18	222°19.7'	3	19	237°22.2'
3	20	252°24.6'	3	21	267°27.1'	3	22	282°29.6'	3	23	297°32.0'
4	0	312°34.5'	4	1	327°37.0'	4	2	342°39.4'	4	3	357°41.9'
4	4	12°44.4'	4	5	27°46.8'	4	6	42°49.3'	4	7	57°51.7'
4	8	72°54.2'	4	9	87°56.7'	4	10	102°59.1'	4	11	118°01.6'
4	12	133°04.1'	4	13	148°06.5'	4	14	163°09.0'	4	15	178°11.5'
4	16	193°13.9'	4	17	208°16.4'	4	18	223°18.9'	4	19	238°21.3'
4	20	253°23.8'	4	21	268°26.2'	4	22	283°28.7'	4	23	298°31.2'
5	0	313°33.6'	5	1	328°36.1'	5	2	343°38.6'	5	3	358°41.0'
5	4	13°43.5'	5	5	28°46.0'	5	6	43°48.4'	5	7	58°50.9'
5	8	73°53.4'	5	9	88°55.8'	5	10	103°58.3'	5	11	119°00.7'
5	12	134°03.2'	5	13	149°05.7'	5	14	164°08.1'	5	15	179°10.6'
5	16	194°13.1'	5	17	209°15.5'	5	18	224°18.0'	5	19	239°20.5'
5	20	254°22.9'	5	21	269°25.4'	5	22	284°27.8'	5	23	299°30.3'
6	0	314°32.8'	6	1	329°35.2'	6	2	344°37.7'	6	3	359°40.2'
6	4	14°42.6'	6	5	29°45.1'	6	6	44°47.6'	6	7	59°50.0'
6	8	74°52.5'	6	9	89°55.0'	6	10	104°57.4'	6	11	119°59.9'
6	12	135°02.3'	6	13	150°04.8'	6	14	165°07.3'	6	15	180°09.7'
6	16	195°12.2'	6	17	210°14.7'	6	18	225°17.1'	6	19	240°19.6'
6	20	255°22.1'	6	21	270°24.5'	6	22	285°27.0'	6	23	300°29.5'
7	0	315°31.9'	7	1	330°34.4'	7	2	345°36.8'	7	3	0°39.3'
7	4	15°41.8'	7	5	30°44.2'	7	6	45°46.7'	7	7	60°49.2'
7	8	75°51.6'	7	9	90°54.1'	7	10	105°56.6'	7	11	120°59.0'
7	12	136°01.5'	7	13	151°04.0'	7	14	166°06.4'	7	15	181°08.9'
7	16	196°11.3'	7	17	211°13.8'	7	18	226°16.3'	7	19	241°18.7'
7	20	256°21.2'	7	21	271°23.7'	7	22	286°26.1'	7	23	301°28.6'
8	0	316°31.1'	8	1	331°33.5'	8	2	346°36.0'	8	3	1°38.4'
8	4	16°40.9'	8	5	31°43.4'	8	6	46°45.8'	8	7	61°48.3'
8	8	76°50.8'	8	9	91°53.2'	8	10	106°55.7'	8	11	121°58.2'
8	12	137°00.6'	8	13	152°03.1'	8	14	167°05.6'	8	15	182°08.0'
8	16	197°10.5'	8	17	212°12.9'	8	18	227°15.4'	8	19	242°17.9'
8	20	257°20.3'	8	21	272°22.8'	8	22	287°25.3'	8	23	302°27.7'
9	0	317°30.2'	9	1	332°32.7'	9	2	347°35.1'	9	3	2°37.6'
9	4	17°40.1'	9	5	32°42.5'	9	6	47°45.0'	9	7	62°47.4'
9	8	77°49.9'	9	9	92°52.4'	9	10	107°54.8'	9	11	122°57.3'
9	12	137°59.8'	9	13	153°02.2'	9	14	168°04.7'	9	15	183°07.2'
9	16	198°09.6'	9	17	213°12.1'	9	18	228°14.5'	9	19	243°17.0'
9	20	258°19.5'	9	21	273°21.9'	9	22	288°24.4'	9	23	303°26.9'
10	0	318°29.3'	10	1	333°31.8'	10	2	348°34.3'	10	3	3°36.7'
10	4	18°39.2'	10	5	33°41.7'	10	6	48°44.1'	10	7	63°46.6'
10	8	78°49.0'	10	9	93°51.5'	10	10	108°54.0'	10	11	123°56.4'
10	12	138°58.9'	10	13	154°01.4'	10	14	169°03.8'	10	15	184°06.3'
10	16	199°08.8'	10	17	214°11.2'	10	18	229°13.7'	10	19	244°16.2'
10	20	259°18.6'	10	21	274°21.1'	10	22	289°23.5'	10	23	304°26.0'
11	0	319°28.5'	11	1	334°30.9'	11	2	349°33.4'	11	3	4°35.9'
11	4	19°38.3'	11	5	34°40.8'	11	6	49°43.3'	11	7	64°45.7'
11	8	79°48.2'	11	9	94°50.6'	11	10	109°53.1'	11	11	124°55.6'
11	12	139°58.0'	11	13	155°00.5'	11	14	170°03.0'	11	15	185°05.4'
11	16	200°07.9'	11	17	215°10.4'	11	18	230°12.8'	11	19	245°15.3'
11	20	260°17.8'	11	21	275°20.2'	11	22	290°22.7'	11	23	305°25.1'
12	0	320°27.6'	12	1	335°30.1'	12	2	350°32.5'	12	3	5°35.0'
12	4	20°37.5'	12	5	35°39.9'	12	6	50°42.4'	12	7	65°44.9'
12	8	80°47.3'	12	9	95°49.8'	12	10	110°52.3'	12	11	125°54.7'
12	12	140°57.2'	12	13	155°59.6'	12	14	171°02.1'	12	15	186°04.6'
12	16	201°07.0'	12	17	216°09.5'	12	18	231°12.0'	12	19	246°14.4'
12	20	261°16.9'	12	21	276°19.4'	12	22	291°21.8'	12	23	306°24.3'
13	0	321°26.7'	13	1	336°29.2'	13	2	351°31.7'	13	3	6°34.1'
13	4	21°36.6'	13	5	36°39.1'	13	6	51°41.5'	13	7	66°44.0'
13	8	81°46.5'	13	9	96°48.9'	13	10	111°51.4'	13	11	126°53.9'

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Aries — GHA of the First Point of Aries, August 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
13	12	141°56.3'	13	13	156°58.8'	13	14	172°01.2'	13	15	187°03.7'
13	16	202°06.2'	13	17	217°08.6'	13	18	232°11.1'	13	19	247°13.6'
13	20	262°16.0'	13	21	277°18.5'	13	22	292°21.0'	13	23	307°23.4'
14	0	322°25.9'	14	1	337°28.4'	14	2	352°30.8'	14	3	7°33.3'
14	4	22°35.7'	14	5	37°38.2'	14	6	52°40.7'	14	7	67°43.1'
14	8	82°45.6'	14	9	97°48.1'	14	10	112°50.5'	14	11	127°53.0'
14	12	142°55.5'	14	13	157°57.9'	14	14	173°00.4'	14	15	188°02.9'
14	16	203°05.3'	14	17	218°07.8'	14	18	233°10.2'	14	19	248°12.7'
14	20	263°15.2'	14	21	278°17.6'	14	22	293°20.1'	14	23	308°22.6'
15	0	323°25.0'	15	1	338°27.5'	15	2	353°30.0'	15	3	8°32.4'
15	4	23°34.9'	15	5	38°37.3'	15	6	53°39.8'	15	7	68°42.3'
15	8	83°44.7'	15	9	98°47.2'	15	10	113°49.7'	15	11	128°52.1'
15	12	143°54.6'	15	13	158°57.1'	15	14	173°59.5'	15	15	189°02.0'
15	16	204°04.5'	15	17	219°06.9'	15	18	234°09.4'	15	19	249°11.8'
15	20	264°14.3'	15	21	279°16.8'	15	22	294°19.2'	15	23	309°21.7'
16	0	324°24.2'	16	1	339°26.6'	16	2	354°29.1'	16	3	9°31.6'
16	4	24°34.0'	16	5	39°36.5'	16	6	54°39.0'	16	7	69°41.4'
16	8	84°43.9'	16	9	99°46.3'	16	10	114°48.8'	16	11	129°51.3'
16	12	144°53.7'	16	13	159°56.2'	16	14	174°58.7'	16	15	190°01.1'
16	16	205°03.6'	16	17	220°06.1'	16	18	235°08.5'	16	19	250°11.0'
16	20	265°13.4'	16	21	280°15.9'	16	22	295°18.4'	16	23	310°20.8'
17	0	325°23.3'	17	1	340°25.8'	17	2	355°28.2'	17	3	10°30.7'
17	4	25°33.2'	17	5	40°35.6'	17	6	55°38.1'	17	7	70°40.6'
17	8	85°43.0'	17	9	100°45.5'	17	10	115°47.9'	17	11	130°50.4'
17	12	145°52.9'	17	13	160°55.3'	17	14	175°57.8'	17	15	191°00.3'
17	16	206°02.7'	17	17	221°05.2'	17	18	236°07.7'	17	19	251°10.1'
17	20	266°12.6'	17	21	281°15.1'	17	22	296°17.5'	17	23	311°20.0'
18	0	326°22.4'	18	1	341°24.9'	18	2	356°27.4'	18	3	11°29.8'
18	4	26°32.3'	18	5	41°34.8'	18	6	56°37.2'	18	7	71°39.7'
18	8	86°42.2'	18	9	101°44.6'	18	10	116°47.1'	18	11	131°49.5'
18	12	146°52.0'	18	13	161°54.5'	18	14	176°56.9'	18	15	191°59.4'
18	16	207°01.9'	18	17	222°04.3'	18	18	237°06.8'	18	19	252°09.3'
18	20	267°11.7'	18	21	282°14.2'	18	22	297°16.7'	18	23	312°19.1'
19	0	327°21.6'	19	1	342°24.0'	19	2	357°26.5'	19	3	12°29.0'
19	4	27°31.4'	19	5	42°33.9'	19	6	57°36.4'	19	7	72°38.8'
19	8	87°41.3'	19	9	102°43.8'	19	10	117°46.2'	19	11	132°48.7'
19	12	147°51.2'	19	13	162°53.6'	19	14	177°56.1'	19	15	192°58.5'
19	16	208°01.0'	19	17	223°03.5'	19	18	238°05.9'	19	19	253°08.4'
19	20	268°10.9'	19	21	283°13.3'	19	22	298°15.8'	19	23	313°18.3'
20	0	328°20.7'	20	1	343°23.2'	20	2	358°25.6'	20	3	13°28.1'
20	4	28°30.6'	20	5	43°33.0'	20	6	58°35.5'	20	7	73°38.0'
20	8	88°40.4'	20	9	103°42.9'	20	10	118°45.4'	20	11	133°47.8'
20	12	148°50.3'	20	13	163°52.8'	20	14	178°55.2'	20	15	193°57.7'
20	16	209°00.1'	20	17	224°02.6'	20	18	239°05.1'	20	19	254°07.5'
20	20	269°10.0'	20	21	284°12.5'	20	22	299°14.9'	20	23	314°17.4'
21	0	329°19.9'	21	1	344°22.3'	21	2	359°24.8'	21	3	14°27.3'
21	4	29°29.7'	21	5	44°32.2'	21	6	59°34.6'	21	7	74°37.1'
21	8	89°39.6'	21	9	104°42.0'	21	10	119°44.5'	21	11	134°47.0'
21	12	149°49.4'	21	13	164°51.9'	21	14	179°54.4'	21	15	194°56.8'
21	16	209°59.3'	21	17	225°01.8'	21	18	240°04.2'	21	19	255°06.7'
21	20	270°09.1'	21	21	285°11.6'	21	22	300°14.1'	21	23	315°16.5'
22	0	330°19.0'	22	1	345°21.5'	22	2	0°23.9'	22	3	15°26.4'
22	4	30°28.9'	22	5	45°31.3'	22	6	60°33.8'	22	7	75°36.2'
22	8	90°38.7'	22	9	105°41.2'	22	10	120°43.6'	22	11	135°46.1'
22	12	150°48.6'	22	13	165°51.0'	22	14	180°53.5'	22	15	195°56.0'
22	16	210°58.4'	22	17	226°00.9'	22	18	241°03.4'	22	19	256°05.8'
22	20	271°08.3'	22	21	286°10.7'	22	22	301°13.2'	22	23	316°15.7'
23	0	331°18.1'	23	1	346°20.6'	23	2	1°23.1'	23	3	16°25.5'
23	4	31°28.0'	23	5	46°30.5'	23	6	61°32.9'	23	7	76°35.4'
23	8	91°37.9'	23	9	106°40.3'	23	10	121°42.8'	23	11	136°45.2'
23	12	151°47.7'	23	13	166°50.2'	23	14	181°52.6'	23	15	196°55.1'
23	16	211°57.6'	23	17	227°00.0'	23	18	242°02.5'	23	19	257°05.0'
23	20	272°07.4'	23	21	287°09.9'	23	22	302°12.3'	23	23	317°14.8'
24	0	332°17.3'	24	1	347°19.7'	24	2	2°22.2'	24	3	17°24.7'

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Aries — GHA of the First Point of Aries, August 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
24	4	32°27.1'	24	5	47°29.6'	24	6	62°32.1'	24	7	77°34.5'
24	8	92°37.0'	24	9	107°39.5'	24	10	122°41.9'	24	11	137°44.4'
24	12	152°46.8'	24	13	167°49.3'	24	14	182°51.8'	24	15	197°54.2'
24	16	212°56.7'	24	17	227°59.2'	24	18	243°01.6'	24	19	258°04.1'
24	20	273°06.6'	24	21	288°09.0'	24	22	303°11.5'	24	23	318°14.0'
25	0	333°16.4'	25	1	348°18.9'	25	2	3°21.3'	25	3	18°23.8'
25	4	33°26.3'	25	5	48°28.7'	25	6	63°31.2'	25	7	78°33.7'
25	8	93°36.1'	25	9	108°38.6'	25	10	123°41.1'	25	11	138°43.5'
25	12	153°46.0'	25	13	168°48.4'	25	14	183°50.9'	25	15	198°53.4'
25	16	213°55.8'	25	17	228°58.3'	25	18	244°00.8'	25	19	259°03.2'
25	20	274°05.7'	25	21	289°08.2'	25	22	304°10.6'	25	23	319°13.1'
26	0	334°15.6'	26	1	349°18.0'	26	2	4°20.5'	26	3	19°22.9'
26	4	34°25.4'	26	5	49°27.9'	26	6	64°30.3'	26	7	79°32.8'
26	8	94°35.3'	26	9	109°37.7'	26	10	124°40.2'	26	11	139°42.7'
26	12	154°45.1'	26	13	169°47.6'	26	14	184°50.1'	26	15	199°52.5'
26	16	214°55.0'	26	17	229°57.4'	26	18	244°59.9'	26	19	260°02.4'
26	20	275°04.8'	26	21	290°07.3'	26	22	305°09.8'	26	23	320°12.2'
27	0	335°14.7'	27	1	350°17.2'	27	2	5°19.6'	27	3	20°22.1'
27	4	35°24.5'	27	5	50°27.0'	27	6	65°29.5'	27	7	80°31.9'
27	8	95°34.4'	27	9	110°36.9'	27	10	125°39.3'	27	11	140°41.8'
27	12	155°44.3'	27	13	170°46.7'	27	14	185°49.2'	27	15	200°51.7'
27	16	215°54.1'	27	17	230°56.6'	27	18	245°59.0'	27	19	261°01.5'
27	20	276°04.0'	27	21	291°06.4'	27	22	306°08.9'	27	23	321°11.4'
28	0	336°13.8'	28	1	351°16.3'	28	2	6°18.8'	28	3	21°21.2'
28	4	36°23.7'	28	5	51°26.2'	28	6	66°28.6'	28	7	81°31.1'
28	8	96°33.5'	28	9	111°36.0'	28	10	126°38.5'	28	11	141°40.9'
28	12	156°43.4'	28	13	171°45.9'	28	14	186°48.3'	28	15	201°50.8'
28	16	216°53.3'	28	17	231°55.7'	28	18	246°58.2'	28	19	262°00.7'
28	20	277°03.1'	28	21	292°05.6'	28	22	307°08.0'	28	23	322°10.5'
29	0	337°13.0'	29	1	352°15.4'	29	2	7°17.9'	29	3	22°20.4'
29	4	37°22.8'	29	5	52°25.3'	29	6	67°27.8'	29	7	82°30.2'
29	8	97°32.7'	29	9	112°35.1'	29	10	127°37.6'	29	11	142°40.1'
29	12	157°42.5'	29	13	172°45.0'	29	14	187°47.5'	29	15	202°49.9'
29	16	217°52.4'	29	17	232°54.9'	29	18	247°57.3'	29	19	262°59.8'
29	20	278°02.3'	29	21	293°04.7'	29	22	308°07.2'	29	23	323°09.6'
30	0	338°12.1'	30	1	353°14.6'	30	2	8°17.0'	30	3	23°19.5'
30	4	38°22.0'	30	5	53°24.4'	30	6	68°26.9'	30	7	83°29.4'
30	8	98°31.8'	30	9	113°34.3'	30	10	128°36.8'	30	11	143°39.2'
30	12	158°41.7'	30	13	173°44.1'	30	14	188°46.6'	30	15	203°49.1'
30	16	218°51.5'	30	17	233°54.0'	30	18	248°56.5'	30	19	263°58.9'
30	20	279°01.4'	30	21	294°03.9'	30	22	309°06.3'	30	23	324°08.8'
31	0	339°11.2'	31	1	354°13.7'	31	2	9°16.2'	31	3	24°18.6'
31	4	39°21.1'	31	5	54°23.6'	31	6	69°26.0'	31	7	84°28.5'
31	8	99°31.0'	31	9	114°33.4'	31	10	129°35.9'	31	11	144°38.4'
31	12	159°40.8'	31	13	174°43.3'	31	14	189°45.7'	31	15	204°48.2'
31	16	219°50.7'	31	17	234°53.1'	31	18	249°55.6'	31	19	264°58.1'
31	20	280°00.5'	31	21	295°03.0'	31	22	310°05.5'	31	23	325°07.9'

Table 11.9: Aries — GHA of the First Point of Aries, September 2026 (UT)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
1	0	340°10.4'	1	1	355°12.9'	1	2	10°15.3'	1	3	25°17.8'
1	4	40°20.2'	1	5	55°22.7'	1	6	70°25.2'	1	7	85°27.6'
1	8	100°30.1'	1	9	115°32.6'	1	10	130°35.0'	1	11	145°37.5'
1	12	160°40.0'	1	13	175°42.4'	1	14	190°44.9'	1	15	205°47.3'
1	16	220°49.8'	1	17	235°52.3'	1	18	250°54.7'	1	19	265°57.2'
1	20	280°59.7'	1	21	296°02.1'	1	22	311°04.6'	1	23	326°07.1'
2	0	341°09.5'	2	1	356°12.0'	2	2	11°14.5'	2	3	26°16.9'
2	4	41°19.4'	2	5	56°21.8'	2	6	71°24.3'	2	7	86°26.8'
2	8	101°29.2'	2	9	116°31.7'	2	10	131°34.2'	2	11	146°36.6'
2	12	161°39.1'	2	13	176°41.6'	2	14	191°44.0'	2	15	206°46.5'
2	16	221°49.0'	2	17	236°51.4'	2	18	251°53.9'	2	19	266°56.3'

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Aries — GHA of the First Point of Aries, September 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
2	20	281°58.8'	2	21	297°01.3'	2	22	312°03.7'	2	23	327°06.2'
3	0	342°08.7'	3	1	357°11.1'	3	2	12°13.6'	3	3	27°16.1'
3	4	42°18.5'	3	5	57°21.0'	3	6	72°23.5'	3	7	87°25.9'
3	8	102°28.4'	3	9	117°30.8'	3	10	132°33.3'	3	11	147°35.8'
3	12	162°38.2'	3	13	177°40.7'	3	14	192°43.2'	3	15	207°45.6'
3	16	222°48.1'	3	17	237°50.6'	3	18	252°53.0'	3	19	267°55.5'
3	20	282°57.9'	3	21	298°00.4'	3	22	313°02.9'	3	23	328°05.3'
4	0	343°07.8'	4	1	358°10.3'	4	2	13°12.7'	4	3	28°15.2'
4	4	43°17.7'	4	5	58°20.1'	4	6	73°22.6'	4	7	88°25.1'
4	8	103°27.5'	4	9	118°30.0'	4	10	133°32.4'	4	11	148°34.9'
4	12	163°37.4'	4	13	178°39.8'	4	14	193°42.3'	4	15	208°44.8'
4	16	223°47.2'	4	17	238°49.7'	4	18	253°52.2'	4	19	268°54.6'
4	20	283°57.1'	4	21	298°59.6'	4	22	314°02.0'	4	23	329°04.5'
5	0	344°06.9'	5	1	359°09.4'	5	2	14°11.9'	5	3	29°14.3'
5	4	44°16.8'	5	5	59°19.3'	5	6	74°21.7'	5	7	89°24.2'
5	8	104°26.7'	5	9	119°29.1'	5	10	134°31.6'	5	11	149°34.0'
5	12	164°36.5'	5	13	179°39.0'	5	14	194°41.4'	5	15	209°43.9'
5	16	224°46.4'	5	17	239°48.8'	5	18	254°51.3'	5	19	269°53.8'
5	20	284°56.2'	5	21	299°58.7'	5	22	315°01.2'	5	23	330°03.6'
6	0	345°06.1'	6	1	0°08.5'	6	2	15°11.0'	6	3	30°13.5'
6	4	45°15.9'	6	5	60°18.4'	6	6	75°20.9'	6	7	90°23.3'
6	8	105°25.8'	6	9	120°28.3'	6	10	135°30.7'	6	11	150°33.2'
6	12	165°35.7'	6	13	180°38.1'	6	14	195°40.6'	6	15	210°43.0'
6	16	225°45.5'	6	17	240°48.0'	6	18	255°50.4'	6	19	270°52.9'
6	20	285°55.4'	6	21	300°57.8'	6	22	316°00.3'	6	23	331°02.8'
7	0	346°05.2'	7	1	1°07.7'	7	2	16°10.1'	7	3	31°12.6'
7	4	46°15.1'	7	5	61°17.5'	7	6	76°20.0'	7	7	91°22.5'
7	8	106°24.9'	7	9	121°27.4'	7	10	136°29.9'	7	11	151°32.3'
7	12	166°34.8'	7	13	181°37.3'	7	14	196°39.7'	7	15	211°42.2'
7	16	226°44.6'	7	17	241°47.1'	7	18	256°49.6'	7	19	271°52.0'
7	20	286°54.5'	7	21	301°57.0'	7	22	316°59.4'	7	23	332°01.9'
8	0	347°04.4'	8	1	2°06.8'	8	2	17°09.3'	8	3	32°11.8'
8	4	47°14.2'	8	5	62°16.7'	8	6	77°19.1'	8	7	92°21.6'
8	8	107°24.1'	8	9	122°26.5'	8	10	137°29.0'	8	11	152°31.5'
8	12	167°33.9'	8	13	182°36.4'	8	14	197°38.9'	8	15	212°41.3'
8	16	227°43.8'	8	17	242°46.2'	8	18	257°48.7'	8	19	272°51.2'
8	20	287°53.6'	8	21	302°56.1'	8	22	317°58.6'	8	23	333°01.0'
9	0	348°03.5'	9	1	3°06.0'	9	2	18°08.4'	9	3	33°10.9'
9	4	48°13.4'	9	5	63°15.8'	9	6	78°18.3'	9	7	93°20.7'
9	8	108°23.2'	9	9	123°25.7'	9	10	138°28.1'	9	11	153°30.6'
9	12	168°33.1'	9	13	183°35.5'	9	14	198°38.0'	9	15	213°40.5'
9	16	228°42.9'	9	17	243°45.4'	9	18	258°47.9'	9	19	273°50.3'
9	20	288°52.8'	9	21	303°55.2'	9	22	318°57.7'	9	23	334°00.2'
10	0	349°02.6'	10	1	4°05.1'	10	2	19°07.6'	10	3	34°10.0'
10	4	49°12.5'	10	5	64°15.0'	10	6	79°17.4'	10	7	94°19.9'
10	8	109°22.4'	10	9	124°24.8'	10	10	139°27.3'	10	11	154°29.7'
10	12	169°32.2'	10	13	184°34.7'	10	14	199°37.1'	10	15	214°39.6'
10	16	229°42.1'	10	17	244°44.5'	10	18	259°47.0'	10	19	274°49.5'
10	20	289°51.9'	10	21	304°54.4'	10	22	319°56.8'	10	23	334°59.3'
11	0	350°01.8'	11	1	5°04.2'	11	2	20°06.7'	11	3	35°09.2'
11	4	50°11.6'	11	5	65°14.1'	11	6	80°16.6'	11	7	95°19.0'
11	8	110°21.5'	11	9	125°24.0'	11	10	140°26.4'	11	11	155°28.9'
11	12	170°31.3'	11	13	185°33.8'	11	14	200°36.3'	11	15	215°38.7'
11	16	230°41.2'	11	17	245°43.7'	11	18	260°46.1'	11	19	275°48.6'
11	20	290°51.1'	11	21	305°53.5'	11	22	320°56.0'	11	23	335°58.5'
12	0	351°00.9'	12	1	6°03.4'	12	2	21°05.8'	12	3	36°08.3'
12	4	51°10.8'	12	5	66°13.2'	12	6	81°15.7'	12	7	96°18.2'
12	8	111°20.6'	12	9	126°23.1'	12	10	141°25.6'	12	11	156°28.0'
12	12	171°30.5'	12	13	186°32.9'	12	14	201°35.4'	12	15	216°37.9'
12	16	231°40.3'	12	17	246°42.8'	12	18	261°45.3'	12	19	276°47.7'
12	20	291°50.2'	12	21	306°52.7'	12	22	321°55.1'	12	23	336°57.6'
13	0	352°00.1'	13	1	7°02.5'	13	2	22°05.0'	13	3	37°07.4'
13	4	52°09.9'	13	5	67°12.4'	13	6	82°14.8'	13	7	97°17.3'
13	8	112°19.8'	13	9	127°22.2'	13	10	142°24.7'	13	11	157°27.2'

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Aries — GHA of the First Point of Aries, September 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
13	12	172°29.6'	13	13	187°32.1'	13	14	202°34.6'	13	15	217°37.0'
13	16	232°39.5'	13	17	247°41.9'	13	18	262°44.4'	13	19	277°46.9'
13	20	292°49.3'	13	21	307°51.8'	13	22	322°54.3'	13	23	337°56.7'
14	0	352°59.2'	14	1	8°01.7'	14	2	23°04.1'	14	3	38°06.6'
14	4	53°09.0'	14	5	68°11.5'	14	6	83°14.0'	14	7	98°16.4'
14	8	113°18.9'	14	9	128°21.4'	14	10	143°23.8'	14	11	158°26.3'
14	12	173°28.8'	14	13	188°31.2'	14	14	203°33.7'	14	15	218°36.2'
14	16	233°38.6'	14	17	248°41.1'	14	18	263°43.5'	14	19	278°46.0'
14	20	293°48.5'	14	21	308°50.9'	14	22	323°53.4'	14	23	338°55.9'
15	0	353°58.3'	15	1	9°00.8'	15	2	24°03.3'	15	3	39°05.7'
15	4	54°08.2'	15	5	69°10.7'	15	6	84°13.1'	15	7	99°15.6'
15	8	114°18.0'	15	9	129°20.5'	15	10	144°23.0'	15	11	159°25.4'
15	12	174°27.9'	15	13	189°30.4'	15	14	204°32.8'	15	15	219°35.3'
15	16	234°37.8'	15	17	249°40.2'	15	18	264°42.7'	15	19	279°45.1'
15	20	294°47.6'	15	21	309°50.1'	15	22	324°52.5'	15	23	339°55.0'
16	0	354°57.5'	16	1	9°59.9'	16	2	25°02.4'	16	3	40°04.9'
16	4	55°07.3'	16	5	70°09.8'	16	6	85°12.3'	16	7	100°14.7'
16	8	115°17.2'	16	9	130°19.6'	16	10	145°22.1'	16	11	160°24.6'
16	12	175°27.0'	16	13	190°29.5'	16	14	205°32.0'	16	15	220°34.4'
16	16	235°36.9'	16	17	250°39.4'	16	18	265°41.8'	16	19	280°44.3'
16	20	295°46.8'	16	21	310°49.2'	16	22	325°51.7'	16	23	340°54.1'
17	0	355°56.6'	17	1	10°59.1'	17	2	26°01.5'	17	3	41°04.0'
17	4	56°06.5'	17	5	71°08.9'	17	6	86°11.4'	17	7	101°13.9'
17	8	116°16.3'	17	9	131°18.8'	17	10	146°21.3'	17	11	161°23.7'
17	12	176°26.2'	17	13	191°28.6'	17	14	206°31.1'	17	15	221°33.6'
17	16	236°36.0'	17	17	251°38.5'	17	18	266°41.0'	17	19	281°43.4'
17	20	296°45.9'	17	21	311°48.4'	17	22	326°50.8'	17	23	341°53.3'
18	0	356°55.7'	18	1	11°58.2'	18	2	27°00.7'	18	3	42°03.1'
18	4	57°05.6'	18	5	72°08.1'	18	6	87°10.5'	18	7	102°13.0'
18	8	117°15.5'	18	9	132°17.9'	18	10	147°20.4'	18	11	162°22.9'
18	12	177°25.3'	18	13	192°27.8'	18	14	207°30.2'	18	15	222°32.7'
18	16	237°35.2'	18	17	252°37.6'	18	18	267°40.1'	18	19	282°42.6'
18	20	297°45.0'	18	21	312°47.5'	18	22	327°50.0'	18	23	342°52.4'
19	0	357°54.9'	19	1	12°57.4'	19	2	27°59.8'	19	3	43°02.3'
19	4	58°04.7'	19	5	73°07.2'	19	6	88°09.7'	19	7	103°12.1'
19	8	118°14.6'	19	9	133°17.1'	19	10	148°19.5'	19	11	163°22.0'
19	12	178°24.5'	19	13	193°26.9'	19	14	208°29.4'	19	15	223°31.8'
19	16	238°34.3'	19	17	253°36.8'	19	18	268°39.2'	19	19	283°41.7'
19	20	298°44.2'	19	21	313°46.6'	19	22	328°49.1'	19	23	343°51.6'
20	0	358°54.0'	20	1	13°56.5'	20	2	28°59.0'	20	3	44°01.4'
20	4	59°03.9'	20	5	74°06.3'	20	6	89°08.8'	20	7	104°11.3'
20	8	119°13.7'	20	9	134°16.2'	20	10	149°18.7'	20	11	164°21.1'
20	12	179°23.6'	20	13	194°26.1'	20	14	209°28.5'	20	15	224°31.0'
20	16	239°33.5'	20	17	254°35.9'	20	18	269°38.4'	20	19	284°40.8'
20	20	299°43.3'	20	21	314°45.8'	20	22	329°48.2'	20	23	344°50.7'
21	0	359°53.2'	21	1	14°55.6'	21	2	29°58.1'	21	3	45°00.6'
21	4	60°03.0'	21	5	75°05.5'	21	6	90°07.9'	21	7	105°10.4'
21	8	120°12.9'	21	9	135°15.3'	21	10	150°17.8'	21	11	165°20.3'
21	12	180°22.7'	21	13	195°25.2'	21	14	210°27.7'	21	15	225°30.1'
21	16	240°32.6'	21	17	255°35.1'	21	18	270°37.5'	21	19	285°40.0'
21	20	300°42.4'	21	21	315°44.9'	21	22	330°47.4'	21	23	345°49.8'
22	0	0°52.3'	22	1	15°54.8'	22	2	30°57.2'	22	3	45°59.7'
22	4	61°02.2'	22	5	76°04.6'	22	6	91°07.1'	22	7	106°09.6'
22	8	121°12.0'	22	9	136°14.5'	22	10	151°16.9'	22	11	166°19.4'
22	12	181°21.9'	22	13	196°24.3'	22	14	211°26.8'	22	15	226°29.3'
22	16	241°31.7'	22	17	256°34.2'	22	18	271°36.7'	22	19	286°39.1'
22	20	301°41.6'	22	21	316°44.0'	22	22	331°46.5'	22	23	346°49.0'
23	0	1°51.4'	23	1	16°53.9'	23	2	31°56.4'	23	3	46°58.8'
23	4	62°01.3'	23	5	77°03.8'	23	6	92°06.2'	23	7	107°08.7'
23	8	122°11.2'	23	9	137°13.6'	23	10	152°16.1'	23	11	167°18.5'
23	12	182°21.0'	23	13	197°23.5'	23	14	212°25.9'	23	15	227°28.4'
23	16	242°30.9'	23	17	257°33.3'	23	18	272°35.8'	23	19	287°38.3'
23	20	302°40.7'	23	21	317°43.2'	23	22	332°45.7'	23	23	347°48.1'
24	0	2°50.6'	24	1	17°53.0'	24	2	32°55.5'	24	3	47°58.0'

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Aries — GHA of the First Point of Aries, September 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
24	4	63°00.4'	24	5	78°02.9'	24	6	93°05.4'	24	7	108°07.8'
24	8	123°10.3'	24	9	138°12.8'	24	10	153°15.2'	24	11	168°17.7'
24	12	183°20.2'	24	13	198°22.6'	24	14	213°25.1'	24	15	228°27.5'
24	16	243°30.0'	24	17	258°32.5'	24	18	273°34.9'	24	19	288°37.4'
24	20	303°39.9'	24	21	318°42.3'	24	22	333°44.8'	24	23	348°47.3'
25	0	3°49.7'	25	1	18°52.2'	25	2	33°54.6'	25	3	48°57.1'
25	4	63°59.6'	25	5	79°02.0'	25	6	94°04.5'	25	7	109°07.0'
25	8	124°09.4'	25	9	139°11.9'	25	10	154°14.4'	25	11	169°16.8'
25	12	184°19.3'	25	13	199°21.8'	25	14	214°24.2'	25	15	229°26.7'
25	16	244°29.1'	25	17	259°31.6'	25	18	274°34.1'	25	19	289°36.5'
25	20	304°39.0'	25	21	319°41.5'	25	22	334°43.9'	25	23	349°46.4'
26	0	4°48.9'	26	1	19°51.3'	26	2	34°53.8'	26	3	49°56.3'
26	4	64°58.7'	26	5	80°01.2'	26	6	95°03.6'	26	7	110°06.1'
26	8	125°08.6'	26	9	140°11.0'	26	10	155°13.5'	26	11	170°16.0'
26	12	185°18.4'	26	13	200°20.9'	26	14	215°23.4'	26	15	230°25.8'
26	16	245°28.3'	26	17	260°30.7'	26	18	275°33.2'	26	19	290°35.7'
26	20	305°38.1'	26	21	320°40.6'	26	22	335°43.1'	26	23	350°45.5'
27	0	5°48.0'	27	1	20°50.5'	27	2	35°52.9'	27	3	50°55.4'
27	4	65°57.9'	27	5	81°00.3'	27	6	96°02.8'	27	7	111°05.2'
27	8	126°07.7'	27	9	141°10.2'	27	10	156°12.6'	27	11	171°15.1'
27	12	186°17.6'	27	13	201°20.0'	27	14	216°22.5'	27	15	231°25.0'
27	16	246°27.4'	27	17	261°29.9'	27	18	276°32.4'	27	19	291°34.8'
27	20	306°37.3'	27	21	321°39.7'	27	22	336°42.2'	27	23	351°44.7'
28	0	6°47.1'	28	1	21°49.6'	28	2	36°52.1'	28	3	51°54.5'
28	4	66°57.0'	28	5	81°59.5'	28	6	97°01.9'	28	7	112°04.4'
28	8	127°06.8'	28	9	142°09.3'	28	10	157°11.8'	28	11	172°14.2'
28	12	187°16.7'	28	13	202°19.2'	28	14	217°21.6'	28	15	232°24.1'
28	16	247°26.6'	28	17	262°29.0'	28	18	277°31.5'	28	19	292°34.0'
28	20	307°36.4'	28	21	322°38.9'	28	22	337°41.3'	28	23	352°43.8'
29	0	7°46.3'	29	1	22°48.7'	29	2	37°51.2'	29	3	52°53.7'
29	4	67°56.1'	29	5	82°58.6'	29	6	98°01.1'	29	7	113°03.5'
29	8	128°06.0'	29	9	143°08.5'	29	10	158°10.9'	29	11	173°13.4'
29	12	188°15.8'	29	13	203°18.3'	29	14	218°20.8'	29	15	233°23.2'
29	16	248°25.7'	29	17	263°28.2'	29	18	278°30.6'	29	19	293°33.1'
29	20	308°35.6'	29	21	323°38.0'	29	22	338°40.5'	29	23	353°43.0'
30	0	8°45.4'	30	1	23°47.9'	30	2	38°50.3'	30	3	53°52.8'
30	4	68°55.3'	30	5	83°57.7'	30	6	99°00.2'	30	7	114°02.7'
30	8	129°05.1'	30	9	144°07.6'	30	10	159°10.1'	30	11	174°12.5'
30	12	189°15.0'	30	13	204°17.4'	30	14	219°19.9'	30	15	234°22.4'
30	16	249°24.8'	30	17	264°27.3'	30	18	279°29.8'	30	19	294°32.2'
30	20	309°34.7'	30	21	324°37.2'	30	22	339°39.6'	30	23	354°42.1'

Table 11.10: Aries — GHA of the First Point of Aries, October 2026 (UT)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
1	0	9°44.6'	1	1	24°47.0'	1	2	39°49.5'	1	3	54°51.9'
1	4	69°54.4'	1	5	84°56.9'	1	6	99°59.3'	1	7	115°01.8'
1	8	130°04.3'	1	9	145°06.7'	1	10	160°09.2'	1	11	175°11.7'
1	12	190°14.1'	1	13	205°16.6'	1	14	220°19.1'	1	15	235°21.5'
1	16	250°24.0'	1	17	265°26.4'	1	18	280°28.9'	1	19	295°31.4'
1	20	310°33.8'	1	21	325°36.3'	1	22	340°38.8'	1	23	355°41.2'
2	0	10°43.7'	2	1	25°46.2'	2	2	40°48.6'	2	3	55°51.1'
2	4	70°53.5'	2	5	85°56.0'	2	6	100°58.5'	2	7	116°00.9'
2	8	131°03.4'	2	9	146°05.9'	2	10	161°08.3'	2	11	176°10.8'
2	12	191°13.3'	2	13	206°15.7'	2	14	221°18.2'	2	15	236°20.7'
2	16	251°23.1'	2	17	266°25.6'	2	18	281°28.0'	2	19	296°30.5'
2	20	311°33.0'	2	21	326°35.4'	2	22	341°37.9'	2	23	356°40.4'
3	0	11°42.8'	3	1	26°45.3'	3	2	41°47.8'	3	3	56°50.2'
3	4	71°52.7'	3	5	86°55.2'	3	6	101°57.6'	3	7	117°00.1'
3	8	132°02.5'	3	9	147°05.0'	3	10	162°07.5'	3	11	177°09.9'
3	12	192°12.4'	3	13	207°14.9'	3	14	222°17.3'	3	15	237°19.8'
3	16	252°22.3'	3	17	267°24.7'	3	18	282°27.2'	3	19	297°29.6'

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Aries — GHA of the First Point of Aries, October 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
3	20	312°32.1'	3	21	327°34.6'	3	22	342°37.0'	3	23	357°39.5'
4	0	12°42.0'	4	1	27°44.4'	4	2	42°46.9'	4	3	57°49.4'
4	4	72°51.8'	4	5	87°54.3'	4	6	102°56.8'	4	7	117°59.2'
4	8	133°01.7'	4	9	148°04.1'	4	10	163°06.6'	4	11	178°09.1'
4	12	193°11.5'	4	13	208°14.0'	4	14	223°16.5'	4	15	238°18.9'
4	16	253°21.4'	4	17	268°23.9'	4	18	283°26.3'	4	19	298°28.8'
4	20	313°31.3'	4	21	328°33.7'	4	22	343°36.2'	4	23	358°38.6'
5	0	13°41.1'	5	1	28°43.6'	5	2	43°46.0'	5	3	58°48.5'
5	4	73°51.0'	5	5	88°53.4'	5	6	103°55.9'	5	7	118°58.4'
5	8	134°00.8'	5	9	149°03.3'	5	10	164°05.7'	5	11	179°08.2'
5	12	194°10.7'	5	13	209°13.1'	5	14	224°15.6'	5	15	239°18.1'
5	16	254°20.5'	5	17	269°23.0'	5	18	284°25.5'	5	19	299°27.9'
5	20	314°30.4'	5	21	329°32.9'	5	22	344°35.3'	5	23	359°37.8'
6	0	14°40.2'	6	1	29°42.7'	6	2	44°45.2'	6	3	59°47.6'
6	4	74°50.1'	6	5	89°52.6'	6	6	104°55.0'	6	7	119°57.5'
6	8	135°00.0'	6	9	150°02.4'	6	10	165°04.9'	6	11	180°07.4'
6	12	195°09.8'	6	13	210°12.3'	6	14	225°14.7'	6	15	240°17.2'
6	16	255°19.7'	6	17	270°22.1'	6	18	285°24.6'	6	19	300°27.1'
6	20	315°29.5'	6	21	330°32.0'	6	22	345°34.5'	6	23	0°36.9'
7	0	15°39.4'	7	1	30°41.9'	7	2	45°44.3'	7	3	60°46.8'
7	4	75°49.2'	7	5	90°51.7'	7	6	105°54.2'	7	7	120°56.6'
7	8	135°59.1'	7	9	151°01.6'	7	10	166°04.0'	7	11	181°06.5'
7	12	196°09.0'	7	13	211°11.4'	7	14	226°13.9'	7	15	241°16.3'
7	16	256°18.8'	7	17	271°21.3'	7	18	286°23.7'	7	19	301°26.2'
7	20	316°28.7'	7	21	331°31.1'	7	22	346°33.6'	7	23	1°36.1'
8	0	16°38.5'	8	1	31°41.0'	8	2	46°43.5'	8	3	61°45.9'
8	4	76°48.4'	8	5	91°50.8'	8	6	106°53.3'	8	7	121°55.8'
8	8	136°58.2'	8	9	152°00.7'	8	10	167°03.2'	8	11	182°05.6'
8	12	197°08.1'	8	13	212°10.6'	8	14	227°13.0'	8	15	242°15.5'
8	16	257°18.0'	8	17	272°20.4'	8	18	287°22.9'	8	19	302°25.3'
8	20	317°27.8'	8	21	332°30.3'	8	22	347°32.7'	8	23	2°35.2'
9	0	17°37.7'	9	1	32°40.1'	9	2	47°42.6'	9	3	62°45.1'
9	4	77°47.5'	9	5	92°50.0'	9	6	107°52.4'	9	7	122°54.9'
9	8	137°57.4'	9	9	152°59.8'	9	10	168°02.3'	9	11	183°04.8'
9	12	198°07.2'	9	13	213°09.7'	9	14	228°12.2'	9	15	243°14.6'
9	16	258°17.1'	9	17	273°19.6'	9	18	288°22.0'	9	19	303°24.5'
9	20	318°26.9'	9	21	333°29.4'	9	22	348°31.9'	9	23	3°34.3'
10	0	18°36.8'	10	1	33°39.3'	10	2	48°41.7'	10	3	63°44.2'
10	4	78°46.7'	10	5	93°49.1'	10	6	108°51.6'	10	7	123°54.1'
10	8	138°56.5'	10	9	153°59.0'	10	10	169°01.4'	10	11	184°03.9'
10	12	199°06.4'	10	13	214°08.8'	10	14	229°11.3'	10	15	244°13.8'
10	16	259°16.2'	10	17	274°18.7'	10	18	289°21.2'	10	19	304°23.6'
10	20	319°26.1'	10	21	334°28.5'	10	22	349°31.0'	10	23	4°33.5'
11	0	19°35.9'	11	1	34°38.4'	11	2	49°40.9'	11	3	64°43.3'
11	4	79°45.8'	11	5	94°48.3'	11	6	109°50.7'	11	7	124°53.2'
11	8	139°55.7'	11	9	154°58.1'	11	10	170°00.6'	11	11	185°03.0'
11	12	200°05.5'	11	13	215°08.0'	11	14	230°10.4'	11	15	245°12.9'
11	16	260°15.4'	11	17	275°17.8'	11	18	290°20.3'	11	19	305°22.8'
11	20	320°25.2'	11	21	335°27.7'	11	22	350°30.2'	11	23	5°32.6'
12	0	20°35.1'	12	1	35°37.5'	12	2	50°40.0'	12	3	65°42.5'
12	4	80°44.9'	12	5	95°47.4'	12	6	110°49.9'	12	7	125°52.3'
12	8	140°54.8'	12	9	155°57.3'	12	10	170°59.7'	12	11	186°02.2'
12	12	201°04.6'	12	13	216°07.1'	12	14	231°09.6'	12	15	246°12.0'
12	16	261°14.5'	12	17	276°17.0'	12	18	291°19.4'	12	19	306°21.9'
12	20	321°24.4'	12	21	336°26.8'	12	22	351°29.3'	12	23	6°31.8'
13	0	21°34.2'	13	1	36°36.7'	13	2	51°39.1'	13	3	66°41.6'
13	4	81°44.1'	13	5	96°46.5'	13	6	111°49.0'	13	7	126°51.5'
13	8	141°53.9'	13	9	156°56.4'	13	10	171°58.9'	13	11	187°01.3'
13	12	202°03.8'	13	13	217°06.3'	13	14	232°08.7'	13	15	247°11.2'
13	16	262°13.6'	13	17	277°16.1'	13	18	292°18.6'	13	19	307°21.0'
13	20	322°23.5'	13	21	337°26.0'	13	22	352°28.4'	13	23	7°30.9'
14	0	22°33.4'	14	1	37°35.8'	14	2	52°38.3'	14	3	67°40.8'
14	4	82°43.2'	14	5	97°45.7'	14	6	112°48.1'	14	7	127°50.6'
14	8	142°53.1'	14	9	157°55.5'	14	10	172°58.0'	14	11	188°00.5'

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Aries — GHA of the First Point of Aries, October 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
14	12	203°02.9'	14	13	218°05.4'	14	14	233°07.9'	14	15	248°10.3'
14	16	263°12.8'	14	17	278°15.2'	14	18	293°17.7'	14	19	308°20.2'
14	20	323°22.6'	14	21	338°25.1'	14	22	353°27.6'	14	23	8°30.0'
15	0	23°32.5'	15	1	38°35.0'	15	2	53°37.4'	15	3	68°39.9'
15	4	83°42.4'	15	5	98°44.8'	15	6	113°47.3'	15	7	128°49.7'
15	8	143°52.2'	15	9	158°54.7'	15	10	173°57.1'	15	11	188°59.6'
15	12	204°02.1'	15	13	219°04.5'	15	14	234°07.0'	15	15	249°09.5'
15	16	264°11.9'	15	17	279°14.4'	15	18	294°16.9'	15	19	309°19.3'
15	20	324°21.8'	15	21	339°24.2'	15	22	354°26.7'	15	23	9°29.2'
16	0	24°31.6'	16	1	39°34.1'	16	2	54°36.6'	16	3	69°39.0'
16	4	84°41.5'	16	5	99°44.0'	16	6	114°46.4'	16	7	129°48.9'
16	8	144°51.3'	16	9	159°53.8'	16	10	174°56.3'	16	11	189°58.7'
16	12	205°01.2'	16	13	220°03.7'	16	14	235°06.1'	16	15	250°08.6'
16	16	265°11.1'	16	17	280°13.5'	16	18	295°16.0'	16	19	310°18.5'
16	20	325°20.9'	16	21	340°23.4'	16	22	355°25.8'	16	23	10°28.3'
17	0	25°30.8'	17	1	40°33.2'	17	2	55°35.7'	17	3	70°38.2'
17	4	85°40.6'	17	5	100°43.1'	17	6	115°45.6'	17	7	130°48.0'
17	8	145°50.5'	17	9	160°53.0'	17	10	175°55.4'	17	11	190°57.9'
17	12	206°00.3'	17	13	221°02.8'	17	14	236°05.3'	17	15	251°07.7'
17	16	266°10.2'	17	17	281°12.7'	17	18	296°15.1'	17	19	311°17.6'
17	20	326°20.1'	17	21	341°22.5'	17	22	356°25.0'	17	23	11°27.4'
18	0	26°29.9'	18	1	41°32.4'	18	2	56°34.8'	18	3	71°37.3'
18	4	86°39.8'	18	5	101°42.2'	18	6	116°44.7'	18	7	131°47.2'
18	8	146°49.6'	18	9	161°52.1'	18	10	176°54.6'	18	11	191°57.0'
18	12	206°59.5'	18	13	222°01.9'	18	14	237°04.4'	18	15	252°06.9'
18	16	267°09.3'	18	17	282°11.8'	18	18	297°14.3'	18	19	312°16.7'
18	20	327°19.2'	18	21	342°21.7'	18	22	357°24.1'	18	23	12°26.6'
19	0	27°29.1'	19	1	42°31.5'	19	2	57°34.0'	19	3	72°36.4'
19	4	87°38.9'	19	5	102°41.4'	19	6	117°43.8'	19	7	132°46.3'
19	8	147°48.8'	19	9	162°51.2'	19	10	177°53.7'	19	11	192°56.2'
19	12	207°58.6'	19	13	223°01.1'	19	14	238°03.5'	19	15	253°06.0'
19	16	268°08.5'	19	17	283°10.9'	19	18	298°13.4'	19	19	313°15.9'
19	20	328°18.3'	19	21	343°20.8'	19	22	358°23.3'	19	23	13°25.7'
20	0	28°28.2'	20	1	43°30.7'	20	2	58°33.1'	20	3	73°35.6'
20	4	88°38.0'	20	5	103°40.5'	20	6	118°43.0'	20	7	133°45.4'
20	8	148°47.9'	20	9	163°50.4'	20	10	178°52.8'	20	11	193°55.3'
20	12	208°57.8'	20	13	224°00.2'	20	14	239°02.7'	20	15	254°05.2'
20	16	269°07.6'	20	17	284°10.1'	20	18	299°12.5'	20	19	314°15.0'
20	20	329°17.5'	20	21	344°19.9'	20	22	359°22.4'	20	23	14°24.9'
21	0	29°27.3'	21	1	44°29.8'	21	2	59°32.3'	21	3	74°34.7'
21	4	89°37.2'	21	5	104°39.7'	21	6	119°42.1'	21	7	134°44.6'
21	8	149°47.0'	21	9	164°49.5'	21	10	179°52.0'	21	11	194°54.4'
21	12	209°56.9'	21	13	224°59.4'	21	14	240°01.8'	21	15	255°04.3'
21	16	270°06.8'	21	17	285°09.2'	21	18	300°11.7'	21	19	315°14.1'
21	20	330°16.6'	21	21	345°19.1'	21	22	0°21.5'	21	23	15°24.0'
22	0	30°26.5'	22	1	45°28.9'	22	2	60°31.4'	22	3	75°33.9'
22	4	90°36.3'	22	5	105°38.8'	22	6	120°41.3'	22	7	135°43.7'
22	8	150°46.2'	22	9	165°48.6'	22	10	180°51.1'	22	11	195°53.6'
22	12	210°56.0'	22	13	225°58.5'	22	14	241°01.0'	22	15	256°03.4'
22	16	271°05.9'	22	17	286°08.4'	22	18	301°10.8'	22	19	316°13.3'
22	20	331°15.8'	22	21	346°18.2'	22	22	1°20.7'	22	23	16°23.1'
23	0	31°25.6'	23	1	46°28.1'	23	2	61°30.5'	23	3	76°33.0'
23	4	91°35.5'	23	5	106°37.9'	23	6	121°40.4'	23	7	136°42.9'
23	8	151°45.3'	23	9	166°47.8'	23	10	181°50.2'	23	11	196°52.7'
23	12	211°55.2'	23	13	226°57.6'	23	14	242°00.1'	23	15	257°02.6'
23	16	272°05.0'	23	17	287°07.5'	23	18	302°10.0'	23	19	317°12.4'
23	20	332°14.9'	23	21	347°17.4'	23	22	2°19.8'	23	23	17°22.3'
24	0	32°24.7'	24	1	47°27.2'	24	2	62°29.7'	24	3	77°32.1'
24	4	92°34.6'	24	5	107°37.1'	24	6	122°39.5'	24	7	137°42.0'
24	8	152°44.5'	24	9	167°46.9'	24	10	182°49.4'	24	11	197°51.9'
24	12	212°54.3'	24	13	227°56.8'	24	14	242°59.2'	24	15	258°01.7'
24	16	273°04.2'	24	17	288°06.6'	24	18	303°09.1'	24	19	318°11.6'
24	20	333°14.0'	24	21	348°16.5'	24	22	3°19.0'	24	23	18°21.4'
25	0	33°23.9'	25	1	48°26.3'	25	2	63°28.8'	25	3	78°31.3'

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Aries — GHA of the First Point of Aries, October 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
25	4	93°33.7'	25	5	108°36.2'	25	6	123°38.7'	25	7	138°41.1'
25	8	153°43.6'	25	9	168°46.1'	25	10	183°48.5'	25	11	198°51.0'
25	12	213°53.5'	25	13	228°55.9'	25	14	243°58.4'	25	15	259°00.8'
25	16	274°03.3'	25	17	289°05.8'	25	18	304°08.2'	25	19	319°10.7'
25	20	334°13.2'	25	21	349°15.6'	25	22	4°18.1'	25	23	19°20.6'
26	0	34°23.0'	26	1	49°25.5'	26	2	64°28.0'	26	3	79°30.4'
26	4	94°32.9'	26	5	109°35.3'	26	6	124°37.8'	26	7	139°40.3'
26	8	154°42.7'	26	9	169°45.2'	26	10	184°47.7'	26	11	199°50.1'
26	12	214°52.6'	26	13	229°55.1'	26	14	244°57.5'	26	15	260°00.0'
26	16	275°02.4'	26	17	290°04.9'	26	18	305°07.4'	26	19	320°09.8'
26	20	335°12.3'	26	21	350°14.8'	26	22	5°17.2'	26	23	20°19.7'
27	0	35°22.2'	27	1	50°24.6'	27	2	65°27.1'	27	3	80°29.6'
27	4	95°32.0'	27	5	110°34.5'	27	6	125°36.9'	27	7	140°39.4'
27	8	155°41.9'	27	9	170°44.3'	27	10	185°46.8'	27	11	200°49.3'
27	12	215°51.7'	27	13	230°54.2'	27	14	245°56.7'	27	15	260°59.1'
27	16	276°01.6'	27	17	291°04.1'	27	18	306°06.5'	27	19	321°09.0'
27	20	336°11.4'	27	21	351°13.9'	27	22	6°16.4'	27	23	21°18.8'
28	0	36°21.3'	28	1	51°23.8'	28	2	66°26.2'	28	3	81°28.7'
28	4	96°31.2'	28	5	111°33.6'	28	6	126°36.1'	28	7	141°38.6'
28	8	156°41.0'	28	9	171°43.5'	28	10	186°45.9'	28	11	201°48.4'
28	12	216°50.9'	28	13	231°53.3'	28	14	246°55.8'	28	15	261°58.3'
28	16	277°00.7'	28	17	292°03.2'	28	18	307°05.7'	28	19	322°08.1'
28	20	337°10.6'	28	21	352°13.0'	28	22	7°15.5'	28	23	22°18.0'
29	0	37°20.4'	29	1	52°22.9'	29	2	67°25.4'	29	3	82°27.8'
29	4	97°30.3'	29	5	112°32.8'	29	6	127°35.2'	29	7	142°37.7'
29	8	157°40.2'	29	9	172°42.6'	29	10	187°45.1'	29	11	202°47.5'
29	12	217°50.0'	29	13	232°52.5'	29	14	247°54.9'	29	15	262°57.4'
29	16	277°59.9'	29	17	293°02.3'	29	18	308°04.8'	29	19	323°07.3'
29	20	338°09.7'	29	21	353°12.2'	29	22	8°14.7'	29	23	23°17.1'
30	0	38°19.6'	30	1	53°22.0'	30	2	68°24.5'	30	3	83°27.0'
30	4	98°29.4'	30	5	113°31.9'	30	6	128°34.4'	30	7	143°36.8'
30	8	158°39.3'	30	9	173°41.8'	30	10	188°44.2'	30	11	203°46.7'
30	12	218°49.1'	30	13	233°51.6'	30	14	248°54.1'	30	15	263°56.5'
30	16	278°59.0'	30	17	294°01.5'	30	18	309°03.9'	30	19	324°06.4'
30	20	339°08.9'	30	21	354°11.3'	30	22	9°13.8'	30	23	24°16.3'
31	0	39°18.7'	31	1	54°21.2'	31	2	69°23.6'	31	3	84°26.1'
31	4	99°28.6'	31	5	114°31.0'	31	6	129°33.5'	31	7	144°36.0'
31	8	159°38.4'	31	9	174°40.9'	31	10	189°43.4'	31	11	204°45.8'
31	12	219°48.3'	31	13	234°50.8'	31	14	249°53.2'	31	15	264°55.7'
31	16	279°58.1'	31	17	295°00.6'	31	18	310°03.1'	31	19	325°05.5'
31	20	340°08.0'	31	21	355°10.5'	31	22	10°12.9'	31	23	25°15.4'

Table 11.11: Aries — GHA of the First Point of Aries, November 2026 (UT)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
1	0	40°17.9'	1	1	55°20.3'	1	2	70°22.8'	1	3	85°25.2'
1	4	100°27.7'	1	5	115°30.2'	1	6	130°32.6'	1	7	145°35.1'
1	8	160°37.6'	1	9	175°40.0'	1	10	190°42.5'	1	11	205°45.0'
1	12	220°47.4'	1	13	235°49.9'	1	14	250°52.4'	1	15	265°54.8'
1	16	280°57.3'	1	17	295°59.7'	1	18	311°02.2'	1	19	326°04.7'
1	20	341°07.1'	1	21	356°09.6'	1	22	11°12.1'	1	23	26°14.5'
2	0	41°17.0'	2	1	56°19.5'	2	2	71°21.9'	2	3	86°24.4'
2	4	101°26.9'	2	5	116°29.3'	2	6	131°31.8'	2	7	146°34.2'
2	8	161°36.7'	2	9	176°39.2'	2	10	191°41.6'	2	11	206°44.1'
2	12	221°46.6'	2	13	236°49.0'	2	14	251°51.5'	2	15	266°54.0'
2	16	281°56.4'	2	17	296°58.9'	2	18	312°01.4'	2	19	327°03.8'
2	20	342°06.3'	2	21	357°08.7'	2	22	12°11.2'	2	23	27°13.7'
3	0	42°16.1'	3	1	57°18.6'	3	2	72°21.1'	3	3	87°23.5'
3	4	102°26.0'	3	5	117°28.5'	3	6	132°30.9'	3	7	147°33.4'
3	8	162°35.8'	3	9	177°38.3'	3	10	192°40.8'	3	11	207°43.2'
3	12	222°45.7'	3	13	237°48.2'	3	14	252°50.6'	3	15	267°53.1'
3	16	282°55.6'	3	17	297°58.0'	3	18	313°00.5'	3	19	328°03.0'

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Aries — GHA of the First Point of Aries, November 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
3	20	343°05.4'	3	21	358°07.9'	3	22	13°10.3'	3	23	28°12.8'
4	0	43°15.3'	4	1	58°17.7'	4	2	73°20.2'	4	3	88°22.7'
4	4	103°25.1'	4	5	118°27.6'	4	6	133°30.1'	4	7	148°32.5'
4	8	163°35.0'	4	9	178°37.5'	4	10	193°39.9'	4	11	208°42.4'
4	12	223°44.8'	4	13	238°47.3'	4	14	253°49.8'	4	15	268°52.2'
4	16	283°54.7'	4	17	298°57.2'	4	18	313°59.6'	4	19	329°02.1'
4	20	344°04.6'	4	21	359°07.0'	4	22	14°09.5'	4	23	29°11.9'
5	0	44°14.4'	5	1	59°16.9'	5	2	74°19.3'	5	3	89°21.8'
5	4	104°24.3'	5	5	119°26.7'	5	6	134°29.2'	5	7	149°31.7'
5	8	164°34.1'	5	9	179°36.6'	5	10	194°39.1'	5	11	209°41.5'
5	12	224°44.0'	5	13	239°46.4'	5	14	254°48.9'	5	15	269°51.4'
5	16	284°53.8'	5	17	299°56.3'	5	18	314°58.8'	5	19	330°01.2'
5	20	345°03.7'	5	21	0°06.2'	5	22	15°08.6'	5	23	30°11.1'
6	0	45°13.6'	6	1	60°16.0'	6	2	75°18.5'	6	3	90°20.9'
6	4	105°23.4'	6	5	120°25.9'	6	6	135°28.3'	6	7	150°30.8'
6	8	165°33.3'	6	9	180°35.7'	6	10	195°38.2'	6	11	210°40.7'
6	12	225°43.1'	6	13	240°45.6'	6	14	255°48.0'	6	15	270°50.5'
6	16	285°53.0'	6	17	300°55.4'	6	18	315°57.9'	6	19	331°00.4'
6	20	346°02.8'	6	21	1°05.3'	6	22	16°07.8'	6	23	31°10.2'
7	0	46°12.7'	7	1	61°15.2'	7	2	76°17.6'	7	3	91°20.1'
7	4	106°22.5'	7	5	121°25.0'	7	6	136°27.5'	7	7	151°29.9'
7	8	166°32.4'	7	9	181°34.9'	7	10	196°37.3'	7	11	211°39.8'
7	12	226°42.3'	7	13	241°44.7'	7	14	256°47.2'	7	15	271°49.7'
7	16	286°52.1'	7	17	301°54.6'	7	18	316°57.0'	7	19	331°59.5'
7	20	347°02.0'	7	21	2°04.4'	7	22	17°06.9'	7	23	32°09.4'
8	0	47°11.8'	8	1	62°14.3'	8	2	77°16.8'	8	3	92°19.2'
8	4	107°21.7'	8	5	122°24.1'	8	6	137°26.6'	8	7	152°29.1'
8	8	167°31.5'	8	9	182°34.0'	8	10	197°36.5'	8	11	212°38.9'
8	12	227°41.4'	8	13	242°43.9'	8	14	257°46.3'	8	15	272°48.8'
8	16	287°51.3'	8	17	302°53.7'	8	18	317°56.2'	8	19	332°58.6'
8	20	348°01.1'	8	21	3°03.6'	8	22	18°06.0'	8	23	33°08.5'
9	0	48°11.0'	9	1	63°13.4'	9	2	78°15.9'	9	3	93°18.4'
9	4	108°20.8'	9	5	123°23.3'	9	6	138°25.8'	9	7	153°28.2'
9	8	168°30.7'	9	9	183°33.1'	9	10	198°35.6'	9	11	213°38.1'
9	12	228°40.5'	9	13	243°43.0'	9	14	258°45.5'	9	15	273°47.9'
9	16	288°50.4'	9	17	303°52.9'	9	18	318°55.3'	9	19	333°57.8'
9	20	349°00.3'	9	21	4°02.7'	9	22	19°05.2'	9	23	34°07.6'
10	0	49°10.1'	10	1	64°12.6'	10	2	79°15.0'	10	3	94°17.5'
10	4	109°20.0'	10	5	124°22.4'	10	6	139°24.9'	10	7	154°27.4'
10	8	169°29.8'	10	9	184°32.3'	10	10	199°34.7'	10	11	214°37.2'
10	12	229°39.7'	10	13	244°42.1'	10	14	259°44.6'	10	15	274°47.1'
10	16	289°49.5'	10	17	304°52.0'	10	18	319°54.5'	10	19	334°56.9'
10	20	349°59.4'	10	21	5°01.9'	10	22	20°04.3'	10	23	35°06.8'
11	0	50°09.2'	11	1	65°11.7'	11	2	80°14.2'	11	3	95°16.6'
11	4	110°19.1'	11	5	125°21.6'	11	6	140°24.0'	11	7	155°26.5'
11	8	170°29.0'	11	9	185°31.4'	11	10	200°33.9'	11	11	215°36.4'
11	12	230°38.8'	11	13	245°41.3'	11	14	260°43.7'	11	15	275°46.2'
11	16	290°48.7'	11	17	305°51.1'	11	18	320°53.6'	11	19	335°56.1'
11	20	350°58.5'	11	21	6°01.0'	11	22	21°03.5'	11	23	36°05.9'
12	0	51°08.4'	12	1	66°10.8'	12	2	81°13.3'	12	3	96°15.8'
12	4	111°18.2'	12	5	126°20.7'	12	6	141°23.2'	12	7	156°25.6'
12	8	171°28.1'	12	9	186°30.6'	12	10	201°33.0'	12	11	216°35.5'
12	12	231°38.0'	12	13	246°40.4'	12	14	261°42.9'	12	15	276°45.3'
12	16	291°47.8'	12	17	306°50.3'	12	18	321°52.7'	12	19	336°55.2'
12	20	351°57.7'	12	21	7°00.1'	12	22	22°02.6'	12	23	37°05.1'
13	0	52°07.5'	13	1	67°10.0'	13	2	82°12.5'	13	3	97°14.9'
13	4	112°17.4'	13	5	127°19.8'	13	6	142°22.3'	13	7	157°24.8'
13	8	172°27.2'	13	9	187°29.7'	13	10	202°32.2'	13	11	217°34.6'
13	12	232°37.1'	13	13	247°39.6'	13	14	262°42.0'	13	15	277°44.5'
13	16	292°46.9'	13	17	307°49.4'	13	18	322°51.9'	13	19	337°54.3'
13	20	352°56.8'	13	21	7°59.3'	13	22	23°01.7'	13	23	38°04.2'
14	0	53°06.7'	14	1	68°09.1'	14	2	83°11.6'	14	3	98°14.1'
14	4	113°16.5'	14	5	128°19.0'	14	6	143°21.4'	14	7	158°23.9'
14	8	173°26.4'	14	9	188°28.8'	14	10	203°31.3'	14	11	218°33.8'

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Aries — GHA of the First Point of Aries, November 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
14	12	233°36.2'	14	13	248°38.7'	14	14	263°41.2'	14	15	278°43.6'
14	16	293°46.1'	14	17	308°48.6'	14	18	323°51.0'	14	19	338°53.5'
14	20	353°55.9'	14	21	8°58.4'	14	22	24°00.9'	14	23	39°03.3'
15	0	54°05.8'	15	1	69°08.3'	15	2	84°10.7'	15	3	99°13.2'
15	4	114°15.7'	15	5	129°18.1'	15	6	144°20.6'	15	7	159°23.0'
15	8	174°25.5'	15	9	189°28.0'	15	10	204°30.4'	15	11	219°32.9'
15	12	234°35.4'	15	13	249°37.8'	15	14	264°40.3'	15	15	279°42.8'
15	16	294°45.2'	15	17	309°47.7'	15	18	324°50.2'	15	19	339°52.6'
15	20	354°55.1'	15	21	9°57.5'	15	22	25°00.0'	15	23	40°02.5'
16	0	55°04.9'	16	1	70°07.4'	16	2	85°09.9'	16	3	100°12.3'
16	4	115°14.8'	16	5	130°17.3'	16	6	145°19.7'	16	7	160°22.2'
16	8	175°24.7'	16	9	190°27.1'	16	10	205°29.6'	16	11	220°32.0'
16	12	235°34.5'	16	13	250°37.0'	16	14	265°39.4'	16	15	280°41.9'
16	16	295°44.4'	16	17	310°46.8'	16	18	325°49.3'	16	19	340°51.8'
16	20	355°54.2'	16	21	10°56.7'	16	22	25°59.2'	16	23	41°01.6'
17	0	56°04.1'	17	1	71°06.5'	17	2	86°09.0'	17	3	101°11.5'
17	4	116°13.9'	17	5	131°16.4'	17	6	146°18.9'	17	7	161°21.3'
17	8	176°23.8'	17	9	191°26.3'	17	10	206°28.7'	17	11	221°31.2'
17	12	236°33.6'	17	13	251°36.1'	17	14	266°38.6'	17	15	281°41.0'
17	16	296°43.5'	17	17	311°46.0'	17	18	326°48.4'	17	19	341°50.9'
17	20	356°53.4'	17	21	11°55.8'	17	22	26°58.3'	17	23	42°00.8'
18	0	57°03.2'	18	1	72°05.7'	18	2	87°08.1'	18	3	102°10.6'
18	4	117°13.1'	18	5	132°15.5'	18	6	147°18.0'	18	7	162°20.5'
18	8	177°22.9'	18	9	192°25.4'	18	10	207°27.9'	18	11	222°30.3'
18	12	237°32.8'	18	13	252°35.3'	18	14	267°37.7'	18	15	282°40.2'
18	16	297°42.6'	18	17	312°45.1'	18	18	327°47.6'	18	19	342°50.0'
18	20	357°52.5'	18	21	12°55.0'	18	22	27°57.4'	18	23	42°59.9'
19	0	58°02.4'	19	1	73°04.8'	19	2	88°07.3'	19	3	103°09.7'
19	4	118°12.2'	19	5	133°14.7'	19	6	148°17.1'	19	7	163°19.6'
19	8	178°22.1'	19	9	193°24.5'	19	10	208°27.0'	19	11	223°29.5'
19	12	238°31.9'	19	13	253°34.4'	19	14	268°36.9'	19	15	283°39.3'
19	16	298°41.8'	19	17	313°44.2'	19	18	328°46.7'	19	19	343°49.2'
19	20	358°51.6'	19	21	13°54.1'	19	22	28°56.6'	19	23	43°59.0'
20	0	59°01.5'	20	1	74°04.0'	20	2	89°06.4'	20	3	104°08.9'
20	4	119°11.4'	20	5	134°13.8'	20	6	149°16.3'	20	7	164°18.7'
20	8	179°21.2'	20	9	194°23.7'	20	10	209°26.1'	20	11	224°28.6'
20	12	239°31.1'	20	13	254°33.5'	20	14	269°36.0'	20	15	284°38.5'
20	16	299°40.9'	20	17	314°43.4'	20	18	329°45.8'	20	19	344°48.3'
20	20	359°50.8'	20	21	14°53.2'	20	22	29°55.7'	20	23	44°58.2'
21	0	60°00.6'	21	1	75°03.1'	21	2	90°05.6'	21	3	105°08.0'
21	4	120°10.5'	21	5	135°13.0'	21	6	150°15.4'	21	7	165°17.9'
21	8	180°20.3'	21	9	195°22.8'	21	10	210°25.3'	21	11	225°27.7'
21	12	240°30.2'	21	13	255°32.7'	21	14	270°35.1'	21	15	285°37.6'
21	16	300°40.1'	21	17	315°42.5'	21	18	330°45.0'	21	19	345°47.5'
21	20	0°49.9'	21	21	15°52.4'	21	22	30°54.8'	21	23	45°57.3'
22	0	60°59.8'	22	1	76°02.2'	22	2	91°04.7'	22	3	106°07.2'
22	4	121°09.6'	22	5	136°12.1'	22	6	151°14.6'	22	7	166°17.0'
22	8	181°19.5'	22	9	196°21.9'	22	10	211°24.4'	22	11	226°26.9'
22	12	241°29.3'	22	13	256°31.8'	22	14	271°34.3'	22	15	286°36.7'
22	16	301°39.2'	22	17	316°41.7'	22	18	331°44.1'	22	19	346°46.6'
22	20	1°49.1'	22	21	16°51.5'	22	22	31°54.0'	22	23	46°56.4'
23	0	61°58.9'	23	1	77°01.4'	23	2	92°03.8'	23	3	107°06.3'
23	4	122°08.8'	23	5	137°11.2'	23	6	152°13.7'	23	7	167°16.2'
23	8	182°18.6'	23	9	197°21.1'	23	10	212°23.6'	23	11	227°26.0'
23	12	242°28.5'	23	13	257°30.9'	23	14	272°33.4'	23	15	287°35.9'
23	16	302°38.3'	23	17	317°40.8'	23	18	332°43.3'	23	19	347°45.7'
23	20	2°48.2'	23	21	17°50.7'	23	22	32°53.1'	23	23	47°55.6'
24	0	62°58.1'	24	1	78°00.5'	24	2	93°03.0'	24	3	108°05.4'
24	4	123°07.9'	24	5	138°10.4'	24	6	153°12.8'	24	7	168°15.3'
24	8	183°17.8'	24	9	198°20.2'	24	10	213°22.7'	24	11	228°25.2'
24	12	243°27.6'	24	13	258°30.1'	24	14	273°32.5'	24	15	288°35.0'
24	16	303°37.5'	24	17	318°39.9'	24	18	333°42.4'	24	19	348°44.9'
24	20	3°47.3'	24	21	18°49.8'	24	22	33°52.3'	24	23	48°54.7'
25	0	63°57.2'	25	1	78°59.7'	25	2	94°02.1'	25	3	109°04.6'

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Aries — GHA of the First Point of Aries, November 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
25	4	124°07.0'	25	5	139°09.5'	25	6	154°12.0'	25	7	169°14.4'
25	8	184°16.9'	25	9	199°19.4'	25	10	214°21.8'	25	11	229°24.3'
25	12	244°26.8'	25	13	259°29.2'	25	14	274°31.7'	25	15	289°34.2'
25	16	304°36.6'	25	17	319°39.1'	25	18	334°41.5'	25	19	349°44.0'
25	20	4°46.5'	25	21	19°48.9'	25	22	34°51.4'	25	23	49°53.9'
26	0	64°56.3'	26	1	79°58.8'	26	2	95°01.3'	26	3	110°03.7'
26	4	125°06.2'	26	5	140°08.6'	26	6	155°11.1'	26	7	170°13.6'
26	8	185°16.0'	26	9	200°18.5'	26	10	215°21.0'	26	11	230°23.4'
26	12	245°25.9'	26	13	260°28.4'	26	14	275°30.8'	26	15	290°33.3'
26	16	305°35.8'	26	17	320°38.2'	26	18	335°40.7'	26	19	350°43.1'
26	20	5°45.6'	26	21	20°48.1'	26	22	35°50.5'	26	23	50°53.0'
27	0	65°55.5'	27	1	80°57.9'	27	2	96°00.4'	27	3	111°02.9'
27	4	126°05.3'	27	5	141°07.8'	27	6	156°10.3'	27	7	171°12.7'
27	8	186°15.2'	27	9	201°17.6'	27	10	216°20.1'	27	11	231°22.6'
27	12	246°25.0'	27	13	261°27.5'	27	14	276°30.0'	27	15	291°32.4'
27	16	306°34.9'	27	17	321°37.4'	27	18	336°39.8'	27	19	351°42.3'
27	20	6°44.7'	27	21	21°47.2'	27	22	36°49.7'	27	23	51°52.1'
28	0	66°54.6'	28	1	81°57.1'	28	2	96°59.5'	28	3	112°02.0'
28	4	127°04.5'	28	5	142°06.9'	28	6	157°09.4'	28	7	172°11.9'
28	8	187°14.3'	28	9	202°16.8'	28	10	217°19.2'	28	11	232°21.7'
28	12	247°24.2'	28	13	262°26.6'	28	14	277°29.1'	28	15	292°31.6'
28	16	307°34.0'	28	17	322°36.5'	28	18	337°39.0'	28	19	352°41.4'
28	20	7°43.9'	28	21	22°46.4'	28	22	37°48.8'	28	23	52°51.3'
29	0	67°53.7'	29	1	82°56.2'	29	2	97°58.7'	29	3	113°01.1'
29	4	128°03.6'	29	5	143°06.1'	29	6	158°08.5'	29	7	173°11.0'
29	8	188°13.5'	29	9	203°15.9'	29	10	218°18.4'	29	11	233°20.8'
29	12	248°23.3'	29	13	263°25.8'	29	14	278°28.2'	29	15	293°30.7'
29	16	308°33.2'	29	17	323°35.6'	29	18	338°38.1'	29	19	353°40.6'
29	20	8°43.0'	29	21	23°45.5'	29	22	38°48.0'	29	23	53°50.4'
30	0	68°52.9'	30	1	83°55.3'	30	2	98°57.8'	30	3	114°00.3'
30	4	129°02.7'	30	5	144°05.2'	30	6	159°07.7'	30	7	174°10.1'
30	8	189°12.6'	30	9	204°15.1'	30	10	219°17.5'	30	11	234°20.0'
30	12	249°22.5'	30	13	264°24.9'	30	14	279°27.4'	30	15	294°29.8'
30	16	309°32.3'	30	17	324°34.8'	30	18	339°37.2'	30	19	354°39.7'
30	20	9°42.2'	30	21	24°44.6'	30	22	39°47.1'	30	23	54°49.6'

Table 11.12: Aries — GHA of the First Point of Aries, December 2026 (UT)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
1	0	69°52.0'	1	1	84°54.5'	1	2	99°57.0'	1	3	114°59.4'
1	4	130°01.9'	1	5	145°04.3'	1	6	160°06.8'	1	7	175°09.3'
1	8	190°11.7'	1	9	205°14.2'	1	10	220°16.7'	1	11	235°19.1'
1	12	250°21.6'	1	13	265°24.1'	1	14	280°26.5'	1	15	295°29.0'
1	16	310°31.4'	1	17	325°33.9'	1	18	340°36.4'	1	19	355°38.8'
1	20	10°41.3'	1	21	25°43.8'	1	22	40°46.2'	1	23	55°48.7'
2	0	70°51.2'	2	1	85°53.6'	2	2	100°56.1'	2	3	115°58.6'
2	4	131°01.0'	2	5	146°03.5'	2	6	161°05.9'	2	7	176°08.4'
2	8	191°10.9'	2	9	206°13.3'	2	10	221°15.8'	2	11	236°18.3'
2	12	251°20.7'	2	13	266°23.2'	2	14	281°25.7'	2	15	296°28.1'
2	16	311°30.6'	2	17	326°33.1'	2	18	341°35.5'	2	19	356°38.0'
2	20	11°40.4'	2	21	26°42.9'	2	22	41°45.4'	2	23	56°47.8'
3	0	71°50.3'	3	1	86°52.8'	3	2	101°55.2'	3	3	116°57.7'
3	4	132°00.2'	3	5	147°02.6'	3	6	162°05.1'	3	7	177°07.5'
3	8	192°10.0'	3	9	207°12.5'	3	10	222°14.9'	3	11	237°17.4'
3	12	252°19.9'	3	13	267°22.3'	3	14	282°24.8'	3	15	297°27.3'
3	16	312°29.7'	3	17	327°32.2'	3	18	342°34.7'	3	19	357°37.1'
3	20	12°39.6'	3	21	27°42.0'	3	22	42°44.5'	3	23	57°47.0'
4	0	72°49.4'	4	1	87°51.9'	4	2	102°54.4'	4	3	117°56.8'
4	4	132°59.3'	4	5	148°01.8'	4	6	163°04.2'	4	7	178°06.7'
4	8	193°09.2'	4	9	208°11.6'	4	10	223°14.1'	4	11	238°16.5'
4	12	253°19.0'	4	13	268°21.5'	4	14	283°23.9'	4	15	298°26.4'
4	16	313°28.9'	4	17	328°31.3'	4	18	343°33.8'	4	19	358°36.3'

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Aries — GHA of the First Point of Aries, December 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
4	20	13°38.7'	4	21	28°41.2'	4	22	43°43.6'	4	23	58°46.1'
5	0	73°48.6'	5	1	88°51.0'	5	2	103°53.5'	5	3	118°56.0'
5	4	133°58.4'	5	5	149°00.9'	5	6	164°03.4'	5	7	179°05.8'
5	8	194°08.3'	5	9	209°10.8'	5	10	224°13.2'	5	11	239°15.7'
5	12	254°18.1'	5	13	269°20.6'	5	14	284°23.1'	5	15	299°25.5'
5	16	314°28.0'	5	17	329°30.5'	5	18	344°32.9'	5	19	359°35.4'
5	20	14°37.9'	5	21	29°40.3'	5	22	44°42.8'	5	23	59°45.3'
6	0	74°47.7'	6	1	89°50.2'	6	2	104°52.6'	6	3	119°55.1'
6	4	134°57.6'	6	5	150°00.0'	6	6	165°02.5'	6	7	180°05.0'
6	8	195°07.4'	6	9	210°09.9'	6	10	225°12.4'	6	11	240°14.8'
6	12	255°17.3'	6	13	270°19.8'	6	14	285°22.2'	6	15	300°24.7'
6	16	315°27.1'	6	17	330°29.6'	6	18	345°32.1'	6	19	0°34.5'
6	20	15°37.0'	6	21	30°39.5'	6	22	45°41.9'	6	23	60°44.4'
7	0	75°46.9'	7	1	90°49.3'	7	2	105°51.8'	7	3	120°54.2'
7	4	135°56.7'	7	5	150°59.2'	7	6	166°01.6'	7	7	181°04.1'
7	8	196°06.6'	7	9	211°09.0'	7	10	226°11.5'	7	11	241°14.0'
7	12	256°16.4'	7	13	271°18.9'	7	14	286°21.4'	7	15	301°23.8'
7	16	316°26.3'	7	17	331°28.7'	7	18	346°31.2'	7	19	1°33.7'
7	20	16°36.1'	7	21	31°38.6'	7	22	46°41.1'	7	23	61°43.5'
8	0	76°46.0'	8	1	91°48.5'	8	2	106°50.9'	8	3	121°53.4'
8	4	136°55.9'	8	5	151°58.3'	8	6	167°00.8'	8	7	182°03.2'
8	8	197°05.7'	8	9	212°08.2'	8	10	227°10.6'	8	11	242°13.1'
8	12	257°15.6'	8	13	272°18.0'	8	14	287°20.5'	8	15	302°23.0'
8	16	317°25.4'	8	17	332°27.9'	8	18	347°30.3'	8	19	2°32.8'
8	20	17°35.3'	8	21	32°37.7'	8	22	47°40.2'	8	23	62°42.7'
9	0	77°45.1'	9	1	92°47.6'	9	2	107°50.1'	9	3	122°52.5'
9	4	137°55.0'	9	5	152°57.5'	9	6	167°59.9'	9	7	183°02.4'
9	8	198°04.8'	9	9	213°07.3'	9	10	228°09.8'	9	11	243°12.2'
9	12	258°14.7'	9	13	273°17.2'	9	14	288°19.6'	9	15	303°22.1'
9	16	318°24.6'	9	17	333°27.0'	9	18	348°29.5'	9	19	3°32.0'
9	20	18°34.4'	9	21	33°36.9'	9	22	48°39.3'	9	23	63°41.8'
10	0	78°44.3'	10	1	93°46.7'	10	2	108°49.2'	10	3	123°51.7'
10	4	138°54.1'	10	5	153°56.6'	10	6	168°59.1'	10	7	184°01.5'
10	8	199°04.0'	10	9	214°06.4'	10	10	229°08.9'	10	11	244°11.4'
10	12	259°13.8'	10	13	274°16.3'	10	14	289°18.8'	10	15	304°21.2'
10	16	319°23.7'	10	17	334°26.2'	10	18	349°28.6'	10	19	4°31.1'
10	20	19°33.6'	10	21	34°36.0'	10	22	49°38.5'	10	23	64°40.9'
11	0	79°43.4'	11	1	94°45.9'	11	2	109°48.3'	11	3	124°50.8'
11	4	139°53.3'	11	5	154°55.7'	11	6	169°58.2'	11	7	185°00.7'
11	8	200°03.1'	11	9	215°05.6'	11	10	230°08.1'	11	11	245°10.5'
11	12	260°13.0'	11	13	275°15.4'	11	14	290°17.9'	11	15	305°20.4'
11	16	320°22.8'	11	17	335°25.3'	11	18	350°27.8'	11	19	5°30.2'
11	20	20°32.7'	11	21	35°35.2'	11	22	50°37.6'	11	23	65°40.1'
12	0	80°42.5'	12	1	95°45.0'	12	2	110°47.5'	12	3	125°49.9'
12	4	140°52.4'	12	5	155°54.9'	12	6	170°57.3'	12	7	185°59.8'
12	8	201°02.3'	12	9	216°04.7'	12	10	231°07.2'	12	11	246°09.7'
12	12	261°12.1'	12	13	276°14.6'	12	14	291°17.0'	12	15	306°19.5'
12	16	321°22.0'	12	17	336°24.4'	12	18	351°26.9'	12	19	6°29.4'
12	20	21°31.8'	12	21	36°34.3'	12	22	51°36.8'	12	23	66°39.2'
13	0	81°41.7'	13	1	96°44.2'	13	2	111°46.6'	13	3	126°49.1'
13	4	141°51.5'	13	5	156°54.0'	13	6	171°56.5'	13	7	186°58.9'
13	8	202°01.4'	13	9	217°03.9'	13	10	232°06.3'	13	11	247°08.8'
13	12	262°11.3'	13	13	277°13.7'	13	14	292°16.2'	13	15	307°18.7'
13	16	322°21.1'	13	17	337°23.6'	13	18	352°26.0'	13	19	7°28.5'
13	20	22°31.0'	13	21	37°33.4'	13	22	52°35.9'	13	23	67°38.4'
14	0	82°40.8'	14	1	97°43.3'	14	2	112°45.8'	14	3	127°48.2'
14	4	142°50.7'	14	5	157°53.1'	14	6	172°55.6'	14	7	187°58.1'
14	8	203°00.5'	14	9	218°03.0'	14	10	233°05.5'	14	11	248°07.9'
14	12	263°10.4'	14	13	278°12.9'	14	14	293°15.3'	14	15	308°17.8'
14	16	323°20.3'	14	17	338°22.7'	14	18	353°25.2'	14	19	8°27.6'
14	20	23°30.1'	14	21	38°32.6'	14	22	53°35.0'	14	23	68°37.5'
15	0	83°40.0'	15	1	98°42.4'	15	2	113°44.9'	15	3	128°47.4'
15	4	143°49.8'	15	5	158°52.3'	15	6	173°54.8'	15	7	188°57.2'
15	8	203°59.7'	15	9	219°02.1'	15	10	234°04.6'	15	11	249°07.1'

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Aries — GHA of the First Point of Aries, December 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
15	12	264°09.5'	15	13	279°12.0'	15	14	294°14.5'	15	15	309°16.9'
15	16	324°19.4'	15	17	339°21.9'	15	18	354°24.3'	15	19	9°26.8'
15	20	24°29.2'	15	21	39°31.7'	15	22	54°34.2'	15	23	69°36.6'
16	0	84°39.1'	16	1	99°41.6'	16	2	114°44.0'	16	3	129°46.5'
16	4	144°49.0'	16	5	159°51.4'	16	6	174°53.9'	16	7	189°56.4'
16	8	204°58.8'	16	9	220°01.3'	16	10	235°03.7'	16	11	250°06.2'
16	12	265°08.7'	16	13	280°11.1'	16	14	295°13.6'	16	15	310°16.1'
16	16	325°18.5'	16	17	340°21.0'	16	18	355°23.5'	16	19	10°25.9'
16	20	25°28.4'	16	21	40°30.9'	16	22	55°33.3'	16	23	70°35.8'
17	0	85°38.2'	17	1	100°40.7'	17	2	115°43.2'	17	3	130°45.6'
17	4	145°48.1'	17	5	160°50.6'	17	6	175°53.0'	17	7	190°55.5'
17	8	205°58.0'	17	9	221°00.4'	17	10	236°02.9'	17	11	251°05.3'
17	12	266°07.8'	17	13	281°10.3'	17	14	296°12.7'	17	15	311°15.2'
17	16	326°17.7'	17	17	341°20.1'	17	18	356°22.6'	17	19	11°25.1'
17	20	26°27.5'	17	21	41°30.0'	17	22	56°32.5'	17	23	71°34.9'
18	0	86°37.4'	18	1	101°39.8'	18	2	116°42.3'	18	3	131°44.8'
18	4	146°47.2'	18	5	161°49.7'	18	6	176°52.2'	18	7	191°54.6'
18	8	206°57.1'	18	9	221°59.6'	18	10	237°02.0'	18	11	252°04.5'
18	12	267°07.0'	18	13	282°09.4'	18	14	297°11.9'	18	15	312°14.3'
18	16	327°16.8'	18	17	342°19.3'	18	18	357°21.7'	18	19	12°24.2'
18	20	27°26.7'	18	21	42°29.1'	18	22	57°31.6'	18	23	72°34.1'
19	0	87°36.5'	19	1	102°39.0'	19	2	117°41.4'	19	3	132°43.9'
19	4	147°46.4'	19	5	162°48.8'	19	6	177°51.3'	19	7	192°53.8'
19	8	207°56.2'	19	9	222°58.7'	19	10	238°01.2'	19	11	253°03.6'
19	12	268°06.1'	19	13	283°08.6'	19	14	298°11.0'	19	15	313°13.5'
19	16	328°15.9'	19	17	343°18.4'	19	18	358°20.9'	19	19	13°23.3'
19	20	28°25.8'	19	21	43°28.3'	19	22	58°30.7'	19	23	73°33.2'
20	0	88°35.7'	20	1	103°38.1'	20	2	118°40.6'	20	3	133°43.1'
20	4	148°45.5'	20	5	163°48.0'	20	6	178°50.4'	20	7	193°52.9'
20	8	208°55.4'	20	9	223°57.8'	20	10	239°00.3'	20	11	254°02.8'
20	12	269°05.2'	20	13	284°07.7'	20	14	299°10.2'	20	15	314°12.6'
20	16	329°15.1'	20	17	344°17.6'	20	18	359°20.0'	20	19	14°22.5'
20	20	29°24.9'	20	21	44°27.4'	20	22	59°29.9'	20	23	74°32.3'
21	0	89°34.8'	21	1	104°37.3'	21	2	119°39.7'	21	3	134°42.2'
21	4	149°44.7'	21	5	164°47.1'	21	6	179°49.6'	21	7	194°52.0'
21	8	209°54.5'	21	9	224°57.0'	21	10	239°59.4'	21	11	255°01.9'
21	12	270°04.4'	21	13	285°06.8'	21	14	300°09.3'	21	15	315°11.8'
21	16	330°14.2'	21	17	345°16.7'	21	18	0°19.2'	21	19	15°21.6'
21	20	30°24.1'	21	21	45°26.5'	21	22	60°29.0'	21	23	75°31.5'
22	0	90°33.9'	22	1	105°36.4'	22	2	120°38.9'	22	3	135°41.3'
22	4	150°43.8'	22	5	165°46.3'	22	6	180°48.7'	22	7	195°51.2'
22	8	210°53.7'	22	9	225°56.1'	22	10	240°58.6'	22	11	256°01.0'
22	12	271°03.5'	22	13	286°06.0'	22	14	301°08.4'	22	15	316°10.9'
22	16	331°13.4'	22	17	346°15.8'	22	18	1°18.3'	22	19	16°20.8'
22	20	31°23.2'	22	21	46°25.7'	22	22	61°28.1'	22	23	76°30.6'
23	0	91°33.1'	23	1	106°35.5'	23	2	121°38.0'	23	3	136°40.5'
23	4	151°42.9'	23	5	166°45.4'	23	6	181°47.9'	23	7	196°50.3'
23	8	211°52.8'	23	9	226°55.3'	23	10	241°57.7'	23	11	257°00.2'
23	12	272°02.6'	23	13	287°05.1'	23	14	302°07.6'	23	15	317°10.0'
23	16	332°12.5'	23	17	347°15.0'	23	18	2°17.4'	23	19	17°19.9'
23	20	32°22.4'	23	21	47°24.8'	23	22	62°27.3'	23	23	77°29.8'
24	0	92°32.2'	24	1	107°34.7'	24	2	122°37.1'	24	3	137°39.6'
24	4	152°42.1'	24	5	167°44.5'	24	6	182°47.0'	24	7	197°49.5'
24	8	212°51.9'	24	9	227°54.4'	24	10	242°56.9'	24	11	257°59.3'
24	12	273°01.8'	24	13	288°04.2'	24	14	303°06.7'	24	15	318°09.2'
24	16	333°11.6'	24	17	348°14.1'	24	18	3°16.6'	24	19	18°19.0'
24	20	33°21.5'	24	21	48°24.0'	24	22	63°26.4'	24	23	78°28.9'
25	0	93°31.4'	25	1	108°33.8'	25	2	123°36.3'	25	3	138°38.7'
25	4	153°41.2'	25	5	168°43.7'	25	6	183°46.1'	25	7	198°48.6'
25	8	213°51.1'	25	9	228°53.5'	25	10	243°56.0'	25	11	258°58.5'
25	12	274°00.9'	25	13	289°03.4'	25	14	304°05.9'	25	15	319°08.3'
25	16	334°10.8'	25	17	349°13.2'	25	18	4°15.7'	25	19	19°18.2'
25	20	34°20.6'	25	21	49°23.1'	25	22	64°25.6'	25	23	79°28.0'
26	0	94°30.5'	26	1	109°33.0'	26	2	124°35.4'	26	3	139°37.9'

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Aries — GHA of the First Point of Aries, December 2026 (UT) (continued)

d	h	GHA γ	d	h	GHA γ	d	h	GHA γ	d	h	GHA γ
26	4	154°40.3'	26	5	169°42.8'	26	6	184°45.3'	26	7	199°47.7'
26	8	214°50.2'	26	9	229°52.7'	26	10	244°55.1'	26	11	259°57.6'
26	12	275°00.1'	26	13	290°02.5'	26	14	305°05.0'	26	15	320°07.5'
26	16	335°09.9'	26	17	350°12.4'	26	18	5°14.8'	26	19	20°17.3'
26	20	35°19.8'	26	21	50°22.2'	26	22	65°24.7'	26	23	80°27.2'
27	0	95°29.6'	27	1	110°32.1'	27	2	125°34.6'	27	3	140°37.0'
27	4	155°39.5'	27	5	170°42.0'	27	6	185°44.4'	27	7	200°46.9'
27	8	215°49.3'	27	9	230°51.8'	27	10	245°54.3'	27	11	260°56.7'
27	12	275°59.2'	27	13	291°01.7'	27	14	306°04.1'	27	15	321°06.6'
27	16	336°09.1'	27	17	351°11.5'	27	18	6°14.0'	27	19	21°16.5'
27	20	36°18.9'	27	21	51°21.4'	27	22	66°23.8'	27	23	81°26.3'
28	0	96°28.8'	28	1	111°31.2'	28	2	126°33.7'	28	3	141°36.2'
28	4	156°38.6'	28	5	171°41.1'	28	6	186°43.6'	28	7	201°46.0'
28	8	216°48.5'	28	9	231°50.9'	28	10	246°53.4'	28	11	261°55.9'
28	12	276°58.3'	28	13	292°00.8'	28	14	307°03.3'	28	15	322°05.7'
28	16	337°08.2'	28	17	352°10.7'	28	18	7°13.1'	28	19	22°15.6'
28	20	37°18.1'	28	21	52°20.5'	28	22	67°23.0'	28	23	82°25.4'
29	0	97°27.9'	29	1	112°30.4'	29	2	127°32.8'	29	3	142°35.3'
29	4	157°37.8'	29	5	172°40.2'	29	6	187°42.7'	29	7	202°45.2'
29	8	217°47.6'	29	9	232°50.1'	29	10	247°52.6'	29	11	262°55.0'
29	12	277°57.5'	29	13	292°59.9'	29	14	308°02.4'	29	15	323°04.9'
29	16	338°07.3'	29	17	353°09.8'	29	18	8°12.3'	29	19	23°14.7'
29	20	38°17.2'	29	21	53°19.7'	29	22	68°22.1'	29	23	83°24.6'
30	0	98°27.0'	30	1	113°29.5'	30	2	128°32.0'	30	3	143°34.4'
30	4	158°36.9'	30	5	173°39.4'	30	6	188°41.8'	30	7	203°44.3'
30	8	218°46.8'	30	9	233°49.2'	30	10	248°51.7'	30	11	263°54.2'
30	12	278°56.6'	30	13	293°59.1'	30	14	309°01.5'	30	15	324°04.0'
30	16	339°06.5'	30	17	354°08.9'	30	18	9°11.4'	30	19	24°13.9'
30	20	39°16.3'	30	21	54°18.8'	30	22	69°21.3'	30	23	84°23.7'
31	0	99°26.2'	31	1	114°28.7'	31	2	129°31.1'	31	3	144°33.6'
31	4	159°36.0'	31	5	174°38.5'	31	6	189°41.0'	31	7	204°43.4'
31	8	219°45.9'	31	9	234°48.4'	31	10	249°50.8'	31	11	264°53.3'
31	12	279°55.8'	31	13	294°58.2'	31	14	310°00.7'	31	15	325°03.1'
31	16	340°05.6'	31	17	355°08.1'	31	18	10°10.5'	31	19	25°13.0'
31	20	40°15.5'	31	21	55°17.9'	31	22	70°20.4'	31	23	85°22.9'

Part IV

Correction & Increment Tables

Chapter 12

Altitude corrections

12.1 Dip of the horizon

Subtract from the sextant altitude. $\text{Dip} = 1.76\sqrt{h'}$, h in metres.

Height (m)	Dip (')	Height (m)	Dip (')
1.0	−1.8	7.0	−4.7
1.5	−2.2	8.0	−5.0
2.0	−2.5	9.0	−5.3
2.5	−2.8	10.0	−5.6
3.0	−3.0	11.0	−5.8
3.5	−3.3	12.0	−6.1
4.0	−3.5	15.0	−6.8
4.5	−3.7	20.0	−7.9
5.0	−3.9	25.0	−8.8
5.5	−4.1	30.0	−9.6
6.0	−4.3		

12.2 Atmospheric refraction

Subtract from the apparent altitude H_a . Bennett's formula; standard atmosphere.

App.Alt (°)	Refr (')	App.Alt (°)	Refr (')
5	−9.9	18	−3.0
6	−8.5	19	−2.9
7	−7.5	20	−2.7
8	−6.6	22	−2.4
9	−5.9	25	−2.1
10	−5.4	30	−1.7
11	−4.9	35	−1.4
12	−4.5	40	−1.2
13	−4.2	50	−0.8
14	−3.9	60	−0.6
15	−3.6	70	−0.4
16	−3.4	80	−0.2
17	−3.2	90	0.0

12.3 Sun semidiameter

Add for a lower-limb sight, subtract for an upper-limb sight.

Date	SD (')	Date	SD (')
Jan 1	16.3	Jul 1	15.7
Jan 15	16.3	Jul 15	15.7
Feb 1	16.2	Aug 1	15.8
Feb 15	16.2	Aug 15	15.8
Mar 1	16.1	Sep 1	15.9
Mar 15	16.1	Sep 15	15.9
Apr 1	16.0	Oct 1	16.0
Apr 15	15.9	Oct 15	16.0
May 1	15.9	Nov 1	16.1
May 15	15.8	Nov 15	16.2
Jun 1	15.8	Dec 1	16.2
Jun 15	15.8	Dec 15	16.3

Chapter 13

GHA increments

To find the GHA between whole-hour entries, add the increment for the elapsed minutes. The Sun's GHA increases at 15° per hour; Aries at 15.041° per hour. For the seconds, add $0.25'$ per second for the Sun and $0.2507'$ per second for Aries.

m	Sun	Aries	m	Sun	Aries
0	$0^\circ 00.0'$	$0^\circ 00.0'$	30	$7^\circ 30.0'$	$7^\circ 31.2'$
1	$0^\circ 15.0'$	$0^\circ 15.0'$	31	$7^\circ 45.0'$	$7^\circ 46.3'$
2	$0^\circ 30.0'$	$0^\circ 30.1'$	32	$8^\circ 00.0'$	$8^\circ 01.3'$
3	$0^\circ 45.0'$	$0^\circ 45.1'$	33	$8^\circ 15.0'$	$8^\circ 16.4'$
4	$1^\circ 00.0'$	$1^\circ 00.2'$	34	$8^\circ 30.0'$	$8^\circ 31.4'$
5	$1^\circ 15.0'$	$1^\circ 15.2'$	35	$8^\circ 45.0'$	$8^\circ 46.4'$
6	$1^\circ 30.0'$	$1^\circ 30.2'$	36	$9^\circ 00.0'$	$9^\circ 01.5'$
7	$1^\circ 45.0'$	$1^\circ 45.3'$	37	$9^\circ 15.0'$	$9^\circ 16.5'$
8	$2^\circ 00.0'$	$2^\circ 00.3'$	38	$9^\circ 30.0'$	$9^\circ 31.6'$
9	$2^\circ 15.0'$	$2^\circ 15.4'$	39	$9^\circ 45.0'$	$9^\circ 46.6'$
10	$2^\circ 30.0'$	$2^\circ 30.4'$	40	$10^\circ 00.0'$	$10^\circ 01.6'$
11	$2^\circ 45.0'$	$2^\circ 45.5'$	41	$10^\circ 15.0'$	$10^\circ 16.7'$
12	$3^\circ 00.0'$	$3^\circ 00.5'$	42	$10^\circ 30.0'$	$10^\circ 31.7'$
13	$3^\circ 15.0'$	$3^\circ 15.5'$	43	$10^\circ 45.0'$	$10^\circ 46.8'$
14	$3^\circ 30.0'$	$3^\circ 30.6'$	44	$11^\circ 00.0'$	$11^\circ 01.8'$
15	$3^\circ 45.0'$	$3^\circ 45.6'$	45	$11^\circ 15.0'$	$11^\circ 16.8'$
16	$4^\circ 00.0'$	$4^\circ 00.7'$	46	$11^\circ 30.0'$	$11^\circ 31.9'$
17	$4^\circ 15.0'$	$4^\circ 15.7'$	47	$11^\circ 45.0'$	$11^\circ 46.9'$
18	$4^\circ 30.0'$	$4^\circ 30.7'$	48	$12^\circ 00.0'$	$12^\circ 02.0'$
19	$4^\circ 45.0'$	$4^\circ 45.8'$	49	$12^\circ 15.0'$	$12^\circ 17.0'$
20	$5^\circ 00.0'$	$5^\circ 00.8'$	50	$12^\circ 30.0'$	$12^\circ 32.1'$
21	$5^\circ 15.0'$	$5^\circ 15.9'$	51	$12^\circ 45.0'$	$12^\circ 47.1'$
22	$5^\circ 30.0'$	$5^\circ 30.9'$	52	$13^\circ 00.0'$	$13^\circ 02.1'$
23	$5^\circ 45.0'$	$5^\circ 45.9'$	53	$13^\circ 15.0'$	$13^\circ 17.2'$
24	$6^\circ 00.0'$	$6^\circ 01.0'$	54	$13^\circ 30.0'$	$13^\circ 32.2'$
25	$6^\circ 15.0'$	$6^\circ 16.0'$	55	$13^\circ 45.0'$	$13^\circ 47.3'$
26	$6^\circ 30.0'$	$6^\circ 31.1'$	56	$14^\circ 00.0'$	$14^\circ 02.3'$
27	$6^\circ 45.0'$	$6^\circ 46.1'$	57	$14^\circ 15.0'$	$14^\circ 17.3'$
28	$7^\circ 00.0'$	$7^\circ 01.1'$	58	$14^\circ 30.0'$	$14^\circ 32.4'$
29	$7^\circ 15.0'$	$7^\circ 16.2'$	59	$14^\circ 45.0'$	$14^\circ 47.4'$

Provenance and verification

Every figure in this book was generated by `almanac/generate_tables.wls` from the package `wolfram/CelestialNavigation.wl` in the repository <https://github.com/mthiel74/SextantNavigation>, whose 186-test suite checks the ephemeris against JPL Horizons and Meeus. Independent sanity checks performed at generation time confirm: the Sun's declination reaches $+23.44^\circ$ at the June solstice and -23.44° at the December solstice; the Sun's GHA advances 15.00° per hour and Aries 15.041° per hour; a bright star's SHA is stable to under $1'$ across the year; and the three worked examples close on the engine's own geometry to $0.0'$. Re-generate with `wolframscript -file almanac/generate_tables.wls` and rebuild with `latexmk -pdf almanac/SextantAlmanac2026.tex`.

Remember the disclaimer: educational use only, approximately one arcminute accuracy, not for sole-means marine navigation.